Section VIII

Pharmacotherapy of Neoplastic Disease

Section Editor: Anton Wellstein

Chapter 69. General Principles in the Pharmacotherapy of Cancer / 1337
Chapter 70. Cytotoxics and Antimetabolites / 1343
Chapter 71. Protein Kinase Inhibitors and Pathway-Targeted Small Molecules / 1381
Chapter 72. Antibodies, CAR T Cells, and Proteins to Treat Cancer / 1415
Chapter 73. Hormones, Hormone Receptor Antagonists, and Related Agents in the Therapy of Cancer / 1435

This page intentionally left blank

Chapter 69

General Principles in the Pharmacotherapy of Cancer

Anton Wellstein

THE CELL CYCLE

CANCER EVOLUTION AND DRUG DISCOVERY

DRUG RESISTANCE

MOLECULAR TESTING TO SELECT APPROPRIATE DRUGS

- Molecular Analysis and Tumor Heterogeneity
- Liquid Biopsies

ACHIEVING THERAPEUTIC INTEGRATION AND EFFICACY

A CAUTIONARY NOTE

Cancer pharmacology has changed dramatically during the recent past, with the improved understanding of cancer biology and an ever-expanding set of new drugs that target vulnerabilities in specific cancers. Effective treatments had been developed earlier for some fatal malignancies, including testicular cancer, lymphomas, and leukemia. Adjuvant chemotherapy and hormonal therapy can extend overall survival and prevent disease recurrence following surgical resection of localized breast, colorectal, and lung cancers. Chemotherapy is also employed as part of the multimodal treatment of locally advanced head and neck, breast, lung, and esophageal cancers; soft-tissue sarcomas; and pediatric solid tumors, thereby allowing for neoadjuvant surgery that is more limited and results in favorable outcomes (Chabner and Roberts, 2005).

In the past 10 years, the ability to harness the power of the immune system in the treatment of cancer has brought about a paradigm shift whereby some of the most feared diseases, such as melanoma and lung cancer and even late-stage metastatic disease, can be eradicated. For some cancers, response rates are surprisingly high: 87% in Hodgkin lymphoma even in heavily pretreated patients (Ansell et al., 2015), and 50% in patients with metastatic melanoma treated with combinations of anti-PD-1 and -CTLA4 immune checkpoint antibodies. Immune checkpoint inhibitors are currently approved for the treatment of over a dozen different cancers that include melanoma and cancers of the bladder, kidneys, liver, lungs as well as Hodgkin lymphoma. In addition to these histologically defined malignancies, any cancers with deficient DNA mismatch repair (e.g., colorectal, ovarian, pancreatic, endometrial cancers) were also approved for treatment.

Despite these major therapeutic successes, few categories of medication have a narrower therapeutic index and greater potential for causing harmful effects than anticancer drugs. A thorough understanding of their mechanisms of action, including clinical pharmacokinetics, drug interactions, and adverse effects, is essential for their safe and effective use. Anticancer drugs are quite varied in structure and mechanism of action. The group includes alkylating agents; antimetabolite analogues of folic acid, pyrimidine, and purine; natural products; hormones and hormone antagonists; and a variety of small-molecule drugs and antibodies directed at specific molecular targets, such as extracellular receptors, intracellular kinases, or the checkpoints of immune surveillance. Figure 69–1 depicts the cellular targets of these classes of drugs, and Chapters 70 to 73 provide information about them.

Anticancer drugs are increasingly used in a variety of nonmalignant diseases and have become treatment standards, for example, for autoimmune diseases (*rituximab*); rheumatoid arthritis (*methotrexate* and *cyclophosphamide*); Crohn's disease (*6-mercaptopurine*); organ transplantation (*methotrexate* and *azathioprine*); sickle cell anemia (*hydroxyurea*); psoriasis (*methotrexate*); and wet macular degeneration (*ranibizumab* and *aflibercept*).

The Cell Cycle

An understanding of the cell cycle is essential for the rational use of cytotoxic anticancer drugs that target proliferating cells (Figure 69–2). Many cytotoxic agents act by damaging DNA. Their efficacy is thus greatest during S phase, the DNA synthetic phase of the cell cycle. Other agents, such as the vinca alkaloids and taxanes, block the formation of a functional mitotic spindle in the M phase. These agents are most effective on cells entering mitosis, the most vulnerable phase of the cell cycle. Accordingly, human cancers most susceptible to chemotherapy are those having a high percentage of proliferating cells. However, normal tissues that proliferate rapidly (bone marrow, hair follicles, and intestinal epithelium) are thus also susceptible to damage from cytotoxic drugs. In addition, slowly growing tumors with a small growth fraction (e.g., carcinomas of the colon or NSCLC) are less responsive to cell cycle–specific drugs.

Although cells from different tumors display differences in the duration of their transit through the cell cycle and in the fraction of cells in active proliferation, all cells display a similar pattern of cell cycle progression (Figure 69–2):

- A phase that precedes DNA synthesis (G_1)
- A DNA synthesis phase (S)
- An interval following the termination of DNA synthesis (G_2)
- The mitotic phase (M) in which the cell, containing a double complement of DNA, divides into two daughter G_1 cells
- A probability of moving into a quiescent state (G_0) for long periods of time

Some anticancer drugs act at specific phases in the cell cycle, mainly at the S and M phases; other drugs are cytotoxic at any point in the cell cycle and are termed *cell cycle phase nonspecific*.

Each transition point in the cell cycle requires the activation of specific cyclin-dependent kinases (*CDKs*), which, in their active forms, couple with corresponding regulatory proteins called *cyclins*. The proliferative impact of CDKs is, in turn, dampened by inhibitory proteins such as $p16^{INK4A}$, a tumor suppressor named for its molecular mass (*p*rotein of *16* kDa) and *in*hibition of CD*K4*. Tumor cells often exhibit changes in cell cycle regulation that lead to relentless proliferation (e.g., mutations or loss of $p16^{INK4A}$ or of other inhibitory components of the so-called retinoblastoma pathway, enhanced cyclin, or enhanced CDK activity).

The CDK family consists of over 20 serine/threonine protein kinases that have been among the first pathway targets pursued for the treatment of cancer. However, different tissue selectivities and distinct cell cycle–specific activity periods of the various CDKs provide a challenge for the development of CDK inhibitors. CDK4/6 have become attractive targets because they control cell cycle progression from the G_1 to the

Abbreviations

ABL: Abelson murine leukemia viral oncogene homolog
ALK: anaplastic lymphoma kinase
BCR: breakpoint cluster region
BRAF: B-Raf proto-oncogene ser/thr protein kinase
CAR(T): chimeric antigen receptor T cell
CDK: cyclin-dependent kinase
ctDNA: circulating, cell-free mutant tumor DNA
CTLA4: cytotoxic T lymphocyte–associated protein 4
EGF(R): epidermal growth factor (receptor) [HER1, ErbB-1]
ER: estrogen receptor
FDA: Food and Drug Administration
GI: gastrointestinal
HER1: human EGFR (ErbB-1)
HER2: human EGFR 2 (ErbB-2)
MEK: mitogen-activated protein kinase kinase
MHC: major histocompatibility class (protein)
NCCN: National Comprehensive Cancer Network
NSCLC: non–small cell lung cancer
PARP: poly(ADP-ribose) polymerase
PD-1: programmed cell death 1
PD-L1: programmed cell death ligand 1
SARS-CoV-2: severe acute respiratory syndrome–coronavirus 2
TMB: tumor mutational burden

S phase. Interaction of cyclin D with CDK4/6 enhances phosphorylation and inactivation of the retinoblastoma (Rb) protein, followed by the transcription of factors that control transition into the S phase. CDK4/6 inhibition will thus cause a G_1 arrest in susceptible cells that utilize this pathway. CDK4/6 inhibitors were recently approved for the treatment of breast cancer (see Chapter 71).

Because of the central importance of DNA to the identity and functionality of a cell, elaborate cellular mechanisms ("cell cycle checkpoints") have evolved to monitor DNA integrity. At each transition point in the cell cycle, specific proteins, such as p53 and chk-1 and-2, monitor the integrity of DNA and, upon detection of DNA damage, may initiate DNA repair processes, or in the presence of massive damage, direct cells down a pathway of cell death (apoptosis). If a cell possesses normal checkpoint function, drug-induced DNA damage will activate apoptosis when the cell reaches the G_1/S or G_2/M boundary. However, if the p53 gene product or other checkpoint proteins are mutated or absent or the checkpoint function fails, damaged cells will not divert to the apoptotic pathway but will proceed through the S phase and mitosis. The cell progeny will then emerge as a mutated and potentially drug-resistant subpopulation (see Figure 69–3A).

Cancer Evolution and Drug Discovery

The rapidly expanding knowledge of cancer biology and the ability to analyze cancer genome alterations in thousands of patient samples have led to a better understanding of the molecular evolution of cancer and the discovery of cancer-specific drug targets: growth factor receptors, intracellular signaling pathways, epigenetic processes, tumor vascularity,

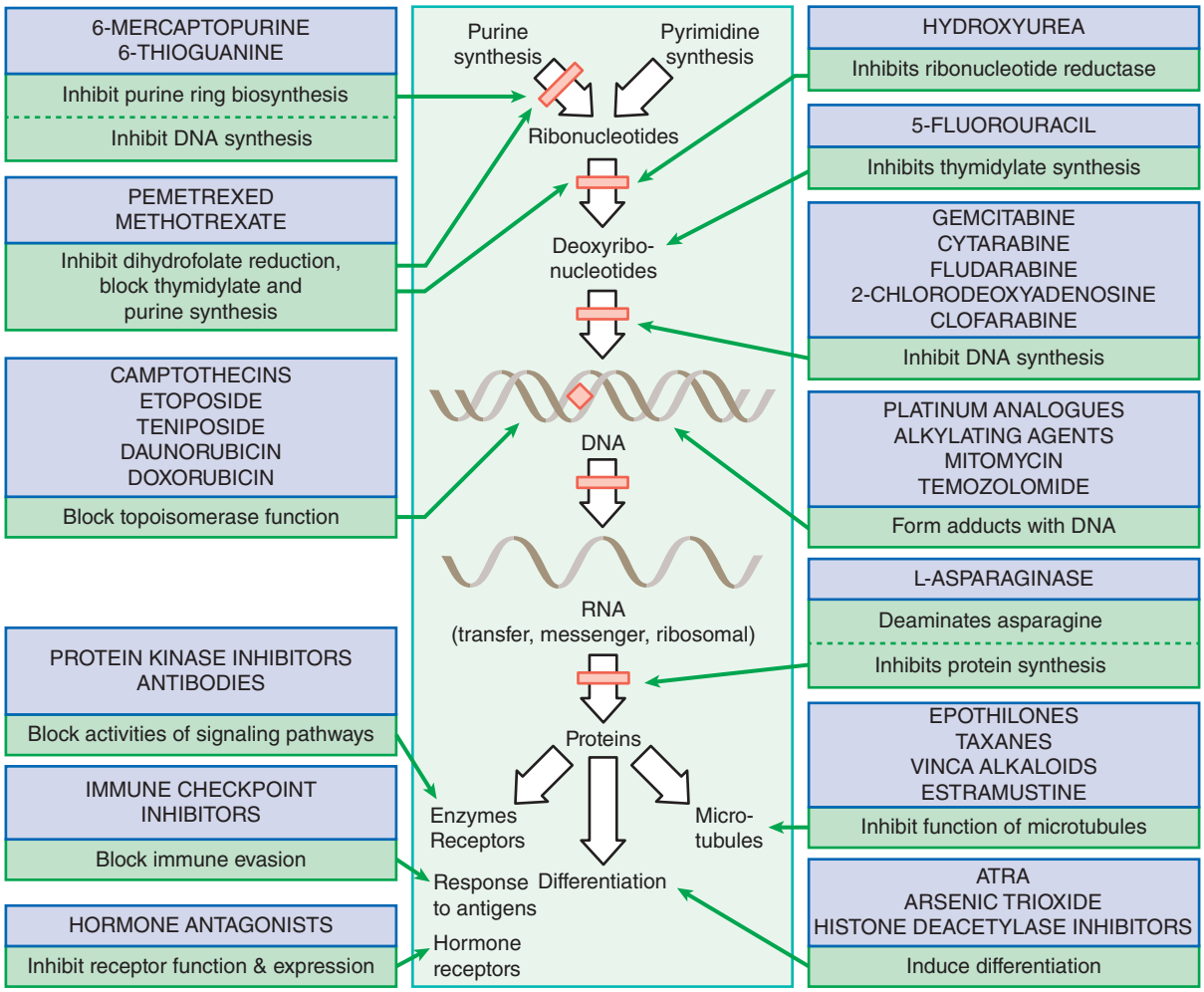


Figure 69–1 *Mechanisms and sites of action of some of the drugs used in the treatment of cancer.*

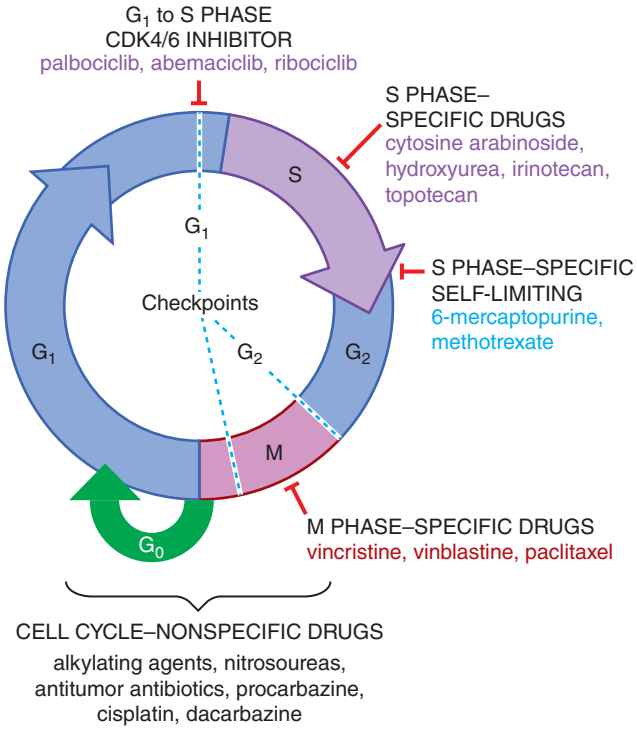


Figure 69–2 *Cell cycle specificity of drugs used in the treatment of cancer.*

DNA repair defects, cell death pathways, and immune escape mechanisms (Hanahan and Weinberg, 2011). Human malignancies are a highly diverse group of diseases that vary even within defined classifications such as organ of origin (lung, breast, prostate, colon, etc.), histology, or molecular marker. Also, the tumor cell population constituting a given cancer at the time of diagnosis has evolved over many years from a few precursor cells that accumulated mutations over time, generating heterogeneity within the primary tumor and at metastatic sites (Figure 69–3).

Clinically detectable cancer lesions represent approximately 1 g of tumor tissue or 10^9 cells and can contain a multitude of subpopulations and a wide variety of genetic alterations (see legend of Figure 69–3). The dynamic evolution of individual cancer genomes and its implications for the development of therapies were established from the analyses of specimens from diverse cancers (Yates and Campbell, 2012). This dynamic was exemplified in a detailed analysis of a series of multiple parallel biopsies from different sites in patients with metastatic melanoma during treatment with inhibitors of mutant BRAF. The genomic analysis of the biopsies revealed complex and distinct evolutionary branching architectures due to the selection of drug-resistant subpopulations during treatment (Shi et al., 2014).

Nevertheless, in many tumors, proliferation and survival of the majority of subpopulations depend on a shared (ancestral) constitutive activity of a single growth factor pathway, or so-called oncogene addiction. Inhibition of that pathway leads to cell death of the sensitive populations. Thus, *imatinib* attacks the unique and specific *bcr-abl* translocation in chronic myelocytic leukemia. *Imatinib* also inhibits *c-kit* and produces extended control of GI stromal tumors that express a mutated and constitutively activated form of *c-kit*. Monoclonal antibodies effectively target tumor-associated antigens such as the amplified *HER2* receptor in breast cancer cells (Slamon et al., 2001). Protein kinase inhibitors targeting mutant EGFR or mutant ALK in lung cancers improve disease outcomes over the use of conventional chemotherapy.

These examples emphasize that new strategies for drug discovery and development, and advances in patient care, will result from new knowledge of cancer biology. One response to the oncogene addiction paradigm has been to group cancers by shared vulnerabilities, include patients in so-called basket trials that evaluate a drug based on its target rather than particular disease entities, and consider sensitivity and resistance to treatments in that context.

However, in the foreseeable future, targeted drugs and cytotoxics will continue to be used in combination. For instance, cytotoxics in combination with monoclonal antibodies such as *trastuzumab* or *bevacizumab* improve efficacy. At the same time, the toxicities of cytotoxic drugs have become more manageable with the development of better antinausea

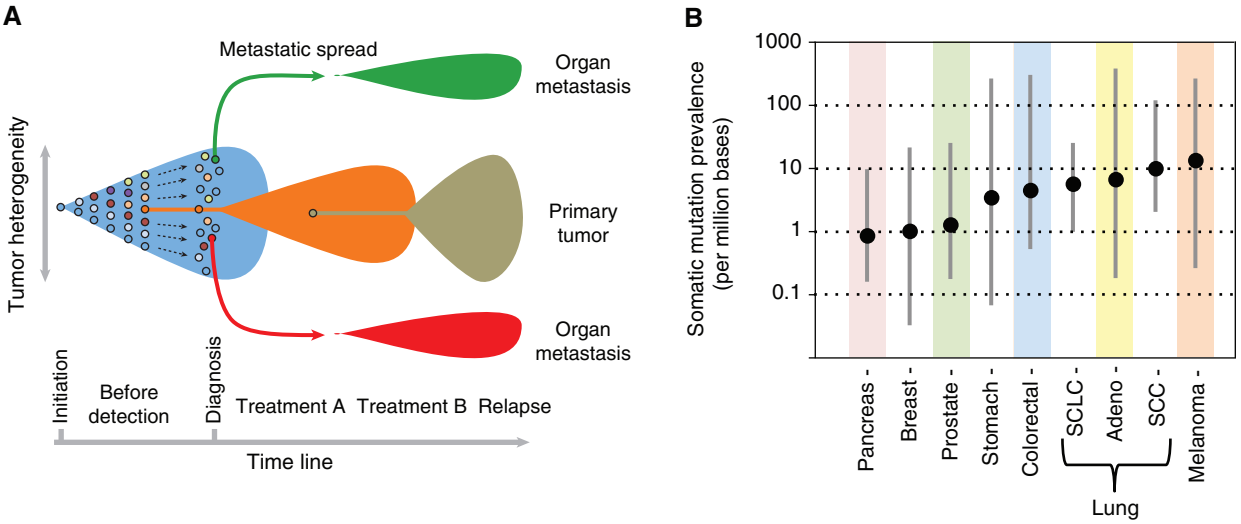


Figure 69–3 *Evolution of treatment resistance; mutational load of human cancers.* **A.** *Treatment resistance.* Cancers accumulate mutations during their evolution. Cancer cell subpopulations are selected based on their growth capacity, adaptation to the tumor microenvironment at the primary or metastatic site, and the evasion of immune surveillance. Drug treatment adds evolutionary pressure and selects for resistant subpopulations. Differently colored dots indicate tumor subpopulations of different genetic or epigenetic makeup. **B.** *Mutational load.* Data are median values ± range of numbers of somatic mutations per million bases observed for some major cancers. Note that the ordinate is a logarithmic scale. In 7042 cancer specimens, between 100 and 1,000,000 mutations were detected per tumor specimen, with a 30- to 1000-fold range among individual specimens from a single cancer type (see original data in Alexandrov et al., 2013). A tumor mutational burden (TMB) of 10 somatic mutations per megabase (= 30,000 mutations in the human genome of 3×10^9 base pairs) results in approximately 150 mutations in amino acid sequences that can alter protein function, drug sensitivity, and antigenicity. In many cancer types, higher TMB is associated with poorer survival. On the other hand, the formation of tumor-specific neoantigens due to DNA mutations permits the immune system to distinguish between tumor and normal cells and contributes to the efficacy of cancer immunotherapy (Schumacher and Schreiber, 2015). Thus, in patients treated with immune checkpoint inhibitors, higher TMB is associated with longer survival. Adeno, adenocarcinoma; SCC, squamous cell carcinoma; SCLC, small cell lung cancer.

medications (see Chapter 54) and with granulocyte colony-stimulating factor to restore bone marrow function (see Chapters 45 and 71).

Finally, targeted drugs are helping to overcome resistance to chemotherapeutic agents by normalizing blood flow, promoting apoptosis, and inhibiting prosurvival signals from growth factor pathways. Tumor angiogenesis leads to increased interstitial pressure and diminishes delivery of drugs to tumor cells; inhibitors of angiogenesis (e.g., *bevacizumab*) normalize blood flow and interstitial pressure, improve drug delivery, and can thus synergize with cytotoxic drugs in the treatment of lung, colon, and other cancers. It is also thought that the combination of cytotoxic drugs or pathway inhibitors can induce tumor cell death and antigen release and thus enhance responses to immune checkpoint inhibitors or other immune modulators. This concept is part of a recommendation for the treatment of patients with melanoma (Kaufman et al., 2013) and should be relevant for a range of cancers (Sharma and Allison, 2015). The ongoing development of activating and inhibitory drugs for additional immune checkpoint pathways (Anderson et al., 2016) will provide new options for drug combinations.

Drug Resistance

Resistance remains the major obstacle to successful cancer treatment. Resistance results from a variety of molecular changes acquired during evolution of a given cancer that can defeat the best-designed treatments. Mechanisms of drug resistance include poor drug absorption and delivery; genetically determined variability in drug transport, activation, and clearance; and mutations, amplifications, or deletions in drug targets (Holohan et al., 2013). The processes of resistance are best understood for pathway-targeted drugs. Tumors developing resistance to *bcr-abl* inhibitors and to inhibitors of the EGFR typically express mutations in the target enzyme. Cells exhibiting drug-resistant mutations preexist in a patient's tumor prior to drug treatment and are selected by drug exposure (see Figure 69–3A). Resistance to inhibitors of the EGFR may also develop through expression of an alternative receptor, *c-met*, which bypasses EGFR blockade and stimulates proliferation (Engelman et al., 2007). Defects in recognition of DNA breaks and overexpression of specific repair enzymes may contribute to resistance to cytotoxic drugs, and a loss of apoptotic pathways can lead to resistance to both cytotoxic and pathway-targeted drugs.

Resistance to immune checkpoint inhibitory drugs appears to follow patterns that are distinct from those of other anticancer drugs, as evidenced by their efficacy in some heavily pretreated patients. One recognized tumor variable of treatment sensitivity is the >30-fold range of the tumor mutational burden (TMB) among patient specimens from a given cancer type (Figure 69–3); TMB is thus used as one marker to assess the likelihood of response or resistance to immunotherapy (Table 69–1). Surprisingly, the composition of the gut microbiome as well as alterations by antibiotic treatment were found to be related to the response of different cancers to immune checkpoint inhibitory drugs, suggesting a potential vulnerability (Finlay et al., 2020; see also Chapter 6). Indeed, recent studies in patients with melanoma suggest that fecal microbiota transplant from appropriate donors may overcome resistance to immunotherapy (Woelk and Snyder, 2021). A major challenge in immunotherapy is the

TABLE 69–1 ■ DIAGNOSTIC TESTS OF CANCER SPECIMENS TO GUIDE TREATMENT DECISIONS

INDIVIDUAL MOLECULAR MARKER (*DNA, mRNA,* PROTEIN)	TARGET: drugs	CANCER INDICATION	CHAPTER
ALK translocation	ALK: alectinib, ceritinib, crizotinib	Lung NSCLC	71
BRAF V600 mutation	BRAF: dabrafenib, vemurafenib	Melanoma	71
BRAF V600 mutation	MEK: trametinib	Melanoma	71
BRCA mutation	PARP: olaparib, rucaparib, talazoparib	Breast, Ovarian, Pancreatic, Prostate	71
EGFR deletion of exon 19 or L858R mutation[a]	EGFR: afatinib, dacomitinib, erlotinib, gefitinib	Lung NSCLC	71
EGFR T790M mutation[a]	EGFR: osimertinib	Lung NSCLC	71
Estrogen receptor (ER) expression	antiestrogens (tamoxifen, raloxifene, fulvestrant) or aromatase inhibitors (anastrozole, letrozole, exemestane)	Breast	73
HER2 amplification; HER2 overexpression	HER2: trastuzumab, ado-trastuzumab emtansine, pertuzumab	Breast, Gastric	71
KRAS wild type	EGFR: cetuximab, panitumumab	Colorectal	71
PD-L1 expression	PD-1, PD-L1: pembrolizumab	Lung NSCLC, etc.	72
Panels of markers			
MSI-H (microsatellite instability high) or dMMR (deficient mismatch repair)	PD-1: pembrolizumab	Solid tumors	72
Tumor mutational burden (TMB) >10 DNA mutations per million bases[b]	PD-1: pembrolizumab	Solid tumors	72
mRNA expression of a panel of genes (Risk of recurrence score)	Chemotherapy	Breast	73
Next Generation DNA sequencing of tumor samples for the presence of gene variants (substitution, insertion, deletion), copy number, rearrangement	Genes sequenced include the above-named targets and their alterations. The respective drugs are listed above	Solid tumors	71

FDA-approved tests continuously updated: www.fda.gov "List of cleared or approved companion diagnostic devices." (Accessed 19 March 2022)
[a]Also detectable as circulating tumor DNA (ctDNA) in blood samples.
[b]See Figure 69–3.

poor predictability, in individual patients, of the immune responses that govern the success of the treatment. mRNA-based vaccines that contain the coding sequences of antigenic mutations detected in a given tumor are the most recently developed approach to elicit an immune response and enhance the efficacy of immune checkpoint inhibitor treatments (Sahin and Türeci, 2018; Sahin et al., 2020). It is noteworthy that this mRNA-based vaccine approach was also used to generate the first successful vaccines against SARS-CoV-2 (COVID-19).

Finally, T cells carrying chimeric antigen receptors (CARs) can be directed against cancer cells that express specific antigens. CARs are engineered to contain an antigen recognition domain of a monoclonal antibody in the extracellular portion and intracellular signaling domains capable of activating T cells independently of the physiological pathway of antigen presentation by an MHC molecule (see Figures 72–6, 38–2, and 38–4). CD19-targeted CAR T cells reached a 70% to 90% response rate in patients with previously treated, relapsed B-cell leukemias (Khalil et al., 2016), indicating a lack of cross-resistance with conventional therapies. This result is consistent with the requirement for effective combination therapies, that is, complementary mechanisms of action and no overlap in major toxicities. Combinations of immune modulating therapies with pathway-targeted drugs and cytotoxic agents are currently evaluated in national and international trials to generate effective treatment combinations (Hughes et al., 2016).

Molecular Testing to Select Appropriate Drugs

Clinical trials and patient treatments increasingly employ results from biomarker analysis to identify patients likely to benefit from particular treatments and individuals at the greatest risk of toxicity. Some of the tests have been FDA-approved as "companion diagnostics" in conjunction with specific drug therapies (see Table 69–1). Pretreatment testing of tumor specimens is standard practice in selecting patients for antihormonal therapy of breast cancer and for treatment with antibodies such as *trastuzumab* (anti-HER2). Detection of a mutated *KRAS* gene indicates that the tumor of a patient with colorectal cancer will not respond to anti-EGFR antibodies; in patients with lung cancers and EGFR mutations, treatment with *erlotinib*, *gefitinib*, or *afatinib* results in response rates of 70%, and in patients with ALK translocations, the response rates are similar for treatment with the ALK inhibitors *crizotinib* and *ceritinib*. A T790M "gatekeeper" mutation in EGFR (Kobayashi et al., 2005) accounts for about 60% of acquired resistance to first- and second-generation inhibitors but is sensitive to *osimertinib*, a third-generation EGFR inhibitor (Thomas et al., 2015). Overall, introduction of molecular analysis and appropriate choice of pathway-targeted inhibitors in the treatment of NSCLC has increased median survival of patients from less than 1 year to over 3 years and the response rate from 30% to 80% (Ke and Wu, 2016).

Inherited differences in protein sequence polymorphisms or levels of RNA expression can also influence toxicity and antitumor response. For example, tandem repeats in the promoter region of the gene encoding thymidylate synthase, the target of *5-fluorouracil*, determine the level of expression of the enzyme. Increased numbers of repeats are associated with increased gene expression, a lower incidence of toxicity, and a decreased rate of response in patients with colorectal cancer (Pullarkat et al., 2001). Polymorphisms of the dihydropyrimidine dehydrogenase gene, the product of which is responsible for degradation of *5-fluorouracil*, are associated with decreased enzyme activity and a significant risk of overwhelming drug toxicity, particularly in the rare individual that is homozygous for the polymorphic genes (Van Kuilenburg et al., 2002).

Gene expression profiling, in which the levels of messenger RNA from thousands of genes are surveyed using gene arrays, has revealed tumor profiles that are highly associated with poor outcomes and that warrant adjuvant chemotherapy (Sotiriou and Pusztai, 2009). As an alternative to this broad analysis, small sets of informative genes can be identified and used clinically. One example is a set of 21 genes used in the analysis of samples from patients with early-stage breast cancer. Based on the known association between the expression pattern of the 21 genes and disease outcomes, the analysis of patient samples can predict the risk of disease relapse. Thus, one can identify the patients at a high risk who will benefit from adjuvant chemotherapy (Paik et al., 2004).

Molecular Analysis and Tumor Heterogeneity

One of the caveats to conclusions drawn from the molecular analysis of tumor tissue specimens is the dynamic evolution of cancers noted above (see Figure 69–3). Clinically important mutations in subclones may be missed due to geographically inadequate sampling and may provide the wrong guidance to treatment decisions. Treatment responses of different subpopulations in a tumor or in different metastatic lesions present a further challenge and would require multiple-site tissue biopsies (Shi et al., 2014). The molecular analysis of serially collected blood samples ("liquid biopsies") provides an alternative approach for treatment monitoring.

Liquid Biopsies

More recent technology advances have made it possible to sequence and quantitate circulating, tumor-derived DNA (ctDNA) in blood samples from patients with cancer ("liquid biopsies"), as first demonstrated for the changing abundance of mutant *KRAS* during colon cancer treatment (Diehl et al., 2008). The analysis of ctDNA has shown that during antiestrogen therapy of breast cancer the appearance of mutant estrogen receptor coincides with subsequent resistance to aromatase inhibitor treatment (Schiavon et al., 2015). Furthermore, mutant *KRAS* DNA in the circulation was increased during the treatment of patients with colon cancer with EGFR antibodies but surprisingly reverted to baseline values after cessation of the treatment. This observation demonstrates the dynamic evolution of cancer subpopulations during drug treatment, as indicated in Figure 69–3 (Siravegna et al., 2015). As a consequence of these technological developments, the FDA approved a test for the presence of mutant *EGFR* DNA in blood samples of patients with NSCLC to select candidates for treatment with *erlotinib* or *osimertinib* and thus circumvent the need for a tissue biopsy. The incorporation of liquid biopsies into treatment monitoring can provide additional molecular insights into drug efficacy and adverse effects and reveal the onset of resistance to a chosen treatment (Kilgour et al., 2020).

Achieving Therapeutic Integration and Efficacy

The clinical benefit of cytotoxic drugs has primarily been measured by radiological assessment of drug effects on tumor size. Pathway-targeted agents, however, may only slow or halt tumor growth, so their effects may be measured in the assessment of time to disease progression; however, for some immune checkpoint inhibitors, tumor lesions may initially increase in size due to cytotoxic lymphocyte infiltration, so called pseudo-progression. Thus, one of the great challenges is to assess efficacy and adjust drug regimens to achieve a therapeutic but nontoxic outcome. Treatment of patients with cancer requires a skillful interdigitation of pharmacotherapy with other modalities of treatment (e.g., surgery and radiation). Each treatment modality carries its own risks and benefits, with the potential for both antagonistic and synergistic interactions between modalities, particularly between drug treatment and radiation.

Individual patient characteristics determine the choice of modalities. Not all patients can tolerate drugs of the primary choice, and not all drug regimens are appropriate for a given patient. Renal and hepatic function, bone marrow reserve, general performance status, and concurrent medical problems all come into consideration in making a therapeutic plan. Other less-quantifiable considerations, such as the natural history of the tumor, the patient's willingness to undergo difficult and potentially dangerous treatments, and the patient's physical and emotional tolerance for adverse effects enter the equation, with the goal of balancing the likely long-term gains and risks in the individual patient. In particular, the long-term adverse effects of cytotoxic drugs have been related to the induction of cellular senescence in different organs that resemble symptoms of premature aging and can adversely affect organ function and the overall well-being of patients long after completion of the treatments (Childs et al., 2015; Couzin-Frankel, 2019). The choice of treatment regimen should

take all of this into account. Finally, in terminally ill patients, the treatment choices must be weighed carefully; the maximal length and highest quality of life may be achieved with palliative care and may reduce the need for chemotherapy (Temel et al., 2010).

A Cautionary Note

Although advances in drug discovery and molecular profiling of tumors offer great promise for improving the outcomes of cancer treatment, a final word of caution regarding all treatment regimen deserves emphasis: *The pharmacokinetics and toxicities of cancer drugs vary amongst individual patients.* It is imperative to *recognize toxicities early*, to *alter doses or discontinue offending medication* to relieve symptoms and reduce risk, and to *provide vigorous supportive care*. Toxicities affecting the heart, lungs, nervous system, or kidneys may be irreversible if recognized late in their course, leading to permanent organ damage or death. Fortunately, such toxicities can be minimized by early recognition and by adherence to standardized protocols and to the guidelines for each drug's use.

A Note on Treatment Regimens

Cancer treatment regimens change to reflect continuous advances in basic and clinical science: new drugs, both small molecules and biologicals; improved methods of targeting and timing of drug delivery; agents with altered pharmacokinetic properties and selectivities; the use of rational multidrug combinations; and greater knowledge of the basic cell biology of tumorigenesis, metastasis, and immune function, among other advances. As a consequence, this chapter and the four that follow present relatively few detailed treatment regimens; rather, we refer the reader to the web-based resources of the U.S. FDA and the National Comprehensive Cancer Network (NCCN). Table 71–1 provides examples that illustrate the complexities of current therapeutic regimens for two cancers.

Acknowledgment: *Paul Calabresi and Bruce A. Chabner contributed to this chapter in earlier editions of this book. We have retained some of their text in the current edition.*

Bibliography

Alexandrov LB, et al. Signatures of mutational processes in human cancer. *Nature*, **2013**, *500*:415–421.

Anderson AC, et al. Lag-3, Tim-3, and TIGIT: co-inhibitory receptors with specialized functions in immune regulation. *Immunity*, **2016**, *44*:989–1004.

Ansell SM, et al. PD-1 blockade with nivolumab in relapsed or refractory Hodgkin's lymphoma. *N Engl J Med*, **2015**, *372*:311–319.

Chabner BA, Roberts TG. Timeline: chemotherapy and the war on cancer. *Nat Rev Cancer*, **2005**, *5*:69–72.

Childs BG, et al. Cellular senescence in aging and age-related disease: from mechanisms to therapy. *Nat Med*, **2015**, *21*:1424–1435.

Couzin-Frankel J. Beyond survival. *Science*, **2019**, *363*:1166–1169.

Diehl F, et al. Circulating mutant DNA to assess tumor dynamics. *Nat Med*, **2008**, *14*:985–990.

Engelman JA, et al. MET amplification leads to gefitinib resistance in lung cancer by activating ERBB3 signaling. *Science*, **2007**, *316*:1039–1043.

Finlay BB, et al. Can we harness the microbiota to enhance the efficacy of cancer immunotherapy? *Nat Rev Immunol*, **2020**, *20*:522–528.

Hanahan D, Weinberg RA. Hallmarks of cancer: the next generation. *Cell*, **2011**, *144*:646–674.

Holohan C, et al. Cancer drug resistance: an evolving paradigm. *Nat Rev Cancer*, **2013**, *13*:714–726.

Hughes PE, et al. Targeted therapy and checkpoint immunotherapy combinations for the treatment of cancer. *Trends Immunol*, **2016**, *37*:462–476.

Kaufman HL, et al. The Society for Immunotherapy of Cancer consensus statement on tumour immunotherapy for the treatment of cutaneous melanoma. *Nat Rev Clin Oncol*, **2013**, *10*:588–598.

Ke EE, Wu YL. EGFR as a pharmacological target in EGFR-mutant non-small-cell lung cancer: where do we stand now? *Trends Pharmacol Sci*, **2016**, *37*:887–903.

Khalil DN, et al. The future of cancer treatment: immunomodulation, CARs and combination immunotherapy. *Nat Rev Clin Oncol*, **2016**, *13*:273–290.

Kilgour E, et al. Liquid biopsy-based biomarkers of treatment response and resistance. *Cancer Cell*, **2020**, *37*:485–495.

Kobayashi S, et al. EGFR mutation and resistance of non-small-cell lung cancer to gefitinib. *N Engl J Med*, **2005**, *352*:786–792.

Paik S, et al. A multigene assay to predict recurrence of tamoxifen-treated, node-negative breast cancer. *N Engl J Med*, **2004**, *351*:2817–2826.

Pullarkat ST, et al. Thymidylate synthase gene polymorphism determines response and toxicity of 5-FU chemotherapy. *Pharmacogenomics J*, **2001**, *1*:69–70.

Sahin U, et al. An RNA vaccine drives immunity in checkpoint-inhibitor-treated melanoma. *Nature*, **2020**, *585*:107–112.

Sahin U, Türeci Ö. Personalized vaccines for cancer immunotherapy. *Science*, **2018**, *359*:1355–1360.

Schiavon G, et al. Analysis of ESR1 mutation in circulating tumor DNA demonstrates evolution during therapy for metastatic breast cancer. *Sci Transl Med*, **2015**, *7*:313ra182.

Schumacher TN, Schreiber RD. Neoantigens in cancer immunotherapy. *Science (New York, NY)*, **2015**, *348*:69–74.

Sharma P, Allison JP. The future of immune checkpoint therapy. *Science (New York, NY)*, **2015**, *348*:56–61.

Shi H, et al. Acquired resistance and clonal evolution in melanoma during BRAF inhibitor therapy. *Cancer Discov*, **2014**, *4*:80–93.

Siravegna G, et al. Clonal evolution and resistance to EGFR blockade in the blood of colorectal cancer patients. *Nat Med*, **2015**, *21*:795–801.

Slamon DJ, et al. Use of chemotherapy plus a monoclonal antibody against HER2 for metastatic breast cancer that overexpresses HER2. *N Engl J Med*, **2001**, *344*:783–792.

Sotiriou C, Pusztai L. Gene-expression signatures in breast cancer. *N Engl J Med*, **2009**, *360*:790–800.

Temel JS, et al. Early palliative care for patients with metastatic non-small-cell lung cancer. *N Engl J Med*, **2010**, *363*:733–742.

Thomas A, et al. Refining the treatment of NSCLC according to histological and molecular subtypes. *Nat Rev Clin Oncol*, **2015**, *12*:511–526.

Van Kuilenburg ABP, et al. High prevalence of the IVS14 + 1G>A mutation in the dihydropyrimidine dehydrogenase gene of patients with severe 5-fluorouracil-associated toxicity. *Pharmacogenetics*, **2002**, *12*:555–558.

Woelk CH, Snyder A. Modulating gut microbiota to treat cancer. *Science*, **2021**, *371*:573–574.

Yates LR, Campbell PJ. Evolution of the cancer genome. *Nat Rev Genet*, **2012**, *13*:795–806.

Chapter 70

Cytotoxics and Antimetabolites

Anton Wellstein and Edward A. Sausville

I. ALKYLATING AGENTS AND PLATINUM COORDINATION COMPLEXES

ACTIONS COMMON TO ALKYLATING DRUGS
- Structure-Activity Relationships
- General Pharmacological Actions
- Mechanisms of Resistance to Alkylating Drugs
- Adverse Effects of Alkylating Drugs

CLINICAL PHARMACOLOGY OF NITROGEN MUSTARDS
- Mechlorethamine
- Cyclophosphamide
- Ifosfamide
- Melphalan
- Chlorambucil
- Bendamustine

ETHYLENEIMINES AND METHYLMELAMINES
- Altretamine
- Thiotepa

ALKYL SULFONATES
- Busulfan

NITROSOUREAS
- Carmustine (BCNU) and Lomustine (CCNU)
- Streptozocin

TRIAZENES
- Dacarbazine (DTIC)
- Temozolomide

METHYLHYDRAZINES
- Procarbazine

PLATINUM COORDINATION COMPLEXES
- Mechanisms of Action
- Resistance to Platinum Analogues
- Cisplatin
- Carboplatin
- Oxaliplatin

II. ANTIMETABOLITES

FOLIC ACID ANALOGUES
- Mechanism of Action
- Selective Toxicity; Rescue
- Cellular Entry and Retention
- Newer Congeners
- Mechanisms of Resistance to Antifolates
- Therapeutic Uses
- Adverse Effects

PYRIMIDINE ANALOGUES
- Cellular Actions of Pyrimidine Antimetabolites
- Fluorouracil, Floxuridine, Capecitabine
- Trifluridine

CYTIDINE ANALOGUES
- Cytarabine (Cytosine Arabinoside; Ara-C)
- Azacitidine (5-Azacytidine); Decitabine
- Gemcitabine

PURINE ANALOGUES
- 6-Thiopurine Analogues
- Fludarabine Phosphate
- Cladribine
- Clofarabine (2-Chloro-2′-fluoro-arabinosyladenine)
- Nelarabine (6-Methoxy-arabinosyl-guanine)
- Pentostatin (2′-Deoxycoformycin)

III. NATURAL PRODUCTS

MICROTUBULE-DAMAGING AGENTS
- Vinca Alkaloids
- Eribulin
- Taxanes
- Estramustine
- Epothilones

CAMPTOTHECIN ANALOGUES
- Chemistry
- Mechanism of Action
- Mechanisms of Resistance
- Topotecan
- Irinotecan

ANTIBIOTICS
- Dactinomycin (Actinomycin D)
- Anthracyclines and Anthracenediones

EPIPODOPHYLLOTOXINS
- Podophyllotoxin Derivatives
- Etoposide
- Teniposide

TRABECTEDIN ANALOGUES
- Trabectedin
- Lurbinectedin

DRUGS OF DIVERSE MECHANISMS OF ACTION
- Bleomycin
- Mitomycin
- Mitotane
- L-Asparaginase
- Hydroxyurea
- Retinoids
- Arsenic Trioxide

Abbreviations

ABVD: ***a***driamycin (doxorubicin), ***b***leomycin, ***v***inblastine, ***d***acarbazine
ADA: adenosine deaminase
ALL: acute lymphoblastic leukemia
AML: acute myeloid leukemia; acute myelocytic leukemia
APL: acute promyelocytic leukemia
Ara-C: cytarabine, cytosine arabinoside
Ara-U: ara-uridine
L-ASP: L-asparaginase
ATO: arsenic trioxide
ATRA: all-*trans* retinoic acid
AUC: area under the curve
BCNU: carmustine [1,3-*bis*-(2-chloroethyl)-1-nitrosourea]
BCRP: breast cancer resistance protein
CCNU: 1-(2-chloroethyl)-3-cyclohexyl-1-nitrosourea (lomustine)
CDK: cyclin-dependent kinase
CHOP: *c*yclophosphamide, doxorubicin (***H***), vincristine (***O***), and *p*rednisone
CL_{Cr}: creatinine clearance
CLL: chronic lymphocytic leukemia
CML: chronic myelocytic leukemia; chronic myelogenous leukemia
CSF: cerebrospinal fluid
CTCL: cutaneous T-cell lymphoma
dCK: deoxycytidine kinase
dFdC: 2′,2′-difluorodeoxycytidine, gemcitabine
5′-dFdU: 5′-deoxyfluorodeoxyuridine
DHFR: dihydrofolate reductase
DPD: dihydropyrimidine dehydrogenase
FdUMP: fluorodeoxyuridine monophosphate
FOLFIRINOX: FOLFOX plus irinotecan
FOLFOX: *fol*inic acid, 5*F*U and *ox*aliplatin
5FU: 5-fluorouracil
FUdR: fluorodeoxyuridine or floxuridine or 5-FUdR
GART: glycinamide ribonucleotide transformylase
G-CSF: granulocyte colony-stimulating factor
HbF: hemoglobin (fetal)
HDM-L: high-dose MTX with leucovorin rescue
HGPRT: hypoxanthine guanine phosphoribosyl transferase
HMG: high-mobility group
HRD: homologous recombination DNA repair
HU: hydroxyurea
IMP: inosine monophosphate
MESNA: 2-***m***ercapto***e***thane***s***ulfonate-***Na***$^+$
MGMT: ***O***^{6}-alkyl, **m**ethyl **g**uanine **m**ethyl**tr**ansferase
MMR: mismatch repair
MOPP: nitrogen ***m***ustard, ***o***ncovin (vincristine), ***p***rocarbazine, and ***p***rednisone
6MP: 6-mercaptopurine
MRP: multidrug resistance-associated protein
MTIC: methyl-triazeno-imidazole-carboxamide
MTX: methotrexate
Nab-paclitaxel: albumin-bound nanoparticle solution of paclitaxel
NER: nucleotide excision repair
PG: polyglutamate
Pgp: P-glycoprotein
PML: promyelocytic leukemia
PRPP: 5-phosphoribosyl-1-pyrophosphate
RAR: retinoic acid receptor
RNR: ribonucleoside diphosphate reductase
ROS: reactive oxygen species
TEM: triethylenemelamine
TEPA: triethylenephosphoramide
6TG: 6-thioguanine
Thiotepa: triethylenethiophosphoramide
TPMT: thiopurine methyltransferase
TS: thymidylate synthase
TTP: thymidine triphosphate
VOD: veno-occlusive disease

A Note on Treatment Regimens

Cancer treatment regimens change to reflect continuous advances in basic and clinical science: new drugs, both small molecules and biologicals; improved methods of targeting and timing of drug delivery; agents with altered pharmacokinetic properties and selectivities; the use of rational multidrug combinations; and greater knowledge of the basic cell biology of tumorigenesis, metastasis, and immune function, among other advances. As a consequence, this chapter presents relatively few detailed treatment regimens; rather, we refer the reader to the web-based resources of the U.S. FDA and the National Comprehensive Cancer Network (NCCN.org). Table 71–1 provides two examples of therapeutic regimens that illustrate the complexity of current cancer drug therapy.

I. Alkylating Agents and Platinum Coordination Complexes

HISTORY OF ALKYLATING ANTICANCER DRUGS

The discovery and initial development of alkylating anticancer drugs are based on observations of the effects of chemical warfare in World War I (Chabner and Roberts, 2005). The pervasively toxic sulfur mustard gas that caused topical burns to skin, eyes, lungs, and mucosa also caused aplasia of the bone marrow and lymphoid tissue and ulceration of the GI tract (Krumbhaar, 1919). In 1942, Louis Goodman and Alfred Gilman, the originators of this text, demonstrated the activity of nitrogen mustards against mouse lymphoma. Their clinical studies of intravenous nitrogen mustards in patients with lymphoma launched the modern era of cancer chemotherapy (Gilman and Philips, 1946).

Actions Common to Alkylating Drugs

Six major types of alkylating agents are used in the chemotherapy of cancer: *nitrogen mustards*, *ethyleneimines*, *alkyl sulfonates, nitrosoureas, triazenes*, and *methylhydrazines*. Because of similarities in their mechanisms of action and resistance, *platinum complexes* are also discussed with classical alkylating agents, even though they do not alkylate DNA but instead form covalent metal adducts with DNA.

The chemotherapeutic alkylating agents have in common the property of forming highly reactive carbonium ion intermediates. These reactive intermediates covalently link to sites of high electron density, such as phosphates, amines, sulfhydryl, and hydroxyl groups. Their chemotherapeutic and cytotoxic effects are directly related to the alkylation of reactive amines, oxygens, or phosphates on DNA. The N7 atom of guanine is particularly susceptible to the formation of a covalent bond with bifunctional alkylating agents and may represent the key target that determines

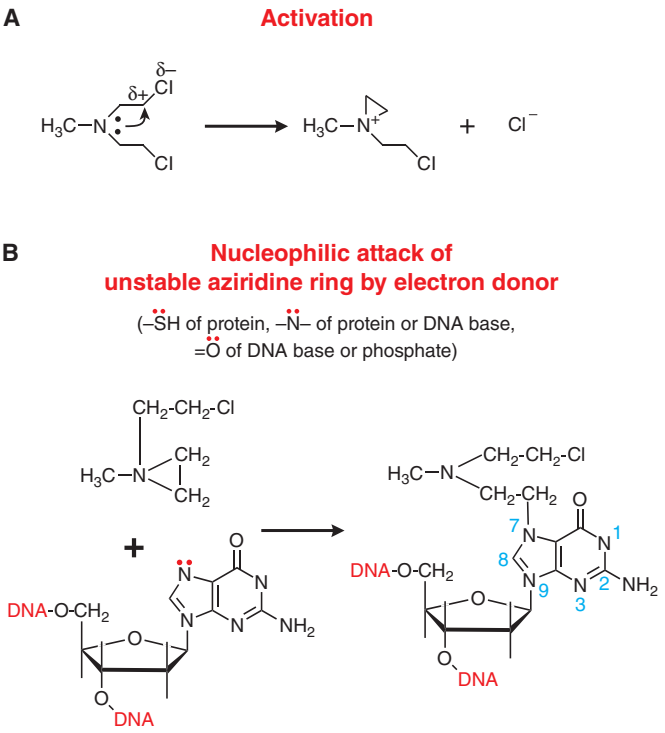


Figure 70–1 *Mechanism of action of alkylating agents of mechlorethamine, a prototypic nitrogen mustard.* **A.** *Activation reaction.* **B.** *Alkylation of N7 of guanine.* With bifunctional drugs such as nitrogen mustards (see Figure 70–2), the second 2-chloroethyl side chain can undergo a similar activation cross-link of two nucleic acid chains or a nucleic acid to a protein.

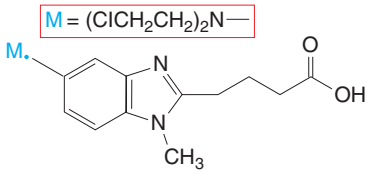
their biological effects (Figure 70–1). Other atoms in the purine and pyrimidine bases of DNA, including N1 and N3 of the adenine ring, N3 of cytosine, and O6 of guanine, react with these agents, as do the amino and sulfhydryl groups of proteins and the sulfhydryls of glutathione. The general mechanisms of alkylating agents on DNA are illustrated in Figure 70–1 using *mechlorethamine* (nitrogen mustard).

With the bifunctional alkylating agents such as nitrogen mustard and its derivatives that are used therapeutically (Figure 70–2), the second 2-chloroethyl side chain can undergo a similar cyclization reaction and alkylate a second guanine residue or another nucleophilic moiety, resulting in the cross-linking of two nucleic acid chains or the linking of a nucleic acid to a protein, alterations that cause a major disruption in nucleic acid function. Any of these effects can contribute to both the mutagenic and the cytotoxic effects of alkylating agents. The extreme cytotoxicity of the bifunctional alkylators correlates closely with interstrand cross-linkage of DNA.

The ultimate cause of cancer cell death related to drug-induced DNA damage is not known. However, cancer cells depend on mitogenic lesions that drive uncontrolled cell cycle progression but also bring them substantially closer to the threshold at which apoptosis can be triggered by cytotoxic drugs. This contrasts with normal cells that receive their growth and survival cues from their orthotopic environment keeping them far above the apoptosis threshold (Lowe et al., 2004). Genomic instability and the consequent acquisition of tumor mutations are recognized as "enablers" of the neoplastic phenotype (Hanahan and Weinberg, 2011). Such defects in DNA repair defects can render cancer cells more susceptible to death after exposure to DNA-damaging drugs (Costa de Almeida et al., 2021). Specific cellular responses to DNA-damaging agents include cell-cycle arrest and attempts to repair DNA. For example, faulty DNA damage repair mechanisms and mutations of p53 can lead to resistance to alkylating drugs (Kastan, 1999). The process of recognizing and repairing DNA generally requires an intact nucleotide excision repair (NER) complex but may differ with each drug and with each tumor. The specific repair enzyme complex utilized will depend on two factors: the chemistry of the adduct formed and the repair capacity of the cell involved. For example, defects in homologous recombination DNA repair (HRD) can convey relative sensitivity to platinum agents (Stover et al., 2020), and characterization of HRD present in clinical tumors has entered clinical practice as a means of selecting patients who might benefit from specific chemotherapeutic agents.

Structure-Activity Relationships

Although alkylating agents share the capacity to alkylate biologically important molecules, modification of the basic backbone structure changes reactivity, lipophilicity, active transport across biological membranes, sites of macromolecular attack, and mechanisms of DNA repair, all of which determine drug activity *in vivo*. Several of the most used agents (e.g., *cyclophosphamide* and ifosfamides) are prodrugs, and the active alkylating moieties are generated *in vivo* through hepatic metabolism (Figure 70–3) (Arnold et al., 1958; Wagner, 1994).



BENDAMUSTINE

The more recently approved alkylating agent, ***bendamustine***, differs from the other cross-linking agents by having the typical chloroethyl reactive groups in nitrogen mustard (M) attached to a benzimidazole backbone. The unique properties and activity of this drug may derive from this purine-like structure; the agent produces slowly repaired DNA cross-links, lacks cross-resistance with other classical alkylators (Leoni et al., 2008), and is approved for treatment of chronic lymphocytic leukemia (CLL) and large-cell lymphomas refractory to standard alkylators. Newer investigational alkylating agents include *fotemustine*, which is approved in Europe, and *evofosfamide,* a hypoxia-activated prodrug (Yan et al., 2020).

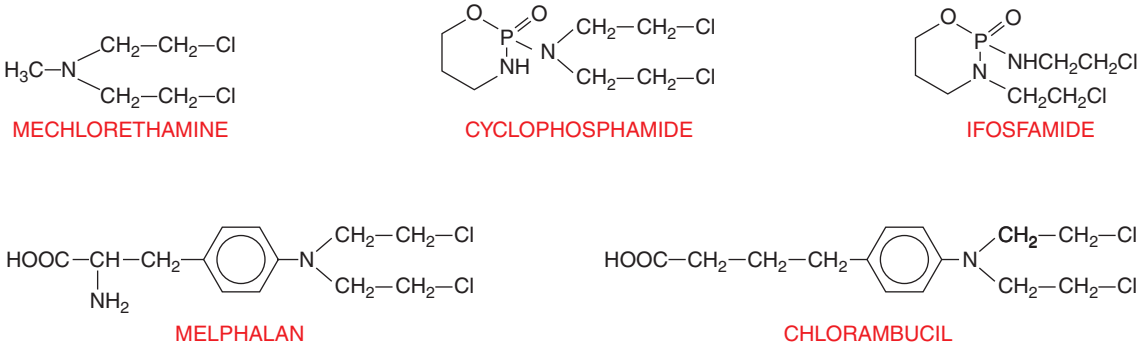


Figure 70–2 *Nitrogen mustards employed in therapy.*

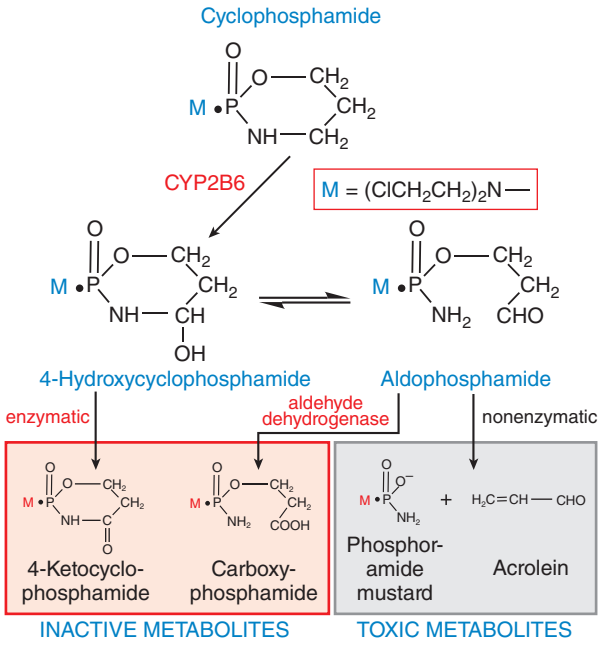


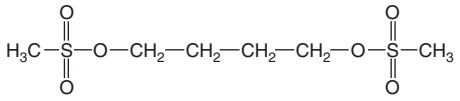
Figure 70–3 *Activation and metabolism of cyclophosphamide. Cyclophosphamide* is activated (hydroxylated) by CYP2B6, with subsequent transport of the activated intermediate to the sites of action. The selectivity of *cyclophosphamide* for some cancer types results from the capacity of normal tissues to degrade the activated intermediates via aldehyde dehydrogenase, glutathione transferase, and other pathways and the lack thereof in cancer cells. *Ifosfamide* is structurally similar to *cyclophosphamide* (see Figure 70–2); whereas *cyclophosphamide* has two chloroethyl groups on the exocyclic nitrogen atom, one of the 2-chloroethyl groups of *ifosfamide* is on the cyclic phosphoramide nitrogen of the oxazaphosphorine ring. *Ifosfamide* is activated by hepatic CYP3A4 and proceeds more slowly than activation of *cyclophosphamide*, with greater production of dechlorinated metabolites and chloroacetaldehyde. These differences in metabolism likely account for the higher doses of *ifosfamide* required for equitoxic effects, the greater neurotoxicity of *ifosfamide*, and perhaps differences in the antitumor spectra of *cyclophosphamide* and *ifosfamide*.

One class of alkylating agents transfers methyl rather than ethyl groups to DNA: The *triazene* derivative 5-(3,3-dimethyl-1-triazeno)-imidazole-4-carboxamide, usually called *dacarbazine* or DTIC, is prototypical of methylating agents. *Dacarbazine* requires initial activation by hepatic CYPs through an *N*-demethylation reaction. In the target cell, spontaneous cleavage of the metabolite, methyl-triazeno-imidazole-carboxamide (MTIC), yields an alkylating moiety, a methyl diazonium ion. A related triazene, *temozolomide*, undergoes spontaneous, nonenzymatic activation to MTIC. It is able to cross the blood-brain barrier and has been shown to have significant activity against gliomas.

The *nitrosoureas,* which include compounds such as 1,3-*bis*-(2-chloroethyl)-1-nitrosourea (*carmustine* [BCNU]), 1-(2-chloroethyl)-3-cyclohexyl-1-nitrosourea (*lomustine* [CCNU]), and its methyl derivative (*semustine* [methyl-CCNU]), as well as the antibiotic *streptozocin*, exert their cytotoxicity through the spontaneous breakdown to an alkylating intermediate, the 2-chloroethyl diazonium ion. As with the nitrogen mustards, interstrand DNA cross-linking appears to be the primary lesion responsible for the cytotoxicity of nitrosoureas. The reactions of the nitrosoureas with macromolecules are shown in Figure 70–4. The use of nitrosoureas is presently limited to brain tumors due to their lipophilicity, except for *streptozocin*, which is uniquely useful in pancreatic islet cell tumors, perhaps reflecting the selective uptake of the carbohydrate portion of the drug into these tumor cells.

Stable ethyleneimine derivatives have antitumor activity; *triethylenemelamine* (TEM) and *thiotepa* have been used clinically. In standard doses, *thiotepa* produces mostly myelosuppression; it is used for high-dose chemotherapy regimens in transplants for hematological malignancies, in which it causes both mucosal and CNS toxicity. *Altretamine* (hexamethylmelamine, HMM) is chemically similar to TEM and has been used to treat ovarian cancer. The methylmelamines are *N*-demethylated by hepatic microsomes with the release of formaldehyde; there is a direct relationship between the degree of the demethylation and their antitumor activity in model systems.

Esters of alkane sulfonic acids alkylate DNA through the release of methyl radicals. Busulfan is of value in high-dose chemotherapy.



BUSULFAN

General Pharmacological Actions

Cytotoxic Actions

The capacity of alkylating agents to interfere with DNA integrity and function and to induce cell death in rapidly proliferating tissues provides the basis for their therapeutic and toxic properties. Acute effects manifest primarily against rapidly proliferating tissues; however, certain alkylating agents may have damaging effects on tissues with normally low mitotic indices (e.g., liver, kidney, and mature lymphocytes); effects in these tissues usually are delayed. Lethality of DNA alkylation depends on the recognition of the adduct, the creation of DNA strand breaks by repair enzymes, and an intact apoptotic response. In nondividing cells, DNA damage activates a checkpoint that depends on the presence of a normal *p53* gene. Cells thus blocked in the G_1/S interface either repair DNA alkylation or undergo apoptosis. Malignant cells with mutant or absent *p53* fail to suspend cell-cycle progression, do not undergo apoptosis, and can exhibit resistance to these drugs.

Distinction Between Mono- and Bifunctional Agents

Although DNA is the ultimate target of all alkylating agents, there is a crucial distinction between the bifunctional agents, in which cytotoxic

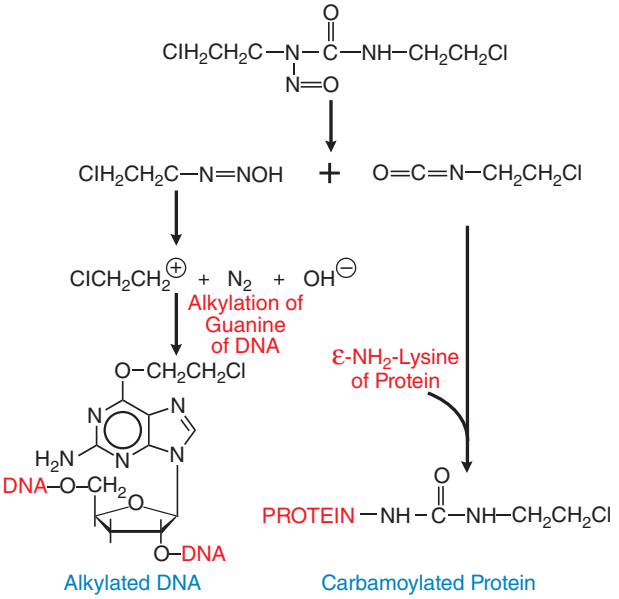


Figure 70–4 *Generation of alkylating and carbamylating intermediates from carmustine (BCNU).* The 2-chloroethyl diazonium ion, a strong electrophile, can alkylate guanine, cytidine, and adenine bases. Displacement of the halogen atom can then lead to interstrand or intrastrand cross-linking of DNA. The formation of cross-links after the initial alkylation reaction proceeds slowly and can be reversed by the DNA repair enzyme MGMT, which displaces the chloroethyl adduct from its binding to guanine in a suicide reaction. The same enzyme, when expressed in human gliomas, produces resistance to nitrosoureas and to other methylating agents, including *dacarbazine*, *temozolomide*, and *procarbazine*.

effects predominate, and the monofunctional methylating agents (*procarbazine*, *temozolomide*), which have greater capacity for mutagenesis and carcinogenesis. This suggests that the cross-linking of DNA strands represents a much greater threat to cellular survival than do other effects, such as single-base alkylation and the resulting depurination and single-chain scission. Conversely, simple methylation may be bypassed by DNA polymerases, leading to mispairing reactions that permanently modify a DNA sequence. These new sequences are transmitted to subsequent generations and may result in mutagenesis or carcinogenesis. Some methylating agents, such as *procarbazine*, are highly carcinogenic.

Adduct recognition systems and DNA repair systems play important roles in removing adducts and thereby determine the selectivity of action against particular cell types and the acquisition of resistance to alkylating agents. Alkylation of a single strand of DNA (monoadducts) is repaired by the NER pathway. The less frequent cross-links require participation of nonhomologous end joining, an error-prone pathway, or the potentially error-free HRD pathway. After drug infusion in humans, monoadducts appear rapidly and peak within 2 h of drug exposure, while cross-links peak at 8 h. The $t_{1/2}$ for repair of adducts varies amongst normal tissues and tumors; in peripheral blood mononuclear cells, both monoadducts and cross-links disappear with a $t_{1/2}$ of 12 to 16 h.

The repair process depends on the presence and accurate functioning of multiple proteins. Their absence or mutation, as in Fanconi anemia or ataxia telangiectasia, leads to extreme sensitivity to DNA cross-linking agents such as *mitomycin*, *cisplatin*, or classical alkylators. Other repair enzymes are specific for removing methyl and ethyl adducts from the O6 of guanine (methyl guanine methyltransferase [MGMT]) and for repair of alkylation of the N3 of adenine and N7 of guanine (3-methyladenine-DNA glycosylase). High expression of MGMT protects cells from cytotoxic effects of nitrosoureas and methylating agents and confers drug resistance, while methylation and silencing of the MGMT gene in brain tumors are associated with clinical response to BCNU and *temozolomide* (Hegi et al., 2008). *Bendamustine* differs from classical chloroethyl alkylators in activating base excision repair, rather than the more complex double-strand break repair or MGMT. It impairs physiological arrest of adduct-containing cells at mitotic checkpoints, leads to mitotic catastrophe rather than apoptosis, and does not require an intact p53 to cause cytotoxicity.

Recognition of DNA adducts is an essential step in promoting attempts at repair and ultimately leading to apoptosis. The Fanconi pathway, consisting of 12 proteins, recognizes adducts and signals the need for repair of a broad array of DNA-damaging drugs and irradiation. Absence or inactivation of components of this pathway leads to increased sensitivity to DNA damage. Conversely, for the methylating drugs, nitrosoureas, *cisplatin* and *carboplatin*, and thiopurine analogues, the mismatch repair (MMR) pathway is essential for cytotoxicity, causing strand breaks at sites of adduct formation, creating mispairing of thymine residues, and triggering apoptosis.

Mechanisms of Resistance to Alkylating Drugs

Resistance to an alkylating agent develops rapidly when it is used as a single agent. Specific biochemical changes implicated in the development of resistance include the following:

- Decreased permeation of actively transported drugs (e.g., *mechlorethamine* and *melphalan*)
- Increased intracellular concentrations of nucleophilic substances, principally thiols such as glutathione, which can conjugate with and detoxify electrophilic intermediates
- Increased activity of DNA repair pathways, which may differ for the various alkylating agents
- Increased rates of metabolic degradation of the activated forms of *cyclophosphamide* and *ifosfamide* to their inactive keto and carboxy metabolites by aldehyde dehydrogenase (see Figure 70–3) and detoxification of most alkylating intermediates by glutathione transferases
- Loss of ability to recognize adducts formed by nitrosoureas and methylating agents, as the result of defective MMR protein capability, confers resistance, as does defective cell-cycle checkpoint function, for virtually all alkylating drugs
- Impaired apoptotic pathways, with overexpression of bcl-2 as an example, confer resistance

Adverse Effects of Alkylating Drugs

Alkylating agents differ in their patterns of the tissue specificity and severity of their adverse effects, which are summarized next.

Bone Marrow

Most alkylating agents cause dose-limiting toxicity to marrow cells and, to a lesser extent, intestinal mucosa. Most alkylating agents (i.e., *melphalan*, *chlorambucil*, *cyclophosphamide*, and *ifosfamide*) cause myelosuppression, with a nadir of the peripheral blood granulocyte count at 6 to 10 days and recovery in 14 to 21 days. *Cyclophosphamide* has lesser effects on peripheral blood platelet counts than do the other agents. *Busulfan* suppresses all blood elements, particularly stem cells, and may produce a prolonged and cumulative myelosuppression lasting months or even years. For this reason, it is used as a preparative regimen in allogenic bone marrow transplantation. *Temozolomide*, *carmustine*, and other chloroethylnitrosoureas cause delayed and prolonged suppression of both platelets and granulocytes, reaching a nadir 4 to 6 weeks after drug administration and reversing slowly thereafter. Both cellular and humoral immunity are suppressed by alkylating agents, which have, therefore, been used to treat various autoimmune diseases. Immunosuppression is reversible at usual doses, but opportunistic infections may occur with extended treatment.

Mucosa, Hair Follicles

Alkylating agents are highly toxic to dividing mucosal cells and to hair follicles, leading to oral mucosal ulceration, intestinal denudation, and alopecia. Mucosal effects are particularly damaging in high-dose chemotherapy protocols associated with bone marrow transplantation, as they predispose to bacterial sepsis arising from the GI tract. In these protocols, *cyclophosphamide*, *melphalan*, and *thiotepa* have the advantage of causing less mucosal damage than the other agents. In high-dose protocols, however, other toxicities become limiting (Table 70–1).

Nervous System

Nausea and vomiting commonly follow administration of nitrogen mustard or BCNU. *Ifosfamide* is the most neurotoxic of the alkylating agents

TABLE 70–1 ■ DOSE-LIMITING EXTRAMEDULLARY TOXICITIES OF SINGLE ALKYLATING AGENTS

DRUG	MTD[a] mg/m²	FOLD INCREASE OVER STANDARD DOSE	MAJOR ORGAN TOXICITIES
Cyclophosphamide	7000	7	Cardiac, hepatic VOD
Ifosfamide	16,000	2.7	Renal, CNS, hepatic VOD
Thiotepa	1000	18	GI, CNS, hepatic VOD
Melphalan	180	5.6	GI, hepatic VOD
Busulfan	640	9	GI, hepatic VOD
Carmustine (BCNU)	1050	5.3	Lung, hepatic VOD
Cisplatin	200	2	PN, renal
Carboplatin	2000	5	Renal, PN, hepatic VOD

GI, gastrointestinal; PN, peripheral neuropathy; VOD, veno-occlusive disease.
[a]Maximum tolerated dose (cumulative) in treatment protocols.

and may produce altered mental status, coma, generalized seizures, and cerebellar ataxia. These side effects result from the release of chloroacetaldehyde from the phosphate-linked chloroethyl side chain of *ifosfamide*. High-dose *busulfan* can cause seizures in up to 10% of patients without prophylaxis.

Other Organs

All alkylating agents, including *temozolomide*, have caused pulmonary fibrosis, usually several months after treatment. In high-dose regimens, particularly those employing *busulfan* or BCNU, vascular endothelial damage may precipitate veno-occlusive disease (VOD) of the liver (see Table 70–1), an often-fatal side effect that is successfully reversed by *defibrotide*. *Defibrotide* is an oligonucleotide mixture with profibrinolytic properties that is FDA-approved for the treatment of adult and pediatric patients with hepatic VOD but may cause hemorrhage as an adverse effect. The nitrosoureas and *ifosfamide*, after multiple cycles of therapy, may lead to renal failure. *Cyclophosphamide* and *ifosfamide* release a nephrotoxic and urotoxic metabolite, acrolein, which causes severe hemorrhagic cystitis in high-dose regimens. This adverse effect can be prevented by coadministration of MESNA (2-mercaptoethanesulfonate-Na^+), which conjugates acrolein in urine. *Ifosfamide* in high doses for hematopoietic stem cell transplant causes a chronic, and often irreversible, renal toxicity; its nephrotoxicity correlates with the total dose of drug received and increases in frequency in children younger than 5 years. The syndrome may be due to chloroacetaldehyde or acrolein excreted in the urine.

The more unstable alkylating agents (e.g., *mechlorethamine* and the nitrosoureas) have strong vesicant properties, damage veins with repeated use, and, if extravasated, produce ulceration. All alkylating agents have toxic effects on the male and female reproductive systems, causing an often-permanent amenorrhea, particularly in perimenopausal women, and an irreversible azoospermia in men.

Leukemogenesis

Alkylating agents have a high capacity to induce leukemia. Acute nonlymphocytic leukemia, induced by treatment, is often associated with a higher incidence of p53 mutations, partial or total deletions of chromosome 5 or 7, and reduced response to chemotherapy. Incidence peaks about 4 years after therapy and may affect up to 5% of patients treated on regimens containing alkylating drugs. Leukemia often is preceded by a period of neutropenia or anemia and by bone marrow morphology consistent with myelodysplasia. *Melphalan*, the nitrosoureas, and the methylating agent *procarbazine* have the greatest propensity to cause leukemia, which is less common after *cyclophosphamide*.

Clinical Pharmacology of Nitrogen Mustards

Mechlorethamine

Mechlorethamine HCl was the first clinically used nitrogen mustard and is the most reactive of the drugs in this class. It is used topically for treatment of cutaneous T-cell lymphoma (CTCL) as a solution that is rapidly mixed and applied to affected areas. It has been largely replaced by *cyclophosphamide*, *melphalan*, and other more stable alkylating agents.

Cyclophosphamide

ADME

Cyclophosphamide is well absorbed orally and is activated to the 4-hydroxy intermediate (see Figure 70–3). Its rate of metabolic activation exhibits significant interpatient variability and increases with successive doses in high-dose regimens but appears to be saturable at concentrations of the parent compound greater than 150 μM. 4-Hydroxycyclophosphamide may be oxidized further by aldehyde oxidase, either in liver or in tumor tissue, to inactive metabolites. 4-Hydroxycyclophosphamide and its tautomer, aldophosphamide, travel in the circulation to tumor cells, where aldophosphamide cleaves spontaneously, generating stoichiometric amounts of phosphoramide mustard and acrolein. Phosphoramide mustard is responsible for antitumor effects, while acrolein causes hemorrhagic cystitis often seen during therapy with *cyclophosphamide*.

Pretreatment with CYP inducers such as *phenobarbital* enhances the rate of activation of the azoxyphosphorenes but does not alter total exposure to active metabolites over time and does not affect toxicity or efficacy. *Cyclophosphamide* can be used in full doses in patients with renal dysfunction because it is eliminated by hepatic metabolism. Patients with significant hepatic dysfunction should receive reduced doses. Maximal plasma concentrations are achieved about 1 h after oral administration; the $t_{1/2}$ of parent drug in plasma is about 7 h.

Therapeutic Uses

Cyclophosphamide is administered orally or intravenously. Recommended doses vary widely, and standard protocols for determining the schedule and doses of *cyclophosphamide* in combination with other chemotherapeutic agents should be consulted. The neutrophil nadir of 500 to 1000 cells/mm^3 generally serves as a lower limit for dosage adjustments in prolonged therapy.

The clinical spectrum of activity for *cyclophosphamide* is broad. It is an essential component of many effective drug combinations for non-Hodgkin lymphomas, other lymphoid malignancies, breast and ovarian cancers, and solid tumors in children. Complete remissions have been reported when *cyclophosphamide* was given as a single agent for Burkitt lymphoma. It frequently is used in combination with *doxorubicin* and a taxane as adjuvant therapy after surgery for breast cancer. Because of its potent immunosuppressive properties, *cyclophosphamide* has been used to treat autoimmune disorders, including Wegener granulomatosis, rheumatoid arthritis, and the nephrotic syndrome. Caution is advised when the drug is considered for nonneoplastic conditions, not only because of its toxic effects but also because of its potential for inducing premature menopause, sterility, teratogenic effects, and leukemia.

Adverse Effects

Phosphoramide mustard is responsible for antitumor effects, while acrolein causes hemorrhagic cystitis often seen during therapy with *cyclophosphamide*. Patients should receive vigorous intravenous hydration during high-dose treatment. Hematuria in a patient receiving daily oral therapy should lead to immediate drug discontinuation. Refractory bladder hemorrhage can become life-threatening and may require cystectomy for control of bleeding. Inappropriate secretion of antidiuretic hormone has been observed (usually at doses greater than 50 mg/kg) and can lead to water intoxication because these patients usually are vigorously hydrated. In addition to the cystitis (counteracted by MESNA and diuresis), GI, pulmonary, renal, hepatic, and cardiac toxicities (hemorrhagic myocardial necrosis) may occur after high-dose therapy with total doses greater than 200 mg/kg.

Ifosfamide

Ifosfamide is an analogue of cyclophosphamide. Severe urinary tract and CNS toxicity initially limited the use of *ifosfamide*, but adequate hydration and coadministration of MESNA have reduced its bladder toxicity. CNS toxicity is handled with methylene blue (Pelgrims et al., 2000).

ADME

Ifosfamide has a plasma elimination $t_{1/2}$ of approximately 1.5 h after doses of 3.8 to 5 g/m^2 and a somewhat shorter $t_{1/2}$ at lower doses; its pharmacokinetics are highly variable due to variable rates of hepatic metabolism (see legend to Figure 70–3).

Therapeutic Uses

Ifosfamide is approved for treatment of patients with relapsed germ cell testicular cancer and is frequently used for first-time treatment of pediatric or adult patients with sarcomas. It is a common component of high-dose chemotherapy regimens with bone marrow or stem cell rescue. In nonmyeloablative regimens, *ifosfamide* is infused intravenously

over at least 30 min at a dose of 1.2 g/m^2 per day or less for 5 days. Intravenous MESNA is given as bolus injections in a dose equal to 20% of the *ifosfamide* dose concomitantly and an additional 20% again 4 and 8 h later, for a total MESNA dose of 60% of the *ifosfamide* dose. Alternatively, MESNA may be given concomitantly in a single dose equal to the *ifosfamide* dose. Patients also should receive at least 2 L of oral or intravenous fluid daily. Treatment cycles are repeated every 3 to 4 weeks.

Adverse Effects

Ifosfamide has a similar toxicity profile as *cyclophosphamide*, although it causes greater platelet suppression, neurotoxicity, nephrotoxicity, and urothelial damage. In high-dose regimens (i.e., total doses of 12–14 g/m^2), it may cause severe neurological toxicity, including hallucinations, coma, and death, with symptoms appearing 12 h to 7 days after beginning the *ifosfamide* infusion. This toxicity may result from a metabolite, chloroacetaldehyde. *Ifosfamide* also causes nausea, vomiting, anorexia, leukopenia, nephrotoxicity, and VOD of the liver.

Melphalan

The alkylating agent *melphalan* primarily is used to treat multiple myeloma and, less commonly, in high-dose chemotherapy with marrow transplantation. The general pharmacological and cytotoxic actions of *melphalan* are similar to those of other bifunctional alkylators.

ADME

Oral *melphalan* is absorbed in an inconsistent manner and, for most indications, is given as an intravenous infusion. The drug has a plasma $t_{1/2}$ of about 45 to 90 min; 10% to 15% of an administered dose is excreted unchanged in the urine. Patients with decreased renal function may develop unexpectedly severe myelosuppression.

Therapeutic Uses and Adverse Effects

Melphalan for multiple myeloma is administered orally for 4 to 7 days every 28 days, with *dexamethasone* or *thalidomide*. Treatment is repeated at 4-week intervals based on response and tolerance. Dosage adjustments should be based on blood cell counts. *Melphalan* also may be used in myeloablative regimens followed by bone marrow or peripheral blood stem cell reconstitution. The toxicity of *melphalan* is mostly hematological and is similar to that of other alkylating agents. Nausea and vomiting are less frequent. The drug causes less alopecia and, rarely, renal or hepatic dysfunction. The drug is not a vesicant.

Chlorambucil

The cytotoxic effects of *chlorambucil* on the bone marrow, lymphoid organs, and epithelial tissues are similar to those observed with other nitrogen mustards. As an orally administered agent, *chlorambucil* is well tolerated in small daily doses and provides flexible titration of blood counts. Nausea and vomiting may result from single oral doses of 20 mg or greater.

ADME

Oral absorption of *chlorambucil* is adequate and reliable. The drug has a $t_{1/2}$ in plasma of about 1.5 h and is hydrolyzed to inactive products.

Therapeutic Uses and Adverse Effects

Chlorambucil historically was used almost exclusively in treating CLL; modern approaches to CLL frequently employ as first-line treatments noncytotoxic approaches such as modulators of B-cell antigen receptor signaling (see Chapter 71). When these agents are not available or not tolerated or as treatment for relapsed disease, *chlorambucil* may be given once daily and continued for 3 to 6 weeks. With a fall in the peripheral total leukocyte count or clinical improvement, the dosage is titrated to maintain neutrophils and platelets at acceptable levels. Maintenance therapy often is required to maintain clinical response. *Chlorambucil* treatment may continue for months or years, achieving its effects gradually and often without significant toxicity to a compromised bone marrow. Marked hypoplasia of the bone marrow may be induced with excessive doses, but the myelosuppressive effects are moderate, gradual, and rapidly reversible. GI discomfort, azoospermia, amenorrhea, pulmonary fibrosis, seizures, dermatitis, and hepatotoxicity rarely may be encountered. A marked increase in the incidence of acute myeloid leukemia (AML) and other tumors was noted when *chlorambucil* was used historically in the treatment of patients with polycythemia vera and as adjuvant chemotherapy after primary removal of breast cancers.

Bendamustine

Bendamustine is approved for treatment of CLL and non-Hodgkin lymphoma. *Bendamustine* is given as a 30-min intravenous infusion on days 1 and 2 of a 28-day cycle. Lower doses may be indicated in heavily pretreated patients. *Bendamustine* is rapidly degraded through sulfhydryl interaction and adduct formation with macromolecules; less than 5% of the parent drug is excreted in the urine intact. *N*-Demethylation and oxidation produce metabolites that have antitumor activity, but less than that of the parent molecule. The parent drug has a plasma $t_{1/2}$ of about 30 min. The clinical toxicity pattern of *bendamustine* is typical of alkylators, with a rapidly reversible myelosuppression and mucositis, both generally tolerable.

Ethyleneimines and Methylmelamines

Although nitrogen mustards containing chloroethyl groups constitute the most widely used class of alkylating agents, other alkylators such as ethyleneimines with greater chemical stability and well-defined activity in specific types of cancer have value in clinical practice.

Altretamine

Altretamine is structurally similar to TEM. Its precise mechanism of cytotoxicity is unknown, although it can alkylate DNA and proteins. It is a palliative treatment of patients with persistent or recurrent ovarian cancer following *cisplatin*-based combination therapy. The usual dosage of *altretamine* as a single agent in ovarian cancer is 260 mg/m^2 per day in four divided doses, for 14 or 21 consecutive days out of a 28-day cycle, for up to 12 cycles.

ADME

Altretamine is well absorbed from the GI tract; its elimination $t_{1/2}$ is 4 to 10 h. The drug undergoes rapid demethylation in the liver; the principal metabolites are pentamethylmelamine and tetramethylmelamine.

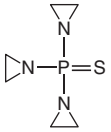
Adverse Effects

The main toxicities of *altretamine* are myelosuppression and neurotoxicity. *Altretamine* causes both peripheral and central neurotoxicity (ataxia, depression, confusion, drowsiness, hallucinations, dizziness, and vertigo). Neurological symptoms abate on discontinuation of therapy. Peripheral blood counts and a neurological examination should be performed prior to the initiation of each course of therapy. Therapy should be interrupted for at least 14 days and subsequently restarted at a lower dosage if the white cell count falls to less than 2000 cells/mm^3 or the platelet count falls to less than 75,000 cells/mm^3 or if neurotoxic or intolerable GI symptoms occur. If neurological symptoms fail to stabilize on the reduced-dose schedule, *altretamine* should be discontinued. Nausea and vomiting also are common side effects and may be dose limiting. Renal dysfunction may necessitate discontinuing the drug. Other rare adverse effects include rashes, alopecia, and hepatic toxicity. Severe, life-threatening orthostatic hypotension may develop in patients who receive monoamine oxidase inhibitors, *amitriptyline*, *imipramine*, or *phenelzine* concurrently with *altretamine*.

Thiotepa

Thiotepa consists of three ethyleneimine groups stabilized by attachment to the nucleophilic thiophosphoryl base. Its current use is primarily for high-dose chemotherapy regimens. Hepatic CYPs rapidly convert

thiotepa to its desulfurated primary metabolite, triethylenephosphoramide (TEPA). Both *thiotepa* and TEPA form DNA cross-links.



ADME

Within hours of *thiotepa* administration, TEPA becomes the predominant form of the drug present in plasma. The parent compound has a plasma $t_{1/2}$ of 1.2 to 2 h; TEPA has a longer $t_{1/2}$, 3 to 24 h. *Thiotepa* pharmacokinetics essentially are the same in children as in adults at conventional doses (≤80 mg/m^2), and drug and metabolite $t_{1/2}$ are unchanged in children receiving high-dose therapy of 300 mg/m^2 per day for 3 days. Less than 10% of the administered drug appears in urine as the parent drug or the primary metabolite.

Adverse Effects

Toxicities include myelosuppression and, to a lesser extent, mucositis. Myelosuppression tends to develop somewhat later than with *cyclophosphamide*, with leukopenic nadirs at 2 weeks and platelet nadirs at 3 weeks. In high doses, *thiotepa* may cause neurotoxic symptoms, including coma and seizures.

Alkyl Sulfonates

Busulfan

Busulfan exerts few pharmacological actions other than myelosuppression at conventional doses and, prior to the advent of *imatinib mesylate*, was a standard agent for patients in the chronic phase of myelocytic leukemia and caused a severe and prolonged pancytopenia in some patients. *Busulfan* now is primarily used in high-dose regimens, in which pulmonary fibrosis, GI mucosal damage, and hepatic VOD are important toxicities.

ADME

Busulfan is well absorbed after oral administration and has a plasma $t_{1/2}$ of 2 to 3 h. The drug is conjugated to glutathione by GSTA1A and further metabolized by CYP-dependent pathways; its major urinary metabolite is methane sulfonic acid. In high doses, children less than 18 years of age clear the drug two to four times faster than adults and tolerate higher doses. *Busulfan* clearance varies considerably among patients. VOD is associated with high area under the curve (AUC; >1500 μM × min) and slow clearance, leading to recommendations for dose adjustment based on drug-level monitoring. A target steady-state concentration of 600 to 900 ng/mL in plasma in adults or AUC of less than 1000 μM × min in children achieves an appropriate balance between toxicity and therapeutic benefit.

Therapeutic Uses

The use of *busulfan* in initially treating Philadelphia chromosome–positive chronic myelocytic leukemia (CML) has been essentially supplanted by inhibitors of the oncogenic protein tyrosine kinase bcr-abl (see Chapter 71). However, where those agents are not available or not tolerated, the initial oral dose of *busulfan* varies with the total leukocyte count and the severity of the disease; daily doses are adjusted to subsequent hematological and clinical responses, with the aim of reducing the total leukocyte count to 10,000 cells/mm^3 or less. A decrease in the leukocyte count usually is not seen during the first 10 to 15 days of treatment, and the leukocyte count may actually increase during this period. Because the leukocyte count may fall for more than 1 month after discontinuing the drug, it is recommended that *busulfan* be withdrawn when the total leukocyte count has declined to about 15,000 cells/mm^3. A normal leukocyte count usually is achieved within 12 to 20 weeks. During remission, daily treatment resumes when the total leukocyte count reaches about 50,000 cells/mm^3.

In high-dose therapy as preparation for stem cell transplants, anticonvulsants must be used concomitantly with busulfan to protect against acute CNS toxicities, including tonic-clonic seizures that may occur several hours after each dose. Although *phenytoin* is a frequent choice, *phenytoin* induces the glutathione S-transferases that metabolize *busulfan*, reducing its AUC by about 20%. In patients requiring concomitant antiseizure medication, non–enzyme-inducing benzodiazepines such as *lorazepam* and *clonazepam* are recommended as an alternative to *phenytoin*. If *phenytoin* is used concurrently, plasma *busulfan* levels should be monitored and the *busulfan* dose adjusted accordingly (Salinger et al., 2010).

Adverse Effects

The toxic effects of *busulfan* are related to its myelosuppressive properties; prolonged thrombocytopenia may occur. Occasionally, patients experience nausea, vomiting, and diarrhea. Long-term use is associated with pulmonary fibrosis (*busulfan* lung) and leads to impotence, sterility, amenorrhea, and fetal malformation. Rarely, patients develop asthenia and hypotension. High-dose *busulfan* causes VOD of the liver in 10% or fewer of patients, as well as seizures, hemorrhagic cystitis, permanent alopecia, and cataracts. The coincidence of VOD and hepatotoxicity is increased by its coadministration with drugs that inhibit CYPs, including imidazoles and *metronidazole*, possibly through inhibition of the clearance of *busulfan* or its toxic metabolites.

Nitrosoureas

The nitrosoureas, except *streptozocin*, have an important role in the treatment of brain tumors and find occasional use in treating lymphomas and in high-dose regimens with bone marrow reconstitution. They function as bifunctional alkylating agents but differ from conventional nitrogen mustards in both pharmacological and toxicological properties.

Carmustine (BCNU) and *lomustine* (CCNU, not discussed in detail) are highly lipophilic and thus readily cross the blood-brain barrier, an important property in the treatment of brain tumors. Unfortunately, with the exception of *streptozocin*, nitrosoureas cause profound and delayed myelosuppression, with recovery 4 to 6 weeks after a single dose. Long-term treatment with the nitrosoureas, especially *semustine* (methyl-CCNU), has resulted in renal failure. As with other alkylating agents, the nitrosoureas are highly carcinogenic and mutagenic. They generate both alkylating and carbamylating moieties (see Figure 70–4).

Carmustine (BCNU) and Lomustine (CCNU)

Carmustine's major action is its alkylation of DNA at the O6-guanine position, an adduct repaired by MGMT. Methylation of the MGMT gene promoter inhibits its expression in about 30% of primary gliomas and is associated with sensitivity to nitrosoureas. In high doses with bone marrow rescue, *carmustine* (BCNU) produces hepatic VOD, pulmonary fibrosis, renal failure, and secondary leukemia. With repeated treatment courses at even conventional doses, its orally administered monochloroethyl analog *lomustine* (CCNU), can be associated with pulmonary dysfunction and prolonged bone marrow toxicity.

ADME

Carmustine (BCNU) is unstable in aqueous solution and in body fluids. After intravenous infusion, it disappears from the plasma with a highly variable $t_{1/2}$ of 15 to 90 min or more. Approximately 30% to 80% of the drug appears in the urine within 24 h as degradation products. The alkylating metabolites enter rapidly into the CSF, and their concentrations in the CSF reach 15% to 30% of the concurrent plasma values.

Therapeutic Uses

Carmustine (BCNU) is administered intravenously over 1 to 2 h and repeated every 6 weeks. *Lomustine* (CCNU) is orally bioavailable. Because of their ability to cross the blood-brain barrier, BCNU and CCNU have been used in the treatment of malignant gliomas. An implantable

carmustine (BCNU) wafer is available for use as an adjunct to surgery for recurrent glioblastoma multiforme.

Streptozocin

Streptozocin (or streptozotocin) has a methylnitrosourea moiety attached to the C2 of glucose. The drug has a high affinity for cells of the islets of Langerhans and causes diabetes in experimental animals.

ADME

Streptozocin is rapidly degraded following intravenous administration. The $t_{1/2}$ of the drug is about 15 min. Only 10% to 20% of a dose is recovered intact in the urine.

Therapeutic Uses

Streptozocin is used in the treatment of human pancreatic islet cell carcinoma and carcinoid tumors. It is administered intravenously once daily for 5 days; this course is repeated every 6 weeks. Alternatively, a higher dose can be given weekly for 2 weeks, and the weekly dose then can be increased as tolerated.

Adverse Effects

Nausea is frequent. Mild, reversible renal or hepatic toxicity occurs in approximately two-thirds of cases; in fewer than 10% of patients, renal toxicity may be cumulative with each dose and may lead to irreversible renal failure. *Streptozocin* should not be given with other nephrotoxic drugs. Hematological toxicities (anemia, leukopenia, thrombocytopenia) occur in 20% of patients.

Triazenes

Dacarbazine (DTIC)

Dacarbazine functions as a methylating agent after metabolic activation in the liver to the monomethyl triazeno metabolite MTIC. It kills cells in all phases of the cell cycle. Resistance has been ascribed to the removal of methyl groups from the O6-guanine bases in DNA by MGMT activity.

ADME

Dacarbazine is administered intravenously. After an initial rapid phase ($t_{1/2}$ of ~20 min), *dacarbazine* is cleared from plasma with a terminal $t_{1/2}$ of about 5 h. The $t_{1/2}$ is prolonged in the presence of hepatic or renal disease. Almost 50% of the compound is excreted intact in the urine by tubular secretion.

Therapeutic Uses

The primary clinical indication for *dacarbazine* is in the chemotherapy of Hodgkin disease. In combination with other drugs for Hodgkin disease, *dacarbazine* is given on days 1 and 15 and repeated every 4 weeks for up to six cycles. It is modestly effective against malignant melanoma and adult sarcomas. *Dacarbazine* for malignant melanoma is given for a 10-day period, repeated every 28 days; alternatively, it can be given daily for 5 days and repeated every 3 weeks. Extravasation of the drug may cause tissue damage and severe pain. Its use in patients with melanoma has largely been replaced by immune checkpoint inhibitors and agents targeting the typically activated mitogen-activated protein kinase pathway (see Chapters 71 and 72).

Adverse Effects

Dacarbazine induces nausea and vomiting in more than 90% of patients; vomiting usually develops 1 to 3 h after treatment and may last up to 12 h. Myelosuppression, with both leukopenia and thrombocytopenia, is mild and readily reversible within 1 to 2 weeks. A flu-like syndrome may occur. Hepatotoxicity, alopecia, facial flushing, neurotoxicity, and dermatological reactions are less common adverse effects.

Temozolomide

Temozolomide is the standard agent, in combination with radiation therapy, for initial treatment of patients with high-grade malignant gliomas. *Temozolomide*, like *dacarbazine*, forms the methylating metabolite MTIC, but without the need for hepatic metabolism, and kills cells in all phases of the cell cycle. It is more active in MGMT-deficient tumors.

ADME

Temozolomide is administered orally and has a bioavailability approaching 100%. Plasma levels of the parent drug decline with a $t_{1/2}$ of 1 to 2 h. The primary active metabolite MTIC reaches a maximum plasma concentration (150 ng/mL) 90 min after a dose and declines with a $t_{1/2}$ of 2 h. Little intact drug is recovered in the urine, the primary urinary metabolite being the inactive imidazole carboxamide.

Adverse Effects

The toxicities of *temozolomide* mirror those of DTIC. Hematological monitoring is necessary to guide dosing adjustments.

Methylhydrazines

Procarbazine

Procarbazine is used in malignant brain tumors and in combination regimens for patients with Hodgkin disease.

Mechanisms of Action

The antineoplastic activity of *procarbazine* results from its conversion by CYP-mediated hepatic oxidative metabolism to highly reactive alkylating species that methylate DNA. Activated *procarbazine* can produce chromosomal damage, including chromatid breaks and translocations, consistent with its mutagenic and carcinogenic actions. Resistance to *procarbazine* develops rapidly when it is used as a single agent; one mechanism of resistance results from increased expression of MGMT, which repairs methylation of guanine.

ADME

The pharmacokinetic behavior of *procarbazine* has not been thoroughly defined. The drug is extensively metabolized by CYPs to azo, methylazoxy, and benzylazoxy intermediates, which are found in the plasma and yield the alkylating metabolites in tumor cells. In patients with brain cancer, the concurrent use of antiseizure drugs that induce hepatic CYPs does not significantly alter the pharmacokinetics of the parent drug.

Therapeutic Uses

Procarbazine is used in combination regimens such as MOPP (*nitrogen mustard*, Oncovin [*vincristine*], *procarbazine*, and *prednisone*) for Hodgkin disease. It is also used in treating gliomas as part of the PVC (*procarbazine*, *vincristine*, and CCNU) regimen.

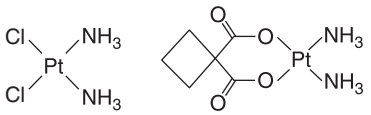
Adverse Effects

The most common toxic effects include leukopenia and thrombocytopenia, which begin during the second week of therapy and reverse within 2 weeks off treatment. GI symptoms such as mild nausea and vomiting occur in most patients; diarrhea and rash are noted in 5% to 10% of cases. Behavioral disturbances also have been reported. Because *procarbazine* augments sedative effects, the concomitant use of CNS depressants should be avoided. The drug is a weak monoamine oxidase inhibitor; it blocks the metabolism of catecholamines, sympathomimetics, and dietary tyramine and thus may provoke hypertension in patients concurrently exposed to these. *Procarbazine* has *disulfiram*-like actions; therefore, the ingestion of alcohol should be avoided. The drug is highly carcinogenic, mutagenic, and teratogenic and is associated with a 5% to 10% risk of acute leukemia in patients treated with MOPP. The highest risk is for patients who also receive radiation therapy. *Procarbazine* is a potent immunosuppressive agent. It causes infertility, particularly in males.

Platinum Coordination Complexes

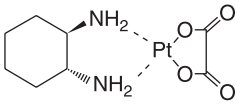
The growth-inhibitory effect of *cisplatin* was discovered during studies of bacterial growth in electrical fields that were generated by platinum electrodes; *cisplatin* generated from corroding platinum electrodes was then found to be an effective anticancer drug in mouse models

(Rosenberg et al., 1969) and gained FDA approval for cancer treatment in 1978. Platinum coordination complexes have broad antineoplastic activity and have become the foundation for treatment of ovarian, head and neck, bladder, esophagus, lung, and colon cancers. Although *cisplatin* and other platinum complexes do not form carbonium ion intermediates like other alkylating agents or formally alkylate DNA, they covalently bind to nucleophilic sites on DNA and share many pharmacological attributes with alkylators.



CISPLATIN

CARBOPLATIN



OXALIPLATIN

Mechanisms of Action

Cisplatin, *carboplatin*, and *oxaliplatin* enter cells by the high-affinity Cu^{2+} transporter CTR1 (SLC31A1, a member of the solute carrier family; see Chapter 5) and in doing so contribute to the degradation of the transporter. The compounds are actively extruded from cells by ATP7A and ATP7B copper exporters and by MRP1; variable expression of these transporters may contribute to clinical resistance, along with a variety of other proteins and epigenetic phenomena (Shen et al., 2012). Inside the cell, the chloride, cyclohexane, or oxalate ligands of the analogues are displaced by water molecules, yielding a positively charged molecule that reacts with nucleophilic sites on DNA and proteins.

Aquation of *cisplatin* is favored at the low concentrations of Cl^- inside the cell and in the urine. High concentrations of Cl^- stabilize the drug, explaining the effectiveness of Cl^- diuresis in preventing nephrotoxicity. The activated platinum complexes can react with electron-rich regions, such as sulfhydryls, and with various sites on DNA, forming both intrastrand and interstrand cross-links. The DNA-platinum adducts inhibit replication and transcription, lead to single- and double-stranded breaks and miscoding, and, if recognized by p53 and other cell-cycle checkpoint proteins, cause induction of apoptosis. Adduct formation is an important predictor of clinical response (Reed, 1998). The analogues differ in the conformation of their adducts and the effects of adduct on DNA structure and function. *Oxaliplatin* and *carboplatin* are slower to form adducts. The *oxaliplatin* adducts are bulkier and less readily repaired, create a different pattern of distortion of the DNA helix, and differ from *cisplatin* adducts in the pattern of hydrogen bonding to adjacent segments of DNA (Sharma et al., 2007).

Unlike the other platinum analogues, *oxaliplatin* exhibits a cytotoxicity that does not depend on an active MMR system, which may explain its greater activity in colorectal cancer. It also seems less dependent on the presence of high-mobility group (HMG) proteins that are required by the other platinum derivatives. Testicular cancers have a high concentration of HMG proteins and are quite sensitive to *cisplatin*. Basal-type breast cancers, such as those with *BRCA1* and *BRCA2* mutations, lack *HER2* amplification and hormone-receptor expression and appear to be uniquely susceptible to *cisplatin* both through their upregulation of apoptotic pathways governed by p63 and p73 (Deyoung and Ellisen, 2007) and defective DNA repair mechanisms including those directed at double-strand breaks, interstrand cross-links, and replication gaps (Li and Heyer, 2008; Panzarino et al., 2021). The cell-cycle specificity of *cisplatin* differs among cell types; the effects of cross-linking are most pronounced during the S phase. The platinum analogues are mutagenic, teratogenic, and carcinogenic. *Cisplatin*- or *carboplatin*-based chemotherapy for ovarian cancer is associated with a 4-fold increased risk of developing secondary leukemia.

Resistance to Platinum Analogues

Resistance to the platinum analogues likely is multifactorial; the compounds differ in their degree of cross-resistance. *Carboplatin* shares cross-resistance with *cisplatin* in most experimental tumors, while *oxaliplatin* does not.

A number of factors influence sensitivity to platinum analogues, including intracellular drug accumulation and intracellular levels of glutathione and other sulfhydryls, such as metallothionein, that bind to and inactivate the drug and rates of repair of DNA adducts. Repair of platinum-DNA adducts requires participation of the NER pathway. Inhibition or loss of NER increases sensitivity to *cisplatin* in patients with ovarian cancer, while overexpression of NER components is associated with poor response to *cisplatin*- or *oxaliplatin*-based therapy in lung, colon, and gastric cancer (Paré et al., 2008).

Resistance to *cisplatin*, but not *oxaliplatin*, appears to be partly mediated through loss of function in the MMR proteins. In the absence of effective repair of DNA-platinum adducts, sensitive cells cannot replicate or transcribe affected portions of the DNA strand. Some DNA polymerases can bypass adducts, possibly contributing to resistance. Overexpression of copper efflux transporters ATP7A and ATP7B correlates with poor survival after *cisplatin*-based therapy for ovarian cancer (Shen et al., 2012).

Cisplatin

ADME

Cisplatin is given only intravenously. To prevent renal toxicity, it is important to establish a chloride diuresis by the infusion of 1 to 2 L of normal saline prior to treatment. The appropriate amount of *cisplatin* then is diluted in a solution containing dextrose, saline, and *mannitol* and administered intravenously over 4 to 6 h. Because aluminum inactivates *cisplatin*, the drug should not come in contact with needles or other infusion equipment that contain aluminum during its preparation or administration. After intravenous administration, *cisplatin* has an initial plasma elimination $t_{1/2}$ of 25 to 50 min; concentrations of total (bound and unbound) drug fall thereafter, with a $t_{1/2}$ of 24 h or more. More than 90% of the platinum in the blood is covalently bound to plasma proteins. High concentrations of *cisplatin* are found in the kidney, liver, intestine, and testes; *cisplatin* penetrates poorly into the CNS. Only a small portion of the drug is excreted by the kidney during the first 6 h; by 24 h, up to 25% is excreted, and by 5 days, up to 43% of the administered dose is recovered in the urine, mostly covalently bound to protein and peptides. Biliary or intestinal excretion of *cisplatin* is minimal.

Therapeutic Uses

Cisplatin, in combination with *bleomycin*, *etoposide*, or with *ifosfamide* and *vinblastine*, cures 90% of patients with testicular cancer. Used with *paclitaxel*, *cisplatin* or *carboplatin* induces complete responses in the majority of patients with carcinoma of the ovary. *Cisplatin* produces responses in cancers of the bladder, head and neck, cervix, and endometrium; all forms of carcinoma of the lung; anal and rectal carcinomas; and neoplasms of childhood. The drug also sensitizes cells to radiation therapy and enhances control of locally advanced lung, esophageal, and head and neck tumors when given with radiation therapy.

Adverse Effects

Cisplatin-induced nephrotoxicity can be ameliorated by forced pretreatment hydration and chloride diuresis. *Amifostine*, a thiophosphate cytoprotective agent, reduces renal toxicity associated with repeated administration of *cisplatin*. Ototoxicity caused by *cisplatin* is unaffected by diuresis and is manifested by tinnitus and high-frequency hearing loss. Marked nausea and vomiting occur in almost all patients and usually can be controlled with serotonin ($5HT_3$) receptor antagonists, neurokinin (NK_1) receptor antagonists (e.g., *aprepitant*), and high-dose corticosteroids (see Chapters 50 and 54).

At higher doses or after multiple cycles of treatment, *cisplatin* causes a progressive peripheral motor and sensory neuropathy that may worsen

after discontinuation of the drug and may be aggravated by subsequent or simultaneous treatment with taxanes or other neurotoxic drugs. *Cisplatin* causes mild-to-moderate myelosuppression, with transient leukopenia and thrombocytopenia. Anemia may become prominent after multiple cycles of treatment. Electrolyte disturbances, including hypomagnesemia, hypocalcemia, hypokalemia, and hypophosphatemia, are common. Hypocalcemia and hypomagnesemia secondary to tubular damage and renal electrolyte wasting may produce tetany if untreated. Routine measurement of Mg^{2+} concentrations in plasma is recommended. Hyperuricemia, hemolytic anemia, and cardiac abnormalities are rare side effects. Anaphylactic-like reactions, characterized by facial edema, bronchoconstriction, tachycardia, and hypotension, may occur within minutes after administration and should be treated by intravenous injection of *epinephrine* and with corticosteroids or antihistamines. *Cisplatin* has been associated with the development of AML, usually 4 years or more after treatment.

Carboplatin

The mechanisms of action and resistance and the spectrum of clinical activity of *carboplatin* are similar to *cisplatin*. However, the two drugs differ significantly in their chemical, pharmacokinetic, and toxicological properties.

ADME

Because *carboplatin* is much less reactive than *cisplatin*, the majority of drug in plasma remains in its parent form, unbound to proteins. Most drug is eliminated via renal excretion, with a $t_{1/2}$ of about 2 h. A small fraction of platinum binds irreversibly to plasma proteins and disappears slowly, with a $t_{1/2}$ of 5 days or more.

Therapeutic Uses

Carboplatin and *cisplatin* are equally effective in the treatment of patients with suboptimally debulked ovarian cancer, non–small cell lung cancer, and extensive-stage small cell lung cancer; however, *carboplatin* may be less effective than *cisplatin* in the treatment of patients with germ cell, head and neck, and esophageal cancers. *Carboplatin* is an effective alternative for responsive tumors in patients unable to tolerate *cisplatin* because of impaired renal function, refractory nausea, significant hearing impairment, or neuropathy, but doses must be adjusted for renal function. In addition, it may be used in high-dose therapy with bone marrow or peripheral stem cell rescue. *Carboplatin* is administered as an intravenous infusion over at least 15 min and is given once every 21 to 28 days; current approaches to defining the overall dose of *carboplatin* use a regimen-specific target AUC (area under the concentration × time elimination curve) that takes into account the patient's creatinine clearance (CL_{Cr}) (Calvert et al., 1989).

Adverse Effects

Carboplatin is relatively well tolerated clinically, causing less nausea, neurotoxicity, ototoxicity, and nephrotoxicity than *cisplatin*. The dose-limiting toxicity of *carboplatin* is myelosuppression, primarily thrombocytopenia. It may cause a hypersensitivity reaction; in patients with a mild reaction, premedication, graded doses of drug, and more prolonged infusion lead to desensitization.

Oxaliplatin

ADME

Oxaliplatin has a short $t_{1/2}$ in plasma, probably as a result of its rapid uptake by tissues and its reactivity; the initial $t_{1/2}$ is about 17 min. No dose adjustment is required for hepatic dysfunction or for patients with a CL_{Cr} greater than or equal to 20 mL/min.

Therapeutic Uses

Oxaliplatin exhibits a range of antitumor activity (colorectal and gastric cancer) that differs from other platinum agents. *Oxaliplatin's* effectiveness in colorectal cancer is perhaps due to its MMR- and HMG-independent effects. It also suppresses expression of thymidylate synthase (TS), the target enzyme of *5-fluorouracil* (5FU) action, which may promote synergy of these two drugs (Fischel et al., 2002). In combination with 5FU, it is approved for treatment of patients with colorectal cancer.

Adverse Effects

The dose-limiting toxicity of *oxaliplatin* is peripheral neuropathy. An acute form, often triggered by exposure to cold liquids, manifests as paresthesias or dysesthesias in the upper and lower extremities, mouth, and throat. A second type relates to cumulative dose and has features similar to *cisplatin* neuropathy; 75% of patients receiving a cumulative dose of 1560 mg/m^2 experience some progressive sensory neurotoxicity, with dysesthesias, ataxia, and numbness of the extremities. Hematological toxicity is mild to moderate, except for rare immune-mediated cytopenias; nausea is well controlled with serotonin ($5HT_3$) receptor antagonists. *Oxaliplatin* may cause leukemia and pulmonary fibrosis months to years after administration. *Oxaliplatin* may cause an acute allergic response with urticaria, hypotension, and bronchoconstriction.

II. Antimetabolites

Folic Acid Analogues

Folic acid is an essential dietary factor that is converted by enzymatic reduction to FH_4 cofactors that provide methyl groups for the synthesis of precursors of DNA (thymidylate and purines) and RNA (purines) (Wilson et al., 2014). Folic acid analogues such as *methotrexate* (MTX) interfere with FH_4 metabolism (Figure 70–5), reducing the cellular capacity for one-carbon transfer and methylation reactions in the synthesis of purine ribonucleotides and TMP (thymidine monophosphate), thereby inhibiting DNA replication.

HISTORY OF ANTIFOLATE CHEMOTHERAPY

Antifolate chemotherapy produced the first striking, although temporary, remissions in leukemia (Farber et al., 1948) and the first cure of a solid tumor, choriocarcinoma (Berlin et al., 1963). Interest in folate antagonists further increased with the development of curative combination therapy for childhood acute lymphocytic leukemia; in this therapy, MTX played a critical role in both systemic treatment and intrathecal therapy. Introduction of high-dose regimens with "rescue" of host toxicity by the reduced folate *leucovorin* (folinic acid, citrovorum factor, 5-formyl tetrahydrofolate, N^5-formyl FH_4) further extended the effectiveness of this drug to both systemic and CNS lymphomas, osteogenic sarcoma, and leukemias. Recognition that MTX, an inhibitor of dihydrofolate reductase (DHFR), also directly inhibits the folate-dependent enzymes of *de novo* purine synthesis and thymidylate synthesis led to development of antifolate analogues that specifically target these other folate-dependent enzymes. New congeners have greater capacity for transport into tumor cells (*pralatrexate*) and exert their primary inhibitory effect on TS (*raltitrexed*), early steps in purine biosynthesis (*lometrexol*), or both (the multitargeted antifolate *pemetrexed*).

Mechanism of Action

The primary target of folic acid analogues, such as MTX, is the enzyme DHFR (Figure 70–6). To function as a cofactor in one-carbon transfer reactions, folate must be reduced by DHFR to FH_4. Inhibitors such as MTX, with a high affinity for DHFR (K_i ~0.01–0.2 nM), cause partial depletion of N^{5-10} methylene FH_4 and N^{10} formyl FH_4 cofactors that are required for the synthesis of thymidylate and purines. In addition, MTX, like cellular folates, undergoes addition of a series of polyglutamates (MTX-PGs) in both normal and tumor cells (see Figure 70–5). These PGs

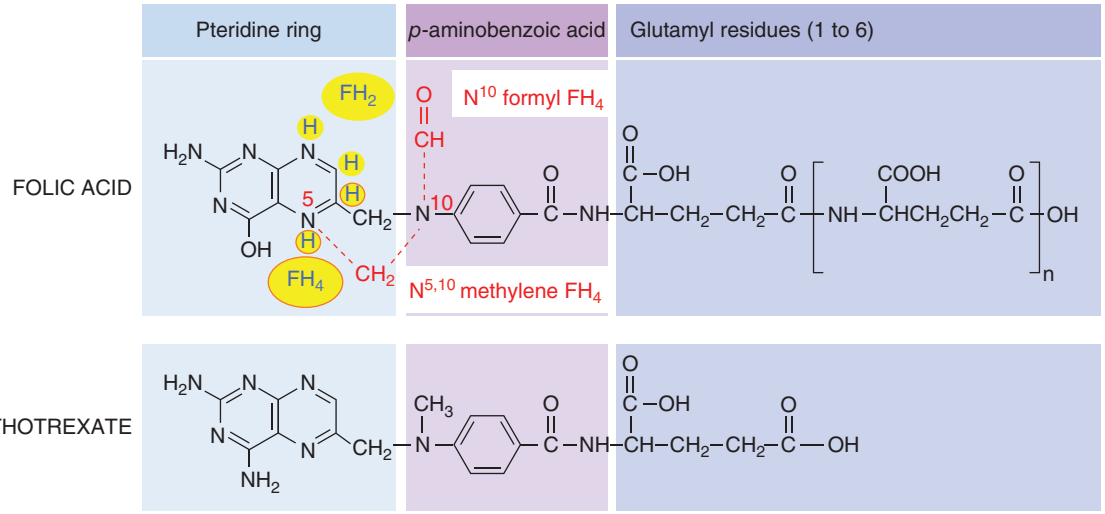


Figure 70–5 *Folic acid, metabolic intermediates, and MTX.* The shading identifies common structural features and areas of modification. Yellow highlights indicate sites modified by DHFR to FH_2 (DHF) and to FH_4 (THF, tetrahydrofolate); red atoms and bonds indicate metabolic intermediates that serve as single-carbon donors (see pathways in Figure 70–6).

constitute intracellular storage forms of folates and folate analogues that dramatically increase inhibitory potency of the analogue for additional targets, including TS and two early enzymes in the purine biosynthetic pathway. The FH_2 PGs that accumulate in cells behind the blocked DHFR reaction also act as inhibitors of TS and other enzymes (see Figure 70–6) (Allegra et al., 1987b).

Selective Toxicity; Rescue

As with most antimetabolites, MTX is only partially selective for tumor cells and kills rapidly dividing normal cells, such as those of the intestinal epithelium and bone marrow. Folate antagonists kill cells during the S phase of the cell cycle and are most effective when cells are proliferating rapidly. The toxic effects of MTX may be terminated by administering *leucovorin*, a fully reduced folate coenzyme that repletes the intracellular pool of FH_4 cofactors (see Figure 70–6). *Levoleucovorin,* approved in 2018, contains the active isomer and shows similar efficacy and adverse effects as racemic *leucovorin*.

Cellular Entry and Retention

Because folic acid and many of its analogues are polar, they cross the blood-brain barrier poorly and require specific transport mechanisms to enter mammalian cells. Three inward folate transport systems are found on mammalian cells:

1. A folate receptor, which has high affinity for folic acid but much lower ability to transport MTX and other analogues
2. The reduced folate transporter, the major transit protein for MTX, *raltitrexed*, *pemetrexed*, and most analogues
3. A transporter that is active at low pH

The reduced folate transporter is highly expressed in the hyperdiploid subtype of ALL, which has extreme sensitivity to MTX (Pui et al., 2004). Once in the cell, additional glutamyl residues are added to the molecule by the enzyme folylpolyglutamate synthase. Because these higher PGs are strongly charged and cross cellular membranes poorly, polyglutamation serves as a mechanism of ion trapping within

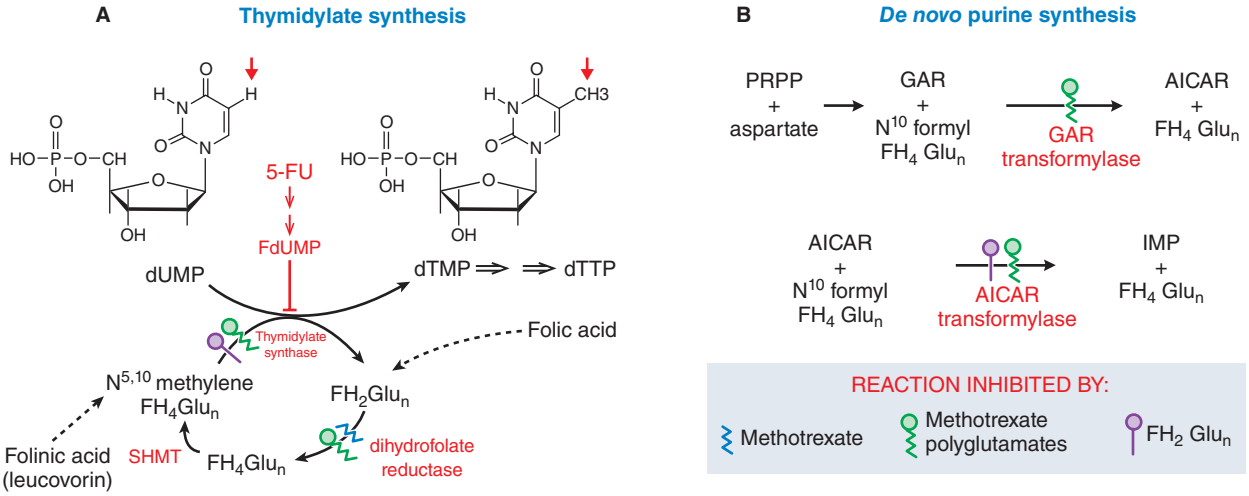


Figure 70–6 *Folate metabolism and the actions of 5FU, MTX, and MTX polyglutamates.* **A.** *Thymidylate synthesis* (for composition of folates, see Figure 70–5). **B.** *De novo purine synthesis.* GAR transformylase provides the C8 and AICAR transformylase the C2 in the purine ring biosynthesis (see Figure 70–1 for numbering of purine ring atoms). AICAR, aminoimidazole carboxamide ribonucleotide is an intermediate in the generation of inosine monophosphate (IMP); dUMP, deoxyuridine monophosphate; FH_2Glu_n, dihydrofolate polyglutamate; FH_4Glu_n, tetrahydrofolate polyglutamate; GAR, glycinamide ribonucleotide; IMP, inosine monophosphate; PRPP, 5-phosphoribosyl-1-pyrophosphate; TMP, thymidine monophosphate.

the cell. The polyglutamation may also account for the prolonged retention of MTX in chorionic epithelium (where it is a potent abortifacient); in tumors derived from this tissue, such as choriocarcinoma cells; and in normal tissues subject to cumulative drug toxicity, such as liver. Polyglutamyl folates and analogues have substantially greater affinity than the monoglutamate form for folate-dependent enzymes that are required for purine and thymidylate synthesis and have at least equal affinity for DHFR.

Newer Congeners

New folate antagonists that are better substrates for the reduced folate carrier have been identified. In efforts to bypass the obligatory membrane transport system and to facilitate penetration of the blood-brain barrier, lipid-soluble folate antagonists also have been synthesized. *Trimetrexate*, a lipid-soluble analogue that lacks a terminal glutamate, has modest antitumor activity, primarily in combination with *leucovorin* rescue. However, it is beneficial in the treatment of *Pneumocystis jirovecii* (*Pneumocystis carinii*) pneumonia, for which *leucovorin* provides differential rescue of the host but not the parasite (Allegra et al., 1987a). The folate analogue, MTA (multitargeted antifolate) or *pemetrexed*, is a pyrrole-pyrimidine structure. It is avidly transported into cells via the reduced folate carrier and is converted to PGs that inhibit TS and glycinamide ribonucleotide transformylase (GART), as well as DHFR (see Figure 70–6). It has activity against ovarian cancer, mesothelioma, and adenocarcinomas of the lung. *Pemetrexed* and its PGs have a somewhat different spectrum of biochemical actions. Like MTX, *pemetrexed* inhibits DHFR, but as a PG, it even more potently inhibits TS and GART. Unlike MTX, it produces little change in the pool of reduced folates, indicating that the distal sites of inhibition (TS and GART) predominate. Its pattern of deoxynucleotide depletion also differs; it causes a greater fall in thymidine triphosphate (TTP) than in other triphosphates. Like MTX, it induces p53 and cell-cycle arrest, but this effect does not depend on induction of p21. A newer congener, *pralatrexate*, is more effectively taken up and polyglutamated than MTX and is approved for treatment of CTCL.

Mechanisms of Resistance to Antifolates

Resistance to MTX can involve alterations in each known step in MTX action, including

- Impaired transport of MTX into cells
- Production of altered forms of DHFR that have decreased affinity for the inhibitor
- Increased concentrations of intracellular DHFR through gene amplification or altered gene regulation
- Decreased ability to synthesize MTX-PGs
- Increased expression of a drug efflux transporter of the MRP class (see Chapter 5)

The DHFR levels in leukemic cells increase within 24 h after treatment of patients with MTX, probably as a result of induction of DHFR synthesis. Unbound DHFR protein may bind to its own message and reduce its own translation, while the DHFR-MTX complex is ineffective in blocking DHFR translation. With longer periods of drug exposure, tumor cell populations emerge that contain markedly increased levels of DHFR. These cells contain multiple gene copies of DHFR either in mitotically unstable double-minute chromosomes (extrachromosomal elements) or in stably integrated, homogeneously staining chromosomal regions or amplicons (Schimke et al., 1978). Similar gene amplification target proteins have been implicated in the resistance to other antitumor agents, including 5FU and *pentostatin* (2′-deoxycoformycin), and observed in patients with lung cancer (Curt et al., 1985).

High doses of MTX may permit entry of the drug into transport-defective cells and may permit the intracellular accumulation of MTX in concentrations that inactivate high levels of DHFR. The understanding of resistance to *pemetrexed* is incomplete. In various cell lines, resistance seems to arise from loss of influx transport, TS amplification, changes in purine biosynthetic pathways, or loss of polyglutamation.

ADME

Methotrexate is readily absorbed from the GI tract at doses of less than 25 mg/m^2; larger doses are absorbed incompletely and are routinely administered intravenously. After intravenous administration, the drug disappears from plasma in a triphasic fashion. The rapid distribution phase is followed by a second phase, which reflects renal clearance ($t_{1/2}$ of about 2–3 h). A third phase has a $t_{1/2}$ of about 8 to 10 h. This terminal phase of disappearance, if prolonged by renal failure, may be responsible for major toxic effects of the drug on the bone marrow, GI epithelium, and skin. Distribution of MTX into body spaces, such as the pleural or peritoneal cavity, occurs slowly. However, if such spaces are expanded (e.g., by ascites or pleural effusion), they may act as a site of storage and slow release of the drug, resulting in prolonged elevation of plasma concentrations and more severe bone marrow toxicity.

Approximately 50% of MTX binds to plasma proteins and may be displaced from plasma albumin by myriad drugs, including sulfonamides, salicylates, *tetracycline*, *chloramphenicol*, and *phenytoin*; caution should be used if these drugs are given concomitantly. Up to 90% of a given dose of MTX is excreted unchanged in the urine, mostly within the first 8 to 12 h. Metabolism of MTX usually is minimal. After high doses, however, metabolites are readily detectable; these include 7-hydroxy-MTX, which is potentially nephrotoxic. Renal excretion of MTX occurs through a combination of glomerular filtration and active tubular secretion. Therefore, the concurrent use of drugs that reduce renal blood flow (e.g., nonsteroidal anti-inflammatory drugs), that are nephrotoxic (e.g., *cisplatin*), or that are weak organic acids (e.g., *aspirin*) can delay drug excretion and lead to severe myelosuppression. In patients with renal insufficiency, the dose should be adjusted in proportion to decreases in renal function, and high-dose regimens should be avoided. Concentrations of MTX in CSF are only 3% of those in the systemic circulation at steady state; hence, neoplastic cells in the CNS probably are not killed by standard dosage regimens. When high doses of MTX are given, cytotoxic concentrations of MTX reach the CNS beyond the blood-brain barrier. MTX is retained in the form of PGs for long periods (e.g., weeks in the kidneys, several months in the liver).

Pharmacogenetics may influence the response to antifolates and their toxicity. The C677T substitution in methylenetetrahydrofolate reductase reduces the activity of the enzyme that generates N^{5-10} methylene FH_4, the cofactor for TS, and thereby increases MTX toxicity (Pullarkat et al., 2001). The presence of this polymorphism in leukemic cells confers increased sensitivity to MTX and might also modulate the toxicity and therapeutic effect of *pemetrexed*, a TS inhibitor. Likewise, polymorphisms in the promoter region of TS affect its expression and, by altering the intracellular levels of TS, modulate the response and toxicity of both antifolates and fluoropyrimidines (Pui et al., 2004).

Therapeutic Uses

Methotrexate is a critical drug in the management of childhood ALL. High-dose MTX is of great value in remission induction and consolidation and in the maintenance of remissions in this highly curable disease. A 6- to 24-h infusion of relatively large doses of MTX may be employed every 2 to 4 weeks but only when *leucovorin* rescue follows within 24 h of the MTX infusion. For maintenance therapy, it is administered at a lower dose, orally every week. Outcome of treatment in children correlates inversely with the rate of drug clearance. During MTX infusion, high steady-state levels are associated with a lower leukemia relapse rate. MTX is of limited value in adults with AML, except for treatment and prevention of leukemic meningitis.

The intrathecal administration of MTX has been employed for treatment or prophylaxis of meningeal leukemia or lymphoma and for treatment of meningeal carcinomatosis. This route of administration achieves high concentrations of MTX in the CSF and also is effective in patients whose systemic disease has become resistant to MTX. The treatment is repeated every 4 days until malignant cells no longer are evident in the CSF. *Leucovorin* may be administered to counteract the potential toxicity of MTX that escapes into the systemic circulation, although this generally

is not necessary. Because MTX administered into the lumbar space distributes poorly over the cerebral convexities, the drug may be given via an intraventricular catheter system in the treatment of active intrathecal disease. MTX is of established value in choriocarcinoma and related trophoblastic tumors of women; cure is achieved in about 75% of advanced cases treated sequentially with MTX and *dactinomycin* and in more than 90% when early diagnosis is made. For choriocarcinoma, MTX is administered intramuscularly every other day for four doses, alternating with *leucovorin*. Courses are repeated at 3-week intervals, toxicity permitting, and urinary β-human chorionic gonadotropin titers are used as a guide for persistence of disease.

Beneficial effects also are observed in the combination therapy of Burkitt and other non-Hodgkin lymphomas. MTX is a component of regimens for carcinomas of the breast, head and neck, ovary, and bladder. High-dose MTX with *leucovorin* rescue (HDM-L) is a standard approach for adjuvant therapy of osteosarcoma and produces a high complete response rate in CNS lymphomas. The administration of HDM-L has the potential for renal toxicity, probably related to the precipitation of the drug, a weak acid, in the acidic tubular fluid. Thus, vigorous hydration and alkalinization of urine pH are required prior to drug administration. If MTX values measured 48 h after drug administration are 1 μM or higher, higher doses (100 mg/m^2) of *leucovorin* must be given until the plasma concentration of MTX falls to less than 50 nM. With appropriate hydration and urine alkalinization, and in patients with normal renal function, the incidence of nephrotoxicity following HDM-L is less than 2%. In patients who become oliguric, intermittent hemodialysis is ineffective in reducing MTX levels. Continuous-flow hemodialysis can eliminate MTX at about 50% of the clearance rate in patients with intact renal function. Alternatively, an MTX-cleaving enzyme, *glucarpidase*, is FDA approved for the treatment of MTX toxicity. *Glucarpidase* is a recombinant bacterial carboxypeptidase G2 that converts MTX into glutamate and 2,4-diamino-N(10)-methylpteroic acid. These metabolites are less toxic and are excreted by the liver. MTX concentrations in plasma fall by 99% or more within 5 to 15 min following enzyme administration. However, systemically administered carboxypeptidase G2 has little effect on MTX levels in the CSF.

Methotrexate is used in the treatment of severe, disabling psoriasis (see Chapter 75), given orally for 5 days, followed by a rest period of at least 2 days, or intravenously weekly. It also is used at low dosage to induce remission in refractory rheumatoid arthritis. MTX inhibits cell-mediated immune reactions and is employed to suppress graft-versus-host disease in allogeneic bone marrow and organ transplantation and for the treatment of dermatomyositis, Wegener granulomatosis, and Crohn's disease (see Chapters 39 and 55). MTX is also used as an abortifacient, generally in combination with a *prostaglandin* (see Chapter 48).

Adverse Effects

The primary toxicities of antifolates are on the bone marrow and the intestinal epithelium. Patients may be at risk for spontaneous hemorrhage or life-threatening infection and may require prophylactic transfusion of platelets and broad-spectrum antibiotics if febrile. Side effects usually reverse completely within 2 weeks, but prolonged myelosuppression may occur in patients with compromised renal function who have delayed drug excretion. The dosage of MTX (and likely *pemetrexed*) must be reduced in proportion to any reduction in CL_{Cr}. Additional toxicities of MTX include alopecia, dermatitis, an allergic interstitial pneumonitis, nephrotoxicity (after high-dose therapy), defective oogenesis or spermatogenesis, abortion, and teratogenesis. Low-dose MTX may lead to cirrhosis after long-term continuous treatment, as in patients with psoriasis. Intrathecal administration of MTX often causes meningismus and an inflammatory response in the CSF. Seizures, coma, and death may occur rarely. Note that *leucovorin* does not reverse neurotoxicity.

Pemetrexed toxicity mirrors that of MTX, with the additional feature of a prominent erythematous and pruritic rash in 40% of patients. *Dexamethasone*, 4 mg twice daily on days –1, 0, and +1, markedly diminishes this toxicity. Unpredictably severe myelosuppression with *pemetrexed*, seen especially in patients with preexisting homocystinemia, largely is eliminated by concurrent administration of low dosages of folic acid, 350 to 1000 mg/day, beginning 1 to 2 weeks prior to *pemetrexed* and continuing while the drug is administered. Patients should receive intramuscular vitamin B_{12} (1 mg) with the first dose of *pemetrexed* to correct possible B_{12} deficiency. These small doses of folate and B_{12} do not compromise the therapeutic effect.

Pyrimidine Analogues

The pyrimidine antimetabolites encompass a diverse group of drugs that inhibit RNA and DNA function. The fluoropyrimidines and certain purine analogues (*6-mercaptopurine* [6MP] and *6-thioguanine* [6TG]) inhibit the synthesis of essential precursors of DNA. Others, such as the cytidine and adenosine nucleoside analogues, become incorporated into DNA and block its further elongation and function (Wilson et al., 2014). Other inhibitory effects of these analogues may contribute to their cytotoxicity and their capacity to induce differentiation.

Cellular Actions of Pyrimidine Antimetabolites

Four bases (Figure 70–7) form DNA: two pyrimidines (thymine and cytosine) and two purines (guanine and adenine). RNA differs in that it incorporates uracil instead of thymine as one of its bases. Strategies for inhibiting DNA synthesis are based on the ability to create analogues of these precursors that readily enter tumor cells and become activated intracellularly. As an example, the pyrimidine analogue 5FU is converted to a fluorodeoxyuridine monophosphate (FdUMP), which in turn blocks TS, an enzyme required for the physiological conversion of deoxyuridine monophosphate (dUMP) to deoxythymidine monophosphate (dTMP),

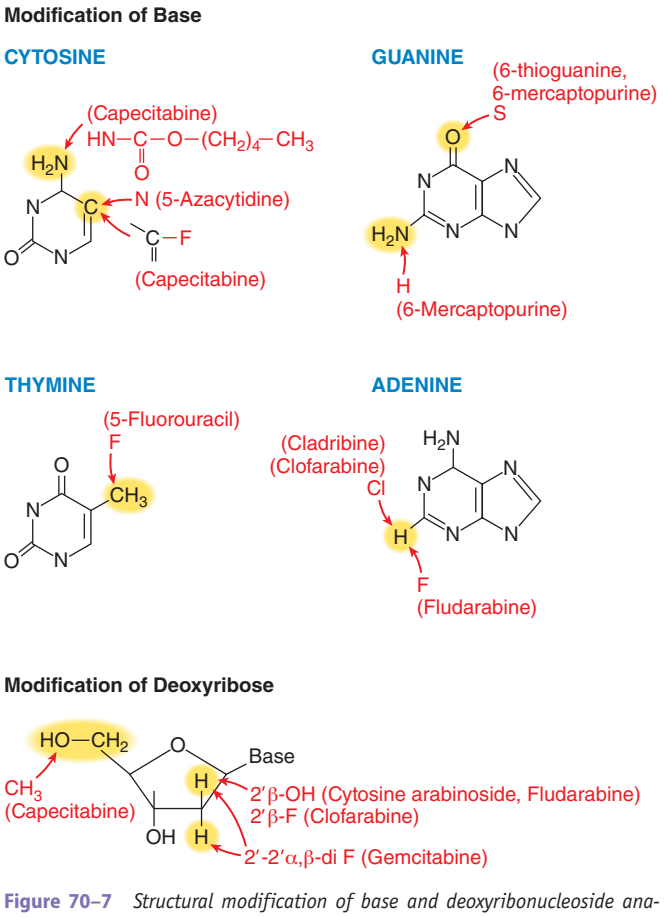


Figure 70–7 *Structural modification of base and deoxyribonucleoside analogues. Yellow ellipses* indicate sites modified to create antimetabolites. Specific substitutions are noted in red for each drug. Modifications occur in the base ring systems, in their amino or hydroxyl side groups, and in the deoxyribose sugar found in deoxyribonucleosides. See structures in Figure 70–8.

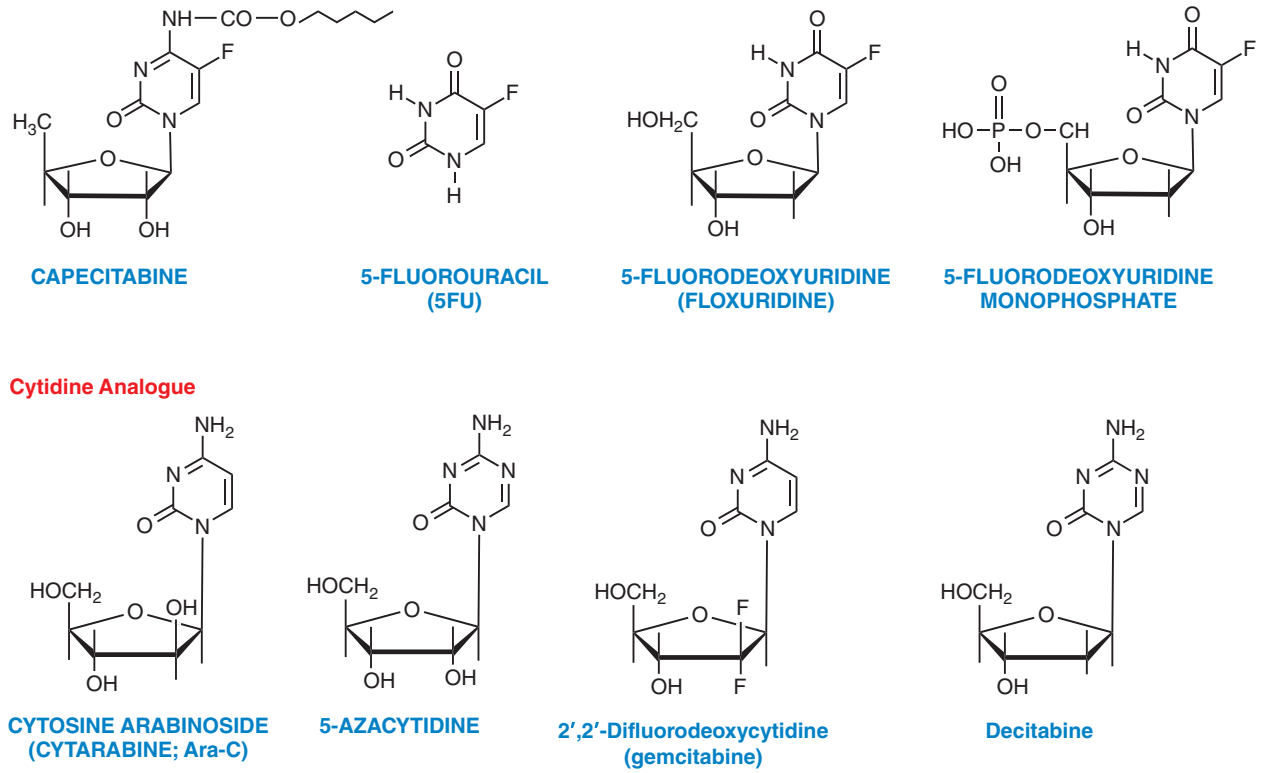


Figure 70–8 *Pyrimidine analogues.*

a component of DNA (see Figure 70–6). Other analogues incorporate into DNA itself and thereby block its function (Wilson et al., 2014).

Cells can make the purine and pyrimidine bases *de novo* and convert them to their active triphosphates (deoxynucleotide triphosphates [dNTPs]), providing substrates for DNA polymerase. Alternatively, cells can salvage free bases or their deoxynucleosides from the bloodstream. Thus, cells can take up uracil, guanine, and their analogues and convert them to (deoxy)nucleotides by the addition of deoxyribose and phosphate groups. Antitumor analogues of these bases (5FU, 6TG) can be formulated as simple substituted bases. Other bases, including cytosine, thymine, and adenine and their analogues can only be used as deoxynucleosides, which are readily transported into cells and activated to deoxynucleotides by intracellular kinase. Thus, *cytarabine* (cytosine arabinoside [Ara-C]), *gemcitabine*, *5-azacytidine*, and adenosine analogues (*cladribine*) (Figures 70–7 and 70–8) are nucleosides readily taken up by cells, converted to nucleotides, and incorporated into DNA.

Fludarabine phosphate, a nucleotide, is dephosphorylated rapidly in plasma, releasing the nucleoside that is readily taken up by cells. Analogues may differ from the physiological bases in a variety of ways: by altering the purine or pyrimidine ring; by altering the sugar attached to the base, as in the arabinoside, Ara-C; or by altering both the base and sugar, as in *fludarabine phosphate* (see Figure 70–7). These alterations produce inhibitory effects on vital enzymatic pathways and prevent DNA synthesis.

Fluorouracil, Floxuridine, Capecitabine

Fluorouracil is available as 5FU, as the derivative fluorodeoxyuridine (FUdR) (not often used in clinical practice), and as a prodrug, *capecitabine*, which is ultimately converted to 5FU.

Mechanisms of Action

5-Fluorouracil requires enzymatic conversion (ribosylation and phosphorylation) to the nucleotide form to exert its cytotoxic activity. As the triphosphate 5-fluorouridine triphosphate (FUTP), the drug is incorporated into RNA. Alternative reactions can produce the deoxy derivative FdUMP; FdUMP inhibits TS and blocks the synthesis of deoxythymidine triphosphate (dTTP), a necessary constituent of DNA (see Figure 70–6). The folate cofactor, 5,10-methylene FH_4, and FdUMP form a covalently bound ternary complex with TS. The physiological complex of TS-folate-dUMP progresses to the synthesis of thymidylate by transfer of the methylene group and two hydrogen atoms from folate to dUMP, but this reaction is blocked in the inhibited complex of TS-FdUMP-folate by the stability of the fluorine carbon bond on FdUMP; sustained inhibition of the enzyme results.

5-Fluorouracil is incorporated into both RNA and DNA. In 5FU-treated cells, both fluorodeoxyuridine triphosphate (FdUTP) and deoxyuridine triphosphate (dUTP; which accumulates behind the blocked TS reaction) incorporate into DNA in place of the depleted physiological TTP. Presumably, such incorporation into DNA calls into action the excision-repair process, which can lead to DNA strand breakage because DNA repair requires TTP, which is lacking as a result of TS inhibition. 5FU incorporation into RNA also causes toxicity as the result of major effects on both the processing and the functions of RNA.

Mechanisms of Resistance

Resistance to the cytotoxic effects of 5FU or FUdR has been ascribed to loss or decreased activity of the enzymes necessary for activation of 5FU, amplification of TS, mutation of TS to a form that is not inhibited by FdUMP, and high levels of the degradative enzymes dihydrouracil dehydrogenase and thymidine phosphorylase. TS levels are finely controlled by an autoregulatory feedback mechanism wherein the unbound enzyme interacts with and inhibits the translational efficiency of its own mRNA, which provides for the rapid TS modulation needed for cellular division. When TS is bound to FdUMP, inhibition of translation is relieved and levels of free TS rise, restoring thymidylate synthesis. Thus, TS autoregulation may be an important mechanism by which malignant cells become insensitive to the effects of 5FU.

Combination with Leucovorin

Some malignant cells appear to have insufficient concentrations of 5,10-methylene FH_4 and thus cannot form maximal levels of the inhibited

ternary complex with TS. Addition of exogenous folate in the form of *leucovorin* increases formation of the complex and enhances responses to 5FU. A number of other agents have been combined with 5FU in attempts to enhance the cytotoxic activity through biochemical modulation. MTX, by inhibiting purine synthesis and increasing cellular pools of 5-phosphoribosyl-1-pyrophosphate (PRPP), enhances the activation of 5FU and increases antitumor activity of 5FU when given prior to but not following 5FU. The combination of *cisplatin* and 5FU has yielded impressive responses in tumors of the upper aerodigestive tract, but the molecular basis of their interaction is unclear. A combination with *oxaliplatin*, which downregulates TS expression, is commonly used with 5FU and *leucovorin* for treating patients with metastatic colorectal cancer; the combination is abbreviated as FOLFOX. Addition of *irinotecan* (see discussion that follows) is abbreviated as FOLFIRINOX and used in the treatment of patients with colorectal or pancreatic cancer. A most important interaction is the enhancement of radiation therapy by fluoropyrimidines, the basis for which is unclear. 5FU with simultaneous radiation is effective in patients with anal cancer and enhances local tumor control in patients with head, neck, cervical, rectal, gastroesophageal, and pancreatic cancer.

ADME

5-Fluorouracil is administered parenterally because absorption after oral ingestion of the drug is unpredictable and incomplete. 5FU is inactivated by reduction of the pyrimidine ring in a reaction carried out by dihydropyrimidine dehydrogenase (DPD), which is found in liver, intestinal mucosa, tumor cells, and other tissues. Inherited deficiency of this enzyme leads to greatly increased sensitivity to the drug (Milano et al., 1999). DPD deficiency can be detected by either enzymatic or molecular assays using peripheral white blood cells or by determining the plasma ratio of 5FU to its metabolite, 5-fluoro-5,6-dihydrouracil.

Plasma clearance is rapid ($t_{1/2}$ about 10–20 min). Only 5% to 10% of a single intravenous dose of 5FU is excreted intact in the urine. The dose does not have to be modified in patients with hepatic dysfunction, presumably because of sufficient degradation of the drug at extrahepatic sites. 5FU enters the CSF in minimal amounts.

Therapeutic Uses

5-Fluorouracil. *5-Fluorouracil* produces partial responses in 10% to 20% of patients with metastatic colon carcinomas, upper GI tract carcinomas, and breast carcinomas but rarely is used as a single agent. 5FU in combination with *leucovorin* and *oxaliplatin* or *irinotecan* (FOLFOX or FOLFIRINOX) in adjuvant therapy is associated with a survival advantage for patients with colorectal cancers. For average-risk patients in good nutritional status with adequate hematopoietic function, the dosage regimen employs *leucovorin* once each week for 6 of 8 weeks. Other regimens use daily doses for 5 days, repeated in monthly cycles. When used with *leucovorin*, doses of daily 5FU for 5 days must be reduced because of mucositis and diarrhea. 5FU increasingly is used as a biweekly infusion, a schedule that has less overall toxicity as well as superior response rates and progression-free survival for patients with metastatic colon cancer. Also, topical application of 5FU is effective in the treatment of premalignant keratoses of the skin and multiple superficial basal cell carcinomas.

Floxuridine. Fluorodeoxyuridine is converted directly to FdUMP by thymidine kinase. The drug is administered primarily by continuous infusion into the hepatic artery for treatment of patients with metastatic carcinoma of the colon or following resection of hepatic metastases; the response rate of intrahepatic infusion (40%–50%) is twice that of intravenous administration. Intrahepatic arterial infusion for 14 to 21 days causes minimal systemic toxicity; however, there is a significant risk of biliary sclerosis if this route is used for multiple cycles of therapy. Treatment should be discontinued at the earliest manifestation of toxicity (usually stomatitis or diarrhea) because the maximal effects of bone marrow suppression and gut toxicity will not be evident until days 7 to 14.

Capecitabine. *Capecitabine*, an orally administered prodrug of 5FU, is approved for the treatment of patients with (1) metastatic breast cancer who have not responded to a regimen of *paclitaxel* and an anthracycline; (2) metastatic breast cancer when used in combination with *docetaxel* in patients who have had a prior anthracycline-containing regimen; and (3) metastatic colorectal cancer.

The recommended dosage is given in two divided doses with food, for 2 weeks, followed by a rest period of 1 week. *Capecitabine* is well absorbed orally. It is rapidly deesterified and deaminated, yielding high plasma concentrations of an inactive prodrug 5′-deoxyfluorodeoxyuridine (5′-dFdU), which disappears with a $t_{1/2}$ of about 1 h. The conversion of 5′-dFdU to 5FU by thymidine phosphorylase occurs in liver tissues, peripheral tissues, and tumors. 5FU levels are less than 10% of those of 5′-dFdU, reaching a maximum of 0.3 mg/L or 1 μM at 2 h. Liver dysfunction delays the conversion of the parent compound to 5′-dFdU and 5FU, but there is no consistent effect on toxicity.

Combination Therapy. Higher response rates are seen when 5FU or *capecitabine* is used in combination with other agents (e.g., with *cisplatin* in head and neck cancer, with *oxaliplatin* or *irinotecan* in colon cancer). The combination of 5FU and *oxaliplatin* or *irinotecan* has become the standard first-line treatment of patients with metastatic colorectal cancer (FOLFOX and FOLFIRINOX). The use of 5FU in combination regimens has improved survival in the adjuvant treatment of breast cancer and, with *oxaliplatin* and *leucovorin*, of colorectal cancer. 5FU also is a potent radiation sensitizer. Beneficial effects have been reported when combined with irradiation as primary treatment of patients with locally advanced cancers of the esophagus, stomach, pancreas, cervix, anus, and head and neck.

Adverse Effects

The clinical manifestations of toxicity caused by 5FU and *floxuridine* are similar. The earliest untoward symptoms during a course of therapy are anorexia and nausea, followed by stomatitis and diarrhea, reliable warning signs that a sufficient dose has been administered. Mucosal ulcerations occur throughout the GI tract and may lead to fulminant diarrhea, shock, and death, particularly in patients who are DPD deficient. The major toxic effects of bolus-dose regimens result from the myelosuppressive action of 5FU. The nadir of leukopenia usually occurs 9 to 14 days after the first injection of drug. Thrombocytopenia and anemia also may occur, as may loss of hair (occasionally progressing to total alopecia), nail changes, dermatitis, and increased pigmentation and atrophy of the skin. Hand-foot syndrome, a particularly prominent adverse effect of *capecitabine*, consists of erythema, desquamation, pain, and sensitivity to touch of the palms and soles. Acute chest pain with evidence of ischemia in the electrocardiogram may result from coronary artery vasospasms during or shortly after 5FU infusion. In general, myelosuppression, mucositis, and diarrhea occur less often with infusional than with bolus regimens, while hand-foot syndrome occurs more often with infusional than with bolus regimens. The significant risk of toxicity with fluoropyrimidines requires close supervision by physicians familiar with the drug's effects and possible hazards.

Capecitabine causes a similar spectrum of toxicities as 5FU (diarrhea, myelosuppression), but the hand-foot syndrome occurs more frequently and may require dose reduction or cessation of therapy.

Trifluridine

Trifluridine is the 5-trifluoromethyl pyrimidine nucleoside analogue of 5-FUdR (*floxuridine*; Figure 70–8). It is used in eye drops for the treatment of herpes simplex virus (see Table 74–5)) and in a fixed combination with *tipiracil*, a thymidine phosphorylase inhibitor, for the treatment of patients with metastatic colorectal cancers who have previously received other standard combination treatments that included FOLFIRINOX. *Trifluridine* and *tipiracil* are formulated together in a single tablet with *tipiracil* added to prevent rapid breakdown of *trifluridine*. The *trifluridine* mechanism of action mimics that of 5FU (Davidson et al., 2016).

Cytidine Analogues

Cytarabine (Cytosine Arabinoside; Ara-C)

Cytarabine is the most important antimetabolite used in the therapy of AML; it is the single most effective agent for induction of remission in this disease.

Mechanisms of Action

Cytarabine is an analogue of 2′-deoxycytidine; the 2′-hydroxyl in a position *trans* to the 3′-hydroxyl of the sugar (see Figures 70–7 and 70–8) hinders rotation of the pyrimidine base around the nucleoside bond and interferes with base pairing. The drug enters cells via ENT1 (SLC29A1). It is then converted to its active form, the 5′-monophosphate ribonucleotide, by deoxycytidine kinase (dCK), an enzyme that shows polymorphic expression among patients (see discussion that follows). Ara-CMP then reacts with deoxynucleotide kinases to form diphosphate and triphosphates (Ara-CDP and Ara-CTP). Ara-CTP competes with deoxycytidine triphosphate (dCTP) for incorporation into DNA by DNA polymerases. The incorporated Ara-CMP residue is a potent inhibitor of DNA polymerase, in both replication and repair synthesis, and blocks the further elongation of the nascent DNA molecule. If DNA breaks are not repaired, apoptosis ensues. Ara-C cytotoxicity correlates with the total Ara-C incorporated into DNA; incorporation of about 5 molecules of Ara-C per 10^4 bases of DNA decreases cellular clonogenicity by about 50% (Kufe et al., 1984).

In infants and adults with ALL and t(4;11) MLL translocation, usefulness of high-dose Ara-C is mediated by the highly expressed nucleoside transporter ENT1, and its expression correlates with sensitivity to Ara-C (Pui et al., 2004). At extracellular drug concentrations greater than 10 μM (levels achievable with high-dose Ara-C), the nucleoside transporter no longer limits drug accumulation, and intracellular metabolism to a triphosphate becomes rate limiting. Patients with particular subtypes of AML derive benefit from high-dose Ara-C treatment; these types include t(8;21), inv(16), t(9;16), and del(16).

Mechanisms of Resistance

Response to Ara-C is strongly influenced by the relative activities of anabolic and catabolic enzymes that determine the proportion of drug converted to Ara-CTP. The rate-limiting activating enzyme dCK produces Ara-CMP. It is opposed by the degradative enzyme, cytidine deaminase, which converts Ara-C to a nontoxic metabolite, ara-uridine (Ara-U). Cytidine deaminase activity is high in many normal tissues, including intestinal mucosa, liver, and neutrophils, but lower in AML cells and other human tumors. A second degradative enzyme, dCMP deaminase, converts Ara-CMP to the inactive metabolite Ara-UMP. Increased synthesis and retention of Ara-CTP in leukemic cells lead to a longer duration of complete remission in patients with AML. The capacity of cells to transport Ara-C also may affect response. Clinical studies have implicated a loss of dCK as the primary mechanism of resistance to Ara-C in AML.

ADME

Due to the presence of high concentrations of cytidine deaminase in the GI mucosa and liver, only about 20% of the drug reaches the circulation after *oral* Ara-C administration; thus, the drug must be given intravenously. Peak concentrations of 2 to 50 μM are measurable in plasma after intravenous injection of 30 to 300 mg/m^2 but fall rapidly ($t_{1/2} \approx 10$ min). Less than 10% of the injected dose is excreted unchanged in the urine within 12 to 24 h; most appears as the inactive deaminated product, Ara-U. Higher concentrations of Ara-C are found in CSF after continuous infusion than after rapid intravenous injection but are 10% or less of concentrations in plasma. After *intrathecal* administration of the drug at a dose of 50 mg/m^2, deamination proceeds slowly, with a $t_{1/2}$ of 3 to 4 h, and peak concentrations of 1 to 2 μM are achieved. CSF concentrations remain above the threshold for cytotoxicity (0.4 μM) for 24 h or longer. A depot liposomal formulation of Ara-C provides sustained release into the CSF. After a standard dose, liposomal Ara-C remains above cytotoxic levels for an average of 12 days, thus avoiding the need for frequent lumbar punctures.

Therapeutic Uses

Continuous inhibition of DNA synthesis for a duration equivalent to at least one cell cycle or 24 h is necessary to expose most tumor cells during the S phase of the cell cycle. The optimal interval between bolus doses of Ara-C is about 8 to 12 h, a schedule that maintains intracellular concentrations of Ara-CTP at inhibitory levels during a multiday cycle of treatment. In general, children tolerate higher doses than adults. The intrathecal administration of liposomal Ara-C every 2 weeks seems equally effective as the regimen every 4 days with the standard drug. Ara-C is indicated for induction and maintenance of remission in AML and is useful in the treatment of patients with other leukemias, such as acute lymphoblastic leukemia (ALL), CML in the blast phase, acute promyelocytic leukemia (APL), and high-grade lymphomas. Because drug concentration in plasma rapidly falls below the level needed to saturate transport and intracellular activation, clinicians have employed high-dose regimens every 12 h for 3 to 4 days to achieve 20 to 50 times higher serum levels, with improved results in remission induction and consolidation for AML. Injection of the liposomal formulation is indicated for the intrathecal treatment of lymphomatous meningitis.

Adverse Effects

Cytarabine is myelosuppressive and can produce acute, severe leukopenia, thrombocytopenia, and anemia with striking megaloblastic changes. Other toxic manifestations include GI disturbances, stomatitis, conjunctivitis, reversible hepatic enzyme elevations, noncardiogenic pulmonary edema, and dermatitis. The onset of dyspnea, fever, and pulmonary infiltrates on chest computed tomographic scans may follow 1 to 2 weeks after high-dose Ara-C and may be fatal in 10% to 20% of patients, especially in patients being treated for relapsed leukemia. No specific therapy, other than Ara-C discontinuation, is indicated. Intrathecal Ara-C, either the free drug or the liposomal preparation, may cause arachnoiditis, seizures, delirium, myelopathy, or coma, especially if given concomitantly with systemic high-dose MTX or systemic Ara-C. Cerebellar toxicity, manifesting as ataxia and slurred speech, and cerebral toxicity (seizures, dementia, and coma) may follow intrathecal administration or high-dose systemic administration, especially in patients older than 50 years or patients with poor renal function.

Azacitidine (5-Azacytidine); Decitabine

5-Azacytidine and *decitabine* (2′-deoxy-5-azacytidine; see Figure 70–8) have antileukemic activity as cytotoxic agents and particularly at lower doses induce leukemic cell differentiation by inhibiting DNA cytosine methyltransferase activity. Both drugs are approved for treatment of myelodysplasia, for which they induce normalization of bone marrow in 15% to 20% of patients and reduce the transfusion requirement in one-third of patients. *5-Azacytidine* improves survival.

Mechanism of Action

The aza-nucleosides enter cells by ENT1 (SLC29A1). The drugs incorporate into DNA, where they become covalently bound to the DNA methyltransferase, depleting intracellular enzyme and leading to global demethylation of DNA that results in tumor cell differentiation and apoptosis. *Decitabine* also induces double-strand DNA breaks, perhaps as a consequence of the effort to repair the protein-DNA adduct.

ADME

After subcutaneous administration, *5-azacytidine* undergoes rapid deamination by cytidine deaminase (plasma $t_{1/2}$ ~20–40 min). Due to the formation of intracellular nucleotides that become incorporated into DNA, the effects of the aza-nucleosides persist for many hours.

Therapeutic Use

The usual treatment regimen for *5-azacytidine* in patients with myelodysplastic syndrome is daily for 7 days every 28 days, while *decitabine* is given intravenously every day for 5 days every 4 weeks. Best responses may become apparent only after two to five courses of treatment.

Adverse Effects

The major toxicities of the aza-nucleosides include myelosuppression and mild GI symptoms. *5-Azacytidine* produces severe nausea and vomiting when given intravenously in large doses (150–200 mg/m^2 per day for 5 days).

Gemcitabine

Gemcitabine, a difluoro analogue of *deoxycytidine* (dFdC; see Figure 70–8), is used in the treatment of patients with pancreatic; nonsquamous, non–small cell lung; ovarian; breast; and bladder cancer.

Mechanism of Action

Gemcitabine enters cells via three distinct nucleoside transporters: ENT1 (SLC29A1; the major route), CNT1 (SLC28A1), and a nucleobase transporter found in malignant mesothelioma cells. Intracellularly, dCK phosphorylates *gemcitabine* to the monophosphate (dFdCMP), which is converted to di- and triphosphates (dFdCDP and dFdCTP, respectively). Although *gemcitabine*'s anabolism and effects on DNA in general mimic those of *cytarabine*, there are distinct differences in kinetics of inhibition, additional enzymatic sites of action, different effects of incorporation into DNA, and a distinct spectrum of clinical activity. Unlike that of *cytarabine*, the cytotoxicity of *gemcitabine* is not confined to the S phase of the cell cycle. The cytotoxic activity may reflect several actions on DNA synthesis. dFdCTP competes with dCTP as a weak inhibitor of DNA polymerase. dFdCDP is a stoichiometric inhibitor of ribonucleoside diphosphate reductase (RNR), resulting in depletion of deoxyribonucleotide pools necessary for DNA synthesis. Incorporation of dFdCTP into DNA causes DNA strand termination (Heinemann et al., 1988) and appears resistant to repair. The capacity of cells to incorporate dFdCTP into DNA is critical for *gemcitabine*-induced apoptosis. *Gemcitabine* is inactivated by cytidine deaminase, which is found both in tumor cells and throughout the body.

ADME

Gemcitabine is administered as an intravenous infusion. The pharmacokinetics of the parent compound are largely determined by deamination in liver, plasma, and other organs, and the predominant urinary elimination product is 2′,2′-difluorodeoxycytidine (dFdU). In patients with significant renal dysfunction, dFdU and its triphosphate accumulate to high and potentially toxic levels. *Gemcitabine* has a short plasma $t_{1/2}$ (~15 min); women and elderly patients clear the drug more slowly.

Therapeutic Uses

The standard dosing schedule for *gemcitabine* is an intravenous infusion on days 1, 8, and 15 of each 21- to 28-day cycle, depending on the indication. Conversion of *gemcitabine* to dFdCMP by dCK is saturated at infusion rates of about 10 mg/m^2/min. To increase dFdCTP formation, the duration of infusion at this maximum concentration has been extended to 100 to 150 min at a fixed rate of 10 mg/min. The 150-min infusion produces a higher level of dFdCTP within peripheral blood mononuclear cells and increases the degree of myelosuppression. The inhibition of DNA repair by *gemcitabine* may increase cytotoxicity of other agents, particularly platinum compounds, and with radiation therapy.

Adverse Effects

The principal toxicity is myelosuppression. Longer duration infusions lead to greater myelosuppression and hepatic toxicity. Nonhematological toxicities include a flu-like syndrome, asthenia, and rarely a posterior leukoencephalopathy syndrome. Mild, reversible elevation in liver transaminases may occur in 40% or more of patients. Interstitial pneumonitis, at times progressing to acute respiratory distress syndrome, may occur within the first two cycles of treatment and usually responds to corticosteroids. Rarely, patients treated for many months may develop a slowly progressive hemolytic uremic syndrome, necessitating drug discontinuation. *Gemcitabine* is a very potent radiosensitizer and should not be used with radiotherapy.

Purine Analogues

The pioneering studies of Hitchings and Elion identified analogues of naturally occurring purine bases with antileukemic and immunosuppressant properties. Figure 70–9 shows structural formulas of several purine analogues, with *adenosine* for comparison. Other purine analogues that have valuable roles in leukemia and lymphoid malignancies include *cladribine* (standard therapy of hairy cell leukemia), *fludarabine phosphate* (standard treatment of CLL), *nelarabine* (pediatric ALL), and *clofarabine* (T-cell leukemia/lymphoma). The apparent selectivity of these agents may relate to their effective uptake, activation, and apoptotic effects in lymphoid tissue.

6-Thiopurine Analogues

6-Mercaptopurine and 6TG are approved agents for human leukemias and function as analogues of the natural purines hypoxanthine and guanine. The substitution of sulfur for oxygen on C6 of the purine ring creates compounds that are readily transported into cells, including activated malignant cells. Nucleotides formed from 6MP and 6TG inhibit *de novo* purine synthesis and also become incorporated into nucleic acids (see Figure 55–5).

Mechanism of Action

Hypoxanthine guanine phosphoribosyl transferase converts 6TG and 6MP to the ribonucleotides 6-thioGMP and 6-thioIMP (T-IMP), respectively. Because T-IMP is a poor substrate for guanylyl kinase (the enzyme that converts GMP to GDP), T-IMP accumulates intracellularly. T-IMP inhibits the new formation of ribosyl-5-phosphate, as well as conversion of IMP to adenine and guanine nucleotides. The most important point of attack seems to be the reaction of glutamine and PRPP to form ribosyl-5-phosphate, the first committed step in the *de novo* pathway. 6TG nucleotide is incorporated into DNA, where it induces strand breaks and base mispairing.

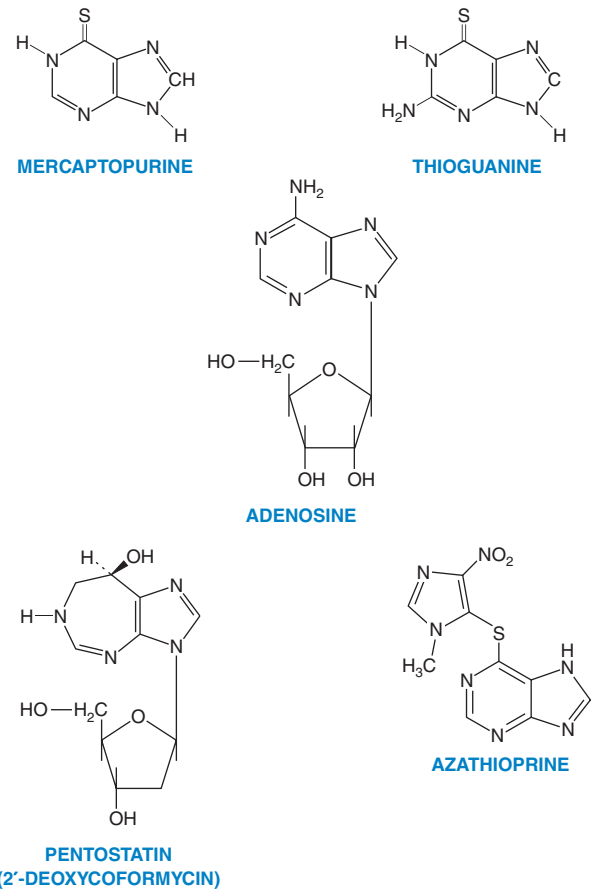


Figure 70–9 *Adenosine and various purine analogues.*

Mechanisms of Resistance

The most common mechanism of 6MP resistance observed *in vitro* is deficiency or complete lack of the activating enzyme HGPRT or increased alkaline phosphatase activity. Other mechanisms for resistance include:

- Decreased drug uptake or increased efflux due to active transporters
- Alteration in allosteric inhibition of ribosylamine 5-phosphate synthase
- Impaired recognition of DNA breaks and mismatches due to loss of a component (MSH6, mutator S homologue 6) of the MMR system (Karran and Attard, 2008)

ADME and Toxicity

Absorption of oral *mercaptopurine* is incomplete (10%–50%); the drug is subject to first-pass metabolism by xanthine oxidase in the liver. Food or oral antibiotics decrease absorption. Oral bioavailability is increased when *mercaptopurine* is combined with high-dose MTX. After an intravenous dose, the $t_{1/2}$ of the drug is about 50 min in adults due to rapid metabolic degradation by xanthine oxidase and by thiopurine methyltransferase (TPMT). Restricted brain distribution of *mercaptopurine* results from an efficient efflux transport system in the blood-brain barrier. In addition to the HGPRT-catalyzed anabolism of *mercaptopurine*, there are two other pathways for its metabolism. The first involves methylation of the sulfhydryl group and subsequent oxidation of the methylated derivatives. Activity of the enzyme TPMT reflects the inheritance of polymorphic alleles; up to 15% of the Caucasian population has decreased enzyme activity. Low levels of erythrocyte TPMT activity are associated with increased drug toxicity in individual patients and a lower risk of relapse. In patients with autoimmune disease treated with *mercaptopurine*, those with polymorphic alleles may experience bone marrow aplasia and life-threatening toxicity. Testing for these polymorphisms prior to treatment is recommended in this patient population.

A relatively large percentage of the administered sulfur appears in the urine as inorganic sulfate. The second major pathway for 6MP metabolism involves its oxidation by xanthine oxidase to 6-thiourate, an inactive metabolite. Oral doses of 6MP should be reduced by 75% in patients receiving the xanthine oxidase inhibitor *allopurinol*; no dose adjustment is required for intravenous dosing.

Therapeutic Uses

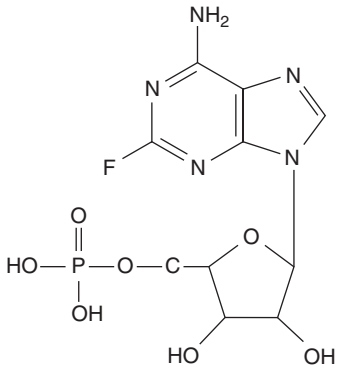
In the maintenance therapy of ALL, an initial daily oral dose of 6MP is adjusted according to white blood cell and platelet counts. The combination of MTX and 6MP appears to be synergistic. By inhibiting the earliest steps in purine synthesis, MTX elevates the intracellular concentration of PRPP, a cofactor required for 6MP activation.

Adverse Effects

The principal toxicity of 6MP is myelosuppression. Thrombocytopenia, granulocytopenia, or anemia may not become apparent for several weeks. Dose reduction usually results in prompt recovery, although myelosuppression may be severe and prolonged in patients with a polymorphism affecting TPMT. Anorexia, nausea, or vomiting is seen in about 25% of adults, but stomatitis and diarrhea are rare; manifestations of GI effects are less frequent in children than in adults. Jaundice and hepatic enzyme elevations occur in up to one-third of adult patients treated with 6MP and usually resolve on discontinuation of therapy. 6MP and its derivative *azathioprine* predispose to opportunistic infection (e.g., reactivation of hepatitis B, fungal infection, and *Pneumocystis* pneumonia) and an increased incidence of squamous cell malignancies of the skin. 6MP is teratogenic during the first trimester of pregnancy, and AML has been reported after prolonged 6MP therapy for Crohn's disease.

Fludarabine Phosphate

Fludarabine phosphate is a fluorinated, deamination-resistant, phosphorylated analogue of the antiviral agent *vidarabine* (9-β-D-arabinofuranosyl-adenine). It is active in CLL and low-grade lymphomas and is also effective as a potent immunosuppressant.



FLUDARABINE PHOSPHATE

Mechanisms of Action and Resistance

The drug is dephosphorylated extracellularly to the nucleoside *fludarabine*, which enters the cell and is rephosphorylated by dCK to the active triphosphate. This antimetabolite inhibits DNA polymerase, DNA primase, DNA ligase, and RNR and becomes incorporated into DNA and RNA. The nucleotide is an effective chain terminator when incorporated into DNA (Kamiya et al., 1996). Incorporation of *fludarabine* into RNA inhibits RNA function, RNA processing, and mRNA translation.

In experimental tumors, resistance to *fludarabine* is associated with decreased activity of dCK (the enzyme that phosphorylates the drug), increased drug efflux, and increased RNR activity. Its mechanism of immunosuppression and paradoxical stimulation of autoimmunity stems from the particular susceptibility of lymphoid cells to purine analogues and the specific effects on the $CD4^+$ subset of T cells, as well as its inhibition of regulatory T-cell responses.

ADME

Fludarabine phosphate is administered both intravenously and orally and is rapidly converted to *fludarabine* in the plasma. The median time to reach maximal concentrations of drug in plasma after oral administration is 1.5 h, and oral bioavailability averages 55% to 60%. The $t_{1/2}$ of *fludarabine* in plasma is about 10 h. The compound is eliminated primarily by renal excretion.

Therapeutic Uses

Fludarabine phosphate is approved for intravenous and oral use and is equally active by both routes. The recommended dose interval is daily for 5 days and may be repeated every 4 weeks. Gradual improvement in CLL usually occurs within two to three cycles. Dosage should be reduced in patients with renal impairment in proportion to the reduction in CL_{Cr}. *Fludarabine phosphate* is highly active alone or with *rituximab* and *cyclophosphamide* for the treatment of patients with CLL; overall response rates in previously untreated patients approximate 80%, and the duration of response averages 22 months. The synergy of *fludarabine* with alkylators may stem from the observation that it blocks the repair of double-strand DNA breaks and interstrand cross-links induced by alkylating agents. It also is effective in follicular B-cell lymphomas refractory to standard therapy. It is increasingly used as a potent immunosuppressive agent in nonmyeloablative allogeneic bone marrow transplantation.

Adverse Effects

Oral and intravenous therapy cause myelosuppression in about 50% of patients, nausea and vomiting in a minor fraction, and, uncommonly, chills and fever, malaise, anorexia, peripheral neuropathy, and weakness. Lymphopenia and thrombocytopenia and cumulative side effects are expected. Depletion of $CD4^+$ T cells with therapy predisposes to opportunistic infections. Tumor lysis syndrome, a rare complication, occurs primarily in previously untreated patients with CLL. Altered mental status, seizures, optic neuritis, and coma have been observed at higher doses and in older patients. Autoimmune events may occur after *fludarabine* treatment. Patients with CLL may develop an acute hemolytic anemia or pure red cell aplasia during or following *fludarabine* treatment. Prolonged cytopenias probably mediated by autoimmunity, also complicate

fludarabine treatment. Myelodysplasia and acute leukemias may arise as late complications. Pneumonitis is an occasional side effect and responds to corticosteroids. In patients with compromised renal function, the initial doses should be reduced in proportion to the reduction in CL_{Cr}.

Cladribine

Cladribine is an adenosine deaminase (ADA)-resistant purine analogue that has potent and curative activity in hairy cell leukemia, CLL, and low-grade lymphomas.

Mechanisms of Action and Resistance

Cladribine enters cells via active nucleoside transport. After phosphorylation by dCK and conversion to cladribine triphosphate, it is incorporated into DNA. It produces DNA strand breaks and depletion of NAD and ATP, leading to apoptosis. It is a potent inhibitor of RNR. The drug does not require cell division to be cytotoxic. Resistance is associated with loss of the activating enzyme dCK; increased expression of RNR; or increased active efflux by ABCG2 or other members of the ATP binding cassette (ABC) family of transporters.

ADME

Cladribine is absorbed orally (55%) but is routinely administered intravenously. It is excreted by the kidneys, with a terminal $t_{1/2}$ in plasma of 6.7 h. *Cladribine* crosses the blood-brain barrier and reaches CSF concentrations of about 25% of those seen in plasma. Doses should be adjusted for renal dysfunction.

Therapeutic Uses

Cladribine is administered daily for 7 days by continuous intravenous infusion. It is the drug of choice in hairy cell leukemia. Eighty percent of patients achieve a complete response after a single course of therapy. The drug also is active in CLL; low-grade lymphomas; Langerhans cell histiocytosis; CTCLs, including mycosis fungoides and the Sézary syndrome; and Waldenström macroglobulinemia.

Adverse Effects

The major dose-limiting toxicity of *cladribine* is myelosuppression. Cumulative thrombocytopenia may occur with repeated courses. Opportunistic infections are common and correlate with decreased $CD4^+$ cell counts. Other toxic effects include nausea, infections, high fever, headache, fatigue, skin rashes, and tumor lysis syndrome.

Clofarabine (2-Chloro-2′-fluoro-arabinosyladenine)

The analogue *clofarabine* (2-chloro-2′-fluoro-arabinosyladenine) incorporates the 2-chloro, glycosylase-resistant substituent of *cladribine* and a 2′-fluoro-arabinosyl substitution, which adds stability and enhances uptake and phosphorylation. The resulting compound is approved for the treatment of pediatric ALL after failure of two prior therapies. *Clofarabine* produces complete remissions in 20% to 30% of these patients. It has activity in pediatric and adult AML and in myelodysplasia. The uptake and metabolic activation of *clofarabine* in tumor cells follow the same path as *cladribine* and the other purine nucleosides, although *clofarabine* is more readily phosphorylated by dCK. *Clofarabine triphosphate* has a long intracellular $t_{1/2}$ (24 h). It incorporates into DNA, where it terminates DNA synthesis and leads to apoptosis; *clofarabine* also inhibits RNR.

Therapeutic Uses and Adverse Effects

In children, *clofarabine* is administered as a 2-h infusion daily for 5 days. The primary elimination $t_{1/2}$ in plasma is 6.5 h. Most of the drug is excreted unchanged in the urine. Doses should be adjusted according to reductions in CL_{Cr}. The primary toxicities are myelosuppression; a clinical syndrome of hypotension, tachyphemia, pulmonary edema, organ dysfunction, and fever, all suggestive of capillary leak syndrome and cytokine release that necessitate immediate discontinuation of the drug; elevated hepatic enzymes and increased bilirubin; nausea, vomiting, and diarrhea; and hypokalemia and hypophosphatemia.

Nelarabine (6-Methoxy-arabinosyl-guanine)

Nelarabine (6-methoxy-arabinosyl-guanine) is the only guanine nucleoside in clinical use. It has selective activity against acute T-cell leukemia (20% complete responses) and the closely related T-cell lymphoblastic lymphoma and is approved for use in patients with relapsed/refractory disease. Its basic mechanism of action closely resembles that of the other purine nucleosides, in that it is incorporated into DNA and terminates DNA synthesis.

ADME

Following infusion, the parent methoxy compound is rapidly activated in blood and tissues by ADA-mediated cleavage of the methyl group, yielding the phosphorylase-resistant Ara-G, which has a plasma $t_{1/2}$ of 3 h. The active metabolite is transported into tumor cells, where it is activated by dCK to Ara-GTP, which incorporates into DNA and terminates DNA synthesis. The drug and its metabolite Ara-G are primarily eliminated by metabolism to guanine, and a smaller fraction is eliminated by renal excretion of Ara-G. The drug should be used with close clinical monitoring in patients with renal impairment (CL_{Cr} <50 mL/min). Adults are given a 2-h infusion on days 1, 3, and 5 of a 21-day cycle, and children are given a lower dose for 5 days, repeated every 21 days.

Adverse Effects

The adverse effects include myelosuppression and liver function test abnormalities, as well as infrequent serious neurological sequelae, such as seizures, delirium, somnolence, peripheral neuropathy, or Guillain-Barré syndrome. Neurological adverse effects may not be reversible.

Pentostatin (2′-Deoxycoformycin)

Pentostatin (2′-deoxycoformycin; see Figure 70–9), a transition-state analogue of the intermediate in the ADA reaction, potently inhibits ADA. Its effects mimic the phenotype of genetic ADA deficiency (severe immunodeficiency affecting T-cell and B-cell functions).

Mechanism of Action

Inhibition of ADA by *pentostatin* leads to accumulation of intracellular adenosine and deoxyadenosine nucleotides, which can block DNA synthesis by inhibiting RNR. Deoxyadenosine also inactivates *S*-adenosyl homocysteine hydrolase. The resulting accumulation of *S*-adenosyl homocysteine is particularly toxic to lymphocytes. *Pentostatin* also can inhibit RNA synthesis, and its triphosphate derivative is incorporated into DNA, resulting in strand breakage. Although the precise mechanism of cytotoxicity is not known, it is probable that the imbalance in purine nucleotide pools accounts for its antineoplastic effect in hairy cell leukemia and T-cell lymphomas.

ADME

Pentostatin is administered intravenously every other week and has a mean terminal $t_{1/2}$ of 5.7 h. After hydration with 500 to 1000 mL of 5% dextrose in half-normal (0.45%) saline, the drug is administered by rapid intravenous injection or by infusion over a period of 30 min or less, followed by an additional 500 mL of fluids. The drug is eliminated almost entirely by renal excretion. Proportional reduction of dosage is recommended in patients with renal impairment as measured by reduced CL_{Cr}.

Therapeutic Use

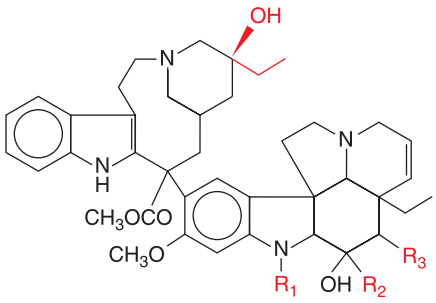
Pentostatin is effective in producing complete remissions (58%) and partial responses (28%) in patients with hairy cell leukemia. It largely has been superseded by *cladribine* (see previous discussion). Toxic manifestations include myelosuppression, GI symptoms, skin rashes, and abnormal liver function studies. Depletion of normal T cells occurs, and neutropenic fever and opportunistic infections may result. Immunosuppression may persist for several years after discontinuation. At high doses, major renal and neurological complications are encountered. *Pentostatin* in combination with *fludarabine phosphate* may result in severe or even fatal pulmonary toxicity.

Microtubule-Damaging Agents

Structural biology analyses have revealed multiple binding sites of microtubule targeting drugs with different mechanisms of action; the major distinct binding sites are for vinca, taxane, and colchicine (Steinmetz and Prota, 2018).

Vinca Alkaloids

Purified alkaloids from the periwinkle plant, including *vinblastine* and *vincristine*, were among the earliest clinical agents for the treatment of patients with leukemias, lymphomas, and testicular cancer. A closely related derivative, *vinorelbine*, has activity against lung and breast cancer.



Basic structure of vinca alkaloids

Mechanism of Action

The vinca alkaloids are cell-cycle–specific agents and, in common with other drugs, such as *colchicine*, *podophyllotoxin*, the taxanes, and the epothilones, block cells in mitosis. The biological activities of the vincas can be explained by their ability to bind to a specific site on β-tubulin and to block its polymerization with α-tubulin into microtubules (Figure 70–10) (Akhmanova and Steinmetz, 2015). The mitotic spindle cannot form, duplicated chromosomes cannot align along the division plate, and cell division arrests in metaphase. Cells blocked in mitosis undergo changes characteristic of apoptosis. Microtubules are found in high concentration in the brain and contribute to other cellular functions, such as movement, phagocytosis, and axonal transport. Adverse effects of the vinca alkaloids, such as their neurotoxicity, may relate to disruption of these functions.

Resistance

Despite their structural similarity, the individual vinca alkaloids have unique patterns of clinical efficacy (see discussion that follows). However, in most experimental systems, they share cross-resistance. Their antitumor effects are blocked by multidrug resistance mediated by P-glycoprotein (Pgp), which confers resistance to a broad range of agents (vinca alkaloids, epipodophyllotoxins, anthracyclines, and taxanes). Pgp, also known as MDR1, ABCB1, and CD243, is an ATP-dependent efflux pump with broad substrate specificity that provides a cellular defense mechanism against potentially harmful substances. Chromosomal abnormalities consistent with gene amplification and markedly increased levels of Pgp have been observed in resistant cells in culture. Other membrane transporters, such as the multidrug resistance-associated protein (MRP) and the closely related breast cancer resistance protein (BCRP), may contribute to resistance. Other forms of resistance to vinca alkaloids stem from mutations in β-tubulin or in the relative expression of isoforms of β-tubulin that prevent the inhibitors from effectively binding to their target.

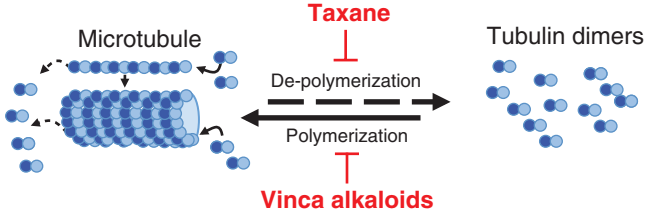


Figure 70–10 *Inhibitors of microtubular function.*

Adverse Effects

The limited myelosuppressive activity of *vincristine* makes it a valuable component of several combination therapy regimens for leukemia and lymphoma, while the lack of severe neurotoxicity of *vinblastine* is a decided advantage in lymphomas and in combination with *cisplatin* against testicular cancer. *Vinorelbine*, which causes mild neurotoxicity and myelosuppression, has an intermediate toxicity profile.

ADME

Hepatic CYPs extensively metabolize all three drugs, and the metabolites are excreted in the bile. Only a small fraction of a dose (<15%) is found unchanged in the urine. In patients with hepatic dysfunction (bilirubin >3 mg/dL), a 50% to 75% reduction in dose of any of the vinca alkaloids is advisable. The elimination $t_{1/2}$ is 20 h for *vincristine*, 23 h for *vinblastine*, and 24 h for *vinorelbine*.

Vinblastine

Therapeutic Uses. *Vinblastine sulfate* is given intravenously; special precautions must be taken against subcutaneous extravasation, which may cause painful irritation and ulceration. The drug should not be injected into an extremity with impaired circulation. After a single dose, myelosuppression reaches its maximum in 7 to 10 days. If a moderate level of leukopenia (~3000 cells/mm^3) is not attained, the weekly dose may be increased gradually. For testicular cancer, *vinblastine* is administered every 3 weeks. Doses should be reduced by 50% for patients with plasma bilirubin greater than 1.5 mg/dL. *Vinblastine* is used with *bleomycin* and *cisplatin* in the curative therapy of patients with metastatic testicular cancer, although it has been mainly supplanted by *etoposide* (see further discussion) or *ifosfamide* (see previous material). It is a component of the standard curative ABVD (***a***driamycin [doxorubicin]), *bleomycin*, *vinblastine*, *dacarbazine*) regimen for Hodgkin lymphoma. *Vinblastine* also is active against Kaposi sarcoma, neuroblastoma, Langerhans cell histiocytosis, bladder cancer, carcinoma of the breast, and choriocarcinoma.

Adverse Effects. Maximal leukopenia occurs within 7 to 10 days, after which recovery ensues within 7 days. Other toxic effects of *vinblastine* include mild neurological manifestations. GI disturbances, including nausea, vomiting, anorexia, and diarrhea, may be encountered. The syndrome of inappropriate secretion of antidiuretic hormone has been reported. Loss of hair, stomatitis, and dermatitis occur infrequently. Extravasation during injection may lead to cellulitis and phlebitis.

Vincristine

Therapeutic Uses. *Vincristine* is a standard component of regimens for treating pediatric patients with leukemias, lymphomas, and solid tumors, such as Wilms tumor, neuroblastoma, and rhabdomyosarcoma. In large-cell non-Hodgkin lymphomas, *vincristine* remains an important agent, particularly when used in the CHOP (*cyclophosphamide*, *doxorubicin*, *vincristine*, and *prednisone*) regimen. *Vincristine*

with glucocorticoids is the treatment of choice to induce remissions in patients with childhood leukemia and in combination with alkylating agents and *anthracycline* for those with pediatric sarcomas. The common adult intravenous dosage for *vincristine* is 1.4 mg/m^2 of body surface area at weekly or longer intervals, with a maximum dose of 2 mg at each periodic administration. *Vincristine* is tolerated better by children than by adults, who may experience severe, progressive neurological toxicity and require lowering of the dose. Administration of the drug more frequently than every 7 days or at higher doses increases the toxic manifestations without proportional improvement in the response rate. Precautions also should be used to avoid extravasation during intravenous administration. Doses should be reduced for patients with elevated plasma bilirubin.

Adverse Effects. The clinical toxicity of *vincristine* is mostly neurological. Severe neurological manifestations may be reversed by suspending therapy or reducing the dosage at the first sign of motor dysfunction. Severe constipation, sometimes resulting in colicky abdominal pain and obstruction, may be prevented by laxatives and hydrophilic (bulk-forming) agents. Reversible alopecia occurs in about 20% of patients. Modest leukopenia may occur. Thrombocytopenia, anemia, and the syndrome of inappropriate secretion of antidiuretic hormone are less common. Inadvertent injection of *vincristine* into the CSF causes a devastating and often-fatal irreversible coma and seizures.

Vinorelbine

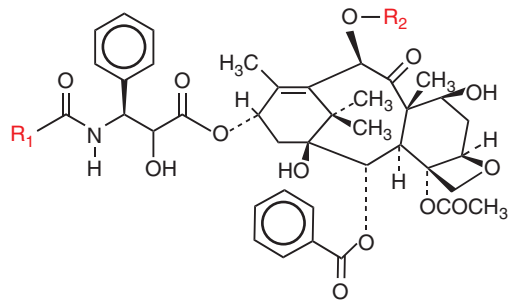
Vinorelbine has activity against non–small cell lung cancer and breast cancer. *Vinorelbine* is administered in normal saline as an intravenous infusion over 6 to 10 min or as an oral capsule formulation. When used alone, it is given intravenously at doses of 30 mg/m^2 either weekly or for 2 of every 3 weeks. When used with *cisplatin* for the treatment of non–small cell lung cancer, it is given at doses of 25 mg/m^2 either weekly or for 3 of every 4 weeks. A lower dose (20–25 mg/m^2) may be required for patients who have received prior chemotherapy and for hematological toxicity. The primary adverse effect of *vinorelbine* is granulocytopenia, with only modest thrombocytopenia and less neurotoxicity than other vinca alkaloids. *Vinorelbine* may cause allergic reactions and mild, reversible changes in liver enzymes. Doses should be reduced in patients with elevated plasma bilirubin.

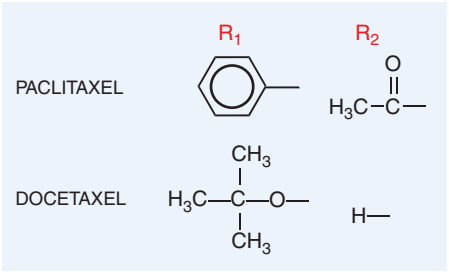
Eribulin

Eribulin is a synthetic analogue of halichondrin, a poly-ether macrolide, originally isolated from the Pacific marine sponge *Halichondria okadai*. The drug binds to the vinca site on β-tubulin and inhibits microtubule assembly. *Eribulin* is a poorer substrate than other microtubule disruptors for the Pgp efflux pump and is effective in drug-resistant tumors that overexpress Pgp. *Eribulin* is approved for the treatment of patients with drug-resistant metastatic breast cancer and liposarcoma. Adverse effects overlap with those of the vinca alkaloids and include neutropenia, neuropathies, and GI toxicities. *Tirbanibulin*, a mitotic inhibitor that binds to the colchicine binding site on β-tubulin, was FDA-approved in 2020 for local treatment of actinic keratosis, a precursor lesion of squamous cell carcinoma.

Taxanes

Paclitaxel was first isolated from the bark of the Western yew tree, *Taxus brevifolia*. *Paclitaxel* and its semisynthetic congeners *docetaxel* and *cabazitaxel* exhibit unique pharmacological properties as inhibitors of mitosis. These taxanes differ from the vinca alkaloids and colchicine derivatives in that they bind to a different site on β-tubulin and promote rather than inhibit microtubule formation (see Figure 70–10), stabilizing tubulin-GDP (Akhmanova and Steinmetz, 2015). The taxanes have a central role in the therapy of patients with ovarian, breast, lung, GI, genitourinary, prostate, and head and neck cancers.





Core structure of taxanes

Mechanism of Action

Paclitaxel binds to the β-tubulin subunit on the inner surface of microtubules and antagonizes their disassembly (see Figure 70–10), with the result that bundles of microtubules and aberrant structures derived from microtubules appear in the mitotic phase of the cell cycle (Schiff et al., 1979). Arrest in mitosis follows. Cell death occurs by apoptosis and depends on both drug concentration and duration of drug exposure. Drugs that block cell-cycle progression prior to mitosis antagonize the toxic effects of taxanes.

Drug interactions have been noted. The sequence of *cisplatin* preceding *paclitaxel* decreases *paclitaxel* clearance and produces greater toxicity than the opposite schedule. *Paclitaxel* decreases *doxorubicin* clearance and enhances its cardiotoxicity; *docetaxel* has no apparent effect on anthracycline pharmacokinetics.

Resistance to taxanes can be a result of decreased cellular drug accumulation due to increased expression of membrane-bound efflux proteins, including MRP1 and Pgp. *Cabazitaxel* is a poor substrate for Pgp and may therefore be useful for treating multidrug-resistant tumors. Other mechanisms of resistance may include an increase in *survivin*, an antiapoptotic factor; an increase in *α-aurora kinase,* which promotes completion of mitosis; an upregulation of the βIII-isoform of tubulin that lacks taxane-binding capacity; or a direct alteration of the drug target by mutation.

ADME

Paclitaxel has limited water solubility and is administered in a vehicle of 50% ethanol and 50% polyethoxylated castor oil. Hepatic CYPs (primarily CYP2C8, secondarily CYP3A4) extensively metabolize the drug. The primary metabolite is 6-OH paclitaxel, which is inactive; multiple additional hydroxylation products are found in plasma; less than 10% of a dose is excreted in the urine intact. Dose reductions in patients with abnormal hepatic function have been suggested, and 50% to 75% of doses of taxanes should be used in the presence of hepatic metastases larger than 2 cm in size or in patients with abnormal serum bilirubin. Drugs that induce CYP2C8 or CYP3A4, such as *phenytoin* and *phenobarbital*, or those that inhibit these CYPs, such as antifungal imidazoles, significantly alter drug clearance and toxicity.

Paclitaxel clearance is nonlinear and decreases with increasing dose or dose rate; the plasma $t_{1/2}$ is 10 to 14 h. The critical plasma concentration for myelosuppression depends on the duration of exposure but likely is 50 to 100 nM. *Paclitaxel* clearance is markedly delayed by *cyclosporine A* and other drugs that inhibit Pgp.

A nanoparticle albumin-bound *paclitaxel* (*nab-paclitaxel*) is soluble and used for infusion. This form of *paclitaxel* has increased cellular uptake via an albumin-specific mechanism. *Nab-paclitaxel* achieves a higher serum concentration than *paclitaxel*, but the increased clearance of *nab-paclitaxel* results in a similar systemic drug exposure. It is somewhat less neurotoxic than *paclitaxel* and is used in combination treatment of metastatic breast, pancreas, and non–small cell lung cancer. Like the other taxanes, *nab*-paclitaxel should not be given to patients with an absolute neutrophil count below 1500 cells/mm^3.

Docetaxel, somewhat more soluble than *paclitaxel*, is administered intravenously in an emulsifier (polysorbate 80). *Docetaxel* pharmacokinetics are similar to those of *paclitaxel*, with an elimination $t_{1/2}$ of about 12 h. Clearance is primarily through CYP3A4- and CYP3A5-mediated hydroxylation, leading to inactive metabolites.

Therapeutic Uses

The taxanes have become central components of regimens for treating patients with metastatic ovarian, breast, lung, GI, genitourinary, and head and neck cancers. These drugs are administered once weekly or once every 3 weeks. *Cabazitaxel* is a poor substrate for Pgp and is approved for hormone-refractory metastatic prostate cancer previously treated with a *docetaxel*-containing regimen.

Adverse Effects

Paclitaxel exerts its primary toxic effects on the bone marrow. Neutropenia usually occurs 8 to 11 days after a dose and reverses rapidly by days 15 to 21. Used with granulocyte colony-stimulating factor (G-CSF), high doses over 24 h are well tolerated, and peripheral neuropathy becomes dose limiting. Many patients experience myalgias after receiving *paclitaxel*. In high-dose schedules or with prolonged use, a stocking-glove sensory neuropathy can be disabling, particularly in patients with underlying diabetic neuropathy or concurrent *cisplatin* therapy. Mucositis is prominent in 72- or 96-h infusions and in the weekly schedule. Hypersensitivity reactions can occur in patients receiving *paclitaxel* infusions of short duration (1–6 h) but are largely averted by pretreatment with *dexamethasone*, *diphenhydramine*, and histamine H_2 receptor antagonists. Premedication is not necessary with 96-h infusions. Many patients experience asymptomatic bradycardia; occasional episodes of silent ventricular tachycardia also occur and resolve spontaneously during 3- or 24-h infusions. *Nab-paclitaxel* produces increased rates of peripheral neuropathy compared to the original *cremophor*-delivered *paclitaxel* but rarely causes hypersensitivity reactions.

Docetaxel causes greater degrees of neutropenia than *paclitaxel* but less peripheral neuropathy and asthenia and less frequent hypersensitivity. Fluid retention is a progressive problem with multiple cycles of *docetaxel* therapy, leading to peripheral edema, pleural and peritoneal fluid, and pulmonary edema in extreme cases. Oral *dexamethasone*, begun 1 day prior to drug infusion and continuing for 3 days, greatly ameliorates fluid retention. In rare cases, *docetaxel* may cause progressive interstitial pneumonitis and respiratory failure if the drug is not discontinued.

Estramustine

Estramustine combines estradiol and normustine (nornitrogen mustard) through a carbamate link. Although the intent of the combination was to enhance the uptake of the alkylating agent into estradiol-sensitive prostate cancer cells, *estramustine* does not function *in vivo* as an alkylating agent; rather, it binds to β-tubulin and microtubule-associated proteins, causing microtubule disassembly and antimitotic actions (Tew and Stearns, 1987).

Therapeutic Use

Estramustine is approved for the palliative treatment of patients with progressive or metastatic prostate cancer.

ADME

Following oral administration, at least 75% of a dose of *estramustine phosphate* is absorbed from the GI tract and rapidly dephosphorylated. The drug undergoes extensive first-pass metabolism by hepatic CYPs to an active 17-keto derivative, estromustine, and to multiple inactive products; the active drug forms accumulate in the prostate. Some hydrolysis of the carbamate linkage occurs in the liver, releasing estradiol, estrone, and the normustine group. *Estramustine* and estromustine have plasma half-lives of 10 and 14 h, respectively, and are excreted as inactive metabolites, mainly in the feces.

Adverse Effects, Drug Interactions

In addition to myelosuppression, *estramustine* also possesses estrogenic side effects (gynecomastia, impotence, elevated risk of thrombosis, and fluid retention); hypercalcemia; acute attacks of porphyria; impaired glucose tolerance; and hypersensitivity reactions, including angioedema. *Estramustine* inhibits the clearance of taxanes.

Epothilones

The epothilones are polyketides discovered as cytotoxic metabolites from a strain of *Sorangium cellulosum*, a myxobacterium isolated from soil on the bank of the Zambezi River in southern Africa (Gerth et al., 1996). One of these, *ixabepilone*, is approved for breast cancer treatment; others are under development (Lee and Swain, 2008).

Ixabepilone

Mechanism of Action; Resistance. Epothilones bind to a β-tubulin pocket located on the luminal side of microtubules, the taxane site (Steinmetz and Prota, 2018), and can trigger microtubule nucleation at multiple sites away from the centriole. This dysfunctional microtubule stabilization triggers cell-cycle arrest at the G_2-M interface and apoptosis. *In vitro* studies suggested that *ixabepilone* is less susceptible to Pgp-mediated export and multidrug resistance than taxanes. Mechanisms implicated in epothilone resistance include mutation of the β-tubulin binding site and upregulation of isoforms of β-tubulin.

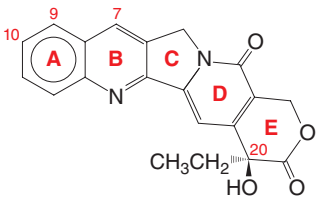
ADME. *Ixabepilone* is administered intravenously. Because of its minimal aqueous solubility, it is delivered in the solubilizing agent *cremophor* (polyoxyethylated castor oil/ethanol), which has been implicated as the cause of infusion reactions; such reactions are infrequent when administration is preceded by premedication with H_1 and H_2 antagonists. The drug is cleared by hepatic CYPs and has a plasma $t_{1/2}$ of 52 h.

Therapeutic Uses. In patients with metastatic breast cancer resistant to or pretreated with anthracyclines and resistant to taxanes, *ixabepilone* plus *capecitabine* provides an improved progression-free survival of 1.6 months compared to *capecitabine* alone. *Ixabepilone* also is indicated as monotherapy for metastatic breast cancer in patients who have previously progressed through treatment with anthracyclines, taxanes, and *capecitabine*. *Ixabepilone* (~40 mg/m^2) is administered as monotherapy or in combination with *capecitabine* over 3 h every 3 weeks. Patients should be premedicated with both an H_1 and an H_2 antagonist before receiving *ixabepilone* to minimize hypersensitivity reactions. In patients with mild-to-moderate hepatic dysfunction who receive *ixabepilone* monotherapy, lower starting doses are recommended due to delayed drug clearance.

Adverse Effects. Epothilones have toxicities similar to those of the taxanes: neutropenia, peripheral sensory neuropathy, fatigue, diarrhea, and asthenia.

Camptothecin Analogues

The camptothecins are potent, cytotoxic antineoplastic agents that target the nuclear enzyme *topoisomerase I*. The lead compound in this class, camptothecin, was isolated from the tree *Camptotheca acuminata*. *Irinotecan* and *topotecan*, currently the only camptothecin analogues approved for clinical use, have activity in colorectal, ovarian, and small cell lung cancer.



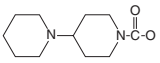
	C-10	C-9	C-7
Camptothecin	H	H	H
Topotecan	OH	$(CH_3)_2NHCH_2$	H
Irinotecan		H	CH_2CH_3
SN-38	OH	H	CH_2CH_3

Figure 70–11 *Camptothecin and its analogues.*

Chemistry

All camptothecins have a fused five-ring backbone that includes a labile lactone ring (Figure 70–11). The hydroxyl group and S conformation of the chiral center at C20 in the lactone ring are required for biological activity. Appropriate substitutions on the A and B rings of the quinoline subunit enhance water solubility and increase potency for inhibiting topoisomerase I. *Topotecan* is a semisynthetic molecule with a basic dimethylamino group that increases its water solubility. *Irinotecan* (CPT-11) differs from *topotecan* in that it is a prodrug. The carbamate bond between the camptothecin moiety and the dibasic bis-piperidine side chain at position C10 (which makes the molecule water soluble) is cleaved by a carboxylesterase to form the active metabolite SN-38 (see Figure 6–6).

Mechanism of Action

The DNA topoisomerases are nuclear enzymes that reduce torsional stress in supercoiled DNA, allowing selected regions of DNA to become sufficiently untangled to permit replication, repair, and transcription (Pommier, 2013). Two classes of topoisomerase (I and II) mediate DNA strand breakage and resealing. Camptothecin analogues inhibit the function of topoisomerase I; other drugs, such as anthracyclines, epipodophyllotoxins, and acridines, inhibit topoisomerase II. The camptothecins bind to and stabilize the normally transient DNA–topoisomerase I cleavable complex. Although the initial cleavage action of topoisomerase I is not affected, the religation step is inhibited, leading to the accumulation of single-stranded breaks in DNA. These lesions are reversible and not by themselves toxic to the cell. However, the collision of a DNA replication fork with this cleaved strand of DNA causes an irreversible double-strand DNA break, ultimately leading to cell death.

Camptothecins are *S phase–specific drugs* because ongoing DNA synthesis is necessary for cytotoxicity. This has important clinical implications. S phase–specific cytotoxic agents generally require prolonged exposures of tumor cells to drug concentrations above a minimum threshold for optimal therapeutic efficacy. In fact, low-dose, protracted administration of camptothecin analogues results in less toxicity and equal or greater antitumor activity than shorter, more intense courses.

Mechanisms of Resistance

Decreased intracellular drug accumulation may underlie resistance. The BCRP/xenobiotic exporter ABCG2/BCRP is overexpressed in cultured cells that have become resistant to *irinotecan* after exposure to camptothecins. Cell lines that lack carboxylesterase activity demonstrate resistance to *irinotecan*, but the liver and red blood cells may have sufficient carboxylesterase activity to convert *irinotecan* to the active metabolite SN-38. Camptothecin resistance also may result from decreased expression or mutation of topoisomerase I. A transient downregulation of topoisomerase I has been demonstrated following prolonged exposure to camptothecins *in vitro* and *in vivo*. Mutations leading to reduced topoisomerase I enzyme catalytic activity or DNA-binding affinity have been associated with experimental camptothecin resistance. Finally, exposure of cells to topoisomerase I–targeted agents upregulates topoisomerase II, an alternative enzyme for DNA strand passage.

Topotecan

ADME

Topotecan is approved for intravenous administration. An oral dosage form in development has a bioavailability of 30% to 40% in patients with cancer. *Topotecan* exhibits linear pharmacokinetics, and it is rapidly eliminated from systemic circulation with a $t_{1/2}$ of about 3.5 to 4.1 h. Only 20% to 35% of the total drug in plasma is found to be in the active lactone form. Within 24 h, 30% to 40% of the administered dose appears in the urine. Doses should be reduced in proportion to reductions in CL_{Cr}. Hepatic metabolism appears to be a relatively minor route of drug elimination. Plasma protein binding of topotecan is low (7%–35%), which may explain its relatively greater CNS penetration.

Therapeutic Uses

Topotecan is indicated for previously treated patients with ovarian and small cell lung cancer. Significant hematological toxicity limits its use in combination with other active agents in these diseases (e.g., *cisplatin*). The recommended dosing regimen of *topotecan* for ovarian cancer and small cell lung cancer is a 30-min infusion for 5 consecutive days every 3 weeks. For the treatment of patients with cervical cancer in conjunction with *cisplatin*, *topotecan* is administered on days 1, 2, and 3, repeated every 21 days. The dose of *topotecan* should be reduced in patients with moderate renal dysfunction (CL_{Cr} of 20–40 mL/min); *topotecan* should not be administered to patients with severe renal impairment (CL_{Cr} <20 mL/min). Hepatic dysfunction does not alter *topotecan* clearance and toxicity. A baseline neutrophil count greater than 1500 cells/mm^3 and a platelet count above 100,000 is necessary prior to *topotecan* administration.

Adverse Effects

The dose-limiting toxicity with all dosing schedules is neutropenia, with or without thrombocytopenia. The incidence of severe neutropenia at 1.5 mg/m^2 daily for 5 days every 3 weeks may be as high as 81%, with a 26% incidence of febrile neutropenia. In patients with hematological malignancies, GI side effects such as mucositis and diarrhea become dose limiting. Other less common and generally mild *topotecan*-related toxicities include nausea and vomiting, elevated liver transaminases, fever, fatigue, and rash.

Irinotecan

ADME

The conversion of *irinotecan* to its active metabolite SN-38 is mediated predominantly by carboxylesterases in the liver (see Figures 5–6 and 5–8). Although SN-38 can be measured in plasma shortly after beginning an intravenous infusion of *irinotecan*, the AUC of SN-38 is only about 4% of the AUC of *irinotecan*, suggesting that only a relatively small fraction of the dose is ultimately converted to the active form of the drug. *Irinotecan* exhibits linear pharmacokinetics. In comparison to *topotecan*, a relatively large fraction of both *irinotecan* and SN-38 are present in plasma as the biologically active intact lactone form. The $t_{1/2}$ of SN-38 is 11.5 h, three times that of *topotecan*. CSF penetration of SN-38 in humans has not been characterized.

In contrast to *topotecan*, hepatic metabolism of *irinotecan* and SN-38 represents an important route of elimination for both. Oxidative metabolites have been identified in plasma, all of which result from CYP3A-mediated reactions directed at the bis-piperidine side chain. These metabolites are not significantly converted to SN-38. The total body clearance of *irinotecan* was found to be two times greater in patients with brain cancer taking antiseizure drugs that induce hepatic CYPs.

UGT1A1 glucuronidates the hydroxyl group at position C10 (resulting from cleavage of the bispiperidine promoiety), producing the inactive glucuronide metabolite SN-38G (see Figures 5–6, 5–8, and 5–9). Biliary excretion

appears to be the primary elimination route of *irinotecan*, SN-38, and metabolites, although urinary excretion also contributes (14%–37%). The extent of SN-38 glucuronidation inversely correlates with the risk of severe diarrhea after *irinotecan* therapy. UGT1A1 polymorphisms associated with familial hyperbilirubinemia syndromes may have a major impact on the clinical use of *irinotecan*. In patients treated with *irinotecan*, there is a positive correlation between baseline serum unconjugated bilirubin concentration and both severity of neutropenia and the AUC of *irinotecan* and SN-38. Severe *irinotecan* toxicity has been observed in patients with cancer with Gilbert syndrome, presumably due to decreased glucuronidation of SN-38. In patients with the syndrome, elevated levels of unconjugated bilirubin in the bloodstream are due to reduced activity of glucuronyltransferase (Figure 5–7). The presence of bacterial glucuronidase in the intestinal lumen potentially can contribute to *irinotecan*'s GI toxicity by releasing unconjugated SN-38 from the inactive glucuronide metabolite.

Therapeutic Uses

Single-agent dosage of *irinotecan* is by weekly infusion for 4 of 6 weeks, with a higher dose given every 3 weeks. In patients with advanced colorectal cancer, *irinotecan* is used as first-line therapy in combination with fluoropyrimidines or as a single agent or in combination with *cetuximab* following failure of a 5FU/*oxaliplatin* regimen. It also has activity in small cell lung cancer. An *irinotecan* liposome injection in combination with 5FU and *leucovorin* is FDA-approved for treatment of metastatic pancreatic cancer after disease progression following *gemcitabine* therapy.

Adverse Effects

The dose-limiting toxicity with all dosing schedules is delayed diarrhea (35%), with or without neutropenia. An intensive regimen of *loperamide* (see Chapter 54) reduces this incidence by more than half. However, once severe diarrhea occurs, standard doses of antidiarrheal agents tend to be ineffective. Diarrhea generally resolves within a week and, unless associated with fever and neutropenia, rarely is fatal.

The second most common *irinotecan*-associated toxicity is myelosuppression. Severe neutropenia occurs in 14% to 47% of the patients treated with a schedule of administration every 3 weeks and is less frequently encountered among patients treated with the weekly schedule. Febrile neutropenia is observed in 3% of patients and may be fatal, particularly when associated with concomitant diarrhea. A cholinergic syndrome resulting from the inhibition of acetylcholinesterase activity by *irinotecan* may occur within the first 24 h after *irinotecan* administration. Symptoms include acute diarrhea, diaphoresis, hypersalivation, abdominal cramps, visual accommodation disturbances, lacrimation, rhinorrhea, and, less often, asymptomatic bradycardia. These effects are short lasting and respond within minutes to *atropine*. Other common toxicities include nausea and vomiting, fatigue, vasodilation or skin flushing, mucositis, elevation in liver transaminases, and alopecia. There have been case reports of dyspnea and interstitial pneumonitis associated with *irinotecan* therapy.

Antibiotics

Dactinomycin (Actinomycin D)

Actinomycins are chromopeptides isolated from *Streptomyces* soil bacteria. Most contain the same chromophore, the planar phenoxazone actinosin, which is responsible for their yellow-red color. The differences among naturally occurring actinomycins are confined to variations in the structure of the amino acids of the peptide side chains. *Actinomycin D* has beneficial effects in the treatment of solid tumors in children and choriocarcinoma in adult women.

Mechanism of Action

The capacity of actinomycins to bind to double-helical DNA is responsible for their biological activity and cytotoxicity (Reich, 1963). The planar phenoxazone ring intercalates between adjacent guanine-cytosine base pairs of DNA, while the polypeptide chains extend along the minor groove of the helix, resulting in a *dactinomycin*-DNA complex with stability sufficient to block the transcription of DNA by RNA polymerase. The DNA-dependent RNA polymerases are much more sensitive to the effects of *dactinomycin* than are the DNA polymerases. In addition, *dactinomycin* causes single-strand breaks in DNA, possibly through a free-radical intermediate or as a result of the action of topoisomerase II. *Dactinomycin* inhibits rapidly proliferating cells of normal and neoplastic origin and is among the most potent antitumor agents known.

ADME

Dactinomycin is administered by intravenous injection. Metabolism of the drug is minimal. The drug is excreted in both bile and urine and disappears from plasma with a terminal $t_{1/2}$ of 36 h. *Dactinomycin* does not cross the blood-brain barrier.

Therapeutic Uses

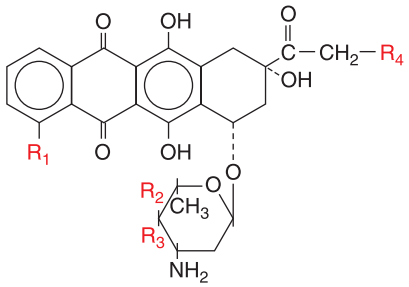
Dactinomycin is given intravenously for 5 days, usually in the range of 10 to 15 μg/kg. If no manifestations of toxicity are encountered, additional courses may be given at intervals of 2 to 4 weeks, although weekly maintenance doses have been used. The main clinical use of *dactinomycin* is in the treatment of rhabdomyosarcoma and Wilms tumor in children, where it is curative in combination with primary surgery, radiotherapy, and other drugs, particularly *vincristine* and *cyclophosphamide*. Ewing, Kaposi, and soft-tissue sarcomas also respond. *Dactinomycin* and MTX form a curative therapy for choriocarcinoma.

Adverse Effects

Toxic manifestations include anorexia, nausea, and vomiting, usually beginning a few hours after administration. Hematopoietic suppression with pancytopenia may occur in the first week after completion of therapy. Proctitis, diarrhea, glossitis, cheilitis, and ulcerations of the oral mucosa are common; dermatological manifestations include alopecia, as well as erythema, desquamation, and increased inflammation and pigmentation in areas previously or concomitantly subjected to X-ray radiation. Severe injury may occur as a result of local drug extravasation; the drug is extremely corrosive to soft tissues.

Anthracyclines and Anthracenediones

Anthracyclines are derived from *Streptomyces peucetius* var. *caesius*. *Idarubicin* and *epirubicin* are analogues of the naturally produced anthracyclines *doxorubicin* and *daunorubicin*, differing only slightly in chemical structure, but having somewhat distinct patterns of clinical activity. *Daunorubicin* and *idarubicin* primarily have been used in acute leukemias, whereas *doxorubicin* and *epirubicin* display broader activity against solid tumors. These agents, which possess the potential for generating free radicals, cause an unusual and often-irreversible cardiomyopathy, the occurrence of which is related to the total dose of the drug. The structurally similar agent *mitoxantrone* has less cardiotoxicity and is useful against prostate cancer and AML and in high-dose chemotherapy.



	DOXORUBICIN	DAUNORUBICIN	EPIRUBICIN	IDARUBICIN
R_1 =	OCH_3	OCH_3	OCH_3	H
R_2 =	H	H	OH	H
R_3 =	OH	OH	H	OH
R_4 =	OH	H	OH	H

Mechanisms of Action and Resistance

Anthracyclines and anthracenediones can intercalate with DNA, directly affecting transcription and replication. More important is their

capacity to form a heterotrimeric complex with topoisomerase II and DNA (Pommier, 2013). Topoisomerase II produces double-strand breaks at the 3′-phosphate backbone, allowing strand passage and uncoiling of supercoiled DNA. Following strand passage, topoisomerase II religates the DNA strands; this enzymatic function is essential for DNA replication and repair. Formation of the ternary complex with anthracyclines or with *etoposide* inhibits the religation of the broken DNA strands, leading to apoptosis. Defects in DNA double-strand break repair sensitize cells to damage by these drugs, while altered DNA damage-sensing and DNA repair capacity may contribute to resistance.

The quinone moieties of anthracyclines can form radical intermediates that react with O_2 to produce superoxide anion radicals, which can generate H_2O_2 and •OH that attack DNA and oxidize DNA bases, leading to apoptosis (Serrano et al., 1999). The production of free radicals is significantly stimulated by the interaction of *doxorubicin* with iron. Enzymatic defenses such as superoxide dismutase and catalase protect cells against the toxicity of the anthracyclines, and these defenses can be augmented by exogenous antioxidants such as α-tocopherol or by the iron chelator *dexrazoxane*, which protects against cardiac toxicity. Multidrug resistance is observed in tumor cell populations exposed to anthracyclines. Anthracyclines also are exported from tumor cells by members of the MRP transporter family and by ABCG2 (BCRP). Other biochemical changes in resistant cells include increased glutathione peroxidase activity, decreased activity or mutation of topoisomerase II, and enhanced ability to repair DNA strand breaks.

ADME

Daunorubicin, *doxorubicin*, *epirubicin*, and *idarubicin* are administered intravenously and are cleared by a complex pattern of hepatic metabolism and biliary excretion. Each anthracycline is converted to an active alcohol intermediate that plays a variable role in the therapeutic activity. The plasma disappearance curves for *doxorubicin* and *daunorubicin* are multiphasic, with a terminal $t_{1/2}$ of 30 h. *Idarubicin* has a $t_{1/2}$ of 15 h, and its active metabolite, idarubicinol, has a $t_{1/2}$ of 40 h. The drugs rapidly enter the heart, kidneys, lungs, liver, and spleen; they do not cross the blood-brain barrier. Clearance is delayed in the presence of hepatic dysfunction; at least a 50% initial reduction in dose should be considered in patients with elevated serum bilirubin levels.

Idarubicin and Daunorubicin

Therapeutic Use. *Idarubicin* (~12 mg/m² per day for 3 days) is administered by slow intravenous injection (10–15 min) to avoid extravasation. *Idarubicin* has less cardiotoxicity than the other anthracyclines. *Daunorubicin* (also named daunomycin or rubidomycin) is administered (at variable doses according to the regimen, 45–90 mg/m² per day) intravenously for 3 days, with care to prevent extravasation. Total doses greater than 550 mg/m² are associated with a high risk of cardiotoxicity. Radiation therapy to the mediastinum may enhance the risk of anthracycline toxicity. *Daunorubicin* may impart a red color to the urine. *Daunorubicin* and *idarubicin* are used in the treatment of patients with AML in combination with Ara-C.

Adverse Effects. Toxic effects of *daunorubicin* and *idarubicin* include bone marrow depression, stomatitis, alopecia, GI disturbances, rash, and cardiac toxicity. Cardiac toxicity of anthracyclines can be acute or chronic and is described in detail next for *doxorubicin*.

Doxorubicin

Therapeutic Uses. The dose (60–75 mg/m²) is administered as a single rapid intravenous infusion that is repeated after 21 days. A *doxorubicin* liposomal product is available for treatment of AIDS-related Kaposi sarcoma and is given intravenously in a dose of 20 mg/m² over 60 min and repeated every 3 weeks. The liposomal formulation also is approved for ovarian cancer for treatment at a dose of 50 mg/m² every 4 weeks and as a treatment of multiple myeloma, where it is given as a 30-mg/m² dose on day 4 of each 21-day cycle. Patients should be advised that the drug may impart a red color to the urine. *Doxorubicin* is effective in malignant lymphomas. In combination with *cyclophosphamide*, vinca alkaloids, and other agents, it is an important ingredient for the successful treatment of lymphomas. It is a valuable component of various regimens adjunct of chemotherapy for and metastatic carcinoma of the breast. The drug also is beneficial in pediatric and adult sarcomas, including osteogenic, Ewing, and soft-tissue sarcomas.

Adverse Effects. Toxicities of *doxorubicin* are similar to those of *daunorubicin*. Myelosuppression is a major dose-limiting complication, with maximal leukopenia usually occurring during the second week of therapy and recovering by the fourth week; thrombocytopenia and anemia follow a similar pattern but usually are less pronounced. Stomatitis, mucositis, diarrhea, and alopecia are common but reversible. Erythematous streaking near the site of infusion ("flare") is a benign local allergic reaction and should not be confused with extravasation. Facial flushing, conjunctivitis, and lacrimation may occur rarely. The drug may produce severe local toxicity in irradiated tissues (e.g., the skin, heart, lung, esophagus, and GI mucosa) even when the two therapies are not administered concomitantly.

Cardiomyopathy is the most important long-term toxicity (Rochette et al., 2015) and may take two forms:

- **An acute form, characterized by abnormal electrocardiographic changes, including ST- and T-wave alterations and arrhythmias.** This is brief and rarely a serious problem. An acute reversible reduction in ejection fraction is observed in some patients in the 24 h after a single dose, and plasma troponin T may increase in a minority of patients in the first few days following drug administration (Lipshultz et al., 2004). Acute myocardial damage, the "pericarditis-myocarditis syndrome," may begin in the days following drug infusion and is characterized by severe disturbances in impulse conduction and frank congestive heart failure, often associated with pericardial effusion.
- **Chronic, cumulative, dose-related toxicity (usually total doses of ≥550 mg/m²) progressing to congestive heart failure.** The mortality rate in patients with congestive heart failure approaches 50%. The risk of cardiomyopathy increases markedly with dose; estimates run as high as 20% at total doses of 550 mg/m² (a total dose limit of 300 mg/m² is advised for pediatric cases). These total dosages should be exceeded only under exceptional circumstances or with the concomitant use of *dexrazoxane*, a cardioprotective iron-chelating agent (Swain et al., 1997), as the mechanism of cardiotoxicity is thought to derive in part from cardiac lipid peroxidation (Myers et al., 1977) promoted by Fe^{2+} (Zweier, 1984). More recent studies have also proposed that *dexrazoxane* can prevent anthracycline-derived DNA damage by interaction with topoisomerase II in cardiac cells (Lyu et al., 2007). Cardiac irradiation, administration of high doses of *cyclophosphamide* or another anthracycline, or concomitant *trastuzumab* increases the risk of cardiotoxicity (Slamon et al., 2001). Late-onset cardiac toxicity, with congestive heart failure years after treatment, may occur in both pediatric and adult populations. In children treated with anthracyclines, there is a 3- to 10-fold elevated risk of arrhythmias, congestive heart failure, and sudden death in adult life. Concomitant administration of *dexrazoxane* may reduce troponin T elevations and avert later cardiotoxicity (Lipshultz et al., 2004).

Epirubicin

The anthracycline *epirubicin* is indicated as a component of adjunctive therapy for treatment of breast cancer. It is administered intravenously in doses of 100 to 120 mg/m² every 3 to 4 weeks. Total doses greater than 900 mg/m² sharply increase the risk of cardiotoxicity. Its toxicity profile is the same as that of *doxorubicin*.

Valrubicin

Valrubicin is a semisynthetic analogue of *doxorubicin* used exclusively for intravesicular treatment of bladder cancer. Once a week for 6 weeks, 800 mg are instilled into the bladder. Less than 10% of instilled drug is absorbed systemically. Side effects relate to bladder irritation.

Mitoxantrone

Mitoxantrone is a synthetic anthracenedione derivative topoisomerase II inhibitor that is approved for use in AML, prostate cancer, and late-stage, secondary progressive multiple sclerosis. *Mitoxantrone* has limited ability to produce quinone-type free radicals and causes less cardiac toxicity than does *doxorubicin*. It produces acute myelosuppression, cardiac toxicity, and mucositis as its major toxicities; the drug also causes nausea, vomiting, and alopecia, although less than *doxorubicin*. *Mitoxantrone* is administered by intravenous infusion. To induce remission in acute nonlymphocytic leukemia in adults, the drug is given in a daily dose of 12 mg/m^2 for 3 days with *cytarabine*. It also is used in advanced hormone-resistant prostate cancer in a dose of 12 to 14 mg/m^2 every 21 days.

Epipodophyllotoxins

Podophyllotoxin Derivatives

Two synthetic derivatives of podophyllotoxins have significant therapeutic activity in pediatric leukemia, small cell carcinomas of the lung, testicular tumors, Hodgkin disease, and large cell lymphomas. These derivatives are *etoposide* (VP-16-213) and *teniposide* (VM-26). Although podophyllotoxin binds to tubulin, *etoposide* and *teniposide* have no effect on microtubular structure or function at usual concentrations.

Mechanisms of Action and Resistance

Etoposide and *teniposide* form ternary complexes with topoisomerase II and DNA and prevent resealing of the break that normally follows topoisomerase binding to DNA. The enzyme remains bound to the free end of the broken DNA strand, leading to an accumulation of DNA breaks and cell death. Cells in the S and G_2 phases of the cell cycle are most sensitive to *etoposide* and *teniposide*. Resistant cells demonstrate (1) amplification of the *MDR1* gene that encodes the Pgp drug efflux transporter, (2) mutation or decreased expression of topoisomerase II, or (3) mutations of the p53 tumor suppressor gene, a required component of the apoptotic pathway (Lowe et al., 1993).

Etoposide

ADME

Oral administration of *etoposide* results in variable absorption that averages about 50%. After intravenous injection, there is a biphasic pattern of clearance with a terminal $t_{1/2}$ of about 6 to 8 h in patients with normal renal function. Approximately 40% of an administered dose is excreted intact in the urine. In patients with compromised renal function, dosage should be reduced in proportion to the reduction in CL_{Cr}. In patients with advanced liver disease, increased toxicity may result from a low serum albumin (decreased drug binding) and elevated bilirubin (which displaces *etoposide* from albumin); guidelines for dose reduction in this circumstance have not been defined. Drug concentrations in the CSF average 1% to 10% of those in plasma.

Therapeutic Uses

The intravenous dose of *etoposide* for testicular cancer in combination therapy (with *bleomycin* and *cisplatin*) is 50 to 100 mg/m^2 for 5 days or 100 mg/m^2 on alternate days for three doses. For small cell carcinoma of the lung, the dosage in combination therapy (with *cisplatin*) is 100 to 200 mg/m^2/d intravenously for 3 days. Cycles of therapy usually are repeated every 3 to 4 weeks. After relapse, oral administration of 50 mg/m^2 per day for 21 days is one treatment option. When given intravenously, the drug should be administered slowly over a 30- to 60-min period to avoid hypotension and bronchospasm, which likely result from the additives used to dissolve *etoposide*.

Etoposide is also active against non-Hodgkin lymphomas, acute nonlymphocytic leukemia, and Kaposi sarcoma associated with AIDS. *Etoposide* has a favorable toxicity profile for dose escalation in that its primary acute toxicity is myelosuppression. In combination with *ifosfamide* and *carboplatin*, it frequently is used for high-dose chemotherapy in total doses of 1500 to 2000 mg/m^2. *Trilaciclib*, a cyclin-dependent kinase (CDK) 4/6 kinase inhibitor, was approved in 2021 for administration prior to cytotoxic chemotherapy to decrease the incidence of myelosuppression in patients receiving a platinum/*etoposide*-containing regimen or *topotecan*-containing regimen for extensive-stage small cell lung cancer (see Chapter 71).

Adverse Effects

The dose-limiting toxicity of *etoposide* is leukopenia (nadir at 10–14 days, recovery by 3 weeks). Thrombocytopenia occurs less often and usually is not severe. Nausea, vomiting, stomatitis, and diarrhea complicate treatment in about 15% of patients. Alopecia is common but reversible. Hepatic toxicity is particularly evident after high-dose treatment. For both *etoposide* and *teniposide*, toxicity increases in patients with decreased serum albumin, an effect related to decreased protein binding of the drug.

A disturbing complication of *etoposide* therapy is the development of an unusual form of acute nonlymphocytic leukemia with a translocation in chromosome 11q23. At this locus is a gene (the *MLL* gene) that regulates the proliferation of pluripotent stem cells. The leukemic cells have the cytological appearance of acute monocytic or monomyelocytic leukemia. Another distinguishing feature of *etoposide*-related leukemia is the short time interval between the end of treatment and the onset of leukemia (1–3 years), compared to the 4- to 5-year interval for secondary leukemias related to alkylating agents, and the absence of a myelodysplastic period preceding leukemia (Pui et al., 1995). Patients receiving weekly or twice-weekly doses of *etoposide*, with cumulative doses greater than 2000 mg/m^2, seem to be at higher risk of leukemia.

Teniposide

Teniposide is administered intravenously. It has a multiphasic pattern of clearance from plasma: After distribution, a $t_{1/2}$ of 4 h and another $t_{1/2}$ of 10 to 40 h are observed. Approximately 45% of the drug is excreted in the urine; in contrast to *etoposide*, as much as 80% is recovered as metabolites. Anticonvulsants such as *phenytoin* increase the hepatic metabolism of *teniposide* and reduce systemic exposure. Dosage need not be reduced for patients with impaired renal function. Less than 1% of the drug crosses the blood-brain barrier. *Teniposide* is available for treatment of refractory ALL in children and is synergistic with *cytarabine*. *Teniposide* is administered by intravenous infusion for 5 days or twice weekly. The drug has limited utility and is given primarily for acute leukemia in children and monocytic leukemia in infants, as well as glioblastoma, neuroblastoma, and brain metastases from small cell carcinomas of the lung. Myelosuppression, nausea, and vomiting are its primary toxic effects.

Trabectedin Analogues

Trabectedin

Trabectedin is derived from the marine invertebrate tunicate *Ecteinascidin turbinate*. *Trabectedin* is an alkylating drug that binds to the minor groove of DNA, allowing the alkylation of the N2 position of guanine and bending the helix toward the major groove. The bulky DNA adduct is recognized by the transcription-coupled NER complex, and these proteins initiate attempts to repair the damaged strand, converting the adduct to a double-stranded break. *Trabectedin* has particular cytotoxic effects on cells that lack components of the Fanconi anemia complex or those that lack the ability to repair double-strand DNA breaks through homologous recombination (Soares et al., 2011). Unlike *cisplatin* and other DNA adduct–forming drugs, the activity of *trabectedin* requires the presence of intact components of NER, including xeroderma pigmentosum endonuclease G, which may be important for initiation of single breaks and attempts at adduct removal.

ADME

Trabectedin is administered as a 24-h infusion of 1.3 mg/m^2 every 3 weeks. It is administered with *dexamethasone*, 4 mg twice daily, starting 24 h before drug infusion to diminish hepatic toxicity. The drug is slowly cleared by CYP3A4, with a plasma $t_{1/2}$ of about 24 to 40 h.

Therapeutic Uses

Trabectedin is approved for the treatment of patients with unresectable or metastatic liposarcoma or leiomyosarcoma after an anthracycline-containing regimen and is under study for the treatment of patients with ovarian and pancreatic cancer.

Adverse Effects

Without *dexamethasone* pretreatment, *trabectedin* causes significant hepatic enzyme elevations and fatigue in at least one-third of patients. With the steroid, the increases in transaminase are less pronounced and rapidly reversible. Other toxicities include mild myelosuppression and, rarely, rhabdomyolysis.

Lurbinectedin

Lurbinectedin is structurally similar to *trabectedin*. It covalently binds to residues in the minor groove of DNA and delays progression through S phase, causing cell cycle arrest in the G2/M phase and cell death. The FDA approved *lurbinectedin* in 2020 for metastatic small cell lung cancer disease progression on or after platinum-based chemotherapy. It has similar adverse effects as *trabectedin*. *Lurbinectedin* can be administered as a 60-min infusion, also every 3 weeks.

Drugs of Diverse Mechanisms of Action

Bleomycin

The bleomycins, an unusual group of DNA-cleaving antibiotics, are fermentation products of *Streptomyces verticillus,* comprising a family of peptide polyketides. The drug currently employed clinically is a mixture of two copper-chelating peptides, bleomycins A_2 and B_2, that differ only in their terminal amino acid. Because their toxicities do not overlap with those of other cytotoxic drugs and because of their unique mechanism of action, *bleomycin* maintains an important role in treating Hodgkin disease and testicular cancer.

Mechanisms of Action and Resistance

Bleomycin's cytotoxicity results from its capacity to cause oxidative damage to DNA. *Bleomycin* is a glycopeptide that has a DNA-binding portion (Chien et al., 1977) and, through a distinct functionality, binds to metal ions. In the presence of O_2, the Fe^{2+}-drug complex becomes activated and transfers electrons from Fe^{2+} to molecular oxygen to produce oxygen radicals, leading to single- and double-stranded breaks with release of free bases from DNA (Sausville et al., 1978). *Bleomycin* causes accumulation of cells in the G_2 phase of the cell cycle, and many of these cells display chromosomal aberrations, including chromatid breaks, gaps, fragments, and translocations.

Bleomycin is degraded by a specific hydrolase found in various normal tissues, including liver. Hydrolase activity is low in skin and lung, perhaps contributing to the serious toxicity. Some *bleomycin*-resistant cells contain high levels of hydrolase activity. In other cell lines, resistance has been attributed to decreased uptake, repair of strand breaks, or drug inactivation by thiols or thiol-rich proteins.

ADME

Bleomycin is administered intravenously, intramuscularly, or subcutaneously or instilled into the bladder for local treatment of bladder cancer. Having a high molecular mass, *bleomycin* crosses the blood-brain barrier poorly. The elimination $t_{1/2}$ is about 3 h. About two-thirds of the drug is excreted intact in the urine. Concentrations in plasma are greatly elevated in patients with renal impairment, and doses of *bleomycin* should be reduced in the presence of a CL_{Cr} less than 60 mL/min.

Therapeutic Uses

Bleomycin is given weekly or twice weekly by the intravenous, intramuscular, or subcutaneous route. A test dose of 2 units or less is recommended for patients with lymphoma. Myriad regimens are employed clinically, with *bleomycin* doses expressed in units. Total courses exceeding 250 mg should be given with caution, and usually only in high-risk testicular cancer treatment, because of a marked increase in the risk of pulmonary toxicity. *Bleomycin* also may be instilled into the pleural cavity in doses of 5 to 60 mg to ablate the pleural space in patients with malignant effusions. *Bleomycin* is highly effective against germ cell tumors of the testis and ovary. In testicular cancer, it is curative when used with *cisplatin* and *vinblastine* or *cisplatin* and *etoposide*. It is a component of the standard curative ABVD regimen for Hodgkin disease.

Adverse Effects

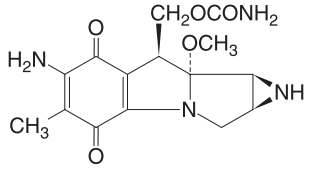
Because *bleomycin* causes little myelosuppression, it has significant advantages in combination with other cytotoxic drugs. However, it does cause a constellation of cutaneous toxicities, including hyperpigmentation, hyperkeratosis, erythema, even ulceration, and rarely, Raynaud phenomenon. Skin lesions may recur when patients are treated with other antineoplastic drugs. Rarely, *bleomycin* causes a flagellate dermatitis consisting of bands of pruritic erythema on the arms, back, scalp, and hands; this rash responds readily to topical corticosteroids.

The most serious adverse reaction to *bleomycin* is pulmonary toxicity, which begins with a dry cough, fine rales, and diffuse basilar infiltrates on X-ray and may progress to life-threatening pulmonary fibrosis. Approximately 5% to 10% of patients receiving *bleomycin* develop clinically apparent pulmonary toxicity, and about 1% die of this complication. Most who recover experience a significant improvement in pulmonary function, but fibrosis may be irreversible. Pulmonary function tests are not of predictive value for detecting early onset of this complication. The risk of pulmonary toxicity is related to total dose, with a significant increase in risk in total doses greater than 250 mg and in patients more than 40 years of age, in those with a CL_{Cr} less than 80 mL/min, and in those with underlying pulmonary disease; single doses of 30 mg/m^2 or more also are associated with an increased risk of pulmonary toxicity. Administration of high O_2 concentrations during anesthesia or respiratory therapy may aggravate or precipitate pulmonary toxicity in patients previously treated with the drug, in keeping with the clear ability of metallobleomycin complexes to generate O_2-derived radicals. There is no known specific therapy for *bleomycin* lung injury except for symptomatic management and standard pulmonary care. Steroids are of variable benefit, with greatest effectiveness in the earliest inflammatory stages of the lesion.

Other toxic reactions to *bleomycin* include hyperthermia, headache, nausea and vomiting, and a peculiar acute fulminant reaction observed in patients with lymphomas. This reaction is characterized by profound hyperthermia, hypotension, and sustained cardiorespiratory collapse; it does not appear to be a classical anaphylactic reaction and may be related to release of an endogenous pyrogen. This reaction has occurred in about 1% of patients with lymphomas or testicular cancer.

Mitomycin

Mitomycin has limited clinical utility, having been replaced by less toxic and more effective drugs, with the exception of the use in patients with anal cancers, for which it is potentially curative.



MITOMYCIN

Mechanisms of Action and Resistance

After intracellular enzymatic or spontaneous chemical alteration, *mitomycin* becomes a bifunctional or trifunctional alkylating agent. The drug inhibits DNA synthesis and cross-links DNA at the N6 position of adenine and at the O6 and N7 positions of guanine. Attempts to repair DNA lead to strand breaks. *Mitomycin* is a potent radiosensitizer, teratogen, and carcinogen in rodents. Resistance has been ascribed to deficient activation, intracellular inactivation of the reduced Q form, and Pgp-mediated drug efflux.

ADME

Mitomycin is administered intravenously. It has a $t_{1/2}$ of 25 to 90 min. The drug distributes widely throughout the body but is not detected in the CNS. Inactivation occurs by hepatic metabolism or chemical conjugation with sulfhydryls. Less than 10% of the active drug is excreted in the urine or the bile.

Therapeutic Uses

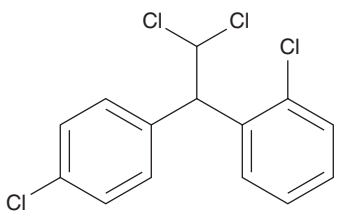
Mitomycin is administered as a single bolus (6–20 mg/m^2) every 6 to 8 weeks. Dosage should be modified based on hematological recovery. *Mitomycin* also may be used by direct instillation into the bladder to treat superficial transitional cell carcinomas. *Mitomycin* is used in combination with 5FU and *cisplatin* for the treatment of anal cancer. *Mitomycin* is also approved for local use as an adjunct to glaucoma surgery to reduce scarring.

Adverse Effects

The major toxic effect is myelosuppression, characterized by marked leukopenia and thrombocytopenia; after higher doses, maximal suppression may be delayed and cumulative, with recovery only after 6 to 8 weeks of pancytopenia. Nausea, vomiting, diarrhea, stomatitis, rash, fever, and malaise also are observed. Patients who have received more than 50 mg/m^2 total dose may acutely develop hemolysis, neurological abnormalities, interstitial pneumonia, and glomerular damage resulting in renal failure. The incidence of renal failure increases to 28% in patients who receive total doses of 70 mg/m^2 or more. There is no effective treatment of the disorder. It must be recognized early, and *mitomycin* must be discontinued immediately. *Mitomycin* causes interstitial pulmonary fibrosis; total doses greater than 30 mg/m^2 have infrequently led to congestive heart failure. *Mitomycin* may potentiate the cardiotoxicity of *doxorubicin*.

Mitotane

Mitotane (*o,p′*-DDD, dichlorodiphenyldichloroethane), a compound chemically similar to the insecticides DDT and DDD, is used in the treatment of adrenal cortex carcinoma. The mechanism of action of *mitotane* has not been elucidated, but its relatively selective destruction of adrenocortical cells, normal or neoplastic, is well established. Administration of the drug causes a rapid reduction in the levels of adrenocorticosteroids and their metabolites in blood and urine, a response that is useful in both guiding dosage and following the course of hyperadrenocorticism (Cushing syndrome) resulting from an adrenal tumor or adrenal hyperplasia. It does not damage other organs.



MITOTANE (*o,p′*-DDD)

ADME

Approximately 40% of *mitotane* is absorbed after oral administration. Plasma concentrations of mitotane are still measurable for 6 to 9 weeks following discontinuation of therapy. Although the drug is found in all tissues, fat is the primary site of storage. A water-soluble metabolite of *mitotane* found in the urine constitutes 25% of an oral or parenteral dose. About 60% of an oral dose is excreted unchanged in the stool.

Therapeutic Uses

Mitotane initial daily oral doses are 2 to 6 g, usually in three or four divided portions. The maximal tolerated dose may vary from 2 to 16 g/day. Treatment should continue for at least 3 months; if beneficial effects are observed, therapy should be maintained indefinitely. *Spironolactone* should not be administered concomitantly because it interferes with the adrenal suppression produced by *mitotane*. Treatment with *mitotane* is indicated for the palliation of inoperable adrenocortical carcinoma, producing symptomatic benefit in 30% to 50% of such patients.

Adverse Effects

Although the administration of *mitotane* produces anorexia and nausea in most patients, somnolence and lethargy in about 34%, and dermatitis in 15% to 20%, these effects do not contraindicate the use of the drug at lower doses. Because this drug damages the adrenal cortex, administration of replacement doses of adrenocorticosteroids is necessary.

L-Asparaginase

Malignant lymphoid cells depend on exogenous sources of L-asparagine. Thus, *L-asparaginase* (L-ASP) has become a standard agent for treating ALL.

Mechanism of Action

Most normal tissues synthesize L-asparagine in amounts sufficient for protein synthesis, but lymphocytic leukemias lack adequate amounts of asparagine synthase and derive the required amino acid from plasma. L-ASP, by catalyzing the hydrolysis of circulating asparagine to aspartic acid and ammonia, deprives these malignant cells of asparagine, leading to cell death. L-ASP is used in combination with other agents, including MTX, *doxorubicin*, *vincristine*, and *prednisone*, for the treatment of ALL and high-grade lymphomas. Resistance arises through induction of asparagine synthetase in tumor cells.

ADME and Therapeutic Use

Asparaginase is given intramuscularly or intravenously. After intravenous administration, *Escherichia coli*–derived L-ASP has a clearance rate from plasma of 0.035 mL/min/kg, a volume of distribution that approximates the volume of plasma in humans, and a $t_{1/2}$ of 1 day. The enzyme is given in doses of 6000 to 10,000 IU every third day for 3 to 4 weeks. *Pegaspargase* is a preparation in which the enzyme is conjugated to 5000-Da units of monomethoxy polyethylene glycol, resulting in a much longer plasma $t_{1/2}$ (6–7 days); it is administered intramuscularly every 14 days, producing rapid and complete depletion of plasma and tumor cell asparagine for 21 days in most patients. *Pegaspargase* has much reduced immunogenicity (<20% of patients develop antibodies) and has been approved for first-line ALL therapy. The conjugated preparation *calaspargase pegol* is administered intravenously every 21 days.

Intermittent dosage regimens and longer durations of treatment increase the risk of inducing hypersensitivity. In hypersensitive patients, neutralizing antibodies inactivate L-ASP. Not all patients with neutralizing antibodies experience clinical hypersensitivity, although the enzyme may be inactivated and therapy may be ineffective. In previously untreated ALL, *pegaspargase* produces more rapid clearance of lymphoblasts from bone marrow than does the *E. coli* preparation and circumvents the rapid antibody-mediated clearance seen with *E. coli* enzyme in relapsed patients. Asparaginase preparations only partially deplete CSF asparagine.

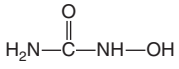
Adverse Effects

The L-ASP toxicities result from its antigenicity as a foreign protein and its inhibition of protein synthesis. Hypersensitivity reactions, including urticaria and full-blown anaphylaxis, occur in 5% to 20% of patients and may be fatal. In these patients, *pegaspargase* is a safe and effective alternative. So-called silent enzyme inactivation by antibodies occurs in a higher percentage of patients than overt hypersensitivity and may be associated with a negative clinical outcome, especially in high-risk patients with ALL.

Other toxicities result from inhibition of protein synthesis in normal tissues (e.g., hyperglycemia due to insulin deficiency, clotting abnormalities due to deficient clotting factors, hypertriglyceridemia due to effects on lipoprotein production, hypoalbuminemia). Pancreatitis also has been observed. The clotting problems may take the form of spontaneous thrombosis or, less frequently, hemorrhagic episodes. Brain magnetic resonance imaging studies should be considered in patients treated with L-ASP who present with seizures, headache, or altered mental status. Intracranial hemorrhage in the first week of L-ASP treatment is an infrequent but devastating complication. L-ASP also suppresses immune function. L-ASP terminates the antitumor activity of MTX when given shortly after the antimetabolite. By lowering serum albumin concentrations, L-ASP may decrease protein binding and accelerate plasma clearance of other drugs.

Hydroxyurea

Hydroxyurea (HU) inhibits the enzyme ribonuleotide reductase (RNR) and has unique and diverse biological effects as an antileukemic drug, radiation sensitizer, and an inducer of fetal hemoglobin (HbF) in patients with sickle cell disease. It is orally administered, and its toxicity is modest and limited to myelosuppression.



HYDROXYUREA

Mechanisms of Action and Resistance

Hydroxyurea inhibits RNR, which catalyzes the reductive conversion of ribonucleotides to deoxyribonucleotides, a rate-limiting step in the biosynthesis of DNA. HU binds non-heme iron atoms that are essential for activation of a tyrosyl radical in the catalytic subunit of RNR. The drug is specific for the S phase of the cell cycle, during which RNR concentrations are maximal. HU produces arrest at or near the G_1-S interface through both p53-dependent and-independent mechanisms. Because cells are highly sensitive to irradiation at the G_1-S boundary, HU and irradiation cause synergistic antitumor effects. Through depletion of deoxynucleotides, HU potentiates the antiproliferative effects of DNA-damaging agents such as *cisplatin*, alkylating agents, or topoisomerase II inhibitors and facilitates the incorporation of antimetabolites such as Ara-C, *gemcitabine*, and *fludarabine* into DNA. It also promotes degradation of the p21 cell-cycle checkpoint and thereby enhances the effects of histone deacetylase (HDAC) inhibitors *in vitro* (Kramer et al., 2008).

Hydroxyurea is the primary drug for improving control of sickle cell (HbS) disease in adults and for inducing HbF in patients with thalassemia HbC and HbC/S. It reduces vaso-occlusive events, painful crises, hospitalizations, and the need for blood transfusions in patients with sickle cell disease. The mechanism of stimulated HbF production is uncertain. HU stimulates nitric oxide production, causing nitrosylation of low-molecular-weight GTPases, a process that stimulates γ-globin production in erythroid precursors. Another property of HU that may be therapeutically relevant is its capacity to reduce L-selectin expression and thereby to reduce adhesion of red cells and neutrophils to vascular endothelium. Also, by suppressing the production of neutrophils, it decreases their contribution to vascular occlusion. Tumor cells become resistant to HU through increased synthesis of the catalytic subunit of RNR, thereby restoring enzyme activity.

ADME

The oral bioavailability of HU is 80% to 100%; comparable plasma concentrations are seen after oral or intravenous dosing. HU disappears from plasma with a $t_{1/2}$ of 3.5 to 4.5 h. The drug readily crosses the blood-brain barrier; significant quantities appear in human breast milk. From 40% to 80% of the drug is recovered in the urine within 12 h after administration. It is advisable to modify initial doses for patients with renal dysfunction.

Therapeutic Uses

In cancer treatment, HU is used alone or in combination with other drugs with two dosage schedules: (1) intermittent therapy administered orally as a single dose (~80 mg/kg) every third day or (2) continuous therapy administered as a single daily dose (20–30 mg/kg). In patients with essential thrombocythemia and in sickle cell disease, HU is given in a daily dose of 15 mg/kg, adjusting that dose upward or downward according to blood counts. The neutrophil count responds within 1 to 2 weeks to discontinuation of the drug. In treating subjects with sickle cell and related diseases, a neutrophil count of at least 2500 cells/mL should be maintained. Treatment typically is continued for 6 weeks to determine effectiveness; if satisfactory results are obtained, therapy can be continued indefinitely, although leukocyte counts at weekly intervals are advisable.

The principal use of HU has been as a myelosuppressive agent in various myeloproliferative syndromes, particularly CML, polycythemia vera, myeloid metaplasia, and essential thrombocytosis, for controlling high platelet or white cell counts. Many of the myeloproliferative syndromes harbor activating mutations of Janus kinase 2 (*JAK2*), a gene that is downregulated by HU. In essential thrombocythemia, it is the drug of choice for patients with a platelet count greater than 1.5 million cells/mm^3 or with a history of arterial or venous thrombosis. In CML, HU has been largely replaced by *imatinib* (see Chapter 71). HU is a potent radiosensitizer as a consequence of its inhibition of RNR and has been incorporated into several treatment regimens with concurrent irradiation (i.e., cervical carcinoma, primary brain tumors, head and neck cancer, non–small cell lung cancer).

Adverse Effects

Leukopenia, anemia, and occasionally thrombocytopenia are the major toxic effects; recovery of the bone marrow is prompt if the drug is discontinued for a few days. Other adverse reactions include a desquamative interstitial pneumonitis, GI disturbances, mild dermatological reactions, and, more rarely, stomatitis, alopecia, and neurological manifestations. Increased skin and fingernail pigmentation may occur, as well as painful leg ulcers, especially in elderly patients or in those with renal dysfunction. HU does not increase the risk of secondary leukemia in patients with myeloproliferative disorders or sickle cell disease. It is a potent teratogen in animals and should not be used in women with childbearing potential.

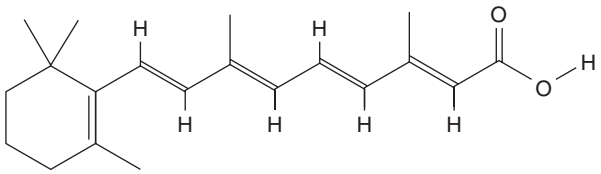
Retinoids

One of the hallmarks of malignant transformation is a block in differentiation. A number of chemical entities (vitamin D and its analogues, retinoids, benzamides and other inhibitors of HDAC [see Chapter 71], various cytotoxic and biological agents, and inhibitors of DNA methylation) can induce differentiation in tumor cell lines. The most important of these for cancer treatment are retinoids, in particular *tretinoin* (ATRA), which induces a high rate of complete remission in patients with APL as a single agent and, in combination with anthracyclines, cures most patients with this disease. The biology and pharmacology of retinoids are discussed in detail in Chapter 75.

Tretinoin (ATRA)

Mechanism of Action. Under physiological conditions, the retinoic acid receptor (RAR) A dimerizes with the retinoid X receptor to form a complex that binds ATRA tightly. ATRA binding displaces a repressor from the transcriptional complex and alters the expression of genes that control differentiation of cells of multiple lineages. Most APLs are

characterized by a chromosomal translocation of the *RARA* gene on chromosome 17 and fusion with the *PML* gene on chromosome 15 that is denoted as t(15;17)(q22;q12). In APL cells, physiological concentrations of retinoid are inadequate to displace the repressor, but pharmacological concentrations are effective in activating the differentiation program and in promoting degradation of the PML-RARA fusion protein (Collins, 2008). The *PML* gene encodes a transcription factor important in inhibiting proliferation and promoting myeloid differentiation. The oncogenic *PML-RARA* gene produces a protein that binds retinoids with much decreased affinity, lacks PML regulatory function, and fails to upregulate transcription factors (C/EBP and PU.1) that promote myeloid differentiation. ATRA also binds and activates RAR-γ and thereby promotes stem cell renewal, and this action may help restore normal bone marrow renewal. Resistance to ATRA arises by further mutation of the fusion gene, abolishing ATRA binding; by induction of the CYP26A1; or by loss of expression of the *PML-RARA* fusion gene.



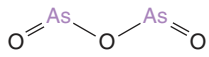
Tretinoin

Therapeutic Use, ADME and Adverse Effects. Oral administration of ATRA is 45 mg/m^2 per day until 30 days after remission is achieved (maximum course of therapy is 90 days). ATRA as a single agent reverses the hemorrhagic diathesis associated with APL and induces a high rate of temporary remission. ATRA in combination with an anthracycline induces remission with 80% or greater relapse-free long-term survival.

ATRA is cleared by a CYP3A4-mediated elimination with a $t_{1/2}$ of less than 1 h. Treatment with inducers of CYP3A4 leads to more rapid drug disappearance and resistance to ATRA. Inhibitors of CYPs, such as antifungal imidazoles, block ATRA degradation and may lead to hypercalcemia and renal failure, which respond to diuresis, bisphosphonates, and ATRA discontinuation. When used as a single agent for remission induction, especially in patients with more than 5000 leukemic cells/mm^3 in the peripheral blood, ATRA induces an outpouring of cytokines and mature-appearing neutrophils of leukemic origin. These cells express high concentrations of integrins and other adhesion molecules on their surface and clog small vessels in the pulmonary circulation, leading to significant morbidity in 15% to 20% of patients. Corticosteroids (e.g., *dexamethasone* 5–10 mg twice daily) and chemotherapy (e.g., hydroxyurea) sharply decrease the occurrence of "retinoic acid differentiation syndrome," which is characterized by fever, dyspnea, weight gain, pulmonary infiltrates, and pleural or pericardial effusions. The syndrome results from the differentiation of leukemia cells and their migration into peripheral tissues, most notably the lungs. Retinoids also cause dry skin, cheilitis, reversible hepatic enzyme abnormalities, bone tenderness, pseudotumor cerebri, hypercalcemia, and hyperlipidemia.

Arsenic Trioxide

Arsenic trioxide (As_2O_3), known as ATO, is a highly effective treatment of relapsed APL, producing complete responses in more than 85% of such patients. The chemistry and toxicity of *arsenic* are considered in Chapter 76.



ARSENIC TRIOXIDE

Mechanism of Action

The basis for ATO's antitumor activity remains uncertain. APL cells have high levels of reactive oxygen species (ROS) and are quite sensitive to further ROS induction. ATO inhibits thioredoxin reductase and thereby generates ROS. It inactivates glutathione and other sulfhydryls that scavenge ROS and thereby aggravates ROS damage. Cells exposed to ATO also upregulate p53, Jun kinase, and caspases associated with the intrinsic pathway of apoptosis and downregulate antiapoptotic proteins such as bcl-2 (see Figure 3–25). ATO's cytotoxic effects are antagonized by cell survival signals emanating from activation of components of the PI3 kinase cell survival pathway, including PKB, S6 kinase, and mTOR (see Figure 71–4). ATO also induces differentiation of leukemic cell lines in experimental and human leukemias, with evidence of enhanced degradation of the PML-RARα fusion protein that acts to inhibit normal differentiation of APL cells (Zhang et al., 2010). Recent studies indicate that ATO binding to p53 DNA binding domain mutants restores wild-type transcriptional function to the mutant p53 proteins (Chen et al., 2021).

ADME

Arsenic trioxide is well absorbed orally but in cancer treatment is administered as a 2-h daily intravenous infusion in dosages of 0.15 mg/kg per day for up to 60 days, until remission is documented. The drug enters cells via one of several glucose transporters. The primary mechanism of elimination is through enzymatic methylation. Multiple methylated metabolites (see Figure 71–9) form rapidly and are excreted in urine. Less than 20% of administered drug is excreted unchanged in the urine. No dose reductions are indicated for hepatic or renal dysfunction.

Adverse Effects

Pharmacological doses of ATO are well tolerated. Patients may experience reversible side effects, including hyperglycemia, hepatic enzyme elevations, fatigue, dysesthesias, and light-headedness. Fewer than 10% of patients experience a leukocyte maturation syndrome similar to that seen with ATRA, including pulmonary distress, effusions, and mental status changes. Oxygen, corticosteroids, and temporary discontinuation of ATO lead to full reversal of this syndrome. Lengthening of the QT interval on the electrocardiogram occurs in 40% of patients, but rarely do patients develop torsade de pointes. Simultaneous treatment with other QT-prolonging drugs should be avoided. Monitoring of serum electrolytes and repletion of serum K^+ in patients with hypokalemia are precautionary measures in patients receiving ATO therapy. In patients exhibiting a significantly prolonged QT (>470 msec), treatment should be suspended, K^+ supplemented, and therapy resumed only if the QT returns to normal.

Drug Facts for Your Personal Formulary: *Cytotoxic Drugs*

Drugs	Therapeutic Uses	Clinical Pharmacology and Tips
Section I: Alkylating Agents and Platinum Coordination Complexes		
Mechanism of action: covalent modification of DNA. Adverse effects of all alkylating drugs: myelosuppression and immunosuppression; toxicity to dividing mucosal cells and hair follicles (e.g., oral mucosal ulceration, intestinal denudation, alopecia); delayed pulmonary fibrosis; reproductive system toxicity (premature menopause, sterility); and leukemogenesis (up to 5%, highest for melphalan, procarbazine, nitrosoureas).		
Nitrogen Mustards: DNA alkylation		
Mechlorethamine	• Hodgkin lymphoma • Topical: cutaneous T-cell lymphoma	• Vascular damage during injection due to vesicant properties
Cyclophosphamide	• Acute and chronic lymphocytic leukemia; Hodgkin lymphoma; non-Hodgkin lymphoma; multiple myeloma; neuroblastoma; breast, ovary, Wilms tumor; soft-tissue sarcoma • Autoimmune disease (Wegener granulomatosis, rheumatoid arthritis, nephrotic syndrome)	• Oral or intravenous administration • Active alkylating moieties generated through hepatic metabolism • Nephrotoxic and urotoxic metabolite, acrolein; severe hemorrhagic cystitis in high-dose regimens; prevented by MESNA • Provide vigorous hydration during high-dose treatment • Elimination not affected by renal dysfunction; reduce dose in patients with hepatic dysfunction
Ifosfamide	• Germ cell testicular cancer • Pediatric and adult sarcoma • High-dose chemotherapy with bone marrow rescue	• See cyclophosphamide • Can cause neurotoxicity (including seizures) • Methylene blue treatment of CNS toxicity possibly useful
Melphalan	• Multiple myeloma	• Oral and intravenous administration
Chlorambucil	• Chronic lymphocytic leukemia	• Oral administration
Bendamustine	• Non-Hodgkin lymphoma • Chronic lymphocytic leukemia	• Lacks cross-resistance with other classical alkylators
Alkyl Sulfonate: DNA alkylation		
Busulfan	• Chronic myelogenous leukemia • High-dose chemotherapy regimen with bone marrow transplantation	• Oral administration • Adverse effects: prolonged (up to years) pancytopenia; suppression of stem cells; seizures; lung fibrosis • Clearance of phenytoin; hepatic VOD
Nitrosoureas: DNA alkylation		
Carmustine (BCNU)	• Malignant gliomas • Hodgkin lymphoma; non-Hodgkin lymphoma	• Vascular damage during injection due to vesicant properties • Profound and delayed myelosuppression
Streptozocin (or streptozotocin)	• Malignant pancreatic insulinoma • Carcinoid	• Frequent renal toxicity, sometimes renal failure
Methylhydrazine Derivatives: Monofunctional DNA alkylation		
Procarbazine (*N*-methylhydrazine, MIH)	• Hodgkin lymphoma • Gliomas	• Greater capacity for mutagenesis and carcinogenesis than bifunctional alkylators (e.g., cyclophosphamide)
Triazenes: Methyl transfer to DNA		
Dacarbazine (DTIC)	• Hodgkin lymphoma; soft-tissue sarcomas • Melanoma	• Intravenous administration • Activation by hepatic CYPs • Adverse effects: nausea, vomiting • Rare hepatotoxicity and neurotoxicity
Temozolomide	• Malignant gliomas	• Oral administration • Combined with radiation therapy • Greater capacity for mutagenesis and carcinogenesis than bifunctional alkylators; more active in MGMT-deficient tumors
Platinum Coordination Complexes: Form covalent metal adducts with DNA		
Cisplatin	• Testicular, ovarian, bladder, esophageal, gastric, lung, head and neck, anal, and breast cancer	• Intravenous administration • Adverse effects: • Nephrotoxicity (reduce by forced pretreatment hydration, diuresis, and use of *amifostine*) • Ototoxicity (tinnitus, high-frequency hearing loss) • Nausea and vomiting (antidote, *aprepitant*) • Peripheral sensory and motor neuropathy (may worsen after discontinuation; may be aggravated by taxane treatment) • Drug resistance due to loss of mismatch repair proteins

Drug Facts for Your Personal Formulary: *Cytotoxic Drugs (continued)*

Drugs	Therapeutic Uses	Clinical Pharmacology and Tips
Carboplatin	• Same as above	• Less nausea, neuro-, oto-, and nephrotoxicity than cisplatin • Dose-limiting toxicity: myelosuppression • May cause hypersensitivity reaction
Oxaliplatin	• Colorectal, gastric, and pancreatic cancer	• Peripheral neuropathy is dose limiting • Some nausea • Efficacy not dependent on intact mismatch repair
Section II: Antimetabolites		
Folic Acid Analogues: Inhibit dihydrofolate reductase		
Methotrexate (amethopterin)	• Acute lymphocytic leukemia; choriocarcinoma; breast, head and neck, ovary, bladder, and lung cancers; osteogenic sarcoma • Noncancer use: psoriasis, rheumatoid arthritis	• Oral, intravenous, or intramuscular administration • Adverse effects: myelosuppression, GI toxicity • Leucovorin can reverse toxic effects; used as "rescue" in high-dose therapy • *Glucarpidase*, a methotrexate-cleaving enzyme, is approved to treat toxicity • ↓ Dose in renal insufficiency
Pemetrexed	• Mesothelioma, lung cancer	• Similar effects and side effects as methotrexate • Attenuate toxicity with folate and vitamin B_{12} supplementation
Pyrimidine Analogues		
5-Fluorouracil (5FU) *Thymidylate synthase inhibitor*	• Breast, colon, esophageal, stomach, anal cancer • In FOLFOX or FOLFIRINOX combination to treat pancreatic or colorectal cancer • Combined with cisplatin in head and neck cancer • Premalignant skin lesion (topical)	• Intravenous administration • Nausea, mucositis, diarrhea, myelosuppression, hand and foot syndrome • Combined with leucovorin to enhance efficacy • Enhanced toxicity with DPD deficiency; may rescue with uridine triacetate
Capecitabine *Thymidylate synthase inhibitor*	• Metastatic breast, colorectal cancer	• Orally administered prodrug of 5FU • Similar adverse effects as 5FU; hand and foot syndrome more frequent than with 5FU
Cytarabine (cytosine arabinoside) *Interferes with base pairing in DNA; inhibits DNA polymerase*	• Acute myelogenous and acute lymphocytic leukemia; non-Hodgkin lymphoma	• Intravenous administration • Myelosuppressive; can cause acute, severe leukopenia, thrombocytopenia, anemia • GI disturbances • Noncardiogenic pulmonary edema • Dermatitis
Gemcitabine (difluoro analogue of deoxycytidine) *Inhibits DNA polymerase; causes strand termination*	• Pancreatic, ovarian, lung, bladder cancer	• Intravenous administration • Female and elderly patients clear the drug more slowly • Myelosuppression, hepatic toxicity • Rare posterior leukoencephalopathy syndrome; sometimes interstitial pneumonitis • Radiosensitizer; should be used with caution in radiotherapy
5-Azacytidine (azacitidine) *Inhibits DNA cytosine methyltransferase*	• Myelodysplasia	• Subcutaneous or intravenous administration • Myelosuppression and mild GI symptoms • After intravenous administration, severe nausea possible
Purine Analogues and Related Inhibitors		
6-Mercaptopurine *Inhibits purine nucleotide synthesis and metabolism*	• Acute lymphocytic and myelogenous leukemia; small cell non-Hodgkin lymphoma • Noncancer: Crohn's disease, ulcerative colitis	• Oral absorption incomplete, thus intravenous administration • Reduce oral dose by 75% in patients receiving allopurinol; no adjustment needed for intravenous administration • Myelosuppression; anorexia, nausea, vomiting; GI side effects less frequent in children than adults • Secondary malignancy: squamous cell skin cancer; AML
Fludarabine *A chain terminator when incorporated into DNA; inhibits RNA function and processing*	• Chronic lymphocytic leukemia • Follicular B-cell lymphoma • Allogeneic bone marrow transplant	• Oral or intravenous administration • Frequently myelosuppression • Less frequent: nausea, vomiting; altered mental status; seizures • Secondary myelodysplasia and acute leukemias • Adjust dose for renal dysfunction
Cladribine *Incorporated into DNA, produces strand breaks; inhibits conversion of ribo- to deoxyribonucleotides*	• Hairy cell leukemia • Chronic lymphocytic leukemia • Low-grade lymphoma • CTCL, Waldenström macroglobulinemia	• Intravenous administration • Adjust dose for renal dysfunction • Myelosuppression, opportunistic infections, nausea, high fever, tumor lysis syndrome

Drug Facts for Your Personal Formulary: *Cytotoxic Drugs* (*continued*)

Drugs	Therapeutic Uses	Clinical Pharmacology and Tips
Clofarabine (mechanism as above)	• Acute myelogenous or lymphocytic leukemia	• Intravenous administration • Adjust dose to creatinine clearance • Myelosuppression • Capillary leak syndrome: discontinue drug • Nausea, vomiting, diarrhea
Nelarabine *Incorporated into DNA, terminates DNA synthesis*	• T-cell leukemia, lymphoma	• Intravenous administration • Myelosuppression; liver function abnormalities; infrequent neurologic sequelae
Pentostatin (2′-deoxycoformycin) *Inhibits adenosine deaminase; causes immunodeficiency (T and B cells)*	• Hairy cell leukemia; chronic lymphocytic leukemia; small cell non-Hodgkin lymphoma	• Intravenous administration • Adjust dose for renal dysfunction • Myelosuppression, GI symptoms, skin rashes, opportunistic infections • Renal, neurologic, pulmonary toxicity
Section III: Natural Products		
Vinca Alkaloids: Inhibit tubulin polymerization and microtubule formation		
Vinblastine	• Hodgkin and non-Hodgkin lymphoma • Breast, bladder, lung, testicular cancer • Kaposi sarcoma, neuroblastoma • Part of ABVD combination with doxorubicin (Adriamycin, bleomycin, dacarbazine) for Hodgkin lymphoma	• Administered IV; extravasation causes irritation and ulceration • Reduce dose in patients with impaired liver function • Least neurotoxic vinca alkaloid • Myelosuppressive • GI side effects nausea, vomiting, diarrhea • Vinca alkaloids are substrates of the Pgp efflux pump
Vinorelbine	• Breast cancer • Non–small cell lung cancer	• Intravenous administration • Reduce dose in patients with impaired liver function • Intermediate neurotoxicity among the vinca alkaloids • Myelosuppressive (granulocytopenia)
Vincristine	• Acute lymphocytic leukemia; neuroblastoma; Wilms tumor; rhabdomyosarcoma; Hodgkin and non-Hodgkin lymphoma • Part of CHOP regimen: cyclophosphamide, doxorubicin (H), vincristine (O), prednisone	• Administered IV; extravasation causes irritation and ulceration • Reduce dose in patients with impaired liver function • Least myelosuppressive vinca alkaloid • Dose-limiting neurotoxicity • Better tolerated by children than adults
Eribulin	• Breast cancer, liposarcoma	• Side effects overlap with those of vinca but less sensitive to extrusion by Pgp
Taxanes: Stabilize microtubules, inhibit depolymerization		
Paclitaxel	• Ovarian, breast, lung, prostate, bladder, head and neck cancer	• Intravenous administration • Metabolized by hepatic CYPs, ↓ dose in patients with hepatic dysfunction • Substrate of Pgp efflux pump • Myelosuppressive, alleviated by G-CSF • Peripheral neuropathy is dose limiting • Mucositis
Docetaxel	• Same as above	• No effect on doxorubicin clearance • Pharmacokinetics similar to paclitaxel's • ↓ Neutropenia, ↓ neuropathy than paclitaxel
Camptothecins: Inhibit topoisomerase I; DNA religation is inhibited: accumulation of single-strand breaks		
Topotecan	• Ovarian cancer; small cell lung cancer	• Intravenous or oral administration • Reduce dose in patients with renal dysfunction • Neutropenia, GI side effects, nausea, vomiting • Substrate for Pgp
Irinotecan	• Colorectal cancer, small cell lung cancer • Part of FOLFIRI (leucovorin, 5FU, and irinotecan) or FOLFIRINOX combination for GI tumors	• Intravenous administration • Prodrug activated in the liver; CYP substrate • Diarrhea and neutropenia • Acetylcholinesterase inhibition results in cholinergic syndrome: treat with atropine

Drug Facts for Your Personal Formulary: *Cytotoxic Drugs (continued)*

Drugs	Therapeutic Uses	Clinical Pharmacology and Tips
Antibiotics		
Dactinomycin (actinomycin D) *Intercalates between GC base pairs of DNA*	• Wilms tumor; rhabdomyosarcoma; Ewing, Kaposi, and other sarcomas; choriocarcinoma	• Intravenous administration; severe injury on extravasation • Nausea, vomiting; myelosuppression; GI side effects; erythema, inflammation of the skin
Anthracyclines and Anthracenediones: Inhibit topoisomerase II and intercalate DNA		
Daunorubicin (daunomycin, rubidomycin)	• Acute myelogenous and acute lymphocytic leukemia	• Intravenous administration • Imparts a red color to the urine • Myelosuppression, GI side effects • Most important long-term side effect is cardiotoxicity, including tachycardia, arrhythmias, congestive heart failure • Alopecia
Doxorubicin	• Soft-tissue, osteogenic, and other sarcomas; Hodgkin and non-Hodgkin lymphoma; acute leukemia; breast, genitourinary, thyroid, and stomach cancer; neuroblastoma	• See Daunorubicin above
Mitoxantrone (an anthracenedione)	• Acute myelocytic leukemia; breast and prostate cancer	• Similar side effects as above • Less cardiotoxic
Epipodophyllotoxins: Inhibit topoisomerase II and religation of cleaved DNA strand		
Etoposide	• Testicular and lung cancer; Hodgkin lymphoma; non-Hodgkin lymphomas; acute myelogenous leukemia; Kaposi sarcoma	• Oral and intravenous administration • Reduce dose in patients with renal dysfunction • Leukopenia, GI side effects; hepatic toxicity after high doses • Secondary leukemia
Teniposide	• Acute lymphoblastic leukemia in children; glioblastoma, neuroblastoma	• Intravenous administration • Myelosuppression, nausea, vomiting
Drugs With Diverse Mechanisms of Action		
Bleomycin *Binds to DNA, generates free radicals, and induces DNA cleavage via deoxyribose ring damage*	• Testicular cancer; Hodgkin and non-Hodgkin lymphoma; local treatment of bladder cancer • Part of the ABVD regimen (doxorubicin [Adriamycin], bleomycin, vinblastine, and dacarbazine)	• IV, IM, or SC administration; instilled into bladder • Reduce dose in patients with renal dysfunction • Most serious: pulmonary toxicity • Cutaneous toxicity (erythema, ulcerations) • Less myelosuppression than other cytotoxics
L-Asparaginase *Hydrolyzes asparagine; deprives leukemia cells that lack asparagine synthase*	• Acute lymphocytic leukemia	• IV and IM administration • Hypersensitivity reactions, anaphylaxis • Hyperglycemia, clotting abnormalities
Hydroxyurea *Inhibits RNR (conversion of ribo- to deoxyribonucleotides)*	• Chronic myelogenous leukemia; polycythemia vera; essential thrombocytosis; sickle cell disease in adults	• Oral administration • Reduce dose in patients with renal dysfunction • Myelosuppression; some GI side effects
Tretinoin (all-*trans* retinoic acid) *Promotes degradation of PML-RARA fusion protein*	• Acute promyelocytic leukemia	• Oral administration • CYP substrate • Leukocyte maturation syndrome, pulmonary distress, effusions, fever, dyspnea • Dry skin, cheilitis • Hypercalcemia and renal failure
Arsenic trioxide *Inhibits thioredoxin and generates reactive oxygen species*	• Acute promyelocytic leukemia	• Oral or intravenous administration • Leukocyte maturation syndrome as above with ATRA • QT prolongation; rare torsade de pointes • See Chapter 76 (section on arsenic)

Note: For drugs that are subject to hepatic metabolism by CYP enzymes, drug exposure of a patient can be affected by coadministration of inhibitors or inducers of CYP3A4 and can then reduce efficacy or increase side effects.

Embryo-fetal toxicity: Consider that all of these drugs can cause fetal harm. Advise women of the potential risk to a fetus and to avoid pregnancy while taking the drug and for 1 month after cessation of therapy. Advise men to avoid fathering a child during the same time period. Avoid lactation during therapies.

Bibliography

Akhmanova A, Steinmetz MO. Control of microtubule organization and dynamics: two ends in the limelight. *Nat Rev Mol Cell Biol*, **2015**, *16*:711–726.

Allegra CJ, et al. Trimetrexate for the treatment of *Pneumocystis carinii* pneumonia in patients with acquired immunodeficiency syndrome. *N Engl J Med*, **1987a**, *317*:978–985.

Allegra CJ, et al. Evidence for direct inhibition of de novo purine synthesis in human MCF-7 breast cells as a principal mode of metabolic inhibition by methotrexate. *J Biol Chem*, **1987b**, *262*:13520–13526.

Arnold H, et al. Chemotherapeutic action of a cyclic nitrogen mustard phosphamide ester (B 518-ASTA) in experimental tumours of the rat. *Nature*, **1958**, *181*:931–931.

Berlin NI, et al. Folic acid antagonist. Effects on the cell and the patient. Combined clinical staff conference at the National Institutes of Health. *Ann Intern Med*, **1963**, *59*:931–956.

Calvert AH, et al. Carboplatin dosage: prospective evaluation of a simple formula based on renal function. *J Clin Oncol*, **1989**, *7*:1748–1756.

Chabner BA, Roberts TG. Timeline: chemotherapy and the war on cancer. *Nat Rev Cancer*, **2005**, *5*:65–72.

Chen S, et al. Arsenic trioxide rescues structural p53 mutations through a cryptic allosteric site. *Cancer Cell*, **2021**, *39*:225–239.

Chien M, et al. Bleomycin-DNA interactions: fluorescence and proton magnetic resonance studies. *Biochemistry*, **1977**, *16*:2641–2647.

Collins SJ. Retinoic acid receptors, hematopoiesis and leukemogenesis. *Curr Opin Hematol*, **2008**, *15*:346–351.

Costa de Almeida L, et al. DNA damaging agents and DNA repair: from carcinogenesis to cancer therapy. *Cancer Genet*, **2021**, *252–253*:6–24.

Curt GA, et al. Determinants of the sensitivity of human small-cell lung cancer cell lines to methotrexate. *J Clin Invest*, **1985**, *76*:1323–1329.

Davidson NE, et al. AACR cancer progress report 2016. *Clin Cancer Res*, **2016**, *22*:S1–S137.

Deyoung MP, Ellisen LW. p63 and p73 in human cancer: defining the network. *Oncogene*, **2007**, *26*:5169–5183.

Farber S, et al. Temporary remissions in acute leukemia in children produced by folic antagonist 4-amethopteroylglutamic acid (aminopterin). *N Engl J Med*, **1948**, *238*:787–793.

Fischel JL, et al. Impact of the oxaliplatin-5 fluorouracil-folinic acid combination on respective intracellular determinants of drug activity. *Br J Cancer*, **2002**, *86*:1162–1168.

Gerth K, et al. Epothilons A and B: antifungal and cytotoxic compounds from *Sorangium cellulosum* (Myxobacteria). Production, physico-chemical and biological properties. *J Antibiot (Tokyo)*, **1996**, *49*:560–563.

Gilman A, Philips FS. The biological actions and therapeutic applications of the β-chlorethylamines and sulfides. *Science*, **1946**, *103*:409–415.

Hanahan D, Weinberg RA. Hallmarks of cancer: the next generation. *Cell*, **2011**, *144*:646–674.

Hegi ME, et al. Correlation of O6-methylguanine methyltransferase (MGMT) promoter methylation with clinical outcomes in glioblastoma and clinical strategies to modulate MGMT activity. *J Clin Oncol*, **2008**, *26*:4189–4199.

Heinemann V, et al. Comparison of the cellular pharmacokinetics and toxicity of 2′,2′-difluorodeoxycytidine and 1-β-D-arabinofuranosyl-cytosine. *Cancer Res*, **1988**, *48*:4024–4031.

Kamiya K, et al. Inhibition of the 3' → 5' exonuclease human DNA polymerase by fludarabine-terminated DNA. *J Biol Chem*, **1996**, *271*:19428–19435.

Karran P, Attard N. Thiopurines in current medical practice: molecular mechanisms and contributions to therapy-related cancer. *Nat Rev Cancer*, **2008**, *8*:24–36.

Kastan MB. Molecular determinants of sensitivity to antitumor agents. *Biochem Biophys Acta*, **1999**, *1424*:R37–R42.

Kramer OH, et al. Histone deacetylase inhibitors and hydroxyurea modulate the cell cycle and cooperatively induce apoptosis. *Oncogene*, **2008**, *27*:732–740.

Krumbhaar EB. Role of the blood and the bone marrow in certain forms of gas poisoning: I. Peripheral blood changes and their significance. *JAMA*, **1919**, *72*:39–41.

Kufe DW, et al. Effects of 1-β-D-arabinofuranosylcytosine incorporation on eukaryotic DNA template function. *Mol Pharmacol*, **1984**, *26*:128–134.

Lee JJ, Swain SM. The epothilones: translating from the laboratory to the clinic. *Clin Cancer Res*, **2008**, *14*:1618–1624.

Leoni LM, et al. Bendamustine (Treanda) displays a distinct pattern of cytotoxicity and unique mechanistic features compared with other alkylating agents. *Clin Cancer Res*, **2008**, *14*:309–317.

Li X, Heyer WD. Homologous recombination in DNA repair and DNA damage tolerance. *Cell Res*, **2008**, *18*:99–113.

Lipshultz SE, et al. The effect of dexrazoxane on myocardial injury in doxorubicin-treated children with acute lymphoblastic leukemia. *N Engl J Med*, **2004**, *351*:145–153.

Lowe SW, et al. p53-dependent apoptosis modulates the cytotoxicity of anticancer agents. *Cell*, **1993**, *74*:957–967.

Lowe SW, et al. Intrinsic tumour suppression. *Nature*, **2004**, *432*:307–315.

Lyu YL, et al. Topoisomerase IIB–mediated DNA double-strand breaks: implications in doxorubicin cardiotoxicity and prevention by dexrazoxane. *Cancer Res*, **2007**, *67*:8839–8846.

Milano G, et al. Dihydropyrimidine dehydrogenase deficiency and fluorouracil-related toxicity. *Br J Cancer*, **1999**, *79*:627–630.

Myers CE, et al. Adriamycin: the role of lipid peroxidation in cardiac toxicity and tumor response. *Science*, **1977**, *197*:165–167.

Panzarino NJ, et al. Replication gaps underlie BRCA-deficiency and therapy response. *Cancer Res*, **2021**, *81*:1388–1397.

Paré L, et al. Pharmacogenetic prediction of clinical outcome in advanced colorectal cancer patients receiving oxaliplatin/5-fluorouracil as first-line chemotherapy. *Br J Cancer*, **2008**, *99*:1050–1055. Erratum in: *Br J Cancer*, **2009**, *100*:1368.

Pelgrims J, et al. Methylene blue in the treatment and prevention of ifosfamide-induced encephalopathy: report of 12 cases and a review of the literature. *Br J Cancer*, **2000**, *82*:291–294.

Pommier Y. Drugging topoisomerases: lessons and challenges. *ACS Chem Biol*, **2013**, *8*:82–95.

Pui CH, et al. Epipodophyllotoxin-related acute myeloid leukemia: a study of 35 cases. *Leukemia*, **1995**, *9*:1990–1996.

Pui CH, et al. Mechanisms of disease: acute lymphoblastic leukemia. *N Engl J Med*, **2004**, *350*:1535–1548.

Pullarkat ST, et al. Thymidylate synthase gene polymorphism determines response and toxicity of 5-FU chemotherapy. *Pharmacogenomics J*, **2001**, *1*:65–70.

Reed E. Platinum-DNA adduct, nucleotide excision repair and platinum based anti-cancer chemotherapy. *Cancer Treat Rev*, **1998**, *24*: 331–344.

Reich E. Biochemistry of actinomycins. *Cancer Res*, **1963**, *23*:1428–1441.

Rochette L, et al. Anthracyclines/trastuzumab: new aspects of cardiotoxicity and molecular mechanisms. *Trends Pharmacol Sci*, **2015**, *36*:326–348.

Rosenberg B, et al. Platinum compounds: a new class of potent antitumour agents. *Nature*, **1969**, *222*:385–386.

Salinger DH, et al. Development of a population pharmacokinetics-based sampling schedule to target daily intravenous busulfan for outpatient clinic administration. *J Clin Pharmacol*, **2010**, *50*:1292–1300.

Sausville EA, et al. Properties and products of the degradation of DNA by bleomycin and iron(II). *Biochemistry*, **1978**, *17*:2746–2754.

Schiff PB, et al. Promotion of microtubule assembly in vitro by taxol. *Nature*, **1979**, *277*:665–667.

Schimke RT, et al. Gene amplification and drug resistance in cultured murine cells. *Science*, **1978**, *202*:1051–1055.

Serrano J, et al. Cardioselective and cumulative oxidation of mitochondrial DNA following subchronic doxorubicin administration. *Biochem Biophys Acta*, **1999**, *1411*:201–205.

Sharma S, et al. Molecular dynamic simulations of cisplatin- and oxaliplatin-d(GG) intrastrand cross-links reveal differences in their conformational dynamics. *J Mol Biol*, **2007**, *373*:1123–1140.

Shen DW, et al. Cisplatin resistance: a cellular self-defense mechanism resulting from multiple epigenetic and genetic changes. *Pharmacol Rev*, **2012**, *64*:706–721.

Slamon DJ, et al. Use of chemotherapy plus a monoclonal antibody against HER2 for metastatic breast cancer that overexpresses HER2. *N Engl J Med*, **2001**, *344*:783–792.

Soares DG, et al. Trabectedin and its C subunit modified analogue PM01183 attenuate nucleotide excision repair and show activity toward platinum-resistant cells. *Mol Cancer Ther*, **2011**, *10*:1481–1489.

Steinmetz MO, Prota AE Microtubule-targeting agents: strategies to hijack the cytoskeleton. *Trends Cell Biol*, **2018**, *28*:1–17.

Stover EH, et al. Clinical assays for assessment of homologous recombination DNA repair deficiency. *Gynecol Oncol*, **2020**, *159*:887–898.

Swain SM, et al. Cardioprotection with dexrazoxane for doxorubicin-containing therapy in advanced breast cancer. *J Clin Oncol*, **1997**, *15*:1318–1332.

Tew KD, Stearns ME. Hormone-independent, non-alkylating mechanism of cytotoxicity for estramustine. *Urol Res*, **1987**, *15*:155–160.

Wagner T. Ifosfamide clinical pharmacokinetics. *Clin Pharmacokinet*, **1994**, *26*:439–456.

Wilson PM, et al. Standing the test of time: targeting thymidylate biosynthesis in cancer therapy. *Nat Rev Clin Oncol*, **2014**, *11*:282–298.

Yan VC, et al. Why great mitotic inhibitors make poor cancer drugs. *Trends Cancer*, **2020**, *6*:924–941.

Zhang X, et al. Arsenic trioxide controls the fate of the PML-RARα oncoprotein by directly binding PML. *Science*, **2010**, *328*:240–243.

Zweier JL. Reduction of O_2 by iron-Adriamycin. *J Biol Chem*, **1984**, *259*:6056–6058.

This page intentionally left blank

Chapter 71

Protein Kinase Inhibitors and Pathway-Targeted Small Molecules

Anton Wellstein and Giuseppe Giaccone

I. INHIBITORS OF RECEPTOR SIGNALING IN CANCER CELLS

TYROSINE KINASE RECEPTORS

- Inhibitors of Human Epidermal Growth Factor Receptor (EGFR/HER1)
- Inhibitors of Human Epidermal Growth Factor Receptor 2 (HER2/Neu/ErbB2)
- ALK and ROS1
- NRTK/TRK Fusions
- Inhibitors of Platelet-Derived Growth Factor Receptor and KIT
- FGFR Inhibitors
- MET/HGFR Inhibitors
- RET Inhibitors

HEDGEHOG PATHWAY INHIBITORS

- SMO Inhibitors
- GLI Inhibitor

II. INHIBITORS OF INTRACELLULAR PROTEIN KINASE SIGNALING IN CANCER CELLS

INHIBITORS OF RAS

- Sotorasib
- Adagrasib

INHIBITORS OF RAF KINASE: VEMURAFENIB, DABRAFENIB, AND ENCORAFENIB

- Combination of BRAF and MEK Inhibitors
- Vemurafenib
- Dabrafenib
- Encorafenib

INHIBITORS OF MEK: TRAMETINIB, COBIMETINIB, BINIMETINIB, AND SELUMETINIB

- Trametinib
- Cobimetinib
- Binimetinib
- Selumetinib

INHIBITORS OF JAK1 AND JAK2

- Ruxolitinib
- Fedratinib

CYCLIN-DEPENDENT KINASE (CDK) 4/6 INHIBITORS

- Palbociclib
- Abemaciclib
- Ribociclib
- Trilaciclib

BRUTON TYROSINE KINASE (BTK) INHIBITORS

- Ibrutinib
- Acalabrutinib
- Zanubrutinib

INHIBITORS OF THE BCR-ABL KINASE

- Imatinib, Dasatinib, and Nilotinib
- Bosutinib
- Ponatinib

INHIBITORS OF THE PI3K/AKT/mTOR PATHWAY

- Background
- Impact of PI3K Inhibitors on the Tumor Microenvironment
- Idelalisib
- Copanlisib
- Duvelisib
- Umbralisib
- Alpelisib
- mTOR Inhibitors: Rapamycin Analogues

MULTIKINASE INHIBITORS: CABOZANTINIB, VANDETANIB, MIDOSTAURIN, AND GILTERITINIB

- Vandetanib
- Midostaurin
- Gilteritinib

III. INHIBITORS OF TUMOR ANGIOGENESIS

INHIBITION VEGF AND THE VEGFR PATHWAY

ANTIANGIOGENIC SMALL-MOLECULE KINASE INHIBITORS

- Sunitinib
- Sorafenib
- Pazopanib
- Axitinib
- Tivozanib
- Lenvatinib
- Regorafenib
- Belzutifan

IV. INHIBITORS OF POLY(ADP-RIBOSE) POLYMERASE (PARP)

- Olaparib
- Rucaparib
- Niraparib
- Talazoparib

V. MODULATORS OF PROTEIN DEGRADATION

TARGETING PROTEIN DEGRADATION

THALIDOMIDE AND LENALIDOMIDE

- Thalidomide
- Lenalidomide
- Adverse Effects of Thalidomide and Lenalidomide
- Pomalidomide

PROTEASOME INHIBITORS

- First Generation
- Second Generation

VI. EPIGENETIC MODULATORS: INHIBITORS OF HDAC, HMT, AND IDH1/2

INHIBITORS OF HISTONE DEACETYLASE

- Panobinostat
- Romidepsin
- Vorinostat
- Belinostat

INHIBITORS OF HISTONE METHYLTRANSFERASE

- Tazemetostat

IDH1/2 INHIBITORS

VII. OTHER INHIBITORS (BCL2, NUCLEAR EXPORT, TRANSLATION, CXCR4)

BCL2 INHIBITORS

- Venetoclax

NUCLEAR EXPORT INHIBITOR

- Selinexor

PROTEIN TRANSLATION INHIBITOR

- Omacetaxine

CXCR4 INHIBITOR

- Plerixafor

Abbreviations

ABC: ATP-binding cassette
ABL: Abelson murine leukemia viral oncogene homolog
ALK: anaplastic lymphoma kinase
ALL: acute lymphoblastic leukemia
AML: acute myelocytic leukemia
APML: acute promyelocytic leukemia
ARNT: aryl hydrocarbon receptor nuclear translocator (also known as HIF-1β)
ATO: arsenic trioxide
AUC: area under the curve
BCC: basal cell carcinoma
BCL: B-cell lymphoma
BCR: breakpoint cluster region (chr22)
BCRP: breast cancer resistance protein
BH: BCL2 homology domain
BIM: BCL2-like protein 11
BRCA: breast cancer susceptibility gene
BRK: breast tumor kinase, PTK6
BTK: Bruton tyrosine kinase
CDK: cyclin-dependent kinase
CLL: chronic lymphocytic leukemia
CML: chronic myelocytic leukemia
CRL4-CRBN: cullin-RING ubiquitin ligase cereblon complex
cuSCC: cutaneous squamous cell carcinoma
CXCR4: C-X-C chemokine receptor 4
DNMT: DNA methyltransferase
ECG: electrocardiogram
EGF(R): epidermal growth factor (receptor) = HER1
EML4: echinoderm microtubule-associated protein-like 4
ER: estrogen receptor
ERK: extracellular signal-related kinase = MAPK
FGF(R): fibroblast growth factor (receptor)
FIP1L1: factor interacting with poly (A) polymerase
FKBP12: immunophilin-binding protein for tacrolimus (FK506)
FL: follicular B-cell non-Hodgkin lymphoma
FLT(1 or 4): fms-related tyrosine kinase (1 or 4) (= VEGFR1 or 3)
FMO3: flavin containing monooxygenase 3
GI: gastrointestinal
GIST: gastrointestinal stromal tumor
HDAC: histone deacetylase
HER1/2: human epidermal growth factor receptor; 1 = EGFR; 2 = erbB2
2-HG: 2-hydroxyglutarate
HGF(R): hepatocyte growth factor (receptor) = cMET
HIF-1 or -2: hypoxia-inducible factor 1 or 2
HMT: histone methyltransferase
HSC: hematopoietic stem cell
IFN: interferon
IGF1R: insulin-like growth factor 1 receptor
IκB: inhibitor of nuclear factor kappa B
IL: interleukin
IMiDs: immunomodulatory imide drugs (e.g., thalidomide)
ITK: inducible T-cell kinase
KDR: kinase insert domain receptor = VEGFR2
KIT: feline sarcoma virus oncogene homolog
LCK: lymphocyte-specific kinase
MAPK: mitogen-activated extracellular signal regulated protein kinase = ERK
MCL: mantle cell lymphoma
MCyR: major cytogenetic response
MDS: myelodysplastic syndrome
MEK: mitogen-activated extracellular signal regulated protein kinase kinase = MKK
MET: mesenchymal-epithelial transition factor (= HGFR)
MM: multiple myeloma
mTOR: mammalian or mechanistic target of rapamycin
NCCN: National Comprehensive Cancer Network
NET: neuroendocrine tumor
NF-κB: nuclear factor-κB
NK: natural killer
***NPM*:** nucleophosmin (gene)
NRAS: neuroblastoma RAS virus homolog
NSCLC: non–small cell lung cancer
OATP: organic anion-transporting polypeptide
PARP: poly(ADP-ribose) polymerase
PDGF(R): platelet-derived growth factor (receptor)
Pgp: P-glycoprotein
PI3K: phosphatidylinositol 3-kinase
PIK3CA: PI3K catalytic subunit α gene; codes for the p110α enzyme
PIP2: phosphatidylinositol 4,5-bisphosphate
PIP3: phosphatidylinositol (3,4,5)-trisphosphate
PNET: peripheral neuroendocrine tumor
PPES: palmar-plantar erythrodysesthesia syndrome

PPI: proton pump inhibitor
PROTAC: proteolysis-targeting chimera
PTEN: phosphatase and tensin homolog
Rb: retinoblastoma (protein)
RCC: renal cell carcinoma
RET: "rearranged during transfection" receptor tyrosine kinase
ROS1: orphan receptor tyrosine kinase
SDF: stromal cell-derived factor
SLL: small lymphocytic lymphoma
TGF: transforming growth factor
TIE: tyrosine kinase with Ig and EGF domains
TKI: tyrosine kinase inhibitor
TSC: tuberous sclerosis complex
VEGF(R): vascular endothelial growth factor (receptor)
VHL: von Hippel-Lindau (pVHL = VHL protein)

A Note on Treatment Regimens

Cancer treatment regimens change to reflect continuous advances in basic and clinical science. As a consequence, this chapter presents relatively few detailed treatment regimens; rather, we refer the reader to the web-based resources of the U.S. FDA and the National Comprehensive Cancer Network (NCCN). Table 71–1 provides two examples of therapeutic regimens that illustrate the complexity of current cancer drug therapy.

Development of targeted therapies for treating cancer relies on the discovery of molecular changes that drive malignant progression of human cancers. A rapidly growing number of drugs are being developed to block oncogenic pathways that lead to dysregulated cancer cell growth and survival and may be combined with cytotoxic cancer drugs described in Chapter 70 to improve efficacy. Growth factor receptors and downstream signaling molecules have been productively targeted by cancer chemotherapeutic agents. Indeed, the use of pathway-targeted therapies has likely contributed to reduced mortality from lung cancer (Howlader et al., 2020). The drivers of cancer growth are oncogenic pathways in malignant cells (e.g., mutant receptors and kinases), the reaction of the tumor microenvironment (e.g., angiogenesis), and the escape of malignant cells from the host's immune surveillance (Hanahan and Weinberg, 2011). The primary tools for targeting oncogenic pathways are *small molecules* (discussed in this chapter) that enter cells and engage intracellular targets (e.g., kinases) and *monoclonal antibodies* (see Chapter 72) that recognize cell surface or shed antigens (e.g., growth factor receptors or receptor ligands). These two classes of drugs have different pharmacological properties (see box titled Two Classes of Pathway-Targeted Drugs Used in Cancer Treatment). The sections of this chapter focus on inhibitory drugs that target these drivers of cancer cell growth and survival:

I. Altered growth factor receptor signaling
II. Activation of intracellular kinases
III. Cancer cell-host interactions and aberrant angiogenesis
IV. Defects in DNA repair
V. Altered protein degradation
VI. Altered epigenetic regulation
VII. Targets that control cancer cell behavior

TABLE 71–1 ■ CANCER TREATMENT REGIMENS: COMPLEXITIES AND RESOURCES

In treating cancers, the selection of the most appropriate treatment(s), dose, and dose intervals and the management of adverse effects require specialized knowledge and a team effort. In addition, cancer treatment regimens are undergoing frequent updates due to findings in ongoing clinical trials. The availability of drugs with new mechanisms of action (e.g., immune checkpoint inhibitors), good target selectivity (e.g., kinase inhibitors), efficacy in specific cancers, and different adverse effect profiles permits the use of new drug combinations and regimens. The chapters on the pharmacotherapy of cancer focus on mechanism of action, ADME, and adverse effects, with less emphasis on treatment regimens. There are several excellent sources of up-to-date treatment information. The U.S. FDA provides authoritative information on approved drugs online (http://www.accessdata.fda.gov/scripts/cder/daf/). Detailed treatment guidelines for most cancers are available from the National Comprehensive Cancer Network (NCCN), a nonprofit alliance of 31 cancer centers that establishes evidence-based guidelines for cancer treatment. The NCCN guidelines are continuously updated to reflect new information that may alter clinical practice. The guidelines can be accessed online (https://www.nccn.org/professionals/physician_gls/f_guidelines.asp).

This table provides examples that illustrate the complexities of current treatment for two cancers: HER2-positive breast cancer and advanced colorectal cancer.

SAMPLE TREATMENT REGIMENS

SUMMARY	DRUGS, SCHEDULE, AND DOSING
HER2-Positive Breast Cancer	
Doxorubicin and cyclophosphamide (AC) followed by paclitaxel, trastuzumab, and pertuzumab	**Day 1:** Doxorubicin 60 mg/m^2 IV **Day 1:** Cyclophosphamide 600 mg/m^2 IV Repeat cycle every 21 days for 4 cycles **Followed by** **Day 1:** Pertuzumab 840 mg IV followed by 420 mg IV **Day 1:** Trastuzumab 8 mg/kg IV followed by 6 mg/kg IV **Days 1, 8, and 15:** Paclitaxel 80 mg/m^2 IV Repeat cycle every 21 days for 4 cycles **Day 1:** Trastuzumab 6 mg/kg IV Repeat cycle every 21 days to complete 1 year of trastuzumab therapy Cardiac monitoring at baseline and 3-month intervals
Advanced or Metastatic Colorectal Cancer RAS Wild-Type	
Oxaliplatin, leucovorin, and 5FU (FOLFOX) plus cetuximab	**Day 1:** Oxaliplatin 85 mg/m^2 IV over 2 h **Day 1:** Leucovorin 400 mg/m^2 IV over 2 h **Days 1–3:** 5FU 400 mg/m^2 IV bolus on day 1, then 1200 mg/m^2 per day × 2 days (total 2400 mg/m^2 over 46–48 h) IV continuous infusion **Repeat cycle every 2 weeks, plus** Cetuximab 400 mg/m^2 IV over 2 h for the first infusion, then 250 mg/m^2 IV over 60 min weekly **OR** **Day 1:** Cetuximab 500 mg/m^2 IV over 2 h every 2 weeks

Source: Data from NCCN.

Two Classes of Pathway-Targeted Drugs Used in Cancer Treatment

Small-molecule inhibitors described in this chapter and **large-molecule drugs** (e.g., monoclonal antibodies) described in Chapter 72 constitute pathway-targeted anticancer drugs. Small molecules (molecular mass <1 kDa) may attack the same targets as monoclonal antibodies but mainly exert their effects by entering cells. A monoclonal antibody (molecular mass of immunoglobulin [Ig] G is ~150 kDa) is generally specific for a single antigen; in contrast, small molecules often inhibit multiple targets (e.g., kinases) with different selectivities and thus are likely to have a broader spectrum of activity and produce a broader spectrum of desired effects, off-target effects, and adverse effects than monoclonal antibodies. Many small-molecule drugs have elimination half-lives of 12 to 24 h and typically require at least daily oral administration, whereas antibodies are typically eliminated with half-lives of many days to weeks and require less frequent administration. For small-molecule drugs that are metabolized by hepatic CYPs, the concomitant use of strong CYP inhibitors (e.g., *ketoconazole*, *nefazodone*, *clarithromycin*, *gemfibrozil*, *cobicistat*) or inducers (e.g., *rifampin*, *phenytoin*, *carbamazepine*, *phenobarbital*, St John's wort) can lead to increased adverse effects due to reduced drug elimination or to reduced efficacy due to accelerated clearance. Syllables contained in the generic names of small-molecule drugs indicate their targeted pathways and are listed in Table 71–2.

I. Inhibitors of Receptor Signaling in Cancer Cells

Tyrosine Kinase Receptors

Inhibitors of Human Epidermal Growth Factor Receptor (EGFR/HER1)

Background

EGFR is essential for the growth and differentiation of epithelial cells. Ligand binding to the extracellular domain of EGFR family members causes receptor dimerization and stimulates the protein tyrosine kinase activity of the intracellular domain, resulting in autophosphorylation of several Tyr residues located in the C-terminal tail of the receptor monomers. These phosphotyrosines provide interaction sites for a variety of adapter proteins, resulting in stimulation of signaling pathways, including the mitogen-activated extracellular signal regulated protein kinase (MAPK) and phosphatidylinositol 3-kinase (PI3K)/Akt pathways (Figure 71–1).

TABLE 71–2 ■ THE NAMING OF SMALL-MOLECULE INHIBITORS

ENDING WITH	INHIBITOR CATEGORY	EXAMPLE
-anib	Angiogenesis	Pazopanib
-ciclib	Cyclin-dependent kinase 4 and 6	Palbociclib
-degib	Hedgehog signaling	Sonidegib
-denib	IDH1 and IDH2	Enasidenib
-lisib	PI3K	Idelalisib
-parib	PARP	Olaparib
-rafenib	BRAF	Vemurafenib
-tinib	Tyrosine kinase	Erlotinib
-zomib	Proteasome	Bortezomib

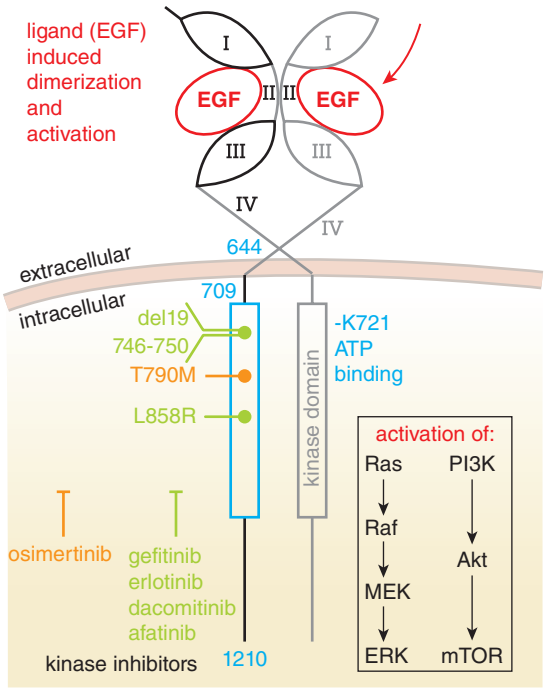


Figure 71–1 *Targeting the EGFR in cancer.* The extracellular portion of the EGFR contains the binding domains for growth factors that include EGF, TGF-α, and amphiregulin. Ligand binding to extracellular domains I and III induces conformational change and receptor dimerization through domains II and IV and activates intracellular signaling by cross-phosphorylation. The intracellular portion contains the tyrosine kinase activity and a C-terminal tail that recruits intracellular substrates after phosphorylation (see Chapter 3). EGFR mutations enhance downstream signaling, which provides a growth and survival advantage for cancer cells and confers sensitivity to treatment with the TKIs *afatinib*, *dacomitinib*, *erlotinib*, *gefitinib*, and *osimertinib*. The most common activating mutations are in-frame deletions in exon 19 (del 19) between E746 and A750 (~45%) and a missense mutation in exon 21 that leads to an L858R substitution (~40%). Lung cancers with these mutations are initially sensitive to first- and second-generation TKIs shown in green font. Resistance develops, however, most commonly via the T790M substitution in exon 20 found in about 50% of resistant tumors. *Osimertinib*, a third-generation inhibitor shown in orange font, is the only TKI that is active against cancers with the T790M mutation. Other insertional mutations in exon 20 are relatively rare and discussed in the text.

Two separate classes of drugs that target the EGFR pathway are important in the therapy of solid tumors: protein tyrosine kinase inhibitors (TKIs) and inhibitors of extracellular ligand binding. EGFR TKIs (*erlotinib*, *gefitinib*, *afatinib*, *osimertinib, dacomitinib*) bind to the intracellular kinase domain and inhibit the enzymatic function of EGFR. Monoclonal antibodies (*cetuximab*, *panitumumab*, *necitumumab*) that recognize the extracellular domain of the EGFR, inhibit ligand-induced signaling, and can elicit an immune response are discussed in Chapter 72. In epithelial cancers, overexpression of the EGFR is a common finding, and mutational activation of the EGFR (e.g., in 10%–40% of non–small cell lung cancers [NSCLCs]) creates a dependence on EGFR signaling. The most common EGFR mutations are in-frame deletions in exon 19 or leucine-to-arginine point mutations (L858R) in exon 21 that cause conformational changes and constitutive activation of the kinase. First-generation (*erlotinib*, *gefitinib*) and second-generation (*afatinib, dacomitinib*) drugs are available that inhibit EGFR kinase activity driven by these mutations. Resistant cancer cell subpopulations that bear the T790M mutation that emerges during treatment with first- and second-generation inhibitors in approximately 50% to 60% of patients are sensitive to the third-generation inhibitor *osimertinib* (see Figure 71–1).

Erlotinib

Mechanism of Action. *Erlotinib* is a reversible inhibitor of the EGFR tyrosine kinase, competitively inhibiting ATP binding at the active site of the kinase. Erlotinib has an IC_{50} of 2 nM for EGFR kinase activity. By comparison, for many protein kinases, the K_d for ATP at the substrate site is in the range of 20 μM.

ADME. *Erlotinib* is about 60% absorbed after oral administration. Peak plasma levels occur after 4 h. *Erlotinib* has a $t_{1/2}$ of 36 h and is metabolized by CYP3A4 and to a lesser extent by CYPs 1A2 and 1A1.

Therapeutic Uses. *Erlotinib* is approved for treatment of patients with advanced or metastatic NSCLC after failure of platinum-based chemotherapy treatment. In newly diagnosed NSCLC, *erlotinib* is approved in patients with EGFR mutations only (see Figure 71–1). *Erlotinib* in combination with *gemcitabine* is also approved for the treatment of patients with locally advanced, unresectable, or metastatic pancreatic cancer.

Adverse Effects and Drug Interactions. The most common adverse reactions (≥20%) are rash, diarrhea, anorexia, fatigue, dyspnea, nail disorders, nausea, and vomiting. Serious adverse reactions include severe rash (>10%). Fatal interstitial lung disease occurs with a frequency of 0.7% to 2.5%, and fatal hepatic failure has been reported, particularly in patients with baseline hepatic dysfunction. Other rare but serious toxicities include GI perforation, renal failure, arterial thrombosis, microangiopathic hemolytic anemia, hand-foot skin reaction, and corneal perforation or ulceration. *Erlotinib* therapy may cause rare cases of Stevens-Johnson syndrome/toxic epidermal necrolysis.

Concurrent use of proton pump inhibitors decreases the bioavailability of *erlotinib* by 50%. Plasma levels can vary due to drug interactions with inducers or inhibitors of CYP3A4. Patients using *warfarin* may experience poorer extrinsic coagulation while taking *erlotinib*. Smoking accelerates metabolic clearance of *erlotinib* and may decrease its antitumor effects.

Gefitinib

Mechanism of Action. *Gefitinib* inhibits the EGFR tyrosine kinase by competitive blockade of ATP binding with an IC_{50} of 15 to 57 nM for the kinase activity.

ADME. Oral bioavailability is about 60%; peak plasma concentrations are achieved within 3 to 7 h. Absorption of *gefitinib* is not significantly altered by food but is reduced by drugs that cause elevations in gastric pH. Metabolism of *gefitinib* is predominantly via CYP3A4, with a terminal $t_{1/2}$ of 41 h. Inducers of CYP3A4 activity decrease *gefitinib* plasma concentrations and efficacy; conversely, CYP3A4 inhibitors increase plasma concentrations.

Therapeutic Uses. *Gefitinib* is approved for treatment of patients with metastatic NSCLC with EGFR mutations (see Figure 71–1).

Adverse Effects and Drug Interactions. Diarrhea and skin reactions occur in more than 20% of patients. In addition to pustular/papular rash, other side effects include dry skin, nail changes, nausea, vomiting, pruritus, anorexia, and fatigue. Most adverse effects occur within the first month of therapy and are manageable with medications. Asymptomatic increases in liver transaminases may necessitate discontinuation of therapy. Respiratory compromise and interstitial lung disease, especially in patients with prior radiation, occurs in fewer than 2% of patients but may have a fatal outcome. Inducers and inhibitors of CYP3A4 will alter plasma concentrations. Patients using *warfarin* should be monitored for poorer extrinsic coagulation while taking *gefitinib*.

Afatinib

Mechanism of Action. *Afatinib* is a second-generation, orally bioavailable, *irreversible* inhibitor of EGFR (HER1) and HER2 receptor kinases with IC_{50} values of 0.5 and 14 nM, respectively.

ADME. The elimination $t_{1/2}$ of *afatinib* is 37 h after repeat dosing in patients with cancer. In patients with severe renal impairment, a dose reduction is recommended. *Afatinib* is a substrate of P-glycoprotein (Pgp), and coadministration of Pgp-modulating drugs can alter drug concentrations of *afatinib*.

Therapeutic Uses. *Afatinib* is approved for the first-line treatment of patients with metastatic NSCLC with EGFR mutations (see Figure 71–1) and patients with metastatic squamous NSCLC that has progressed subsequent to platinum-based chemotherapy.

Adverse Effects and Drug Interactions. Most common adverse effects are diarrhea and skin rash/acneiform dermatitis, stomatitis, and hand-foot skin reactions and nail changes. Also observed are interstitial lung disease, liver function abnormalities, left ventricular dysfunction, and rarely, gastrointestinal perforation (0.2% of patients).

Dacomitinib

Dacomitinib, approved in 2018, is an orally available, second-generation irreversible inhibitor of EGFR indicated for first-line treatment of patients with metastatic NSCLC with EGFR exon 19 deletion or exon 21 L858R substitution mutations. Plasma $t_{1/2}$ of *dacomitinib* is 70 h. Hepatic metabolism is the main route of clearance of *dacomitinib*. There is no impact of hepatic or renal impairment on drug elimination. Adverse effects are similar to *afatinib*, with diarrhea and skin rashes being the most common.

Mobocertinib

Mobocertinib is an irreversible kinase inhibitor that was approved in 2021 for the treatment of patients with advanced NSCLC with EGFR exon 20 insertional mutations progressing after chemotherapy. Exon 20 insertional mutation in EGFR are relatively infrequent (approximately 2%) and are not sensitive to first-, second-, and third-generation EGFR inhibitors. The response rate to mobocertinib is lower (28%) than that of EGFR inhibitors targeting NSCLC with the more common mutations (50%–80%). Adverse effects are similar to the second-generation EGFR inhibitors, in particular skin rash and diarrhea.

Resistance to Gefitinib, Erlotinib, Afatinib, and Dacomitinib

The majority of tumors with activating mutations in the EGFR initially respond to the kinase inhibitors of the first (*gefitinib*, *erlotinib*) and second generation (*afatinib, dacomitinib*). Eventually, however, the tumors progress. Approximately 60% of NSCLCs acquire a second EGFR mutation in the so-called EGFR gatekeeper residue in exon 20, T790M (see Figure 71–1), that reduces binding of these inhibitors to the kinase domain. The second-generation irreversible inhibitors that bind covalently to the kinase can overcome resistance in preclinical models, but their activity against wild-type EGFR exacerbates adverse effects such as skin rash and diarrhea and thus has limited their clinical use. *Osimertinib*, a third-generation inhibitor designed to recognize the T790M-mutant EGFR, is described below. Other mechanisms of resistance include mutational activation of downstream signaling molecules such as *KRAS* or activation of parallel signaling pathways, for example, through *MET* mutation or amplification, *EML4-ALK* translocation (Figure 71–2), and transformation to small cell lung cancer. Anaplastic lymphoma kinase (ALK) and mesenchymal-epithelial transition factor (MET) kinase inhibitors are described further below (Smit and Baas, 2015).

Osimertinib

Mechanism of Action. *Osimertinib* is a third-generation, orally bioavailable, *irreversible* inhibitor of T790M-mutant EGFR, with an IC_{50} value of 10 nM. In addition, *osimertinib* is also a very potent inhibitor of the EGFR mutants that are sensitive to first- and second-generation EGFR inhibitors. The T790M gatekeeper mutation in Exon 20 is often acquired after prior treatment with other EGFR kinase inhibitors (see previous discussion). Very sensitive detection methods can identify the presence of T790M in up to 30% of untreated patients, suggesting preexisting resistant subpopulations (see Chapter 69, Figure 69–3).

ADME. The mean $t_{1/2}$ of *osimertinib* is 48 h. *Osimertinib* is a substrate of CYP3A. The drug is primarily eliminated in the feces and to a lesser extent in the urine. Concurrent administration of inducers of CYP3A4 may necessitate an increase in the dose.

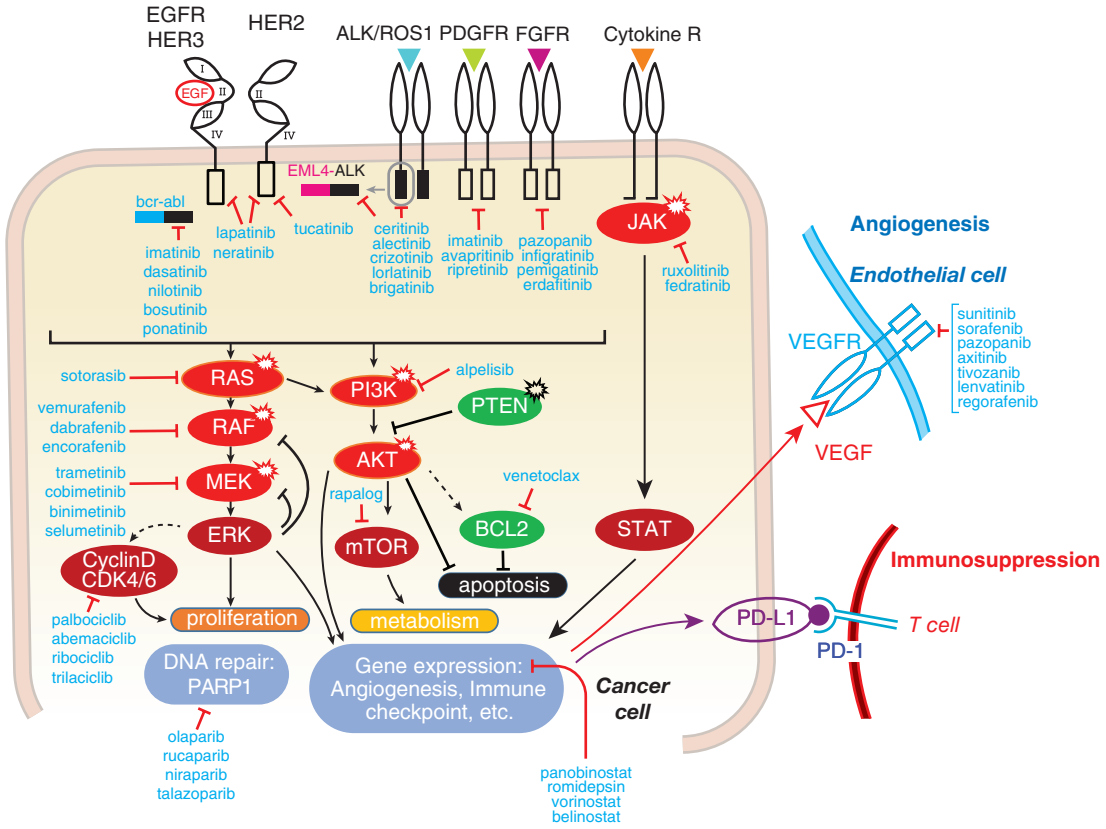


Figure 71–2 *Cancer cell signaling pathways and drug targets.* Extracellular binding of agonist ligands to transmembrane growth factor receptors causes receptor dimerization and activation of a C-terminal protein kinase that initiates intracellular signaling cascades that regulate gene expression and control cancer cell proliferation, apoptosis, metabolism, and metastasis. Crosstalk between cancer cells and the host stroma (including the vasculature and immune cells) is modulated through the release of angiogenic factors and the expression of immune checkpoint proteins. Cancer cell signaling can be altered through increased expression of receptors (e.g., HER2) or ligands, activating mutations in receptors (e.g., ALK, EGFR, FGFR) or in intracellular kinases (e.g., RAS, RAF, MEK, PI3K, AKT, JAK), gene translocations resulting in activated fusion proteins (e.g., bcr-abl, EML4-ALK), and the loss of function of an inhibitor of a pathway (e.g., the PTEN phosphatase) through mutation, gene deletion, or promoter methylation. Inhibitors are shown acting on a cancer cell and an endothelial cell. A T cell is included to represent the crosstalk with the immune system and effects of combination therapies. Inhibitors (in blue) are depicted adjacent to their intracellular target proteins. Proteins within red ellipses provide activating signals, those in green ellipses, inhibitory signals. Star bursts (✸) indicate signaling components in which mutations are common. More details of mechanisms of action are shown below for CDK4/6 inhibitors (Figure 71–3), BTK and PI3Kδ inhibitors in B cells (Figure 71–4), the mTOR pathway (Figure 71–5), HIF-2α inhibition (Figure 71–6), PARP inhibition (Figure 71–7), *thalidomide* congeners (Figures 71–8 and 71–9), and BH3 mimics (Figure 71–10). Immune suppression by mTOR inhibition during organ transplants is discussed in Chapter 39.

Therapeutic Uses. *Osimertinib* is approved for first-line treatment of patients with metastatic NSCLC and adjuvant therapy after resection of NSCLC with EGFR exon 19 deletions or exon 21 L858R mutations; it is also approved in patients with metastatic NSCLC that progressed after prior EGFR TKI treatment and is positive for the EGFR T790M mutation. Genotyping of tumors for the T790M mutation is required after resistance develops and, besides a tumor biopsy, can be done noninvasively from blood samples analyzing DNA shed from tumor cells into the bloodstream (Oxnard et al., 2016). Response rates are greater than 60% in patients with metastatic NSCLC. *Osimertinib* improved survival in patients with NSCLC EGFR T790M mutations and in first-line treatment compared to first-generation inhibitors and thus has become the standard of care irrespective of the presence of the EGFR T790M mutation (Cohen et al., 2021). Acquired resistance after *osimertinib* treatment is manifested by the appearance of a third mutation, C797S, in less than 20% of tumors. The mutation of Cys^{797} to Ser^{797} prevents formation of the potency-conferring covalent bond of the inhibitor (Thress et al., 2015); however, MET amplification is a more common resistance mechanism along with several other mechanisms that have been described. Inhibitors of C797S-mutant EGFR are under development for combination treatment with an EGFR-targeted antibody (Jia et al., 2016).

Adverse Effects and Drug Interactions. The most common adverse effects after *osimertinib* treatment are diarrhea and skin rash; however, these are much less common and severe than for first- and second-generation inhibitors, because the activity on wild-type EGFR is reduced with the third-generation inhibitor. Also observed are interstitial lung disease, QTc prolongation, and left ventricular dysfunction.

Inhibitors of Human Epidermal Growth Factor Receptor 2 (HER2/Neu/ErbB2)

Human epidermal growth factor receptor 2 (also named Neu or ErbB2; see History box) is a member of the HER family, which also includes EGFR (HER1), HER3, and HER4 (see Figure 71–2). The fixed conformation of the extracellular domain of HER2 resembles the ligand-activated state of the other HER family members and explains its unique function as a coreceptor that does not require ligand activation. In addition, due to this conformation, the overexpression of wild-type HER2 is sufficient to activate the intracellular tyrosine kinase and oncogenic signaling even in the absence of activating mutations, coreceptors, or ligands. Overexpression of HER2 is found in 20% to 30% of human breast cancers due to gene amplification on chromosome 17 and results in more aggressive tumors, lower response rates to hormonal therapies, and higher risk of disease recurrence after treatment.

HISTORY **EGFR/HER PATHWAY-TARGETED CANCER THERAPY**

The discovery and analysis of the EGFR (HER1) in 1984 (Ullrich et al., 1984) revealed a remarkable homology of its protein kinase domain to an oncogenic protein in the avian erythroblastosis retrovirus, v-erb (Downward et al., 1984). The first oncogenic protein isolated from the erb virus is a homologue of the thyroid hormone receptor and was named erbA; the second oncogenic protein isolated from v-erb was named erbB, an alternate name for members of the EGFR/HER gene family members. The *HER2* gene was discovered based on its homology to v-erbB in human breast cancer and its oncogenic function in rat ***neu***ral tumors; hence the different names HER2/ErbB2/Neu (King et al., 1985; Schechter et al., 1985). The association of HER2 overexpression with poor prognosis in breast cancer (Slamon et al., 1987) provided the impetus for the development of HER2-targeted therapies and a new concept in the treatment of HER2-positive breast cancers. The most recent addition in 2020 to drugs that target HER2 is *tucatinib*, an HER2-selective, small-molecule kinase inhibitor approved for use in HER2-positive breast cancer.

Lapatinib

Mechanism of Action; ADME. *Lapatinib* is an orally bioavailable, small-molecule inhibitor of the EGFR and HER2 tyrosine kinases with an IC_{50} of about 10 nM for both kinases. *Lapatinib* is metabolized by CYP3A4 with a plasma $t_{1/2}$ of 14 h. Concurrent administration of inducers and inhibitors of CYP3A4 may necessitate adjustment of the dose.

Therapeutic Uses; Adverse Effects. *Lapatinib* in combination with *capecitabine* is approved for the treatment of patients with metastatic HER2-positive *trastuzumab*-refractory breast cancer. *Lapatinib* is also used in combination with the aromatase inhibitor *letrozole* (see Chapter 73) to treat postmenopausal women with hormone receptor–positive, metastatic breast cancer that overexpresses HER2. Frequent adverse effects include acneiform rash, diarrhea, cramping, and exacerbation of gastroesophageal reflux. When *lapatinib* is combined with *capecitabine*, diarrhea is worsened. Cardiac toxicity appears less pronounced than with *trastuzumab*, although *lapatinib* should be used with caution in combination with cardiotoxic drugs. A dose-dependent prolongation of the QT interval has been reported; thus, careful monitoring of patients with heart disease is recommended. Hepatotoxicity that may be severe has been observed.

Neratinib

Neratinib is an orally bioavailable, *irreversible* inhibitor of the HER2 and EGFR protein tyrosine kinase with IC_{50} values of 59 nM and 92 nM, respectively. It is approved as a single agent in early-stage HER2-positive breast cancer following *trastuzumab*–based therapy and in advanced or metastatic HER2-positive breast cancer after two or more anti-HER2–based regimens in combination with *capecitabine*. Diarrhea is the most common and frequent adverse effect, with grade 3 or 4 diarrhea occurring in over one-third of patients (Park et al., 2016); prophylaxis with *loperamide* is recommended. The metabolism is via CYP3A in the liver with an elimination $t_{1/2}$ of 7 to 17 h; concomitant use of *ketoconazole*, a strong inhibitor of CYP3A4 and Pgp, increased maximal plasma concentration by over 2-fold.

Tucatinib

Tucatinib is an orally bioavailable, reversible, highly selective TKI of HER2 with minimal inhibition of EGFR (HER1). *Tucatinib* demonstrated efficacy compared with placebo in progression-free and overall survival of heavily pretreated patients with HER2-positive metastatic breast cancer, including patients with brain metastases (Murthy et al., 2020). In this study, addition of *tucatinib* to the anti-HER2 antibody *trastuzumab* improved outcomes, demonstrating that targeting of the intracellular kinase with a small-molecule inhibitor in addition to targeting the extracellular domain with a monoclonal antibody can be beneficial. The FDA approved its use in 2020 for the treatment of patients with unresectable or metastatic HER2-positive breast cancers, including patients with brain metastases and pretreated patients.

Tucatinib is metabolized mostly by CYP2C8 and less by CYP3A and has an elimination $t_{1/2}$ of 8.5 h. The most common adverse effects are diarrhea and hepatotoxicity. Skin toxicity that is frequently dose limiting and associated with EGFR inhibition is significantly less common with *tucatinib*.

ALK and ROS1

Background

ALK is a membrane monospan with an extracellular domain and an intracellular protein tyrosine kinase domain (see Figure 71–2). The importance of this protein kinase in human cancers is the capacity of the *ALK* gene to form fusion genes that become oncogenic drivers. A 2;5 chromosomal translocation yields an oncogenic fusion gene in which a 3′ portion of *ALK* (chromosome 2) contributes coding for the catalytic domain and is joined to a 5′ region of the *NPM* gene. The extracellular and transmembrane sequences of ALK are absent in the *ALK-NPM* fusion gene, which is the oncogenic driver of malignant progression in anaplastic large cell lymphoma. A similar translocation of the ALK kinase and fusion with EML4 creates the oncogenic driver in a subset of NSCLC (Martelli et al., 2010), as shown in Figure 71–2. *ALK* also fuses with numerous additional 5′ partners (Roskoski, 2017). *Alectinib*, *brigatinib, crizotinib*, *ceritinib*, and *lorlatinib* are small-molecule inhibitors of ALK kinase that are used in the treatment of ALK translocation–positive cancers.

The overall organization of ROS1 (orphan receptor tyrosine kinase) is similar to ALK, and the intracellular kinase portion of ROS1 also can become part of chimeric oncoproteins due to different chromosomal translocation partners most frequently found in glioblastoma and NSCLC but also in other cancers. Several of the kinase inhibitors discussed below also target the ROS1 kinase due to protein sequence homologies in the kinase domains of approximately 70% (reviewed by Drilon et al., 2021). ROS1 inhibition correlates with increased efficacy and improved survival in patients with ROS1-positive NSCLCs (Nokin et al., 2020).

Crizotinib, originally developed as a MET inhibitor, was found to also target ALK and ROS1 and became the first inhibitor to be approved for ALK-positive cancers. The next-generation ALK inhibitors have mostly replaced *crizotinib* due to their improved clinical efficacy in ALK-positive NSCLC.

Crizotinib

Mechanism of Action. *Crizotinib* is an orally bioavailable inhibitor of receptor tyrosine kinases that include ROS1, HGFR/cMET, ALK, and ROS1. IC_{50} values are 0.6, 11, and 24 nM for ROS1, hepatocyte growth factor receptor (HGFR)/cMet, and ALK, respectively.

ADME. *Crizotinib* is predominantly metabolized by CYP3A4/5 and shows a terminal $t_{1/2}$ of 42 h. The typical twice daily dosing is reduced to once daily in patients with severe renal impairment and a creatinine clearance below 30 mL/min.

Therapeutic Uses. *Crizotinib* was initially developed and approved for the treatment of patients with locally advanced or metastatic NSCLC harboring *ROS1* or *ALK* gene rearrangements described above (see Figure 71–2) (Thomas et al., 2015).

Adverse Effects. The most common adverse reactions are GI toxicities including nausea, diarrhea, and vomiting as well as hepatic, respiratory, and ocular toxicity and neuropathy. Also observed are QT prolongation and bradycardia.

Alectinib

Mechanism of Action. *Alectinib* is an orally bioavailable inhibitor of ALK kinase activity with an IC_{50} of 1.9 nM. *Alectinib* shows a higher selectivity for ALK than *crizotinib* and is active against mutant forms of ALK that are found in tumors resistant to *crizotinib*.

ADME. *Alectinib* has an elimination half-life of 33 h and is metabolized by CYP3A4. Coadministration of *alectinib* with CYP3A inhibitors or inducers will alter plasma concentrations.

Therapeutic Uses. *Alectinib* is approved for the treatment of patients with advanced or recurrent NSCLC harboring the ALK fusion gene (ALK^+). Patients with ALK^+ NSCLC who have progressed on treatment with *crizotinib* or are intolerant to *crizotinib* also qualify. *Alectinib* is also approved for first-line treatment of ALK-positive advanced NSCLC.

Adverse Effects. The most common adverse effects are fatigue, constipation, edema, and myalgia. Pneumonitis, GI toxicity, hepatotoxicity, bradycardia, and prolonged QT intervals are also observed (Katayama et al., 2015).

Ceritinib

Ceritinib is an orally bioavailable, competitive inhibitor of the ALK kinase with an IC_{50} of 0.2 nM and is 40-fold selective relative to the homologous insulin-like growth factor 1 receptor (IGF1R) kinase. *Ceritinib* is approved for the treatment of ALK-positive metastatic NSCLC unresponsive to *crizotinib*. *Ceritinib* is metabolized by CYP3A and eliminated with a $t_{1/2}$ of 41 h. Common adverse effects are GI toxicity and hepatotoxicity. Bradycardia and prolonged QT intervals also occur.

Brigatinib

Brigatinib is an inhibitor of the ALK and ROS1 (IC_{50} 0.9 nM) and of IGF1R and FLT3 tyrosine kinases at lower potency. *Brigatinib* is approved for the treatment of ALK-positive metastatic NSCLC, where treatment resulted in a longer progression-free survival and higher response rate (78%) against brain metastases than *crizotinib* (29%) (Camidge et al., 2018). *Brigatinib* is approved for first-line treatment. *Brigatinib* is metabolized by CYP3A and excreted through the urine. Reduced dosing is recommended with hepatic or renal impairment. The elimination $t_{1/2}$ is 25 h. Adverse effects can include the induction of interstitial lung disease/pneumonitis, hypertension, and bradycardia. Blood pressure and heart rate monitoring are recommended.

Lorlatinib

Lorlatinib (originally named PF-06463922) is an orally active, third-generation inhibitor of the ALK and ROS1 kinase with an IC_{50} of less than 1 nM. It was approved in 2018 for the treatment of metastatic, ALK-positive NSCLC who failed 1 or 2 prior ALK inhibitors. In 2021 it was also approved for first-line treatment.

Lorlatinib is metabolized by CYP3A4 and eliminated with a $t_{1/2}$ of 24 h. CNS adverse effects including seizures, cognitive dysfunction, and psychoses or mood changes are a risk during therapy. Potential induction of hypertension requires monitoring of blood pressure. Hepatotoxicity is a further risk.

NRTK/TRK Fusions

Chromosomal rearrangements resulting in in-frame fusions of the kinases with different partners can result in constitutively active chimeric TRK that act as oncogenic drivers, promoting proliferation and survival in tumor cells. Fusions involving one of three TRKs (A, B, or C) occur in diverse cancers (e.g., salivary gland, soft-tissue sarcoma, fibrosarcoma) and define a unique molecular subgroup of advanced tumors. TRK refers to tropomyosin receptor kinase, one of a family of receptor tyrosine kinases encoded by neurotrophic tyrosine receptor kinase (NTRK) gene fusions. Assays to detect TRK fusions are used to identify cancers that contain such a drug-sensitive driver (Nokin et al., 2020).

Larotrectinib

Larotrectinib is an orally available inhibitor of the TRKA, TRKB, and TRKC kinases with IC_{50} values of 5 to 11 nM that showed antitumor activity against constitutively active TRK fusion proteins or TRK overexpression. *Larotrectinib* was not active against mutant TRKA or TRKC. It was approved in 2018 for the treatment of any cancers with TRK gene fusions without resistance mutation (Drilon et al., 2018). Adverse effects include neurotoxicity and hepatotoxicity. It is metabolized by CYP3A and eliminated with a $t_{1/2}$ of 2.9 h.

Entrectinib

Entrectinib is an orally active inhibitor of the TRKA, TRKB, TRKC, ROS1, and ALK kinases with IC_{50} values of 0.1 to 2 nM. It was approved in 2019 for the treatment of any cancers with TRK gene fusions without resistance mutation and for metastatic NSCLC positive for ROS1. Adverse effects include an increased risk for congestive heart failure, cognitive impairment, hepatotoxicity, and prolongation of QT intervals requiring periodic monitoring. It is metabolized by CYP3A with an elimination $t_{1/2}$ of approximately 20 h.

Inhibitors of Platelet-Derived Growth Factor Receptor and KIT

Signaling by the platelet-derived growth factor receptor (PDGFR) plays a significant part in mesenchymal biology, including stem cell growth, and is involved in oncogenesis through aberrant cancer cell signaling, modulation of the tumor microenvironment, and facilitation of angiogenesis and metastasis. Small-molecule protein kinase inhibitors, including *imatinib* (see discussion below), have been used in the treatment of patients with gastrointestinal stromal tumors (GISTs) that carry mutations in *KIT* (feline sarcoma virus oncogene homolog) or *PDGFRA* (see Figure 71–2).

Ripretinib

Ripretinib is an orally bioavailable kinase inhibitor of the PDGFR, wild-type and mutant KIT, and other kinases including VEGFR2 and tyrosine kinase with Ig and EGF domains (TIE) 2 with IC_{50} values of 4 to 18 nM. It was approved in 2020 for the treatment of patients with advanced GIST who have received prior treatment with three or more kinase inhibitors, including *imatinib*. Adverse effects include palmar-plantar erythrodysesthesia syndrome (PPES), nausea, vomiting, constipation, diarrhea, and alopecia. Hypertension and cardiac dysfunction have also been reported. It is metabolized by CYP3A and eliminated with a $t_{1/2}$ of 14.8 h.

Avapritinib

Avapritinib is an orally bioavailable inhibitor of KIT and PDGFRA mutant kinases with IC_{50}s of less than 25 nM. It was approved in 2020 for the treatment of metastatic GIST harboring *PDGFRA* mutations and for advanced systemic mastocytosis and mast cell leukemia. Adverse events include risk of intracranial hemorrhage and cognitive impairment. *Avapritinib* is primarily metabolized by CYP3A. The elimination $t_{1/2}$ ranges from 20 to 57 h.

FGFR Inhibitors

Physiological fibroblast growth factor (FGF) receptor (FGFR) signaling is modulated by 22 human protein ligands that bind to four transmembrane tyrosine kinase receptors expressed as seven distinct splice variant receptors. In approximately 7% of cancers, FGFR genes are mutated or amplified or become part of translocations that lead to constitutively active gene fusions and thus contribute to cancer initiation and malignant progression. *FGFR1* is amplified in NSCLC and breast cancer and mutated in gliomas. FGFR2 is activated by gene fusions in approximately 20% of cholangiocarcinomas, leading to recent approval of kinase inhibitors. The first approval of FGFR kinase inhibitors was based on FGFR3 mutations in metastatic urothelial carcinomas but FGFR3 fusions are also found in brain tumors and bladder cancers. Unusual for tyrosine kinase receptors, the most common FGFR3 mutations occur in the ligand-binding domain, mimicking a constitutive extracellular signaling. A more detailed overview is provided by Facchinetti et al. (2020).

Erdafitinib

Erdafitinib is a selective pan-FGFR kinase inhibitor that also inhibits other kinases including RET, CSF1R, PDGFR, KIT, and VEGFR2. It was approved in 2019 for the treatment of patients with locally advanced or metastatic urothelial carcinoma with FGFR2 or FGFR3 genetic alterations

that has progressed after chemotherapy. Adverse effects include ocular disorders that require monthly ophthalmological monitoring. Hyperphosphatemia as a consequence of FGFR inhibition was observed in 76% of patients with a median onset time of 20 days. About one-third of patients received phosphate binders during treatment. *Erdafitinib* is metabolized by CYP2C9 and CYP3A4 and eliminated with a $t_{1/2}$ of 59 h.

Pemigatinib

Pemigatinib is an orally bioavailable selective FGFR1–3 kinase inhibitor (IC_{50} <1.2 nM; FGFR4 IC_{50} is 30 nM). It was approved in 2020 for the treatment of pretreated, locally advanced, or metastatic cholangiocarcinoma with an FGFR2 fusion or rearrangement. Adverse events include diarrhea, ocular toxicity, and hyperphosphatemia as a consequence of FGFR inhibition, resulting in soft-tissue mineralization and vascular and myocardial calcification. It is metabolized by CYP3A and eliminated with a $t_{1/2}$ of 15.4 h.

Infigratinib

Infigratinib (previously named BGJ398) is an orally bioavailable selective FGFR1–3 kinase inhibitor (IC_{50} <2 nM) approved in 2021 for the treatment of pretreated, locally advanced, or metastatic cholangiocarcinoma with an FGFR2 fusion or rearrangement. Adverse events include diarrhea, ocular toxicity, and hyperphosphatemia as a consequence of FGFR inhibition, resulting in soft-tissue mineralization and vascular and myocardial calcification. It is metabolized by CYP3A and eliminated with a $t_{1/2}$ of 33.5 h.

MET/HGFR Inhibitors

Under physiological conditions, the MET/HGFR transmembrane receptor is activated by HGF ligand binding. Malignant transformation observed in NSCLC and other cancers can lead to alternate MET activation through gene rearrangements that code for constitutively active cytosolic fusion proteins (Nokin et al., 2020). Also, somatic mutations in MET are known to result in aberrant mRNA splicing resulting in increased MET activity: Exon 14 codes for a binding site for the E3 ubiquitin ligase CBL that is responsible for ubiquitination of the MET protein and for its degradation. Skipping of exon 14 thus leads to decreased ubiquitination and results in increased protein levels and activity of MET (Guo et al., 2020) that can be targeted therapeutically (Paik et al., 2015).

Capmatinib

Capmatinib is an orally bioavailable selective inhibitor of the MET kinase (IC_{50} of 0.13 nM) that was approved in 2020 for the treatment of patients with metastatic NSCLC with a mutation that leads to MET exon 14 skipping (see above). Adverse effects include peripheral edema, dyspepsia, and fatigue. *Capmatinib* is predominantly metabolized by CYP3A4 and eliminated with a $t_{1/2}$ of 6.5 h.

Tepotinib

Tepotinib is an orally bioavailable inhibitor of the MET kinase (IC_{50} of 4 nM) approved in 2021 for the treatment of patients with metastatic NSCLC with a mutation that leads to MET exon 14 skipping (see above). Adverse effects include peripheral edema, dyspepsia, fatigue, and musculoskeletal pain. *Tepotinib* is predominantly metabolized by CYP3A4 and eliminated with a $t_{1/2}$ of 32 h.

RET Inhibitors

Physiologically, the RET (rearranged during transfection) transmembrane tyrosine kinase receptor is activated through extracellular interactions with co-receptors and ligands from the glial cell–derived neurotrophic factor family and is involved in development of different organ systems including hematopoiesis, the CNS and the gastrointestinal (GI) tract. Activating RET mutations or gene fusions can become oncogenic drivers in NSCLC (1%–2%), thyroid cancer (10%–20%), and other cancers (Thein et al., 2021).

Selpercatinib

Selpercatinib is an orally bioavailable inhibitor of wild-type and mutated RET isoforms (IC_{50} of 1–4 nM) as well as VEGFRs and FGFRs at higher concentrations. It was approved in 2020 for the treatment of patients with metastatic RET fusion–positive NSCLC and advanced or metastatic RET-mutant medullary thyroid cancer. Adverse effects include the risk of hepatotoxicity that requires close monitoring every 2 weeks; hypertension requiring monitoring; QTc prolongation; impaired wound healing risk after surgery; and tumor lysis syndrome. It is metabolized predominantly by CYP3A4 and eliminated with a $t_{1/2}$ of 32 h.

Pralsetinib

Pralsetinib is an orally bioavailable inhibitor of wild-type and oncogenic RET fusions and mutations (IC_{50} <0.5 nM) that is over 10-fold selective relative to other kinases. It was approved in 2020 for the treatment of patients with metastatic RET fusion–positive NSCLC and advanced or metastatic RET-mutant medullary thyroid cancer. Adverse effects include the risk of severe interstitial lung disease/pneumonitis (10% of patients; 0.5% with fatal reactions) and hypertension (29% of patients; 14% with grade 3) that needs medications. It is metabolized predominantly by CYP3A; elimination $t_{1/2}$ is 15.7 h.

Hedgehog Pathway Inhibitors

The hedgehog signaling pathway was discovered by genetic studies in fruit flies: Mutations resulting in stubby and hairy *Drosophila melanogaster* larvae inspired the naming of the signaling pathway as "hedgehog" or Hh, which is the name of the pathway's polypeptide ligand. The hedgehog pathway controls embryonic cell differentiation in tissues of different species through distinct gradients of hedgehog signaling proteins that are key regulators during development. In adults, hedgehog signaling plays roles in stem cell regulation and tissue regeneration. In mammals, binding of one of three Hh ligands, named Sonic, Indian, and Desert Hh, to Patched-1 (PTCH1), a 12-transmembrane domain cell surface receptor protein, releases the PTCH1-mediated suppression of the seven-transmembrane receptor protein SMO (smoothens hedgehog), resulting in its activation of the glioma-associated (GLI) family of transcription factors that control the expression of Hh target genes. Mutations in PTCH1 are found in patients with Gorlin's syndrome predisposed to basal cell carcinoma (BCC), medulloblastoma, and rhabdomyosarcoma due to increased Hh pathway signaling. Overexpression of Hh ligands occurs in glioblastoma and GI and breast cancers, and abnormal activation of the pathway is seen in BCC and in some hematological malignancies, resulting in increased self-renewal capacity of leukemic cells (Jamieson et al., 2020). Hh pathway inhibitors induce differentiation of stem cells and can thus be used in combination with chemotherapy to target both dormant stem cells and growing cancer cells in the bulk of the tumor (Amakye et al., 2013). Clinical trials in solid tumors and hematological malignancies are ongoing to test the efficacy of such combinations (Jamieson et al., 2020). Three inhibitors of SMO and arsenic trioxide, an inhibitor of GLI transcription factors, are currently FDA-approved for cancer treatment and discussed next.

SMO Inhibitors

Vismodegib

Vismodegib, a 32% orally bioavailable first-in-class hedgehog signaling pathway inhibitor, is approved for the treatment of adult patients with BCC. It acts as a competitive antagonist of SMO and is indicated for patients with metastatic or relapsed BCC. Common adverse effects include GI disorders (nausea, vomiting, diarrhea, constipation), muscle spasms, fatigue, hair loss, and dysgeusia (distortion of the sense of taste). These adverse effects are mostly mild to moderate. *Vismodegib* can cause embryo-fetal death and severe birth defects and thus must not be used during pregnancy. Also, patients are advised not to donate blood for 24 months after the final dose and males not to donate semen during and for 3 months after therapy. The estimated elimination $t_{1/2}$ is 4 days after continuous dosing.

Sonidegib

Sonidegib is a less than 10% orally bioavailable hedgehog pathway inhibitor that binds to and inhibits SMO signal transduction similar to

vismodegib. *Sonidegib* is approved for the treatment of patients with locally advanced BCC that has recurred following surgery or radiation therapy. Warnings and adverse effects are similar to those of *vismodegib*. It is predominantly metabolized by CYP3A and has an elimination $t_{1/2}$ of approximately 28 days.

Glasdegib

Glasdegib is a 77% orally active hedgehog pathway inhibitor that was approved in 2018 for the treatment of patients 75 years of age or older with newly diagnosed acute myelocytic leukemia (AML) in combination with low-dose *cytarabine* if comorbidities preclude use of intensive induction chemotherapy. In addition to the warnings and adverse effects in common with *vismodegib*, *glasdegib* can cause QTc prolongation and thus requires electrocardiogram (ECG) monitoring. *Glasdegib* is metabolized by CYP3A4 and has a mean elimination $t_{1/2}$ of 17.4 h.

GLI Inhibitor

Arsenic Trioxide

Arsenic trioxide (ATO) is administered IV and binds to GLI1 and GLI2, resulting in inhibition of hedgehog target gene expression. ATO is approved in combination with *tretinoin* for the treatment of acute promyelocytic leukemia (APML) with a t(15;17) translocation that fuses PML to the retinoic acid receptor RAR-α. Patients with newly diagnosed, low-risk APML or patients after relapse from retinoid or chemotherapy are eligible. Adverse effects and warnings associated with the treatment include the occurrence of differentiation syndrome, which may be life-threatening and requires high-dose corticosteroids; risk of QTc interval prolongation; complete atrioventricular block; and torsades de pointes that can be fatal and requires ECG and electrolyte monitoring. Patients with ventricular arrhythmias or prolonged QTc should be excluded from treatment. Encephalopathy, including Wernicke's, can occur and requires parenteral thiamine treatment. Hepatic function should be monitored due to potential hepatotoxicity, and patients should be surveyed for new primary malignancies due to the carcinogenic effect of ATO. Patients should also be advised on the potential risk to a fetus and use effective contraception. Other symptoms indicative of adverse effects are nausea, abdominal pain, dyspnea, and pruritus. The elimination $t_{1/2}$ of ATO is 10 to 14 h. Chapter 76 presents aspects of the toxicology of ATO.

II. Inhibitors of Intracellular Protein Kinase Signaling in Cancer Cells

Inhibitors of RAS

The RAS GTPase family is a major switch that transduces extracellular signals from transmembrane receptors to multiple intracellular effectors, including the RAF family of kinases and PI3 kinase. This initiates a cascade of coordinated cellular events required for cell growth and maintenance, including regulation of gene expression and metabolic steady state. The cascade is visualized in Figure 71–2. GTP-bound RAS activates downstream signaling by binding to the RAS-binding domain of effector proteins with the GTPase activity limiting the signal strength and duration. Mutations in codons for glycines G12 or G13 or for glutamine Q61, impact GTP binding and disrupt GTPase activity of RAS proteins. Mutations in codons 12 and 61 decrease the rate of GTP hydrolysis, and mutations in codon 61 also accelerate the rate of GDP-GTP exchange. This results in elevated RAS^{GTP} and dysregulated activation of the downstream effectors characteristic of malignant transformation. Notably, almost all of the components of the downstream pathway can be found mutated in different cancers, and inhibitors are described in this section. RAS mutations occur in a tumor-specific manner for the K-, N-, and HRAS isoforms (reviewed in Moore et al., 2020). *KRAS* is one of the most highly mutated oncogenes in human cancer, occurring in 95% of pancreatic, 50% of colorectal, and 32% of lung adenocarcinomas (Rosen, 2021). Ostrem et al. (2013) developed drugs that bind covalently to the cysteine in the G12C mutant *KRAS* and disrupt downstream signaling of mutant and not wild-type protein. The recently approved KRAS G12C inhibitor *sotorasib* inhibits activation of mutant KRAS by preventing its regeneration from the inactive RAS-GDP state. The promise of mutant RAS inhibition in cancer treatment is supported by the results from the initial *sotorasib* trial. In previously treated patients with NSCLC harboring G12C KRAS mutations, a complete or partial response was observed in 37.1% of patients and disease control was observed in 80.6% of patients (Skoulidis et al., 2021).

HISTORY **THE RAS-RAF-MEK-ERK PATHWAY**

HRAS and *KRAS*, named after ***H***arvey and ***K***irsten ***ra***t ***s***arcoma viruses and discovered in the early 1980s, were the first oncogenes discovered in human tumors. *NRAS* was identified in human ***n***euroblastoma. These oncogenes were found to be mutated and constitutively active in approximately 20% of all cancers, up to 95% in some cancers (pancreatic adenocarcinoma). RAS proteins are GTPases that toggle between the active, GTP-bound and inactive, GDP-bound state. The first *RAF* gene was identified from a murine retrovirus named "***r***apidly ***a***ccelerated ***f***ibrosarcoma" due to its oncogenic effects. Following the discovery of the first human *RAF* gene (*RAF1*, also named *cRAF* = normal cellular *RAF*), the *RAF* genes *ARAF* and *BRAF* were identified; their gene products are ser/thre protein kinases. The signaling cascade of RAS-RAF-MEK-ERK was established by the early 1990s. This pathway plays a central role in cell proliferation and differentiation, is activated in many human cancers, and controls the crosstalk of cancer cells with the tumor microenvironment (see Figure 71–2). Mutations in *RAF* genes can lead to one (BRAF) or two (cRAF, BRAF) changes in amino acids, rendering these gene products oncogenic and resulting in continuous cell proliferation and the development of cancer. Activating mutations of *BRAF* were detected in melanoma and other tumors in 2002. A small-molecule inhibitor targeting mutated BRAF, *vemurafenib*, was approved by the FDA in 2011 for treatment of advanced melanoma; the first inhibitor of MEK, *trametinib*, was approved in 2013; the first inhibitor of mutant $KRAS^{G12C}$, *sotorasib*, was approved in 2021. The role of this pathway for immune crosstalk led to combination trials with the intent of triggering cancer cell death by pathway-targeted inhibitors and promoting cytotoxic T-cell efficacy. Ongoing combination trials have been reviewed a workshop of the National Academy of Sciences (Balogh and Nass, 2019), Hughes et al. (2016), and Petroni et al. (2021).

Sotorasib

Sotorasib (initially named AMG-510) is a first-in-class covalent inhibitor of the RAS GTPase family that targets G12C-mutant oncogenic *KRAS* and was approved in 2021 for the treatment of patients with locally advanced or metastatic NSCLC who have received at least one prior systemic therapy. Approximately 13% of NSCLCs carry mutant KRAS G12C, along with 3% of colorectal cancers and other cancers at lower frequencies. *Sotorasib* forms an irreversible, covalent bond with the unique cysteine in mutant KRAS G12C, locking the protein in an inactive state that prevents downstream signaling. *Sotorasib* showed minimal off-target activity and was associated with antitumor immunity in KRAS G12C preclinical models. Adverse effects include diarrhea, nausea, low-grade hepatotoxicity, and rarely, pneumonitis and dyspnea. The main metabolic pathways of *sotorasib* are nonenzymatic conjugation and metabolism by CYP3A. The mean elimination $t_{1/2}$ is 5 h.

Adagrasib

Adagrasib is a covalent inhibitor of KRAS G12C currently in clinical trials in NSCLC and colorectal cancer (Awad et al., 2021).

Emerging resistance mechanisms to KRAS G12C inhibition by *adagrasib*, an investigational drug still in clinical trials, showed a multitude of resistance pathways in patients with NSCLC and colorectal cancer (Awad et al., 2021). Treatment-resistant cancers contained secondary mutations or amplifications in the targeted KRAS, activation of the signaling pathway without alterations of KRAS itself, and histologic transformation from lung adenocarcinoma to squamous cell carcinoma. Trials exploring various combinations are ongoing to take advantage of this novel pathway inhibitor.

Inhibitors of RAF Kinase: Vemurafenib, Dabrafenib, and Encorafenib

BRAF belongs to the RAF/Mil family of serine/threonine protein kinases and plays a role in regulating signaling via MAPK (also named extracellular signal-related kinase [ERK]) (see Figure 71–2). *RAF* is the most frequently altered gene (7%–10% of all cancers) in the MAPK pathway downstream of RAS.

In melanoma, MAPK signaling can be constitutively activated through alterations in membrane receptors or through mutations of RAS or BRAF. BRAF mutations confer constitutive activation of the kinase pathway independent of RAS activation. BRAF is mutated in about 55% of melanomas, and 90% of these mutations display a substitution of valine to glutamic acid (V600E) or to lysine (V600K), which results in constitutive protein kinase activation (Vultur and Herlyn, 2013). Approximately 8% of other cancers, colon cancer and NSCLC among them, also carry an activating mutation in *BRAF*. The BRAF inhibitors *vemurafenib* and *dabrafenib* are superior to traditional chemotherapy in improving survival in patients with mutant BRAF melanoma and are discussed in the material that follows.

Combination of BRAF and MEK Inhibitors

Since the introduction of BRAF inhibitors in 2011 and MEK inhibitors in 2013 (see below), their combination has shown improved efficacy and is now standard of care in melanoma and NSCLC including *dabrafenib* plus *trametinib*, *vemurafenib* plus *cobimetinib*, and *encorafenib* plus *binimetinib* (Curti and Faries, 2021). Notably, in patients with BRAF-mutant colon cancers, no therapeutic benefit was observed, indicating that colon cancers do not depend on the activated mutant BRAF as a rate-limiting oncogenic driver.

Vemurafenib

Mechanism of Action

Vemurafenib is an orally bioavailable inhibitor of mutated BRAF (V600E). The name *vemuraf*enib is derived from the target, that is, ***V***600***E mu***tated B***RAF***. *Vemurafenib* also is effective against the less common BRAF V600K mutation. Melanoma cells lacking these mutations are not inhibited by *vemurafenib*.

ADME

Following oral administration, the elimination $t_{1/2}$ for *vemurafenib* is 57 h. *Vemurafenib* is a substrate of CYP3A4 and a substrate and an inhibitor of the Pgp exporter. *Vemurafenib* can increase concentrations of CYP1A2 substrates. Thus, concomitant use with the centrally acting α_2 adrenergic receptor agonist *tizanidine*, which has a narrow therapeutic window and is used as a muscle relaxant, should be avoided.

Therapeutic Uses

Vemurafenib is approved for the treatment of patients with metastatic melanoma harboring activating BRAF (V600E/K) mutations. It is not indicated for the treatment of patients with melanoma carrying wild-type BRAF. It is also approved for treatment of patients with Erdheim-Chester disease, a rare blood cancer (600–700 patients worldwide). Half of this slow-growing, histiocyte malignancy carries mutant BRAF as a driver.

Adverse Effects

The most common adverse effects (30%–60% of patients) are cutaneous events. Arthralgia, fatigue, and nausea are also observed but at a lower frequency. Cutaneous squamous cell carcinomas (cuSCCs) and keratoacanthomas that may need surgical removal appear in more than 20% of patients. The median time to the first appearance of these skin lesions is 7 to 8 weeks, and they are caused by a paradoxical stimulation of wild-type BRAF. Increased photosensitivity reactions can be prevented with sunblock. QT prolongation has been observed, necessitating controlling for risk factors and monitoring ECG and electrolytes.

Drug Resistance

Melanoma is one of the most aggressive malignancies, with a high mutation rate that renders these tumors highly heterogeneous and thus prone to the quick development of resistance to drug treatments. After the initial response of melanoma lesions to treatment with *vemurafenib*, resistant cancer cell subpopulations are selected, typically in less than 6 months. Constitutively active mutant *NRAS*; downstream mutations in MEK; overexpression of the growth receptor PDGFR-B; activation of parallel pathways such as PI3K/Akt, mammalian or mechanistic target of rapamycin (mTOR), and STAT3 signaling; or stromal cell secretion of HGF can drive this drug resistance (Chapman et al., 2011). Combination treatment with a MEK inhibitor (see below) that acts downstream of BRAF (see Figure 71–2) can delay development of resistance to single-agent BRAF inhibitors (Robert et al., 2015).

Dabrafenib

Mechanism of Action

Dabrafenib is an orally bioavailable, small-molecule inhibitor of mutated forms of BRAF kinases. *Dabrafenib* can inhibit the proliferation of tumor cells that contain a constitutively active mutated BRAF with *in vitro* IC_{50} values of 0.65, 0.5, and 1.84 nM for the V600E, V600K, and V600D mutations, respectively. At about 5-fold higher concentrations, *dabrafenib* also inhibits wild-type BRAF and cRAF (IC_{50} values of 3.2 and 5.0 nM, respectively). *Dabrafenib*, in combination with *trametinib* (see discussion that follows), is approved for the treatment of patients with unresectable or metastatic melanoma and NSCLC with BRAF mutations. The drug is not indicated for the treatment of patients with wild-type BRAF melanoma.

ADME

The mean terminal $t_{1/2}$ of *dabrafenib* is 8 h after oral administration. *Dabrafenib* is metabolized primarily by CYP2C8 and CYP3A4. Patients should be monitored closely for adverse reactions and for loss of efficacy if treated with drugs that induce CYP2C8 and CYP3A4 activity.

Therapeutic Uses

Dabrafenib was FDA approved as a single-agent treatment of patients with BRAF V600E mutation–positive advanced melanoma in 2013. Resistance to *dabrafenib* and other BRAF inhibitors occurs within about 6 months, but a combination with the MEK inhibitor *trametinib* can delay the development of resistance (see discussion that follows). Thus, the FDA has approved the combination of *dabrafenib* and *trametinib* for the treatment of patients with BRAF V600E/K–mutant metastatic melanoma and BRAF V600E–mutant NSCLC.

Adverse Effects and Drug Interactions

Adverse effects, in order of decreasing frequency, are hyperkeratosis, headache, pyrexia, arthralgia, papilloma, alopecia, and PPES. An increased incidence of cuSCC is expected in about 10% of patients, with a median time of 2 months to the first diagnosis. About one-third of those patients develop more than one lesion with continued administration of *dabrafenib*. With the *dabrafenib-trametinib* combination, the appearance of cuSCC is delayed to a median time of 7 months and occurs in only 3% of patients. Most common adverse reactions (≥20%) for *dabrafenib* in combination with *trametinib* are pyrexia, rash, chills, headache, arthralgia, and cough. Cardiomyopathy has also been reported.

Encorafenib

Encorafenib is an orally active protein kinase inhibitor that targets BRAF V600E, as well as wild-type BRAF and CRAF. It was approved in 2018 in combination with *binimetinib* (a MEK inhibitor) for the treatment of patients with unresectable metastatic melanoma with BRAF mutations. A combination of *encorafenib* and *cetuximab* (anti-EGFR monoclonal antibody; see Chapter 72) is approved for the treatment of patients with metastatic colorectal cancer with a BRAF V600E mutation in whom treatment increased progression-free and overall survival of patients. Adverse events include fatigue, nausea, diarrhea, acneiform dermatitis, abdominal pain, the risk of cuSCC, and dose-dependent QTc interval prolongation. *Encorafenib* is mostly metabolized by CYP3A4, and dosage adjustment is recommended in patients with mild to moderate renal or hepatic impairment. The elimination $t_{1/2}$ of *encorafenib* is 3.5 h.

Inhibitors of MEK: Trametinib, Cobimetinib, Binimetinib, and Selumetinib

The MKKs or MEKs (mitogen-activated extracellular signal regulated protein kinase kinases) are serine/threonine kinases that act downstream of RAF. They provide additional targets for the inhibition of the RAS-RAF-MEK-ERK pathway that is frequently activated in cancer and controls cell survival, differentiation, and growth (see Figure 71–2). Most MEK inhibitors are allosteric kinase inhibitors and their use as a single drug is susceptible to adaptive feedback reactivation of MAPK signaling that can reduce their efficacy (Yaeger and Corcoran, 2019).

Trametinib

Trametinib was the first inhibitor of the MEK family to be FDA-approved.

Mechanism of Action

Trametinib is an orally bioavailable, reversible, allosteric inhibitor of ATP binding to the MEK1/2 kinases. *Trametinib* inhibits MEK1/2 kinase activities with an IC_{50} of about 2 nM.

ADME

Bioavailability of oral *trametinib* is 72% and is reduced if taken with a high-calorie meal. Peak plasma concentrations are achieved 1.5 h postdose. The drug has a relatively long $t_{1/2}$ of 4 to 5 days. *Trametinib* is not a substrate of CYP enzymes, and mild renal or hepatic impairment does not affect exposure to the drug significantly.

Therapeutic Uses

The FDA approved *trametinib* as a monotherapy for patients with mutant BRAF V600E/K melanoma due to improved survival relative to standard therapy. *Trametinib* in combination with the BRAF inhibitor *dabrafenib* is also approved for mutant V600E/K metastatic melanoma and mutant V600E metastatic NSCLC as well as metastatic anaplastic thyroid cancer. *Trametinib* treatment lacks efficacy in patients who received prior treatment with BRAF inhibitors. This suggests that resistance to BRAF and MEK inhibitors involves similar mechanisms (Kim et al., 2013).

Adverse Effects

The most frequent adverse effects are cutaneous rash, acneiform dermatitis, and diarrhea. Fatigue, nausea, and lymphedema also occur. Serious grade 3 to 4 skin toxicity is found in 6% of patients. Other serious adverse effects are cardiomyopathy, hypertension, hemorrhage, interstitial lung disease, and ocular toxic effects. In contrast to the BRAF inhibitors, *trametinib* does not cause cuSCC. Because *trametinib* can cause fetal harm when administered to a pregnant woman, this risk must be weighed against the potential benefit.

Drug Resistance

Activation of alternative signaling pathways that include PI3K/Akt, mTOR, and STAT3 signaling can bypass MEK inhibition and result in drug-resistant tumors. Also, mutations in the allosteric binding pocket and activation loop of MEK1 can inhibit the binding of the inhibitor and cause resistance to *trametinib*. Resistance to single-agent *trametinib* occurs within 6 to 7 months of the initiation of treatment at a rate approaching 50%. As a strategy to overcome resistance, *trametinib* has been combined with the BRAF inhibitor *dabrafenib* for the treatment of patients with BRAF V600E/K–mutant metastatic melanoma (Flaherty et al., 2012). This combination is FDA-approved.

Cobimetinib

Mechanism of Action; ADME

Cobimetinib is an orally bioavailable, reversible inhibitor of MEK1/2 protein kinase activity with an IC_{50} of about 4 nM. Its elimination $t_{1/2}$ ranges from 1 to 3 days. *Cobimetinib* is a substrate for CYP3A, and concomitant treatment with inducers or inhibitors of CYP3A will reduce or increase the systemic exposure of patients.

Therapeutic Uses

A combination of *cobimetinib* and *vemurafenib* is approved for the treatment of patients with unresectable or metastatic melanoma with a BRAF V600E/K mutation.

Adverse Effects

The most common adverse effects are diarrhea, photosensitivity reaction, nausea, pyrexia, and vomiting. Major hemorrhagic events can occur with *cobimetinib*; patients should be monitored for signs and symptoms of bleeding. The drug also increases the risk of cardiomyopathy. The safety in patients with decreased left ventricular ejection fraction has not been established.

Binimetinib

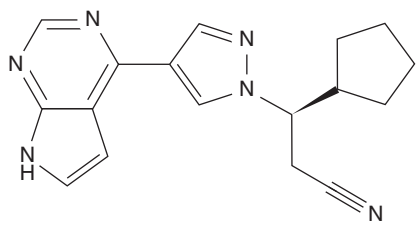
Binimetinib is an orally bioavailable MEK1/2 kinase inhibitor with an IC_{50} of 12 nM that was approved in 2018, in combination with *encorafenib* (see above), for the treatment of patients with unresectable or metastatic melanoma with a BRAF V600E or V600K mutation. After moderate to severe hepatic impairment, the area under the curve (AUC) is increased 2-fold. Impaired renal function does not impact drug elimination. The elimination $t_{1/2}$ is 3.5 h.

Selumetinib

Selumetinib is an orally bioavailable inhibitor of MEK1 (IC_{50}, 14 nM) and MEK2 with a lower affinity (IC_{50}, 530 nM). *Selumetinib* was approved in 2020 for the treatment of pediatric patients with neurofibromatosis type 1 and inoperable, plexiform neurofibromas. Warnings include the risk of cardiomyopathy requiring assessment of ejection fraction prior to treatment, ocular and skin toxicity, and possible GI toxicity requiring antidiarrheal treatment and increased fluid intake. Additional adverse reactions (≥40%) are vomiting, rash, and abdominal pain. *Selumetinib* is primarily metabolized by CYP3A4 and eliminated with a $t_{1/2}$ of 6.2 h.

Inhibitors of JAK1 and JAK2

***J**anus **a**ssociated **k**inases* (JAKs) mediate the signaling of cytokines and growth factors in hematopoiesis and immune function (see, for instance, Figure 39–2). Intracellular JAK signaling recruits STATs to cytokine membrane receptors (see Figure 71–2). After STAT activation and subsequent localization to the nucleus, STATs modulate gene expression. JAK signaling is dysregulated in myelofibrosis and polycythemia vera, which prompted the development of JAK inhibitors (JAKinibs). *Ruxolitinib* is the first drug in this class.



Ruxolitinib

Ruxolitinib

Mechanism of Action; ADME

Ruxolitinib is an orally bioavailable (95%) analogue of ATP that inhibits the protein kinase activities of JAK1 and JAK2 (see Figure 72–4) with an IC_{50} of 3 nM and more than 100-fold selectivity over JAK3. The elimination $t_{1/2}$ of *ruxolitinib* is about 3 h; its metabolism is mainly via hepatic CYP3A4, with products excreted largely in the urine. In patients with renal or hepatic impairment, a dose reduction is recommended.

Therapeutic Uses

Ruxolitinib is indicated for the treatment of patients with polycythemia vera who have had an inadequate response to *hydroxyurea* and for the treatment of myelofibrosis and steroid-refractory, acute graft-versus-host disease (>12 years of age).

Adverse Effects and Drug Interactions

The most common adverse reactions, thrombocytopenia and anemia, may require a dose reduction or discontinuation of treatment. Infections should be resolved before treatment with *ruxolitinib*. In a few patients, BCC or squamous cell skin carcinoma may occur, warranting periodic skin examinations during treatment. Strong CYP3A4 inhibitors (e.g., *ketoconazole*, *fluconazole*) increase exposure to *ruxolitinib* and prolong its $t_{1/2}$ to 6 h. Due to the adverse hematopoietic effects, postpartum patients should discontinue nursing during drug treatment.

Fedratinib

Fedratinib is an orally bioavailable JAK2-selective inhibitor (IC_{50} is 3 nM; >35-fold selective vs. JAK1 and JAK3) with activity against wild-type and mutationally activated JAK2 as well as FLT3 (IC_{50} is 15 nM) and RET (IC_{50} is 48 nM). Abnormal activation of JAK2 is associated with myeloproliferative disease and polycythemia vera. *Fedratinib* was approved in 2019 for the treatment of patients with myelofibrosis. Warnings include the risk of serious and fatal encephalopathy requiring monitoring and possibly replenishment of thiamine levels, anemia and thrombocytopenia, as well as GI and hepatic toxicity. Adverse reactions include diarrhea and nausea. The metabolism is through CYP3A with a terminal elimination $t_{1/2}$ of 114 h.

Cyclin-Dependent Kinase (CDK) 4/6 Inhibitors

The CDKs are a family of over 20 serine/threonine protein kinases that modulate intracellular signaling during cell cycle progression (see Figure 71–2). Given their pivotal role in cellular proliferation, CDKs are prime targets for development of inhibitors. However, different tissue selectivity and distinct cell cycle–specific activity periods of different CDKs provide a challenge.

CDK4/6 are attractive targets because they control cell cycle progression from the G_0/G_1 to S phase. Interaction of cyclin D with CDK4/6 enhances phosphorylation of the Rb (retinoblastoma) tumor suppressor protein, inactivating Rb and permitting the transcription of factors that control transition into the S phase. Inhibition of CDK4/6 will inhibit the phosphorylation and inactivation of Rb and cause a G_1 arrest in susceptible cells that utilize this pathway, thereby providing the rationale for the use of inhibitors of CDK4/6 (Figure 71–3). This cell cycle checkpoint is frequently compromised in cancers due to cyclin D amplification, loss of Rb function, or loss of negative regulators of CDK4/6, such as $p16^{INK4A}$, a tumor suppressor named for its molecular mass (it is a small protein of 16 kDa) and its capacity to inhibit CDK4. The class of CDK inhibitory drugs is identified by the syllable *ciclib*. CDK4/6 inhibitors with inhibitory profiles similar to that of *palbociclib* include *abemaciclib*, *ribociclib*, and *trilaciclib*, which are now approved for the treatment mostly of patients with breast cancer and are in clinical trials for other cancers. The efficacy and adverse effects of these inhibitors overlap, although differences in the selectivity for the targeted kinases result in different adverse effect profiles and approved uses of these inhibitors (O'Leary et al., 2016).

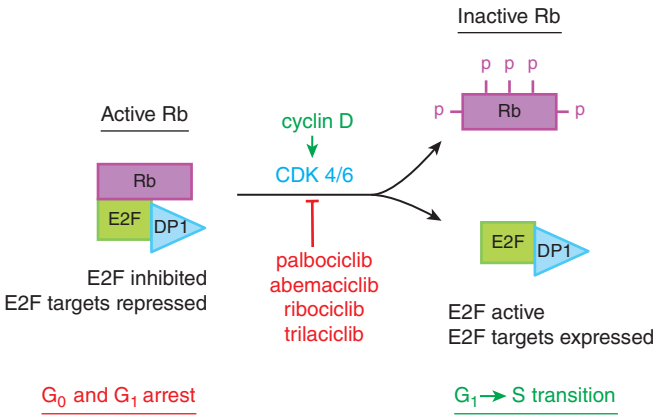


Figure 71–3 *CDK4/6 inhibitors: retinoblastoma protein, cyclin-dependent protein kinases, and the regulation of cell cycle progression.* In quiescent and differentiated cells (in G_0 or in cells arrested in G_1), the Rb protein is active and interacts with a heterodimer of the transcription factor E2F and its dimerization partner DP1, repressing transcription of promoters regulated by E2F. As a cell begins a cycle of division, cyclins activate CDKs that phosphorylate Rb, disrupting the Rb-E2F/DP complexes and permitting accumulation of active E2F complexes that drive transcription. This cell cycle checkpoint is frequently compromised in cancers due to cyclin D amplification, loss of Rb function, or loss of negative regulators of CDK4/6. Hyperphosphorylation of Rb can occur via mutations in Rb or expression of viral oncoproteins targeting Rb, placing the cell in a state of extensive proliferation, with reduced capacity to exit the cell cycle. Inhibition of CDK4/6 can cause G_1 arrest in susceptible cells (Dyson, 2016). CDK inhibitors (in red lettering) are identified by the disyllabic suffix "ciclib."

Palbociclib

Mechanism of Action

Palbociclib is an orally bioavailable, small-molecule inhibitor of CDK4 and CDK6, with IC_{50} values of 11 and 16 nM, respectively, and without inhibitory activity against CDK1, 2, or 5.

ADME

The mean plasma elimination $t_{1/2}$ is 29 h. *Palbociclib* is a substrate and an inhibitor of CYP3A; thus, doses of *palbociclib* or of substrates of CYP3A4 may need to be reduced when they are given concurrently, with special attention to coadministered agents with narrow therapeutic indexes (see the Adverse Effects and Drug Interactions section). The sulfotransferase SULT2A1 also participates in *palbociclib* metabolism.

Therapeutic Uses

Palbociclib is the first CDK inhibitor approved by the FDA (in 2015) for the treatment of advanced or metastatic breast cancer that is estrogen receptor (ER) positive and HER2 negative. *Palbociclib* is used in combination with *letrozole*, an aromatase inhibitor, as initial endocrine-based therapy in postmenopausal women or with the antiestrogen *fulvestrant* in women with disease progression after endocrine-based therapy. *Letrozole* and *fulvestrant* are described in Chapter 73. The inclusion of *palbociclib* with different endocrine-based treatments almost doubles the progression-free survival of patients (Cristofanilli et al., 2016; Finn et al., 2016).

Adverse Effects and Drug Interactions

The most common adverse effects of *palbociclib* are neutropenia, leukopenia, infections, stomatitis, fatigue, nausea, anemia, headache, diarrhea, and thrombocytopenia. The most common grade 3 and 4 adverse effects are neutropenia (>60%), leukopenia (~25%), and anemia (~5%). Patients should avoid breastfeeding while taking *palbociclib*. *Letrozole* and *fulvestrant* do not alter the pharmacokinetics of *palbociclib*. No effect on the QTc interval was observed.

Patients should avoid concomitant use of strong CYP3A inhibitors (e.g., *clarithromycin, indinavir, itraconazole, ketoconazole, lopinavir/ritonavir, nefazodone, nelfinavir, posaconazole, ritonavir, saquinavir, telaprevir, telithromycin*, and *voriconazole*). Coadministration of a strong

CYP3A inhibitor (e.g., *itraconazole*) can markedly increase exposure to *palbociclib* (by 87% in healthy subjects). Patients should also avoid grapefruit or grapefruit juice. If *palbociclib* coadministration with a strong CYP3A inhibitor cannot be avoided, the dose of *palbociclib* should be reduced.

Coadministration of a strong CYP3A inducer will decrease exposure to *palbociclib*. Thus, concomitant use of *phenytoin*, *rifampin*, *carbamazepine*, *enzalutamide*, and St. John's wort should be avoided.

Palbociclib may increase exposure to coadministered CYP3A substrates with narrow therapeutic indexes (e.g., *midazolam, alfentanil, cyclosporine, dihydroergotamine, ergotamine, everolimus, fentanyl, pimozide, quinidine, sirolimus*, and *tacrolimus*), requiring reduction of *palbociclib* dose.

Abemaciclib

Abemaciclib is an orally bioavailable inhibitor of CDK4/6 approved in 2017 in the treatment of patients with ER-positive and HER2-negative, advanced or metastatic breast cancer. One indication is in postmenopausal women in combination with an aromatase inhibitor (see Chapter 73); a second indication is in patients whose disease progressed following endocrine therapy in combination with the antiestrogen *fulvestrant* (see Chapter 73); a third indication is in patients with disease progression following endocrine therapy and prior chemotherapy in the metastatic setting. Most common adverse reactions (incidence ≥20%) are diarrhea, neutropenia, nausea, abdominal pain, infections, fatigue, anemia, decreased appetite, vomiting, headache, alopecia, and thrombocytopenia.

Abemaciclib is metabolized by CYP3A, and it is recommended to avoid concomitant use of inhibitors (see discussion above on *palbociclib* metabolism). The elimination $t_{1/2}$ of *abemaciclib* is 18.3 h.

Ribociclib

Ribociclib, an orally bioavailable inhibitor of CDK4/6, was approved in 2017 for the treatment of patients with ER-positive and HER2-negative advanced or metastatic breast cancer. One indication is in combination with an aromatase inhibitor (see Chapter 73) for the treatment of pre-, peri-, or postmenopausal patients as initial endocrine-based therapy; a second indication is for the treatment of postmenopausal patients in combination with the antiestrogen *fulvestrant* (see Chapter 73) as initial endocrine-based therapy or following disease progression on endocrine therapy. Recent warnings indicate an increased risk of interstitial lung disease/pneumonitis during treatment. Monitoring of patients for pulmonary symptoms is thus recommended. Other risks are concentration-dependent increases in the QTc interval, with pretreatment and continuous ECG and electrolyte monitoring recommended; severe cutaneous adverse reactions, including Stevens-Johnson syndrome and toxic epidermal necrolysis; and neutropenia, with pretreatment and every-2-week monitoring of CBC counts recommended. Common adverse effects (incidence ≥20%) are nausea, fatigue, diarrhea, leukopenia, vomiting, alopecia, headache, rash, and cough. *Ribociclib* is metabolized through CYP3A, and concomitant use of interacting drugs should be avoided (see above discussion for *palbociclib*). The elimination $t_{1/2}$ of *ribociclib* ranges from 29.7 to 54.7 h.

Trilaciclib

Trilaciclib is a CDK4/6 kinase inhibitor that was approved in 2021 to decrease the incidence of chemotherapy-induced myelosuppression in patients receiving a platinum/*etoposide*-containing regimen or *topotecan*-containing regimen (see Chapter 70) for extensive-stage small cell lung cancer. *Trilaciclib* is administered as a single IV 30-min infusion within 4 h prior to chemotherapy on each day of chemotherapy. *Trilaciclib* transiently maintains immune cells and hemopoietic stem and progenitor cells in G_1 cell cycle arrest to reduce bone marrow toxicity of chemotherapy. The elimination $t_{1/2}$ is approximately 14 h, and bone marrow cell cycle arrest lasts for up to 32 h after a single dose. After that time, bone marrow progenitor subsets resume proliferation.

An intriguing possibility is that the intermittent administration of *trilaciclib* may preserve anticancer immune reactivity and contribute to increased efficacy of chemotherapy. This is suggested from a study using *gemcitabine* and *carboplatin* chemotherapy in patients with hormone receptor–negative and HER2-negative ("triple negative") metastatic breast cancer that is not considered responsive to CDK4/6 inhibitors (Tan et al., 2019). An editorial in *Lancet Oncology* (Goel and Tolaney, 2019) speculates that *trilaciclib* before chemotherapy may preserve immune cell function.

Bruton Tyrosine Kinase (BTK) Inhibitors

The BTK protein tyr kinase plays important roles in the signal transduction and normal function of B cells as well as their malignant transformation. B-cell receptor signaling recruits SYK followed by the activation of BTK and PI3Kδ, leading to further downstream responses and multiple cellular effects (Figure 71–4). Integrin signaling and chemokine receptor signaling can modulate the activity of the pathway. A SYK kinase inhibitor, *fostamatinib*, was approved for the treatment of patients with chronic immune thrombocytopenia. BTK inhibitors are discussed next, and inhibitors of the PI3Kδ isoform that is frequently activated in B-cell malignancies are discussed further below. BTK and PI3K inhibitors can disrupt oncogenic B-cell receptor signaling and accelerate tumor cell death (reviewed by Burger and O'Brien, 2018).

Ibrutinib

Mechanism of Action

Ibrutinib is an orally bioavailable, small-molecule inhibitor that inactivates BTK through covalent binding to cys^{481} near the ATP-binding domain.

ADME

Administration with food (vs. on an empty stomach) roughly doubles absorption. The elimination $t_{1/2}$ of *ibrutinib* is 4 to 6 h. It is a substrate of CYP3A; CYP2D6 participates to a minor degree. Metabolites are excreted mainly via the feces; renal excretion is minor. Hepatic impairment is likely to increase *ibrutinib* exposure. *Ibrutinib* is not a substrate for Pgp.

Therapeutic Uses

Ibrutinib inhibits malignant B-cell proliferation and is indicated for the treatment of patients with mantle cell lymphoma (MCL) and marginal zone lymphoma who have received at least one prior therapy and patients with chronic lymphocytic leukemia (CLL), small lymphocytic lymphoma (SLL), Waldenström's macroglobulinemia, and chronic graft-versus-host disease after failure of systemic therapy.

Adverse Effects and Drug Interactions

The most common adverse effects (≥20%) in patients with B-cell malignancies are neutropenia, pyrexia, thrombocytopenia, hemorrhage, anemia, diarrhea, nausea, musculoskeletal pain, rash, and fatigue. The onset of hypertension has been observed within less than a month and up

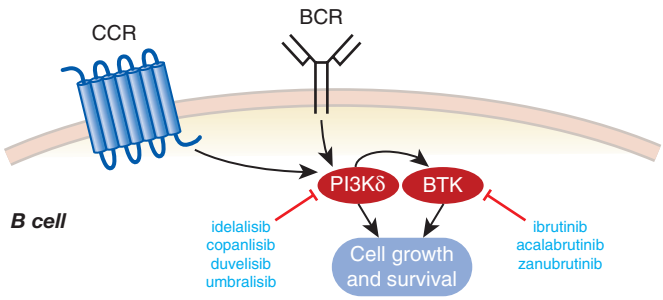


Figure 71–4 *Inhibitors of signaling in B cells.* The signaling by B-cell receptors (BCRs) and chemokine receptors (CCRs) enables the recruitment of intracellular signal transducers, which is followed by the activation of BTK and PI3Kδ. The BTK and PI3K inhibitors indicated can disrupt oncogenic BCR signaling and thus lead to cell death.

to 2 years after the start of *ibrutinib* treatment. Thus, blood pressure should be monitored and antihypertensive treatment initiated or modified. Atrial fibrillation is observed in up to 7% of patients; this requires monitoring and treatment. Second primary malignancies occur in up to 16% of patients, most of them nonmelanoma skin cancers.

Strong (e.g., azole antifungals) and moderate (e.g., *diltiazem, erythromycin*) CYP inhibitors will increase *ibrutinib*'s AUC; thus, dose reduction is recommended when using CYP3A inhibitors concomitantly. The reverse is true for strong and moderate CYP inducers (e.g., *rifampin*, *efavirenz*), which will decrease *ibrutinib* exposure.

Ibrutinib may cause fetal harm. Pregnant patients and those at risk for pregnancy should be apprised of the drug's hazardous potential.

Acalabrutinib

Acalabrutinib is a second-generation, orally bioavailable, irreversible BTK inhibitor that shows higher selectivity for BTK versus other kinases (e.g., EGFR, ITK [inducible T-cell kinase]) than *ibrutinib*. *Acalabrutinib* was first approved in 2017 and is indicated for the treatment of patients with MCL who have received at least one prior therapy and patients with CLL and SLL. Adverse effects include the risk of infections, hemorrhage, cytopenias, second primary malignancies including nonmelanoma skin cancers, and cardiac arrhythmias. The elimination $t_{1/2}$ is 1 h for *acalabrutinib* and 3.5 h for an active metabolite. Still, dosing every 12 h generates BTK occupancy of 95% or greater in peripheral blood and inactivation of BTK likely due to the covalent binding to a cysteine residue in the active site of the kinase.

Zanubrutinib

Zanubrutinib is an orally bioavailable inhibitor of BTK that forms a covalent bond with a cysteine residue in the BTK active site and shows a higher selectivity for BTK versus other kinases (e.g., ITK, EGFR), resulting in less off-target effects. *Zanubrutinib* was approved in 2019 for the treatment of patients with MCL who have received at least one prior therapy. Adverse events and warnings overlap with those of *acalabrutinib* (above) and include skin cancers as second primary malignancies, hemorrhage, and arrhythmias. The elimination $t_{1/2}$ is 2 to 4 h. Still, BTK occupancy in peripheral mononuclear cells was maintained at 100% over 24 h and at 94% to 100% in lymph nodes.

Inhibitors of the BCR-ABL Kinase

A single molecular event, the Philadelphia chromosome translocation t(9;22), leads to expression of ABL (Abelson murine leukemia viral oncogene homolog) and BCR. This fusion generates a constitutively active protein kinase, BCR-ABL, resulting in continuous and uncontrollable cell division. BCR-ABL drives the malignant phenotype of chronic myelocytic leukemia (CML) (see Figure 71–2).

HISTORY **BCR-ABL KINASE INHIBITORS**

Imatinib mesylate, originally named "sti 571" (signal transduction inhibitor 571), was the first protein kinase inhibitor designed to target a driver mutation in a cancer and to receive FDA approval. It was approved in 2001 under the name *Gleevec* and is now designated as an essential drug by the World Health Organization. *Imatinib* targets the BCR-ABL tyrosine kinase fusion protein that drives CML. Imatinib-resistant ABL mutations were uncovered in 2002 and led to the development of next-generation inhibitors *dasatinib* and *nilotinib*, which overcome the resistance.

Imatinib and the second-generation inhibitors *dasatinib* and *nilotinib* induce clinical and molecular remissions in more than 90% of patients with CML in the chronic phase of disease. Due to its inhibition of other kinases, *imatinib* is also effective in some other tumors, including GISTs (driven by a c-KIT mutation), hypereosinophilia syndrome, and dermatofibrosarcoma protuberans (all driven by activating mutations in the PDGFR; see Figure 71–2).

Imatinib, Dasatinib, and Nilotinib

Mechanisms of Action

These are orally bioavailable, small-molecule kinase inhibitors. *Imatinib* was identified through high-throughput screening against the BCR-ABL kinase. *Dasatinib*, a second-generation BCR-ABL inhibitor, also inhibits the Src kinase, and unlike *imatinib*, it binds both the open (active) and closed (inactive) configurations of the BCR-ABL kinase. *Nilotinib* was designed to have increased potency and specificity compared to *imatinib*. Its structure overcomes mutations that cause *imatinib* resistance. *Imatinib* and *nilotinib* bind to a segment of the kinase domain that fixes the enzyme in a closed or nonfunctional state, in which the protein is unable to bind its substrate/phosphate donor, ATP. These three BCR-ABL kinase inhibitors differ in their inhibitory potencies, binding specificities, and susceptibility to resistance mutations in the target enzyme (reviewed by Braun et al., 2020). *Dasatinib* (IC_{50} ~1 nM) and *nilotinib* (IC_{50} ~20 nM) inhibit BCR-ABL kinase more potently than *imatinib* (IC_{50} ~100 nM).

Mechanisms of Resistance

Resistance to these TKIs arises from point mutations in three separate segments of the kinase domain. The contact points between *imatinib* and the enzyme become sites of mutations in drug-resistant leukemic cells; these mutations prevent tight binding of the drug and lock the enzyme in its open configuration, in which it has access to substrate and is enzymatically active. *Nilotinib* retains inhibitory activity in the presence of most point mutations that confer resistance to *imatinib*. Other mutations affect the phosphate-binding region and the "activation loop" of the domain with varying degrees of associated resistance. Some mutations, such as those at amino acids 351 and 355, confer low levels of resistance to *imatinib*, possibly explaining the clinical response of some resistant tumors to dose escalation of *imatinib*.

Molecular studies have detected cells with resistance-mediating kinase mutations *prior* to initiation of therapy, particularly in patients with Ph^+ acute lymphoblastic leukemia (ALL) or CML in blast crisis. This finding indicates that drug-resistant cells arise through spontaneous mutation and expand under the selective pressure of drug exposure. Mechanisms other than BCR-ABL kinase mutations play a minor role in resistance to *imatinib*. Amplification of the wild-type kinase gene, leading to overexpression of the enzyme, has been identified in tumor samples from patients resistant to treatment. The *MDR* gene confers resistance experimentally but has not been implicated in clinical resistance. Ph^- clones lacking the BCR-ABL translocation and displaying the karyotype of myelodysplastic cells may emerge in patients receiving imatinib for CML; these may progress to myelodysplastic syndrome (MDS) and AML. Their origin is unclear.

A comparison among secondary mutations in the EGFR, ABL, and KIT kinases following treatment with BCR-ABL kinase inhibitors shows striking differences: The T790M gatekeeper mutation in EGFR (see Figure 71–1) accounts for about 90% of secondary mutations. In contrast, secondary resistance mutations in ABL or KIT after treatment of CML or GIST with *imatinib* are found across the kinase domain and rarely at the analogous gatekeeper residue in ABL (T^{315}) or in KIT (T^{670}).

ADME

Imatinib. *Imatinib* is well absorbed after oral administration and reaches maximal plasma concentrations within 2 to 4 h. The elimination $t_{1/2}$ of *imatinib* and its major active metabolite, the *N*-desmethyl derivative, are about 18 and 40 h, respectively. Food does not influence the pharmacokinetic profile. Doses of more than 300 mg/day achieve trough levels of 1 μM, which correspond to *in vitro* levels required to kill BCR-ABL–expressing cells. In the treatment of GISTs, higher doses may improve response rates. CYP3A4 is the major metabolizer of *imatinib*; thus, drugs that induce or interact with CYP3A4 can alter the pharmacokinetics of *imatinib*. Coadministration of *imatinib* and *rifampin*, an inducer of

CYP3A4, lowers the plasma *imatinib* AUC by 70%. *Imatinib*, as a competitive CYP3A4 substrate, inhibits the metabolism of *simvastatin* and increases its plasma AUC by 3.5-fold.

Dasatinib. Oral *dasatinib* is well absorbed; its bioavailability is significantly reduced at neutral gastric pH (i.e., after antacids and H_2 blockers) but is unaffected by food. The plasma $t_{1/2}$ of *dasatinib* is 3 to 5 h. *Dasatinib* exhibits dose-proportional increases in AUC, and its clearance is constant over the dose range of 15 to 240 mg/day. *Dasatinib* is metabolized primarily by CYP3A4, with minor contributions by FMO3 (flavin containing monooxygenase 3) and UGT. The major metabolite is equipotent to the parent drug but represents only 5% of the AUC. Plasma concentrations of *dasatinib* are affected by inducers and inhibitors of CYP3A4 in a similar fashion to *imatinib*.

Nilotinib. Approximately 30% of an oral dose of *nilotinib* is absorbed after administration, with peak concentrations in plasma 3 h after dosing. Unlike the other BCR-ABL inhibitors, *nilotinib*'s bioavailability increases significantly in the presence of food. The drug has a plasma $t_{1/2}$ about 17 h, and plasma concentrations reach a steady state only after 8 days of daily dosing. *Nilotinib* is metabolized by CYP3A4, with predictable alteration by inducers, inhibitors, and competitors of CYP3A4. *Nilotinib* is a substrate and an inhibitor of Pgp.

Therapeutic Uses

These protein TKIs have efficacy in diseases in which ABL, *KIT*, or PDGFR has dominant roles in driving tumor growth, reflecting the presence of a mutation that results in constitutive activation of the kinase. *Imatinib* shows therapeutic benefits in patients with chronic-phase CML (BCR-ABL) or GIST and a subset of patients with mucosal or acral lentiginous melanoma (*KIT* mutation positive), chronic myelomonocytic leukemia (EVT6-PDGFR translocation), hypereosinophilia syndrome (FIP1L1-PDGFR), and dermatofibrosarcoma protuberans (constitutive production of the ligand for PDGFR). It is the agent of choice for patients with metastatic GIST and is used as adjuvant therapy of *c-KIT*–positive GIST. *Dasatinib* is approved for patients with newly diagnosed CML with resistance to or intolerance of *imatinib* in both chronic and advanced phases of disease and for use combined with cytotoxic chemotherapy in patients with Ph^+ ALL who are resistant or intolerant to prior therapies.

Adverse Effects

Imatinib, *dasatinib*, and *nilotinib* cause GI symptoms (diarrhea, nausea, and vomiting) that are usually readily controlled. All three drugs promote fluid retention, edema, and periorbital swelling. Myelosuppression occurs infrequently but may require transfusion support and dose reduction or discontinuation of the drug. These drugs can be associated with hepatotoxicity. *Dasatinib* may cause pleural effusions and pulmonary hypertension in a small subset of patients. *Nilotinib* and *dasatinib* may prolong the QT interval; *nilotinib* has been associated with cardiac and vascular events, including ischemia. Most nonhematological adverse reactions are self-limited and respond to dose adjustments. After the adverse reactions have resolved, the drug may be reinitiated and titrated back to effective doses.

Bosutinib

Bosutinib is another second-generation, orally bioavailable BCR-ABL kinase inhibitor (IC_{50} = 1.2 nM). *Bosutinib* also inhibits activity of Src (IC_{50} = 1 nM) and other members of the Src family. It is FDA approved for the treatment of patients with chronic, accelerated, or blast-phase Ph^+ CML with resistance or intolerance to prior therapy. The most common adverse reactions (incidence >20%) are diarrhea, nausea, thrombocytopenia, vomiting, abdominal pain, rash, anemia, pyrexia, and fatigue.

Ponatinib

Ponatinib is a third-generation BCR-ABL kinase inhibitor. *Imatinib* (first-generation) lacks efficacy for the more advanced disease phases, and cells with mutations in the BCR-ABL tyrosine kinase domain are resistant. Second-generation inhibitors (*nilotinib*, *dasatinib*, *bosutinib*) were developed to address these weaknesses, although they do not inhibit the T315I mutant of BCR-ABL that is found in 15% to 20% of patients with CML.

Mechanism of Action

Ponatinib is a third-generation, orally bioavailable kinase inhibitor that is active against the T315I mutant BCR-ABL and is also effective toward all other known BCR-ABL1 kinase mutants. IC_{50} values (nM) for different kinases are as follows: ABL, 0.37; PDGFR, 1.1; VEGFR2, 1.5; FGFR1, 2.2; and Src, 5.4.

ADME

Peak *ponatinib* concentrations are observed within 6 h and are unaffected by food or fasting. Drug solubility is pH dependent, with higher pH resulting in decreased solubility. Accordingly, drugs that affect gastric pH (e.g., H_2 antagonists, antacids, and proton pump inhibitors [PPIs]) may significantly alter *ponatinib* bioavailability. *Ponatinib* is highly (99%) bound to plasma proteins and is a weak substrate for Pgp and ABCG2 transporters. Metabolism is primarily via CYP3A4 and to a lesser extent by CYPs 2C8, 2D6, and 3A511; esterases; and amidases. Coadministration of CYP3A inhibitors significantly increases *ponatinib* concentrations. The $t_{1/2}$ of *ponatinib* is about 24 h. Unmetabolized *ponatinib* is primarily excreted in the feces (87%), with a small portion (5%) in the urine.

Therapeutic Uses

Ponatinib is approved for resistant CML and Ph^+ ALL. Clinical cytogenetics measurements are a key tool for monitoring the response to therapy, reported as the percentage of Ph^+ karyotypes in 20 bone marrow metaphases (major cytogenetic response [MCyR]). *Ponatinib* produces an overall 54% MCyR in patients with early-phase CML and a 70% MCyR in patients with early-phase CML with the T315I mutation. Because of *ponatinib*'s potency and efficacy against all known BCR-ABL single mutants, trials are under way that may advance this therapy to a first-line treatment.

Adverse Effects

Arterial thrombosis and hepatotoxicity are major adverse effects; thus, appropriate precautions, dose reduction, monitoring, or discontinuation is recommended. Dose-limiting toxicities include elevated lipase and amylase levels and pancreatitis.

Inhibitors of the PI3K/Akt/mTOR Pathway

Background

Activation of PI3K and the consequent activation of signaling events through protein kinase B (Akt) and the mTOR pathway are important in cell growth and survival and in the regulation of cell metabolism (Figures 71–2 and 71–5). Excessive signaling through the PI3K pathway is a frequent aberration in many human cancers; thus, this pathway is an attractive target in cancer therapy. There are three classes of PI3Ks, with the class I being the most important in cancer. The class I PI3K enzymes are heterodimers of a p85 regulatory subunit and a p110 catalytic subunit. The p110 catalytic subunit exists in four identified isoforms: α, β, γ, and δ. The α and β isoforms are ubiquitously expressed, whereas expression of the γ and δ isoforms is restricted to cells of immune lineage. The p110 subunit of the enzyme is a lipid phosphokinase that catalyzes synthesis of phosphatidylinositol (3,4,5)-trisphosphate (PIP3) from the phosphatidylinositol 4,5-bisphosphate (PIP2) substrate, an abundant lipid at the plasma membrane. The different catalytic subunits have distinct roles in normal and malignant cells. The gene coding for PI3Kα (*PIK3CA*) is the most frequently mutated PI3K in human cancer (Fruman and Rommel, 2014). In B-cell malignancies, the PI3Kδ isoform is frequently activated. A PI3Kδ inhibitor, *idelalisib*, is described below followed by the discussion of additional inhibitors of PI3Kδ, *copanlisib*, *duvelisib*, and *umbralisib*, that are approved for the treatment of B-cell lymphoma. An inhibitor of PI3Kα, *alpelisib*, shows a distinct efficacy profile, was approved in 2019 for the treatment of patients with advanced breast cancer, and is described at the end of this section.

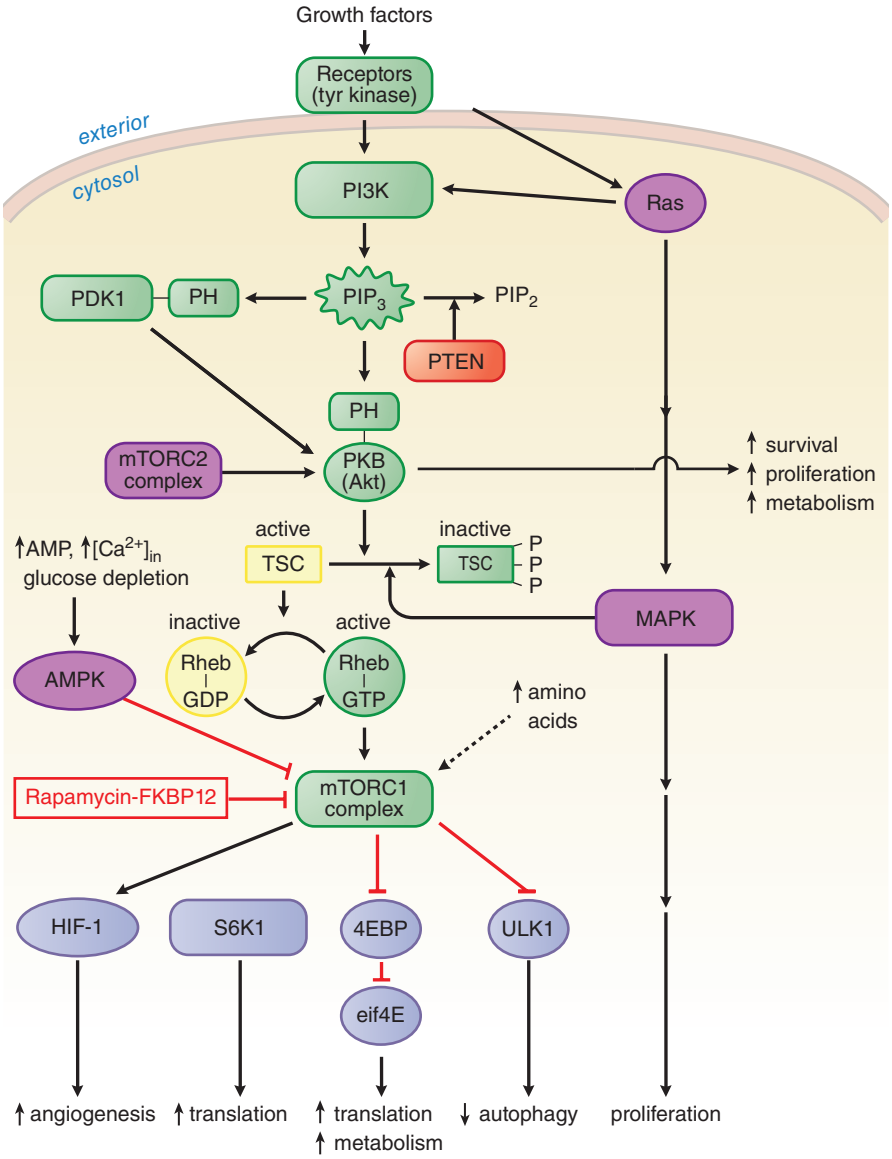


Figure 71–5 *Caveat mTOR: effect of rapamycin on growth factor signaling.* The PI3K-mTOR pathway responds to extracellular signals, the metabolic and nutrient status, and the energy charge of the cell, transmitting signals downstream that drive maintenance and proliferation (Hall, 2016). mTOR activity appears to contribute to resistance to many targeted cancer therapies (Guri and Hall, 2016), and dysregulation of the pathway is implicated in diabetes, cancer, and other pathologies (Dibble and Cantley, 2015). In this simplified version of the PI3K-mTOR pathway, the green shapes show the action pathway from membrane receptor to the mTOR multiprotein complex, mTORC1. Substrates of the mTOR ser/thr protein kinase activity and integrated responses are shown in blue across the bottom of the figure. Red T bars represent inhibitory influence. Signaling through the pathway is initiated by agonist activation of membrane receptors such as IGF1R and stimulation of their tyrosine kinase activities, leading to activation of PI3K, which phosphorylates the inositol ring of the PIP2 in the membrane to produce PIP3. PIP3 recruits to the membrane several proteins containing PH domains, including PKD1 and Akt. Akt, also known as PKB, phosphorylates TSC-2 (a subunit of TSC) at multiple sites. TSC is a GTPase-activating protein (GAP) for Rheb. Phosphorylation of TSC inhibits its GAP activity, thereby reducing the GTPase activity of Rheb and increasing the fraction of Rheb in the GTP-liganded state. Rheb, in its GTP-liganded configuration, activates mTORC1. TSC phosphorylation and PIP3 phosphorylation can be reversed by the phosphatase PTEN. mTORC1 phosphorylates substrates that mediate downstream effects; among them are the following:

- HIF-1, leading to increased angiogenesis. HIF-1 and -2 are overexpressed or upregulated in many cancers.
- S6K1 (ribosomal protein S6 kinase β1), enhancing protein synthesis. Dephospho S6K1 exists in a complex with eIF3; phosphorylation by mTOR dissociates the complex, freeing S6K1 to phosphorylate its targets that promote translation.
- 4EBP, enhancing transcription. The phosphorylation of 4EBP by mTOR reduces the capacity of 4EBP to inhibit eIF4E and slow transcription.
- ULK1, whose phosphorylation reduces autophagic activity. mTOR activity prevents ULK1 activation by disrupting its interaction with AMPK, a key promoter of autophagy. Conversely, inhibition of mTOR will stimulate autophagy.

Rapamycin (*sirolimus*) and the "rapalogs" *everolimus* and *temsirolimus* inhibit mTOR by associating with the intracellular protein FKBP12. The FKBP12-rapalog complex binds directly to mTOR and inhibits its functions in the multiprotein complex mTORC1. mTORC2 remains relatively unaffected by mTORC1 inhibitors and still can respond by upregulating Akt activity.

Impact of PI3K Inhibitors on the Tumor Microenvironment

Tumor angiogenesis is dependent on PI3Kα signaling and its inhibition may contribute to the treatment effects. Also, immune recognition of tumor cells may be enhanced by inhibition of myeloid suppressive cells that are sensitive to PI3Kδ/γ inhibition and macrophages that require PI3Kγ signaling for their suppression of T-cell activation. For further details, see the review by Castel et al. (2021).

Idelalisib

Mechanisms of Action

Idelalisib is an orally bioavailable inhibitor of the PI3Kδ isoform with an IC_{50} of about 19 nM. It is 6-fold selective versus the PI3Kγ and over 200-fold more selective versus the PI3Kα and β isoforms. PI3Kδ is expressed in normal and malignant B cells. *Idelalisib* induces apoptosis and inhibits proliferation in cell lines derived from malignant B cells and in primary tumor cells. *Idelalisib* inhibits several cell signaling pathways, including BCR, C-X-C chemokine receptor (CXCR) 4, and CXCR5 signaling, which are involved in trafficking and homing of B cells to the lymph nodes and bone marrow. Treatment of lymphoma cells with *idelalisib* resulted in inhibition of chemotaxis and adhesion and reduced cell viability.

ADME

Food does not affect absorption of *idelalisib*. The drug is metabolized by aldehyde oxidase, CYP3A, and UGT1A4, yielding an elimination $t_{1/2}$ of 8.2 h. *Idelalisib* inhibits CYP3A and is a substrate of Pgp and of the efflux transporter breast cancer resistance protein (BCRP)/ATP-binding cassette (ABC) G2 *in vitro*. *Idelalisib* increases plasma levels of *midazolam* (a CYP3A substrate) but does not affect plasma levels of *digoxin* (a Pgp substrate) or *rosuvastatin* (an OATP1B1/B3 substrate). Patients receiving strong CYP inhibitors may be at increased risk for *idelalisib* toxicity. Hepatic impairment increases systemic exposure, putting patients at risk for toxicity.

Clinical Use

Idelalisib is approved for the treatment of relapsed or refractory B-cell malignancies in patients who have received at least two prior systemic therapies. This includes CLL in combination with rituximab, follicular B-cell non-Hodgkin lymphoma (FL), and SLL. *Idelalisib* is not indicated and is not recommended for first-line treatment.

Adverse Effects

Idelalisib may cause serious and sometimes fatal adverse effects, including liver toxicity in 16% to 18%, fatal diarrhea or colitis in 14% to 20%, pneumonitis in 4%, or serious infections in 21% to 48% of patients. Patients taking *idelalisib* should avoid other hepatotoxic drugs. Other common side effects include fever, chills, cough, pneumonia, fatigue, nausea, rash, hyperglycemia, and elevated levels of triglycerides and liver enzymes. *Idelalisib* may cause embryofetal toxicity. Breastfeeding during therapy is contraindicated because of the potential for adverse reactions in nursing infants.

Copanlisib

Copanlisib is an inhibitor of PI3Kα (IC_{50} of 0.5 nM) and PI3Kδ (IC_{50} of 0.7 nM) isoforms that are expressed in malignant B cells. Parenterally administered, *copanlisib* was approved in 2017 for the treatment of patients with relapsed FL who have received at least two prior systemic therapies. The efficacy of *copanlisib* has been linked to inhibition of nuclear factor-κB (NF-κB) signaling and reduced release of interleukin (IL)-6 and IL-10. Adverse effects include hyperglycemia, hypertension that may require treatment, pneumonitis, diarrhea, and decreased general strength and energy. *Copanlisib* is a CYP3A substrate and eliminated with a $t_{1/2}$ of 39.1 h.

Duvelisib

Duvelisib is an orally bioavailable inhibitor of PI3Kδ (IC_{50} of 2.5 nM) and PI3Kγ (IC_{50} of 27 nM), which are expressed in normal and malignant B cells. The potency of *duvelisib* against the β (IC_{50} of 85 nM) and α isoforms (IC_{50} of 1602 nM) is lower. *Duvelisib* was approved in 2018 for the treatment of patients with relapsed or refractory CLL, SLL, or FL after at least two prior therapies. Adverse effects include the risk of infections in 31% of patients, diarrhea and colitis in 18%, and pneumonitis in 5%. Metabolism of *duvelisib* is through CYP3A4. The elimination $t_{1/2}$ is 4.7 h.

Umbralisib

Umbralisib is an orally bioavailable inhibitor of PI3Kδ (IC_{50} of 6.2 nM) with a greater than 100-fold selectivity versus PI3Kα, β, or γ. It also inhibits casein kinase ε. *Umbralisib* was approved in 2021 for the treatment of patients with marginal zone lymphoma after at least one prior therapy and for relapsed or refractory FL after three prior therapies. It is in trials for the treatment of other hematological malignancies. Adverse effects include neutropenia, increased risk of infections, diarrhea, noninfectious colitis, and hepatotoxicity. *Umbralisib* is metabolized by CYP2C9, CYP3A4, and CYP1A2. The elimination $t_{1/2}$ is 91 h.

Alpelisib

Alpelisib is an orally bioavailable inhibitor of PI3Kα (IC_{50} of 4.6 nM) with a greater than 50-fold selectivity relative to the other isoforms of PI3K. PIK3CA gene mutations are found in approximately 40% of patients with hormone receptor–positive, HER2-negative advanced breast cancer and indicate a worse prognosis and may mediate resistance to endocrine therapy. *Alpelisib* was approved in 2019 in combination with *fulvestrant* for the treatment of postmenopausal women, and men, with hormone receptor–positive, HER2-negative, PIK3CA-mutated, advanced breast cancer following progression on or after endocrine-based therapy (André et al., 2019). The most frequent adverse events of higher grade are hyperglycemia (36%), rash (10%), and diarrhea (7%). *Alpelisib* is metabolized by CYP3A and has an elimination $t_{1/2}$ of 8 to 9 h.

mTOR Inhibitors: Rapamycin Analogues

Rapamycin (*sirolimus*) is a product from a strain of the soil bacterium *Streptomyces* found on Rapa Nui (Easter Island). It inhibits a serine/threonine protein kinase in mammalian cells named mTOR. The PI3K/PKB(Akt)/mTOR pathway responds to a variety of signals from growth factors. The activation of the PI3K pathway is opposed by the phosphatase activity of the tumor suppressor PTEN (phosphatase and tensin homolog; see Figure 71–2). Activating mutations and amplification of genes in the receptor-PI3K pathway and loss-of-function alterations in PTEN occur frequently in cancer cells, with the result that PI3K signaling is exaggerated and cells exhibit enhanced survival. *Rapamycin* (*sirolimus*) and its congener rapalog *everolimus* are first-line drugs in posttransplant immunosuppression (see Chapter 39) and are used in coronary artery stents to prevent fibrotic growth. Rapalogs have also been approved for the treatment of patients with renal cancer; neuroendocrine tumors (NETs) of pancreatic, GI tract or lung origin; anti–hormone refractory, hormone receptor–positive breast cancer; and tuberous sclerosis complex (TSC)–related tumors.

Mechanism of Action

The rapalogs inhibit an enzyme complex, mTORC1, which occupies a downstream position in the PI3K pathway (see Figures 71–2 and 71–5). mTOR forms the mTORC1 complex with FKBP12 (immunophilin-binding protein for *tacrolimus* [FK506]), a member of the FK506-binding protein family. The antitumor actions of the rapalogs result from their binding to FKBP12 and preventing activation of mTOR, as detailed in Figure 71–5 and its legend.

ADME

For renal cell cancer, *temsirolimus* is given in weekly intravenous doses. It is metabolized by CYP3A4, with the parent drug elimination $t_{1/2}$ of 30 h and the primary active metabolite *sirolimus* (*rapamycin*) elimination $t_{1/2}$ of 53 h. Because *sirolimus* has equivalent activity as an inhibitor of mTORC1 and has a greater AUC, *sirolimus* is likely the more important contributor to antitumor action in patients. *Everolimus* is administered orally and is metabolized by CYP3A4. *Everolimus* has an elimination $t_{1/2}$

of 30 h and maintains inhibition of mTORC1 for 7 days in white blood cells. Both drugs are susceptible to interactions with other agents that affect CYP3A4 activity. For *everolimus*, the dose should be reduced for patients with moderate hepatic impairment; guidelines for dose reduction of *temsirolimus* in such patients have not been established. The drugs' pharmacokinetics do not depend on renal function, and hemodialysis does not hasten *temsirolimus* clearance.

Therapeutic Uses

Both *temsirolimus* and *everolimus* are approved for treatment of patients with advanced renal cell carcinoma. *Everolimus* prolongs survival in patients who had failed initial treatment with antiangiogenic drugs. *Everolimus* is also approved for the treatment of postmenopausal women with advanced hormone receptor–positive, HER2-negative breast cancer in combination with the aromatase inhibitor *exemestane* after failure of treatment with *letrozole* or *anastrozole*. Other indications of *everolimus* are peripheral neuroendocrine tumors (PNETs) and progressive, well-differentiated, nonfunctional NETs of GI or lung origin as well as TSC-related tumors.

Adverse Effects; Resistance

The rapamycin analogues cause similar patterns of adverse effects: a mild maculopapular rash, mucositis, anemia, and fatigue (30%–50%). A few patients develop leukopenia or thrombocytopenia, effects that are reversible if therapy is discontinued. Other, less common side effects include hyperglycemia, hypertriglyceridemia, and, rarely, pulmonary infiltrates and interstitial lung disease. Pulmonary infiltrates emerge in 8% of patients receiving *everolimus* and in a smaller percentage of those treated with *temsirolimus*. If symptoms such as cough or shortness of breath develop or radiological changes progress, the drug should be discontinued. *Prednisone* may hasten the resolution of radiological changes and symptoms.

Resistance to mTOR inhibitors is incompletely understood but may arise through the action of a second mTOR complex, mTORC2, which is unaffected by rapalogs and which can upregulate Akt activity (see Figure 71–5).

Multikinase Inhibitors: Cabozantinib, Vandetanib, Midostaurin, and Gilteritinib

Selectivity of small-molecule protein kinase inhibitors for their targets is dependent on the similarity of the targeted site with sites in other kinases and the chemical composition of the inhibitor. Most inhibitors interact with the ATP-binding site of their target that is relatively conserved within a kinase family; thus, such inhibitors have only relative specificity and will cross-react at higher concentrations with closely related kinases. Nonetheless, inhibition of distantly related kinases is possible. The selectivity of inhibitors for a range of targets in the kinome is determined experimentally using assays with recombinant kinase proteins and *in vivo* using intact cells that express the kinases (Elkins et al., 2015). Inhibitors that target multiple kinase families within the clinically used dose range can be therapeutically efficacious with tolerable adverse effects and are discussed in the material that follows.

Cabozantinib

Mechanism of Action

Cabozantinib is an orally bioavailable, small-molecule inhibitor of several tyrosine kinases; as ranked by the IC_{50} values (in nM) from in vitro assays, these protein kinases are VEGFR2, 0.035; MET, 1.3; RET, 4; KIT, 4.6; AXL, 7; FLT3, 11.3; VEGFR1, 12; and TIE2, 14.3. These receptor tyrosine kinases control normal cellular function and pathologic processes that include maintenance of the tumor microenvironment, tumor angiogenesis, and metastatic spread.

ADME

The elimination half-life of *cabozantinib* is about 99 h. *Cabozantinib* is a substrate of CYP3A4; thus, the dose of *cabozantinib* should be reduced in patients with mild-to-moderate hepatic impairment as well as for concurrent administration of CYP3A4 inhibitors. Conversely, strong inducers of CYP3A4 will reduce exposure to the drug and necessitate increased dosage of *cabozantinib* unless the CYP inducers can be avoided.

Therapeutic Uses

Cabozantinib is indicated for the treatment of patients with advanced RCC and as a first- line treatment in combination with nivolumab; for patients with HCC who have been previously treated with sorafenib and for patients with with progressive, metastatic medullary thyroid cancer.

Adverse Effects

The most commonly reported (≥25%) adverse reactions are diarrhea, fatigue, nausea, PPES, hypertension, vomiting, weight loss, and constipation. Cabozantinib should not be administered to patients with a recent history of bleeding and should be discontinued in patients with thrombotic events, GI perforations and fistulas. Blood pressure should be monitored regularly for the onset of hypertension. Due to the risk of fetal harm, women of childbearing age should use contraception during treatment and for 4 months thereafter.

Vandetanib

Mechanism of Action; ADME

Vandetanib is an orally bioavailable, small-molecule multikinase inhibitor of VEGFRs, the EGFR/HER family, RET, BRK (breast tumor kinase), TIE2, and members of the ephrin receptor and SRC kinase families. These receptor tyrosine kinases are involved in both normal cellular functions and pathological processes (see Figure 71–2). *Vandetanib* is a substrate of CYP3A4. Its median elimination $t_{1/2}$ is 19 days.

Therapeutic Uses

Vandetanib is indicated for the treatment of symptomatic or progressive medullary thyroid cancer that is unresectable, locally advanced, or metastatic. In patients with asymptomatic or slowly progressing disease, *vandetanib* is only indicated after careful consideration of the treatment-related risks.

Adverse Effects and Drug Interactions

The most common adverse effects (>20%) are diarrhea/colitis, rash, acneiform dermatitis, hypertension, nausea, headache, upper respiratory tract infections, decreased appetite, and abdominal pain. Less common are QT prolongation, torsades de pointes, and sudden death. *Vandetanib* should not be used in patients with hepatic impairment or congenital long QT syndrome.

Midostaurin

Midostaurin is an orally bioavailable multikinase inhibitor with activity against mutant FLT3, a driver in AML, as well as other kinases including KIT (wild-type and D816V mutant), PDGFR, SRC, and PKC. It was approved in 2017 for the treatment of patients with newly diagnosed FLT-mutant AML in combination with *cytarabine* and *daunorubicin*. Other indications include systemic mastocytosis with associated leukemia. Major adverse effects of treatment include a risk of prolonged severe leukopenia and thrombocytopenia, mucositis, upper respiratory tract infection, diarrhea, musculoskeletal pain, and QT prolongation. It is metabolized by CYP3A. The parent *midostaurin* has an elimination $t_{1/2}$ of 19 h. Two active metabolites that contribute 28% and 38% of circulating drug are eliminated much more slowly, with $t_{1/2}$ of 32 h and 482 h, respectively.

Gilteritinib

Gilteritinib is an orally bioavailable inhibitor of multiple receptor tyrosine kinases, including wild-type and mutant FLT3 (IC_{50} of 0.29 nM). *Gilteritinib* was approved in 2018 for the treatment of patients with relapsed or refractory AML with an FLT3 mutation. Resistance of leukemic cells to the treatment can be provided by the normal bone marrow microenvironment or by cancer cell intrinsic changes (Joshi et al., 2021) and may

require combinatorial treatment. Serious adverse effects include the risk of posterior reversible encephalopathy syndrome (1% of patients) and differentiation syndrome (3% of patients) that is associated with rapid proliferation and differentiation of myeloid cells and may be life-threatening or fatal if not treated with corticosteroids. Symptoms may be fever, dyspnea, or pulmonary infiltrates. Differentiation syndrome (previously termed "retinoic acid syndrome") is also a risk of ATO treatment (see above). *Gilteritinib* is metabolized through CYP3A, and an increase in the AUC was observed when coadministered with an antifungal, *itraconazole*, that is also a strong CYP3A inhibitor. The elimination $t_{1/2}$ is 113 h.

III. Inhibitors of Tumor Angiogenesis

Cancer cells secrete angiogenic factors that induce the formation of new blood vessels and guarantee the flow of nutrients to the tumor cells to permit growth and metastasis. Many tumor types overexpress these angiogenic factors, turning on an "angiogenic switch" whereby the tumor cells adopt an invasive phenotype favoring proliferation of endothelial cells and neovascularization. In 1971, Judah Folkman hypothesized that the growth of solid tumors was dependent on angiogenesis and that blockade of the effects of putative angiogenic factors would be a good treatment modality for human cancers (Folkman, 1971). Folkman's hypothesis proved to be correct and led to the characterization of a number of secreted angiogenic factors, including VEGF, FGF, transforming growth factor (TGF)-β, and PDGF. Furthermore, inhibitors of these angiogenic factors have, indeed, become useful therapeutic agents against certain cancers. VEGF is a major driver of angiogenesis, and inhibitors of VEGF signaling comprise an important class of antitumor agents. Beyond nourishing an expanding cancerous lesion, the control of the vascular steady state by growth factors also impacts diseased tissue surveillance by circulating immune cells. Hence, combination therapies of immune checkpoint inhibitors and antiangiogenic drugs are investigated, and some have received approval as indicated below (e.g., *pembrolizumab* and *lenvatinib*) (Huinen et al., 2021). Recent research revealed the modulation of "angiogenic switch" genes by the activities of hypoxia-inducible transcription factors (HIF) that are controlled by oxygen tension in tissues. Indeed, cellular oxygen-sensing mechanisms are essential for the maintenance of complex multicellular life forms (Hammarlund et al., 2020). William G. Kaelin, Jr, Peter J. Ratcliffe, and Gregg L. Semenza shared the 2019 Nobel Prize in Medicine/Physiology "for their discoveries of how cells sense and adapt to oxygen availability." The HIF-2α inhibitor belzutifan, approved in 2021 for the treatment of renal cell carcinoma (RCC), is based on these discoveries (Choueiri and Kaelin, 2020).

Inhibition VEGF and the VEGFR Pathway

Vascular endothelial growth factor initiates endothelial cell proliferation and vascular permeability when it binds to a member of the VEGFR family, a group of highly homologous receptors with intracellular tyrosine kinase domains; these receptors include VEGFR1 (FLT1), VEGFR2 (KDR [kinase insert domain receptor]), and VEGFR3 (FLT4) (see Figure 71–2). The binding of VEGF to its receptors activates the intracellular VEGFR tyrosine kinase activity and initiates mitogenic and antiapoptotic signaling pathways (Nagy et al., 2007). Antibodies targeting VEGF, such as *bevacizumab*, sterically hinder the interaction of VEGF with its receptor (see Chapter 72). *Aflibercept* acts as a VEGF trap; it is a recombinant molecule that uses the VEGFR1-binding domain to sequester VEGF, basically acting as a "soluble decoy receptor" for VEGF. Several small-molecule drugs that inhibit the protein tyrosine kinase function of VEGFR (*pazopanib*, *sorafenib*, *sunitinib*, and *axitinib*) as well as monoclonal antibodies that target the receptor (*ramucirumab*) have been approved for clinical use. The inhibition of endothelial function by these different approaches results in a similar spectrum of adverse effects.

Antiangiogenic Small-Molecule Kinase Inhibitors

Sunitinib

Mechanism of Action

Sunitinib is a small-molecule, orally bioavailable inhibitor of multiple kinases, including KIT and VEGFR2.

ADME

The typical treatment cycle of *sunitinib* is 4 weeks on treatment followed by 2 weeks off. The dosage and schedule of *sunitinib* can be decreased in patients with adverse effects. More recently, a 2-week-on/1-week-off schedule has proven to be better tolerated and as effective as the original treatment schedule. *Sunitinib* is metabolized by CYP3A4 to produce an active metabolite, SU12662, the $t_{1/2}$ of which is 80 to 110 h; steady-state levels of the metabolite are reached after about 2 weeks of daily administration of the parent drug. Further metabolism results in the formation of inactive products. The pharmacokinetics of *sunitinib* are not affected by food intake.

Therapeutic Uses

Sunitinib is approved in metastatic renal cell cancer, producing a higher response rate and a longer progression-free survival than seen with *interferon* (IFN) or *bevacizumab*. *Sunitinib* is also approved for the treatment of pancreatic NETs and of GIST in patients who have developed resistance to *imatinib* as a consequence of *c-KIT* mutations. Specific *c-KIT* mutations correlate with the degree of response to *sunitinib* (e.g., patients with *c-KIT* exon 9 mutations have a response rate of 37%; patients with *c-KIT* exon 11 mutations have a 5% response rate).

Adverse Effects

The main adverse effects of *sunitinib* are shared by antiangiogenic inhibitors that target VEGF and VEGFR signaling and thus impact endothelial cell survival signaling: hypertension, proteinuria, and, uncommonly, bleeding, arterial thromboembolic events, and intestinal perforation. However, because *sunitinib* is a multitargeted TKI, it has a broader adverse effect profile than the monoclonal antibody *bevacizumab* that selectively targets VEGF (see Chapter 72). Fatigue affects 50% to 70% of patients and may be disabling. Hypothyroidism occurs in 40% to 60% of patients. Bone marrow suppression and diarrhea also are common side effects; severe neutropenia (neutrophils <1000/mL) develops in 10% of patients. Less common side effects include hepatotoxicity, congestive heart failure as a consequence of hypertension, and PPES. It is essential to check blood counts and thyroid function at regular intervals. Monitoring of blood pressure and periodic echocardiograms also are recommended.

Sorafenib

Mechanism of Action and Therapeutic Use

Sorafenib, like *sunitinib*, is an orally bioavailable inhibitor of multiple protein kinases. *Sorafenib* is approved for treatment of patients with thyroid and hepatocellular carcinoma as well as patients with metastatic renal cell cancer, but *sunitinib* and *pazopanib* are generally preferred first-line therapies (see previous discussion and below).

ADME

Sorafenib is given orally every day without treatment breaks. *Sorafenib* is metabolized to inactive products by CYP3A4 with a $t_{1/2}$ of 20 to 27 h; with repeated administration, steady-state concentrations are reached within 1 week.

Adverse Effects

Sorafenib-treated patients can experience the vascular toxicities seen with other antiangiogenic medications described under *sunitinib*. More common adverse effects include fatigue, nausea, diarrhea, anorexia, rash, and palmar-plantar erythrodysesthesias; uncommonly, bone marrow suppression, GI perforation, and cardiomyopathy occur.

Pazopanib

Pazopanib is an orally bioavailable kinase inhibitor of VEGFR-1, -2, and -3 as well as FGFRs, KIT, LCK (lymphocyte-specific kinase), PDGFR,

and other kinases implicated in pathological angiogenesis and cancer progression. *Pazopanib* is approved for the treatment of patients with advanced RCC and advanced soft-tissue sarcoma after prior chemotherapy. It has less toxicity and equivalent efficacy as *sunitinib* in patients with treatment-naïve metastatic renal cancer and therefore has become the preferred first-line treatment. Major adverse effects include hypertension, thrombotic and hemorrhagic events, GI perforation, QT prolongation, and cardiomyopathy. *Pazopanib* carries a black-box warning indicating that it can produce severe and life-threatening hepatotoxicity; as a consequence, *pazopanib* should not be used in elderly patients or those with preexisting liver function test abnormalities. *Pazopanib* is metabolized mostly by CYP3A and eliminated with a $t_{1/2}$ of 31 h.

Axitinib

Axitinib is an orally bioavailable kinase inhibitor of VEGFR-1, -2, and -3, which are implicated in pathological angiogenesis, tumor growth, and cancer progression. *Axitinib* is approved for the treatment of patients with advanced RCC after failure of one prior systemic therapy. Major adverse effects include hypertension, thrombotic and hemorrhagic events, and GI perforation, similar to those of the other VEGFR pathway–targeting drugs (see above). *Axitinib* is metabolized mainly by CYP3A4/5 and eliminated with a variable $t_{1/2}$ of 2.5 to 6.1 h.

Tivozanib

Tivozanib is an orally bioavailable kinase inhibitor of VEGFR-1, -2, and -3 that also inhibits other kinases including c-KIT and PDGFR. *Tivozanib* was approved in 2021 for the treatment of patients with relapsed or refractory advanced RCC following two or more prior systemic therapies. Adverse effects are typical for inhibitors of the VEGFR pathway and include hypertension that requires continuous monitoring and potential treatment as well as monitoring of cardiac function, thromboembolic and hemorrhaging events, diarrhea, decreased appetite, nausea, and dysphonia. *Tivozanib* is mostly metabolized by CYP3A4; it showed an increased AUC with moderately impaired hepatic function and is eliminated with a $t_{1/2}$ of 111 h.

Lenvatinib

Lenvatinib is an orally bioavailable kinase inhibitor of VEGFRs, FGFRs, PDGFR, KIT, and RET and other kinases that are implicated in pathological angiogenesis and cancer progression. *Lenvatinib* is approved for the treatment of patients with recurrent or metastatic differentiated thyroid cancer, in combination with *everolimus* for advanced RCC following one prior antiangiogenic therapy, and for first-line treatment of unresectable hepatocellular carcinoma. In 2021, approval was granted for treatment of patients with endometrial carcinoma that progressed after systemic treatment, in combination with the immune checkpoint inhibitor *pembrolizumab*, even in absence of high levels of microsatellite instability or deficient mismatch repair, which are typical requirements for *pembrolizumab* treatment (see Chapter 72). Adverse effects and warnings are typical for inhibitors of the VEGFR signaling pathway including hypertension, cardiac dysfunction, thromboembolism, and impaired wound healing. Osteonecrosis of the jaw is a rarer adverse effect of treatment. *Lenvatinib* is metabolized by CYP3A and eliminated with a $t_{1/2}$ of 28 h.

Regorafenib

Regorafenib is an orally bioavailable kinase inhibitor of VEGFR1–3, PDGFR, KIT, RET, and RAF1 with IC_{50} values in the low nanomolar range and other kinases including FGFR1/2, TIE2, and ABL. Several of these kinases are implicated in pathological angiogenesis and malignant progression. *Regorafenib* is approved for the treatment of patients with metastatic colorectal cancer after previous chemotherapy, anti-VEGF therapy, or anti-EGFR therapy; for patients with hepatocellular carcinoma who have been previously treated with *sorafenib*; and for patients with locally advanced, unresectable, or metastatic GIST after *imatinib* or *sunitinib*. Major adverse effects include hypertension, thrombotic and hemorrhagic events, hepatotoxicity, GI perforation, QT prolongation, and wound-healing complications. *Regorafenib* is metabolized mostly by CYP3A and eliminated with a $t_{1/2}$ of about 28 h.

Belzutifan

Belzutifan is an orally bioavailable inhibitor of HIF-2α that was approved in 2021 for the treatment of patients with von Hippel-Lindau (VHL) disease and associated RCC, CNS hemangioblastomas, or pancreatic NET. HIF-2α is a transcription factor that plays a role in cellular responses to oxygen levels. Under hypoxic conditions, HIF-2α accumulates in the cytosol and translocates into the nucleus where it heterodimerizes with HIF-1β, resulting in the induction of genes associated with cellular growth and angiogenesis. *Belzutifan* binds to HIF-2α and blocks its heterodimerization with HIF-1β and activation of target genes. Under normoxic conditions, the VHL protein targets HIF-2α for proteasomal degradation, thus preventing its activity. VHL is a subunit of the E3 ubiquitin ligase complex, which mediates the proteasomal degradation of HIF-2α (Figure 71–6). The VHL syndrome is a hereditary condition associated mostly with hemangioblastomas that can arise in multiple organs. Loss of VHL is a crucial oncogenic event in approximately 90% of clear cell RCC, resulting in the accumulation of HIF-2α and induction of cell growth and angiogenesis. HIF-2α also plays a role in the function of immune cells, and combination therapy of *belzutifan* with other cancer therapies is being investigated (Choueiri and Kaelin, 2020) and has shown some promising efficacy in patients with RCC pretreated with immune checkpoint inhibitor and antiangiogenic therapy (Choueiri et al., 2021). Adverse effects of *belzutifan* treatment include anemia due to the reduction of erythropoietin, which occurs within 2 weeks of dosing and can return to baseline values within 3 months of treatment. Monitoring for anemia before initiation and throughout treatment to keep the hemoglobin level at 9 g/dL or greater is required. Blood transfusions may be considered. Other adverse effects include fatigue, headache, and dizziness. *Belzutifan* is primarily metabolized by UGT2B17 and CYP2C19 and to a lesser extent by CYP3A4 and is eliminated with a $t_{1/2}$ of 14 h.

IV. Inhibitors of Poly(ADP-Ribose) Polymerase (PARP)

DNA damage-repair genes are frequently inactivated in human cancer. PARP1 is the product of one such gene. PARP1 is a nuclear protein that transfers ADP-ribose from NAD^+ to target proteins, and this poly(ADP-ribosyl)ation (or PARylation) of nuclear proteins by PARP1 plays a significant role in the DNA damage response (Tallis et al., 2014). Inactive PARP1 associates with chromatin and helps to create a compact chromatin structure in the nucleosome. DNA damage (e.g., strand breaks) mobilizes the enzyme, promoting PARP-mediated PARylation, with the result that chromatin relaxes in the area of damage. The mechanisms of relaxation are the PARylation of PARP1 itself and of histones and other chromatin-associated proteins, resulting in their dissociation from DNA. PARylation produces branching chains up to 200 ADP-ribose units in length. These chains provide docking centers for the localized recruitment of PAR-binding factors, DNA repair enzymes (e.g., DNA polβ, DNA ligase III), and proteins that help to maintain an inactive environment in the open chromatin region while repairs are made to the damaged DNA. The synthesis and degradation of PAR chains *in vivo* are tightly regulated, with the chains having half-lives measured in minutes (Wei and Yu, 2016). Excessive activation of PARP1 leads to depletion of the cellular pool of NAD^+ and to cell death; teleologically, this could be viewed as a means of preserving genomic integrity by removing cells with heavily damaged DNA.

Deficient PARP1 activity leads to defective DNA repair. However, PARP-deficient cells are still able to carry out DNA repair through a different mechanism—homologous recombination. The protein machinery

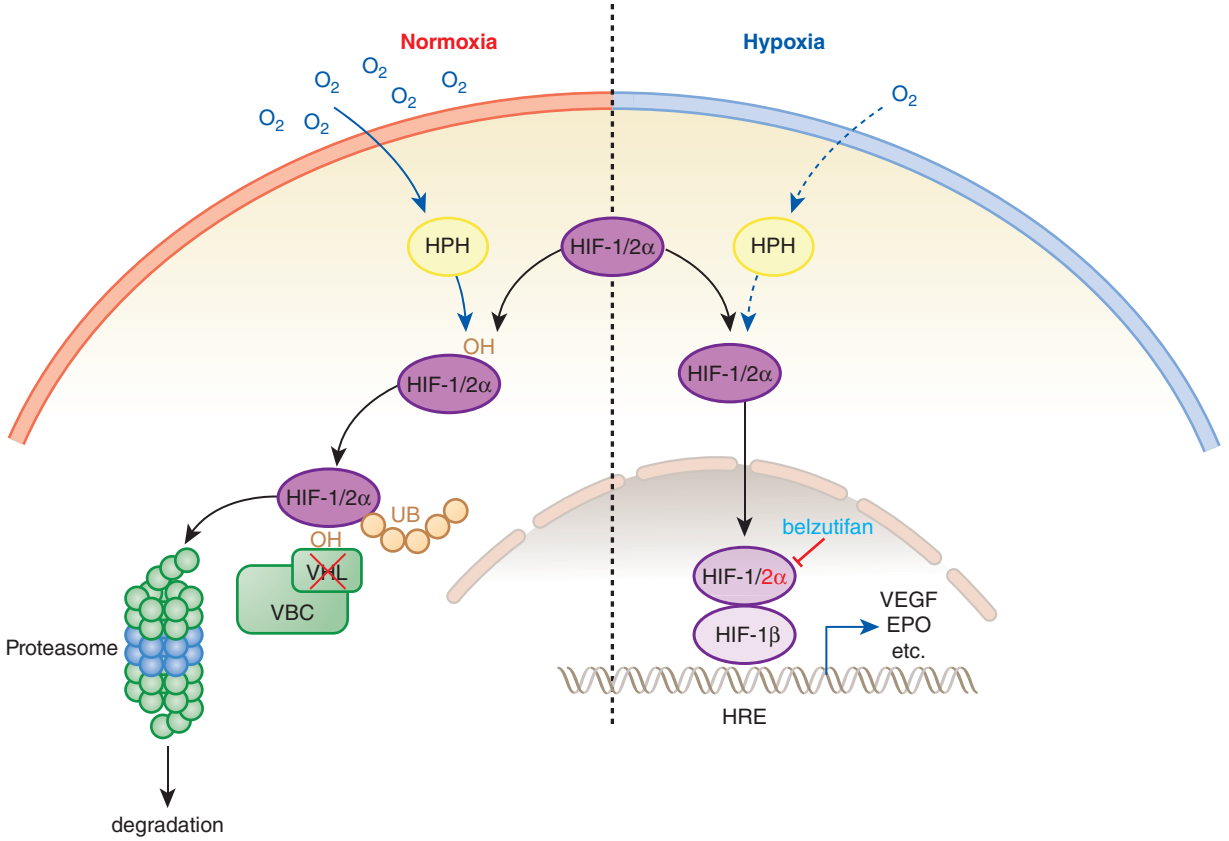


Figure 71–6 *Oxygen-sensing and hypoxia-inducible genes targeted.* Under normoxic conditions, HIF-1α and -2α (HIF-1/2α in short) are hydroxylated (symbolized by the "OH") on proline residues by HIF-prolyl hydroxylases (HPHs) in an oxygen-dependent manner. This allows the VHL protein to recognize and target HIF-1/2α for rapid degradation. VHL is part of an E3 ubiquitin ligase complex of VHL-Elongin B/C-CUL2 (VBC) that transfers ubiquitin (Ub). The ubiquitinated HIF is transported to the proteasome for degradation. In its canonical function depicted here, the VHL protein serves as the substrate recognition component of the VBC protein complex. Under hypoxic conditions, HIF-1/2α are not recognized by VHL and degraded but translocate into the nucleus. In the nucleus HIF-1/2α heterodimerize with HIF-1β (also named aryl hydrocarbon receptor nuclear translocator [ARNT]). The heterodimer binds to hypoxia response element (HRE) DNA sequences and recruits co-activators to drive transcription of hypoxia-inducible genes such as EPO (erythropoietin) and VEGF.

Loss of VHL, symbolized by the red X, is observed in the majority of clear cell RCC and is associated with a pseudohypoxic state that results in the accumulation of HIF-2α and upregulation of HRE target genes that include VEGF. Indeed, inhibition of VEGF and its pathway has shown efficacy and been approved for treatment of clear cell RCC as described in the text. Identification of a binding pocket in HIF-2α led to the development of small molecules that cause conformational changes of HIF-2α and disrupt its heterodimerization with HIF-1β. After successful clinical trials, the HIF-2α inhibitor *belzutifan* was approved in 2021 for the treatment of patients with VHL-deficient cancers (Choueiri et al., 2021).

for homologous recombination repair includes the breast cancer susceptibility genes *BRCA1* and *BRCA2*, suggesting that cells with diminished or absent BRCA1/2 function are unusually susceptible to the inhibition of PARP activity. BRCA1/2 mutations predispose carriers to breast, ovarian, and other cancers that can be resistant to chemotherapy. BRCA-deficient cells are more dependent on PARP1 and base excision repair to maintain genomic integrity. BRCA1/2-deficient cancer cells are less able to carry out repair and ultimately die; inhibition of PARP could accelerate this process (Curtin and Szabo, 2020) and is described in Figure 71–7. PARP inhibitors thus represent a successful example of a molecular medicine application to treatment that takes advantage of DNA repair defects in cancer cells and enhances cancer cell death. Inhibitors of PARP are designed to compete with NAD^+ at its binding site on the PARP enzyme. Catalytic activity is required for repulsion between auto-PARylated PARP1 and DNA, and thus, inhibition of PARP catalytic activity will trap PARP1 and PARP2 on DNA, contributing to the efficacy but also adverse effects and genotoxicity of these inhibitors (Slade, 2020). PARPs catalyze PARylation of a number of proteins that regulate cellular processes beyond DNA damage repair; these extra-chromatin actions may contribute to the clinical efficacy and adverse effects of PARP inhibitors. Different selectivities, subtle differences between the inhibitors, and future drug development in the field were reviewed by Slade (2020). *Olaparib*, *rucaparib*, *niraparib*, and *talazoparib* are currently approved for the treatment of different subsets of patients with ovarian, breast, prostate, and pancreatic cancer, discussed in the below sections.

Olaparib

Olaparib is an orally bioavailable inhibitor of PARP1, -2, and -3 enzymes and was initially approved in 2014 as a monotherapy in patients with BRCA-mutated ovarian cancer who have been treated with three or more prior lines of chemotherapy. The indications have been expanded in 2020 to patients with breast, pancreatic, and prostate cancers with mutant BRCA or DNA repair deficiency. Treatment of these ovarian cancer patients is now approved after first-line platinum-based chemotherapy. Other approvals are in patients with HER2-negative metastatic breast cancer after chemotherapy or after appropriate endocrine therapy for hormone receptor–positive tumors; metastatic pancreatic adenocarcinoma after platinum-based chemotherapy; and metastatic castration-resistant prostate cancer after endocrine therapy with *enzalutamide* or *abiraterone* (see Chapter 73). Adverse effects of treatment include the risk of MDS and AML with mostly fatal outcome in approximately 1.5% of patients after monotherapy. Monitoring of patients for hematological toxicity at baseline and during treatment is required. Pneumonitis with some risk of fatal outcome is another severe risk in 0.8% of patients and requires

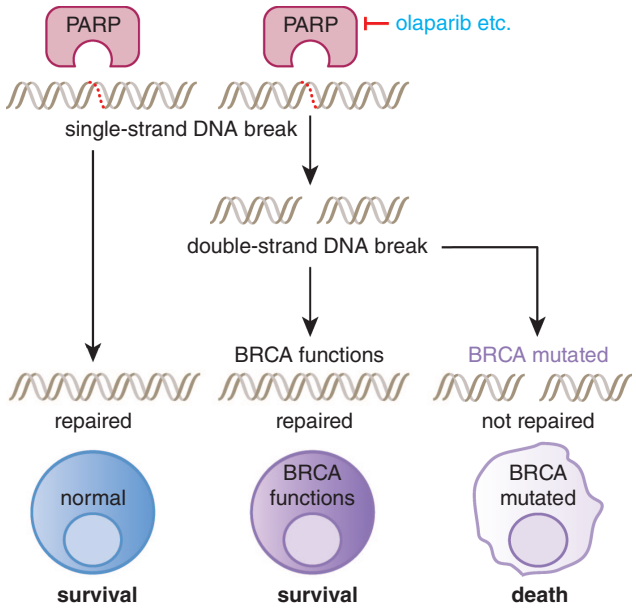


Figure 71–7 *Synthetic lethality from PARP inhibitors in cancer cells that lost BRCA function.* In normal cells, PARP functions in the repair of single-strand breaks in DNA. Without this function of PARP, double-strand DNA breaks accumulate during DNA replication. Cells with functional BRCA can repair these double-strand breaks in DNA through homology-directed repair and thus survive. In contrast, cancer cells with loss of function of BRCA cannot effectively repair double-strand DNA breaks and will become genomically unstable and undergo apoptosis. PARP inhibitors are approved for the treatment of patients with ovarian, breast, prostate, and pancreatic cancer.

monitoring. There is also risk to a fetus, and use of effective contraception is indicated while on treatment. Monitoring for potential venous thromboembolic events including pulmonary embolism is also required. Other adverse effects include nausea and vomiting, loss of appetite, fatigue, muscle and joint pain, low blood cell counts, and anemia. *Olaparib* is rapidly absorbed after oral administration and metabolized primarily by CYP3A4 and eliminated with a $t_{1/2}$ of 14.9 h. Drug exposure is increased or reduced when administered in combination with a CYP3A4 inhibitor or inducer, respectively.

Rucaparib

Rucaparib is an orally bioavailable inhibitor of PARP1–3. *Rucaparib* was approved in 2016 for the treatment of patients with advanced ovarian cancer with BRCA mutations and is used in the maintenance treatment of these patients after a partial response to platinum-based chemotherapy. Other uses include patients with BRCA-mutated, metastatic, castration-resistant prostate cancer after endocrine and taxane-based therapy. Warnings, adverse effects, and precautions overlap with those for *olaparib* (see previous paragraph) and include the risk for potentially fatal MDS and AML. *Rucaparib* is metabolized primarily by CYP2D6 and to a lesser extent by CYP1A2 and CYP3A4 and can impact metabolism of drugs that are substrates of CYP1A2, CYP3A, CYP2C9, or CYP2C19. If coadministered with *warfarin* (a CYP2C9 substrate), increased monitoring of coagulation parameters is required. The mean terminal $t_{1/2}$ of *rucaparib* is 25.9 h.

Niraparib

Niraparib is an orally bioavailable inhibitor of PARP1 and PARP2, was approved in 2017, and is used for the treatment of patients with BRCA-mutated or homologous recombination–deficient, advanced ovarian cancer after first-line platinum-based chemotherapy as maintenance treatment. Warnings and adverse effects include the risk of MDS, AML, and posterior reversible encephalopathy syndrome, which require liscon inu ation. Potential hyp(rte ision an(cardiovascular effe ts due to the inhibition of the dopamine, norepinephrine, and serotonin transporters require monitoring of blood pressure and heart rate and appropriate treatment or dose adjustment. Nausea, thrombocytopenia, anemia, leukopenia, dyspnea, and diarrhea are common adverse events. *Niraparib* can cause fetal harm. *Niraparib* is metabolized and inactivated by carboxylesterases and eliminated with a $t_{1/2}$ of 36 h.

Talazoparib

Talazoparib is an orally bioavailable inhibitor of PARP1 and PARP2 that was approved in 2018 for the treatment of patients with BRCA-mutated, HER2-negative, locally advanced or metastatic breast cancer. *Talazoparib* is the strongest PARP trapper on DNA among the currently approved PARP inhibitors and has the highest occurrence rate of anemia.

Adverse effects are similar to the other PARP inhibitors and include the risk for MDS and AML (0.3% of patients), myelosuppression with anemia, neutropenia, thrombocytopenia, and potential risk to the fetus. *Talazoparib* is mostly excreted in the urine with minimal hepatic metabolism and eliminated with a $t_{1/2}$ of 90 h.

V. Modulators of Protein Degradation

Targeting Protein Degradation

The selective degradation of target proteins is an attractive approach to expand the druggable proteome. Targeted protein degradation uses proteolysis-targeting chimeras (PROTACs), which bind to a protein of interest and link that to an E3 ligase (Dale et al., 2021). An established example of this concept is the antiestrogen *fulvestrant*, which functions as a monovalent degrader of ERα by occupying the ligand-binding pocket of the receptor, exposing hydrophobic residues with its long aliphatic tail and thus tagging ERα for proteolysis (reviewed by Besten and Lipford, 2020). In 2013, the immunomodulatory imide drugs (IMiDs) *thalidomide* and *lenalidomide*, which contain a CO-NH-CO structure as their signature (see figure with structures and highlight), were discovered to exert their therapeutic effects in a similar manner (Bartlett et al., 2004; Jan et al., 2021). They act as a molecular glue that recruits disease-relevant proteins to an E3 ubiquitin ligase, resulting in proteasomal degradation. *Thalidomide* and its analogues bind to cereblon (CRBN), a component of a cullin-RING E3 ligase (CRL) complex, and their binding to CRBN mediates interaction with transcription factors of the Ikaros zinc finger family (IKZF), leading to their ubiquination and subsequent degradation. Figure 71–8 outlines the mechanism of action. Based on the protein targeted by this class of drugs, they may also be referred to as cereblon modulators (Kannt and Đikić, 2021). Proteasomal degradation of these transcription factors kills multiple myeloma cells. *Thalidomide* is associated with significant toxicity and was mostly replaced by *lenalidomide*, a more potent and less toxic IMiD used in the treatment of multiple myeloma (MM). *Pomalidomide*, a more potent, third-generation IMiD, shows efficacy even in patients with *lenalidomide*-refractory cancers. Also, in del(5q) MDS, *lenalidomide* induces the degradation of CK1α, which preferentially affects del(5q) cells because they express this gene at haploinsufficient levels.

Thalidomide and Lenalidomide

Thalidomide and *lenalidomide* have a most unusual history and a multiplicity of biological and immunological effects (see Chapters 39, 65, and 75). *Thalidomide* originally was used for the treatment of pregnancy-associated morning sickness but was withdrawn from the market due to teratogenicity and dysmelia (stunted limb growth). It reentered clinical practice for treatment of erythema nodosum leprosum (see Chapter 60). Further research revealed its antiangiogenic and immunomodulatory effects. At least four distinct mechanisms have been proposed to explain its antitumor activity, which are summarized in Figure 71–9 and enumerated in its legend.

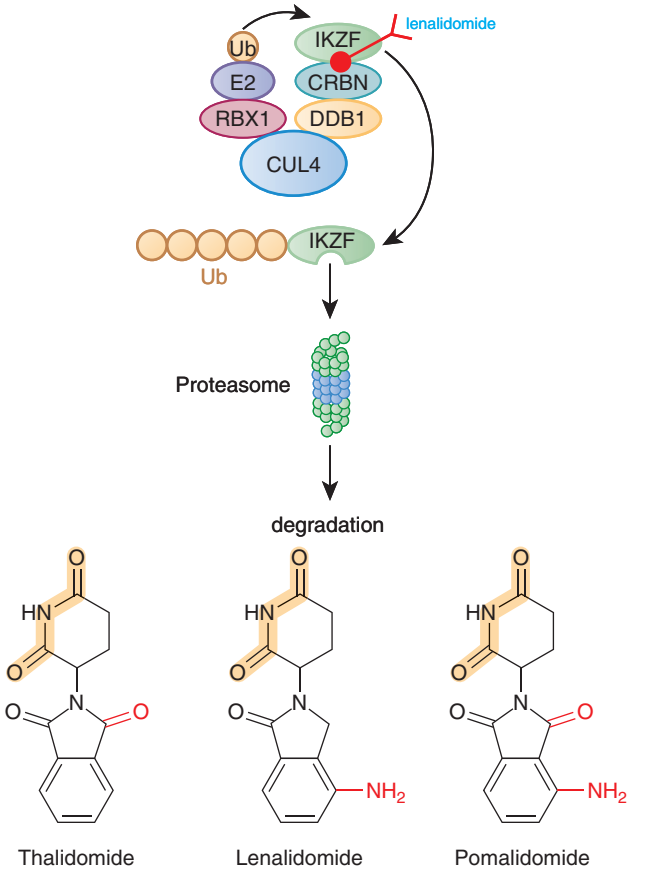


Figure 71–8 *Recruitment of target substrates by lenalidomide.* *Lenalidomide* functions as a molecular "glue" for the recruitment of proteins to the cereblon (CRBN) receptor component of the cullin–RING E3 ubiquitin ligase complex 4 (CRL4) comprising cullin 4 (CUL4), DNA damage-binding protein 1 (DDB1), and E3 ubiquitin–protein ligase RBX1. This recruitment results in protein ubiquitylation and degradation. Due to the structural differences, thalidomide, and its congeners, interact with and lead to the depletion of overlapping but distinct sets of proteins that include IKZF1/3 (Ikaros family zinc finger protein) and CK1α (casein kinase I isoform-α) (Jan et al., 2021).

Both *thalidomide* and *lenalidomide* possess potent activity in patients with newly diagnosed and heavily pretreated relapsed/refractory MM. *Lenalidomide* is also approved for its activity in the 5q– subset [or del(5q) subset] of MDS. A specific gene array profile identifies patients with MDS who lack the 5q– abnormality but respond to *lenalidomide*. A more recently added derivative of *thalidomide* is *pomalidomide*, approved for treatment of patients with MM resistant to *lenalidomide*.

Thalidomide

ADME

Thalidomide exists at physiological pH as a racemic mixture of cell-permeable and rapidly interconverting nonpolar S(–) and R(+) isomers. Of these mirror image molecules, the *S*-enantiomer is associated with the teratogenic, phocomelia-inducing, and growth control activities, while the *R*-enantiomer accounts for the sedative properties of *thalidomide*. It is noteworthy that the isomers interconvert under biological conditions. When treating MM, doses usually are escalated by 200 mg/day every 2 weeks until dose-limiting side effects (sedation, fatigue, constipation, or a sensory neuropathy) appear. With extended treatment, the neuropathy may necessitate dose reduction or discontinuation of treatment for a period of time. *Thalidomide* absorption from the GI tract is slow and highly variable. It distributes throughout most tissues and organs, without significant binding to plasma proteins. The enantiomers are eliminated with a $t_{1/2}$ of about 6 h, mainly due to spontaneous hydrolysis in all body fluids; the *S*-enantiomer is cleared more rapidly than the *R*-enantiomer. *Thalidomide* and its metabolites are excreted in the urine, while the non-absorbed portion of the drug is excreted unchanged in feces. The inactive hydrolysis products undergo CYP-mediated metabolism. Longer plasma $t_{1/2}$ is reported at the highest doses (1.2 g daily). No dose adjustment is necessary in the presence of renal failure.

Lenalidomide

Lenalidomide constitutes the lead compound of immunomodulatory *thalidomide* derivatives. *Lenalidomide* induces the ubiquitination and degradation of target proteins by the E3 ubiquitin ligase CRL4-CRBN (Fink and Ebert, 2015). Target proteins in MM cells are IKZF1/3, which are crucial for cell survival (Lu et al., 2014) (Figure 71–9). In MDS, casein kinase 1A1 is the target protein (Krönke et al., 2015). Cellular effects of *lenalidomide* include direct suppression of tumor cell growth in culture, T-cell and natural killer (NK)-cell activation, suppression of tumor necrosis factor-α and other cytokines, antiangiogenesis, and promotion of HSC differentiation.

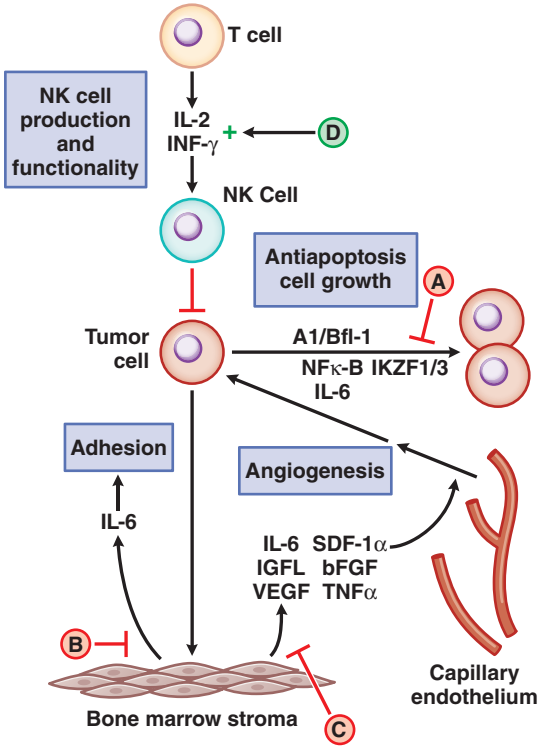


Figure 71–9 *Overview of proposed mechanisms of antimyeloma activity of thalidomide and its derivatives.* Some biological hallmarks of the malignant phenotype are indicated in blue *boxes*. The proposed sites of action for *thalidomide* (letters inside red and green circles) are hypothesized to also be operative for *thalidomide* derivatives. **A.** Direct anti-MM effect on tumor cells, including G_1 growth arrest or apoptosis, even against MM cells resistant to conventional therapy. This is due to the disruption of the antiapoptotic effect of BCL2 family members, blocking NF-κB signaling, and inhibition of the production of IL-6. Mechanistic studies show that *lenalidomide* can kill MM cells by inducing E3 ubiquitin ligase CRL4-CRBN–mediated ubiquitination and degradation of the crucial IKZF1 and -3 (zinc finger transcription factors). **B.** Inhibition of MM cell adhesion to bone marrow stromal cells partially due to the reduction of IL-6 release. **C.** Decreased angiogenesis due to the inhibition of cytokine and growth factor production and release. **D.** Enhanced T-cell production of cytokines, such as IL-2 and IFN-γ, that increase the number and cytotoxic functionality of NK cells.

ADME

The drug is rapidly absorbed following oral administration and eliminated with a $t_{1/2}$ of 9 h. Approximately 70% of the orally administered dose of *lenalidomide* is excreted intact by the kidney. Dose adjustments for reduced creatinine clearance are recommended.

Therapeutic Use

Lenalidomide exhibits antitumor activity in MM, MDS, and CLL; it causes fewer adverse effects and lacks the teratogenicity of *thalidomide*.

Adverse Effects of Thalidomide and Lenalidomide

Thalidomide is well tolerated at doses less than 200 mg daily. Common adverse effects are sedation and constipation. The most serious adverse effect is peripheral sensory neuropathy, which occurs in 10% to 30% of patients with MM or other malignancies in a dose- and time-dependent manner. *Thalidomide*-related neuropathy is an asymmetrical, painful, peripheral paresthesia with sensory loss, commonly presenting with numbness of toes and feet, muscle cramps, weakness, signs of pyramidal tract involvement, and carpal tunnel syndrome. Although symptoms improve upon drug discontinuation, long-standing sensory loss may not reverse. Particular caution should be applied in patients with preexisting neuropathy (e.g., related to diabetes) or prior exposure to drugs that can cause peripheral neuropathy (e.g., vinca alkaloids, *bortezomib*). *Thalidomide* enhances the sedative effects of barbiturates and alcohol and the catatonic effects of *chlorpromazine*. Conversely, CNS stimulants (such as *methamphetamine* and *methylphenidate*) counteract the depressant effects of *thalidomide*.

Adverse effects of *lenalidomide* are less severe; it causes little sedation, constipation, or neuropathy. *Lenalidomide* depresses bone marrow function and is associated with significant leukopenia (20% of patients). Hepatotoxicity and renal dysfunction are rare. In some patients with CLL, *lenalidomide* causes dramatic lymph node swelling and tumor lysis (tumor flare reaction). Patients with renal dysfunction are prone to this reaction; thus, patients with CLL should be started at lower doses of 10 mg/day, with escalation as tolerated. Patients with CLL should receive pretreatment hydration and *allopurinol* to avoid the consequences of tumor swelling and tumor lysis syndrome. A negative interaction with *rituximab*, an anti-CD20 antibody, may result from *lenalidomide*'s downregulation of CD20, an interaction that has clinical implications for their combined use in lymphoid malignancies.

Thalidomide and *lenalidomide* increase the risk for thromboembolic events, including deep vein thrombosis, stroke, and myocardial infarction, which occur with increased frequency in combination with glucocorticoids and with anthracyclines. Anticoagulation and antiplatelet medication reduce this risk and are indicated in patients with risk factors for clotting.

Pomalidomide

Pomalidomide, a third-generation *thalidomide* congener, was first approved in 2013 and is indicated in combination with *dexamethasone* for the treatment of patients with MM who have received at least two prior therapies including *lenalidomide*. It was also approved in 2020 for the treatment of AIDS-related Kaposi sarcoma. Adverse effects are similar to those of *thalidomide*. *Pomalidomide* is contraindicated during pregnancy and in females who may become pregnant. As a consequence, this drug is available only under a pregnancy control system to ensure that recipients are not pregnant and are using effective birth control.

Proteasome Inhibitors

First Generation

Bortezomib

Bortezomib is a first-generation inhibitor of proteasome-mediated protein degradation that has a central role in the treatment of MM.

Mechanism of Action

Bortezomib binds to the β5 subunit of the 20S core of the 26S proteasome and reversibly inhibits its chymotrypsin-like activity. This event disrupts multiple intracellular signaling cascades, leading to apoptosis. An important consequence of proteasome inhibition is its effect on NF-κB, a transcription factor that promotes cell damage response and cell survival. Most cellular NF-κB is cytosolic and bound to IκB (inhibitor of NF-κB); in this form, NF-κB is restricted to the cytosol and cannot enter the nucleus to regulate transcription. In response to stress signals resulting from hypoxia, chemotherapy, and DNA damage, IκB becomes ubiquitinated and then degraded via the proteasome. Its degradation releases NF-κB, which enters the nucleus and transcriptionally activates a host of genes involved in cell survival (e.g., cell adhesion proteins) and proliferation (e.g., cyclin D1) or antiapoptosis (e.g., cIAPs, BCL-2). NF-κB is highly expressed in many human tumors, including MM, and may be a key factor in tumor cell survival in a hypoxic environment and during chemotherapy. *Bortezomib* blocks proteasomal degradation of IκB, thereby preventing the transcriptional activity of NF-κB and downregulating survival responses.

Bortezomib also disrupts the ubiquitin-proteasomal degradation of p21, p27, p53, and other key regulators of the cell cycle and initiators of apoptosis. *Bortezomib* activates the cell's stereotypical "unfolded protein response," in which abnormal protein conformation activates adaptive signaling pathways in the cell. The composite effect leads to irreversible commitment of MM cells to apoptosis.

ADME

The recommended starting dose of *bortezomib* is 1.3 mg/m^2 given as an intravenous bolus on days 1, 4, 8, and 11 of every 21-day cycle (with a 10-day rest period per cycle). At least 72 h should elapse between doses. Drug administration should be withheld until resolution of any grade 3 nonhematological toxicity or grade 4 hematological toxicity, and subsequent doses should be reduced by 25%. The drug exhibits a terminal $t_{1/2}$ in plasma of 5.5 h. Peak proteasome inhibition reaches 60% within 1 h and declines thereafter, with a $t_{1/2}$ of about 24 h. *Bortezomib* clearance results from the deboronation of the parent compound (90%), followed by hydroxylation of the boron-free product by CYPs 3A4 and 2D6; administration of this drug with potent inducers or inhibitors/substrates of CYP3A4 requires caution. No dose adjustment is required for patients with renal dysfunction.

Therapeutic Uses

Bortezomib is used as initial therapy for patients with MM and as therapy for patients with MM after relapse from other drugs. It is also approved for treatment of patients with relapsed or refractory MCL. The drug is active in myeloma, including the induction of complete responses in up to 30% of patients when used in combination with other drugs (i.e., *thalidomide*, *lenalidomide*, liposomal *doxorubicin*, or *dexamethasone*).

Adverse Effects

Bortezomib toxicities include thrombocytopenia (28%), fatigue (12%), peripheral neuropathy (12%), neutropenia, anemia, vomiting, diarrhea, limb pain, dehydration, nausea, and weakness. Peripheral neuropathy, the most chronic of the toxicities, develops most frequently in patients with a prior history of neuropathy secondary to prior drug treatment (e.g., *thalidomide*) or diabetes or with prolonged use. Dose reductions or discontinuation of *bortezomib* ameliorates the neuropathic symptoms. Injection of *bortezomib* may precipitate hypotension, especially in volume-depleted patients, in those who have a history of syncope, or in patients taking antihypertensive medications. Cardiac toxicity is rare, but congestive heart failure and prolonged QT interval have been reported.

Second Generation

Carfilzomib is a second-generation selective proteasome inhibitor based on a tetrapeptide epoxyketone. It is FDA-approved as a single agent or in combination with *dexamethasone* or *lenalidomide* plus *dexamethasone*

for the treatment of patients with relapsed or refractory MM who have received at least one prior therapy. Frequent adverse effects (>20% of patients) include anemia, thrombocytopenia, diarrhea, dyspnea, and peripheral edema. There is a risk for cardiac, renal, pulmonary, and hepatic toxicity as well as hypertension.

Ixazomib is an orally bioavailable second-generation peptide analogue proteasome inhibitor that interacts with subunit beta type 5 (PSMB5) of the 20S proteasome complex. *Ixazomib* has an elimination $t_{1/2}$ of 9.5 days and is approved for use in combination with *lenalidomide* and *dexamethasone* for the treatment of patients with MM after at least one prior therapy (Moreau et al., 2016). *Ixazomib* can cause diarrhea, peripheral neuropathy, and hepatotoxicity.

VI. Epigenetic Modulators: Inhibitors of HDAC, HMT, and IDH1/2

Inhibitors of Histone Deacetylase

Histone deacetylases (HDACs) are a class of enzymes that catalyze the removal of acetyl groups from acetylated lysine amino acids in histones, thereby altering the transcriptional activation of cellular genes. Thus, HDAC inhibitors produce a broad spectrum of epigenetic effects (Bates, 2020). HDACs can also deacetylate other proteins, including transcription factors. Overexpression of HDACs found in some cancers or aberrant recruitment of HDACs to oncogenic transcription factors can cause hypoacetylation of core nucleosomal histones. This hypoacetylation results in a condensed chromatin structure and repression of gene transcription. In contrast, inhibition of HDAC activity leads to the accumulation of acetyl groups on the histone lysine residues, an open chromatin structure, and activation of target genes that are selectively repressed in tumors. The result of HDAC inhibition can be differentiation with the emergence of a more normal cellular phenotype or cell cycle arrest with expression of endogenous regulators of cell cycle progression. Due to the diversity of the HDAC family of enzymes and their tissue distribution, HDAC inhibitors with different selectivity can induce a wide variety of cellular effects. Beyond the cellular effects, the impact of HDAC inhibitors on the crosstalk of cancer cells and the immune system is currently evaluated in trials using combination treatments with immune checkpoint inhibitors (Hogg et al., 2020). Four HDAC inhibitors, *panobinostat*, *romidepsin*, *vorinostat*, and *belinostat*, have gained FDA approval and are discussed below. An updated review of other epigenetic modulators is provided by Bates (2020).

Panobinostat

Panobinostat is an orally bioavailable, nonselective pan-HDAC inhibitor. The inhibitory activity leads to apoptosis of malignant cells via multiple pathways. *Panobinostat* is approved for the treatment of patients with MM who have received at least two previous treatments, including *bortezomib* and an immunomodulatory agent.

ADME

The oral bioavailability is about 21%. *Panobinostat* is metabolized through CYP3A. It is recommended to avoid concomitant use of strong CYP3A4 inducers or inhibitors; otherwise, dose adjustments will be necessary. The elimination $t_{1/2}$ is approximately 37 h.

Adverse Effects

The most common adverse effects are diarrhea (severe in 25% of patients), fatigue, nausea, peripheral edema, decreased appetite, pyrexia, and vomiting. The most common nonhematologic laboratory abnormalities (incidence ≥40%) are hypophosphatemia, hypokalemia, hyponatremia, and increased creatinine. Common hematological laboratory abnormalities (incidence ≥60%) are thrombocytopenia, lymphopenia, leukopenia, neutropenia, and anemia. Fatal cardiac ischemic events, severe arrhythmias, and ECG changes have occurred in patients receiving *panobinostat*. Arrhythmias may be exacerbated by electrolyte abnormalities. Avoid concomitant use of antiarrhythmic or QT-prolonging drugs. Close cardiac monitoring during treatment is recommended.

Romidepsin

Romidepsin is an HDAC inhibitor used in the treatment of cutaneous and peripheral T-cell lymphoma. Recently, its indication in peripheral T-cell lymphoma has been withdrawn. *Romidepsin* is a natural product obtained from the bacteria *Chromobacterium violaceum* and is also referred to as *depsipeptide*. *Romidepsin* functions as a prodrug. Within cells, a zinc-binding thiol of the drug is reduced and interacts with a zinc atom in the binding pocket of HDAC to block its activity.

ADME

After intravenous administration, the major metabolism is through CYP3A4. Thus, it is recommended to monitor for toxicities related to increased *romidepsin* exposure when coadministering strong CYP3A4 inhibitors. *Rifampin* and other strong CYP3A4 inducers should be avoided. The terminal $t_{1/2}$ is about 3 h.

Adverse Effects

The most commonly observed adverse effects are nausea and vomiting, anemia, thrombocytopenia, and leukopenia, as well as abnormal electrolyte levels and ECG changes.

Vorinostat

Vorinostat is a small-molecule, orally bioavailable HDAC inhibitor; it is also known as SAHA based on its chemical name ***s***uber***a***nilo***h***ydroxamic ***a***cid.

Mechanism of Action

Vorinostat binds to the active site of HDACs and chelates zinc ions in the active site. The resulting inhibition of HDACs causes the accumulation of acetylated histones and other acetylated proteins, among which are transcription factors crucial for cell differentiation. *Vorinostat* inhibits the enzymatic activities of HDAC1, -2, and -3 (class I) and HDAC6 (class II) at nanomolar concentrations (IC_{50} <100 nM). *In vitro*, *vorinostat* induces cell cycle arrest or apoptosis of some cancer cells.

ADME

The absorption of *vorinostat* is slightly improved when taken with a meal. Metabolism is mostly through glucuronidation and hydrolysis. The elimination $t_{1/2}$ is about 2 h.

Therapeutic Use; Adverse Effects

Vorinostat is approved for the treatment of patients with cutaneous T-cell lymphoma with persistent or recurrent disease after two systemic therapies. The most common adverse reactions are diarrhea, fatigue, nausea, thrombocytopenia, anorexia, and dysgeusia. Patients with severe hepatic impairment should be excluded from treatment. Women should be apprised of the potential harm to the fetus.

Belinostat

Belinostat is an HDAC inhibitor developed for the treatment of hematological malignancies and solid tumors. It is approved to treat peripheral T-cell lymphoma and is administered by intravenous infusion. Warnings and adverse effects include the risk of hematological toxicity, infections, and hepatotoxicity. Treatment can also cause fetal harm.

Inhibitors of Histone Methyltransferase

Tazemetostat

Tazemetostat is an orally available, S-adenosyl methionine (SAM) competitive inhibitor of histone methyltransferase (HMT) EZH2 (wild-type and mutant) that was approved in 2020 for the treatment of patients with

metastatic or locally advanced epithelioid sarcoma not eligible for complete resection. Inhibition of EZH2 prevents the methylation of histone H3 lysine 27 (H3K27), alters gene expression patterns associated with cancer pathways, and inhibits growth of EZH2-mutated cancer cells. EZH2 is overexpressed or mutated in a variety of cancer cells and plays a key role in tumor cell proliferation. The most common adverse effects include fatigue, nausea, and decreased appetite. There is an increased risk of developing secondary malignancies and of fetal harm. *Tazemetostat* is metabolized by CYP3A, and coadministration of inhibitors can increase the AUC. The elimination $t_{1/2}$ is 3.1 h.

IDH1/2 Inhibitors

IDH1 and IDH2 mutations result in preferential production of 2-hydroxyglutarate (2-HG) over alpha-ketoglutarate, mimicking the loss-of-function of the ten-eleven translocation (TET) gene that controls DNA demethylation and resulting in DNA hypermethylation; simultaneously, inhibition of lysine demethylases by 2-HG leads to increased histone lysine methylation (Yang et al., 2012). Mutations in DNMT3A, TET2, IDH1, and IDH2 are found in 28%, 14%, 9%, and 10% of AMLs, respectively, with a similar mutation pattern in peripheral T-cell lymphoma. DNMT3A and TET2 mutations have been detected only at lower frequency in solid tumors, whereas IDH1 mutations were detected at higher frequency, including in 78% of low-grade gliomas, 14% of cholangiocarcinomas, and 4% of cutaneous melanomas. Two drugs targeting mutants IDH1 and IDH2 are currently approved and have shown some efficacy in the treatment of AML.

Enasidenib is an orally bioavailable IDH2 inhibitor that was approved in 2017 for the treatment of patients with relapsed or refractory AML with an IDH2 mutation. *Ivosidenib* is an orally bioavailable IDH1 inhibitor approved in 2018 for the treatment of AML with a susceptible IDH1 mutation (DiNardo et al., 2018; Issa and DiNardo, 2021) and recently for cholangiocarcinoma with IDH1 mutations. For both drugs, there is a risk of differentiation syndrome, which can be fatal and requires corticosteroid therapy and hemodynamic monitoring if suspected. For *ivosidenib*, risks include QTc prolongation and Guillain-Barré syndrome.

VII. Other Inhibitors (BCL2, Nuclear Export, Translation, CXCR4)

BCL2 Inhibitors

The B-cell lymphoma (BCL) 2 protein family comprises more than 20 proteins that govern mitochondrial outer membrane permeabilization and control programmed cell death (apoptosis). Proteins in this family can be either *pro-* or *anti*apoptotic depending on their content of BCL2 homology domains (BH1–4). *Proapoptotic* proteins contain a BH3 domain that is necessary for dimerization with other proteins of the BCL2 family. *Antiapoptotic* proteins contain BH1 and BH2 domains. The balance of these interacting proteins controls mitochondrial outer membrane permeabilization, cytochrome c release, and the activation of caspases that leads to apoptosis (Figure 71–10). BCL2 promotes cellular survival by inhibiting proapoptotic proteins like BIM (BCL2-like protein 11), BAX, and BAK and is overexpressed in CLL and some other tumors where it can support tumor cell survival and resistance to cancer treatments.

Venetoclax

Mechanism of Action

Venetoclax is a first-in-class, orally bioavailable, small-molecule inhibitor of BCL2. It was designed as a BH3 mimetic that inhibits BCL2 interaction with the proapoptotic BH3-only family members, such as BIM, BID, and BAD. *In vitro*, *venetoclax* inhibits BCL2 action with an IC_{50} of less than 0.01 nM and with a selectivity of two to three orders of magnitude over other BCL2 family members. In the presence of *venetoclax*, BH3-only proteins can translocate to mitochondria and initiate BAX/BAK-dependent apoptosis. Overexpression of BCL2 occurs in CLL cells, where it supports tumor cell survival and resistance to chemotherapeutic agents. Thus, *venetoclax* helps restore apoptosis in these cells.

Therapeutic Use; Adverse Effects; ADME

Venetoclax is approved for the treatment of patients with CLL or SLL and in combination with *azacitidine*, *decitabine*, or low-dose

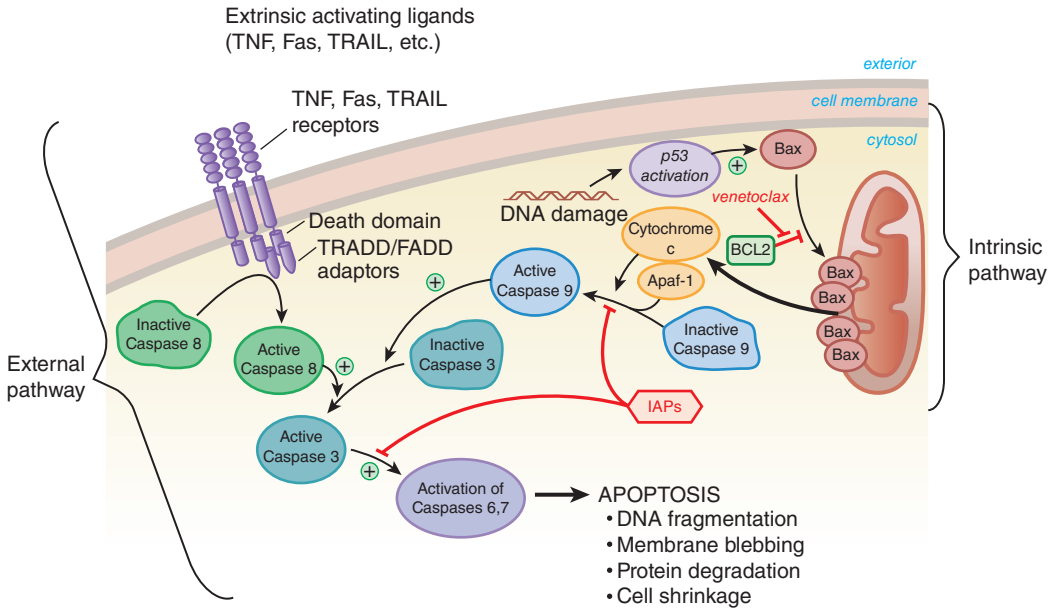


Figure 71–10 *BH3 mimetics enhance apoptosis.* Permeability of the outer mitochondrial membrane to cytochrome c is regulated by proteins required for forming pores, including proapoptotic BAX and BAK. BAK and BAX are rendered inactive by binding to the BH3 domains of antiapoptotic proteins such as BCL2. BH3 mimetic drugs (*venetoclax*) target BCL2 and its interaction with BAX/BAK, reducing the inhibitory effect of BCL2 on apoptosis progression, enhancing cytochrome c release, and sensitizing the cell to apoptosis. Cancer cells may be resistant to apoptosis by overexpression of BCL2; BH3 mimetics such as *venetoclax* elease he esultant blockade an pr omote ap ptosis.

cytarabine for the treatment of newly diagnosed AML in patients 75 years or older or with comorbidities that preclude the use of chemotherapy. The most common adverse effects are neutropenia, diarrhea, nausea, anemia, upper respiratory tract infection, thrombocytopenia, and fatigue. *Venetoclax* may cause fetal harm. Oral absorption is significantly improved 3- to 5-fold by administration with a meal. *Venetoclax* is metabolized by CYP3A4/5 and has an elimination $t_{1/2}$ of 18 to 26 h.

Nuclear Export Inhibitor

Selinexor

Selinexor is an orally bioavailable reversible inhibitor of exportin 1 (XPO1). It was approved in 2019 in combination with *bortezomib* and *dexamethasone* for the treatment of patients with MM who have received at least one prior therapy, and for the treatment of patients with relapsed or refractory diffuse large B-cell lymphoma after at least two lines of systemic therapy. Warnings and risks of treatment that require monitoring and management include thrombocytopenia, neutropenia, hyponatremia, and infection. Also, there is a risk of fetal harm. The most common adverse effects are fatigue, nausea, vomiting, decreased appetite, diarrhea, peripheral neuropathy, upper respiratory tract infection, decreased weight, and cataract. *Selinexor* is metabolized by CYP3A4 and eliminated with a $t_{1/2}$ of 6 to 8 h.

Protein Translation Inhibitor

Omacetaxine

Omacetaxine is administered by SC injection and is approved for the treatment of patients with CML with resistance or intolerance to two or more kinase inhibitors. The mechanisms of action include inhibition of protein translation by preventing the initial elongation step of protein synthesis, thereby depleting the cell of short-lived proteins. Warnings include the risk of myelosuppression, bleeding, and fetal toxicity. Adverse effects also include nausea, fatigue, and infections. *Omacetaxine* is hydrolyzed by plasma esterases and eliminated with a $t_{1/2}$ of 14.6 h.

CXCR4 Inhibitor

Plerixafor

Plerixafor is a small-molecule inhibitor of the CXCR4 chemokine receptor and is used in combination with granulocyte colony-stimulating factor to mobilize hematopoietic stem cells (HSC) to the peripheral blood for collection and subsequent autologous transplantation in patients with non-Hodgkin's lymphoma and MM. *Plerixafor* blocks binding of stromal cell-derived factor (SDF)-1α to CXCR4. CXCR4 activity supports homing of HSCs to the bone marrow, and *plerixafor* mobilizes the release and entry of HSCs into the circulation. The combination of granulocyte colony-stimulating factor with *plerixafor* increases the efficacy of HSC mobilization.

Drug Facts for Your Personal Formulary: *Pathway-Targeted Therapies*

Drug	Therapeutic Use	Clinical Pharmacology and Tips
Section I: Inhibitors of Growth Factors and Receptors		
Epidermal Growth Factor Receptor Inhibitors		
Small-Molecule EGFR Kinase Inhibitors: Oral Administration		
Erlotinib	• Advanced NSCLC with mutant EGFR (del exon 19; L858R) • Advanced pancreatic cancer in combination with gemcitabine	• Skin rash, stomatitis, diarrhea, interstitial lung disease • CYP3A4 substrate • Anticoagulant effect of warfarin enhanced • Concurrent use of PPI ↓ bioavailability
Gefitinib	• Advanced NSCLC with mutant EGFR (del exon 19; L858R)	• Side effects similar to erlotinib, but bioavailability not affected by PPI
Afatinib	• Irreversible inhibitor EGFR > HER2 • Advanced NSCLC with mutant EGFR (del exon 19; L858R) • Squamous cell carcinoma of the lung progressing after systemic therapy	• Side effects stronger than gefitinib; can cause hepatotoxicity • Not affected by CYP3A4 modulation • ↓ Dose with renal impairment or Pgp inhibitors
Osimertinib	• Advanced NSCLC with EGFR mutations • Advanced NSCLC that is resistant to other EGFR kinase inhibitors and positive for exon 20 T790M-mutant EGFR	• Similar to gefitinib but less intense side effects; can ↑ QTc interval
Human Epidermal Growth Factor Receptor 2 Inhibitors		
Small-Molecule HER2 Kinase Inhibitors: Oral Administration, Also Inhibit EGFR		
Lapatinib	• HER2-positive breast cancer in combination with capecitabine • HER2-positive, hormone receptor–positive breast cancer in combination with letrozole (aromatase inhibitor)	• Skin rash, diarrhea • Cardiotoxicity (less than trastuzumab), QT interval prolongation • CYP3A4 substrate
Neratinib	• Irreversible inhibitor of EGFR and HER2 • HER2-positive breast cancer in addition to chemotherapy	• Diarrhea is major adverse effect, with one-third of patients with severe grade 3–4 diarrhea

Drug Facts for Your Personal Formulary: *Pathway-Targeted Therapies (continued)*

Drug	Therapeutic Use	Clinical Pharmacology and Tips
Section II: Inhibitors of Intracellular Protein Kinases		
Mutant B-RAF Kinase Inhibitors		
Vemurafenib	• BRAF V600E/K mutant melanoma	• Cutaneous adverse effects up to 60%, with squamous cell carcinoma in >20% • Arthralgia, fatigue, nausea less frequently • Can cause QT prolongation • CYP3A4 substrate; CYP1A2 inhibitor
Dabrafenib	• BRAF V600E/K mutant melanoma also in combination with the MEK inhibitor trametinib • BRAF V600E mutant NSCLC with trametinib	• Cutaneous adverse effects include hyperkeratosis and papilloma • cuSCC in ~10% of patients • Combination with trametinib ↓ incidence of cuSCC to 3% and delays onset • CYP2C8 and CYP3A substrate
Mitogen-Activated Protein Kinase Kinase Inhibitors		
Cobimetinib	• BRAF V600E/K mutant melanoma	• Diarrhea, photosensitivity, nausea common • Risk of hemorrhage, cardiomyopathy • CYP3A4 substrate
Trametinib	• BRAF V600E/K mutant melanoma and BRAF V600E mutant NSCLC: with dabrafenib • Ineffective in patients who developed resistance to BRAF inhibitor treatment	• Cutaneous rash, acneiform dermatitis, diarrhea most frequent; serious skin toxicity in 6% • Risk of cardiomyopathy, hypertension, hemorrhage, interstitial lung disease • Reduced absorption after high-calorie meal
Janus-Associated Protein Kinase Inhibitors		
Ruxolitinib	• Polycythemia vera • Myelofibrosis	• Thrombocytopenia, anemia most frequent • Rare basal cell carcinoma or squamous cell carcinoma • ↓ Dose in renal or hepatic impairment • CYP3A4 substrate
Cyclin-Dependent Kinase 4/6 Inhibitors		
Palbociclib	• Advanced ER-positive, HER2-negative breast cancer • Combination with aromatase inhibitor or antiestrogen (fulvestrant)	• Common: neutropenia, leukopenia, infections, stomatitis, anemia, thrombocytopenia, nausea, diarrhea • Substrate and inhibitor of CYP3A4
Abemaciclib and ribociclib	• Same as for palbociclib	• Side effects similar to palbociclib
Bruton Tyrosine Kinase Inhibitors		
Ibrutinib	• Mantle cell lymphoma, CLL, SLL, Waldenström's macroglobulinemia	• Neutropenia, thrombocytopenia, diarrhea, anemia, musculoskeletal pain • Slow onset of hypertension possible: monitor blood pressure • Atrial fibrillation: monitor and treat • Secondary malignancies mostly skin, nonmelanoma • CYP3A4 substrate
BCR-ABL, PDGFR, KIT Kinase Inhibitors		
Imatinib	• Chronic-phase CML; mucosal and acral lentiginous melanoma (KIT mutation positive), GIST (KIT mutation positive), dermatofibrosarcoma protuberans, chronic myelomonocytic leukemia	• GI tract adverse effects: diarrhea, nausea, vomiting • Fluid retention, edema • Rare myelosuppression, hepatotoxicity • CYP3A4 substrate
Dasatinib	• CML resistant to imatinib after prior therapy	• Adverse effects: diarrhea, nausea, vomiting • Fluid retention, edema, pleural effusions • Rare myelosuppression, hepatotoxicity • Bioavailability ↓ after antacids or H_2 blockers • CYP3A4 substrate
Nilotinib	• CML resistant to imatinib after prior therapy	• Adverse effects: diarrhea, nausea, vomiting, fluid retention, edema • May ↑ QT interval; beware vascular events, including ischemia • Rare myelosuppression, hepatotoxicity • Bioavailability ↑ in the presence of food • CYP3A4 and Pgp substrate • Inhibitor of the Pgp
Bosutinib	• CML resistant to prior therapy	• Diarrhea, nausea, thrombocytopenia, vomiting, rash
Ponatinib	• Resistant CML and Ph^+ ALL	• Major adverse effects: thrombosis, hepatotoxicity, pancreatitis • Absorption ↓ by elevated gastric pH (H_2 antagonists, antacids, PPIs) • CYP3A4 substrate

Drug Facts for Your Personal Formulary: *Pathway-Targeted Therapies (continued)*

Drug	Therapeutic Use	Clinical Pharmacology and Tips
Anaplastic Lymphoma Kinase Inhibitors		
Alectinib	• Advanced NSCLC with ALK kinase fusion gene • Advanced NSCLC with ALK kinase fusion gene and disease progression on crizotinib	• Fatigue, constipation, edema, and myalgia are common adverse effects • Pneumonitis, GI and hepatic toxicity; bradycardia and prolonged QT intervals observed
Ceritinib, crizotinib	• As above	• As above
PI3K (Phosphatidylinositol-4,5-Bisphosphate 3-Kinase) Inhibitors		
Idelalisib	• Relapsed or refractory B-cell malignancies: CLL (with rituximab), FL, SLL • Not indicated for first-line treatment	• Serious and sometimes fatal adverse effects: hepatotoxicity, colitis, pneumonitis, intestinal perforations, skin toxicity • CYP3A substrate
mTOR (Mechanistic or Mammalian Target of Rapamycin) Inhibitors		
Temsirolimus	• Advanced renal cell carcinoma	• Adverse effects: frequent rash, mucositis, anemia, fatigue (30%–50%); rare leukopenia, thrombocytopenia, interstitial lung disease • Metabolized to a longer-lived active metabolite (sirolimus) by CYP3A4: avoid CYP3A4 inhibitors, including grapefruit juice
Everolimus	• Renal cell carcinoma • Breast cancer: advanced ER positive, HER2 negative in combination with aromatase inhibitor exemestane after failure of other aromatase inhibitors (letrozole, anastrozole) • PNETs • Progressive, well-differentiated, nonfunctional neuroendocrine GIST	• Adverse effects overlap with temsirolimus (see above) • CYP3A4 substrate
Multikinase Inhibitors		
Cabozantinib	• Advanced renal cell carcinoma	• Diarrhea, fatigue, nausea, abdominal pain • Hypertension: monitor blood pressure • Not indicated in patients with recent bleeding history or thromboembolic event • Discontinue in patients with GI perforation, fistulas • CYP3A4 substrate; ↓ dose with hepatic impairment
Vandetanib	• Progressing, locally advanced medullary thyroid cancer	• Diarrhea, colitis, rash • QT interval prolongation, torsades des pointes, sudden death • Do not use in patients with hepatic impairment, long QT syndrome • CYP3A4 substrate
Section III: Inhibitors of Tumor Angiogenesis		
Inhibitor of Hypoxia-Inducible Factor (HIF) Activity		
Belzutifan	• Inhibition of heterodimerization of HIF-2α with HIF-1β transcription factors • Clear cell RCC	Adverse effects: • Anemia due to reduction of erythropoietin • Hypertension
Inhibitors of Vascular Endothelial Growth Factor (VEGFR) and Intracellular Kinases Participating in Angiogenesis		
Sunitinib	• VEGFR2 and multiple other kinases inhibited • Metastatic RCC • GIST after imatinib resistance • Pancreatic neuroendocrine tumors	• Adverse effects shared with anti-VEGF: bleeding, hypertension, proteinuria (frequent); thromboembolism, GI perforations (rare) • Adverse events distinct from anti-VEGF: • Fatigue (50%–70%), hypothyroidism (40%–60%) • Common: bone marrow suppression and diarrhea • Less common: hepatotoxicity, congestive heart failure • Check blood pressure, blood counts, and thyroid functions at regular intervals • Elimination $t_{1/2}$ ~4 days: regimen in some cancers is 4 weeks on, 2 weeks off vs. continuous daily administration for other cancers
Sorafenib	• Hepatocellular carcinoma • Metastatic RCC (however, sunitinib is the first choice)	• Adverse vascular effects match with those of sunitinib • More common: fatigue, diarrhea, anorexia, rash • Less common: bone marrow suppression, GI perforation, cardiomyopathy
Axitinib	• Inhibition of VEGFR1–3 • Advanced renal cell cancer after failure of prior systemic therapy	• Adverse effects overlap with those of anti-VEGF: hypertension, thrombotic and hemorrhagic events, GI perforation • CYP3A4/5 substrate

Drug Facts for Your Personal Formulary: *Pathway-Targeted Therapies (continued)*

Drug	Therapeutic Use	Clinical Pharmacology and Tips
Inhibitors of Vascular Endothelial Growth Factor (VEGFR) and Intracellular Kinases Participating in Angiogenesis (cont.)		
Lenvatinib	• Inhibition of VEGFR1–3, FGFRs, PDGFR • Recurrent or metastatic differentiated thyroid cancer • RCC in combination with everolimus • Endometrial cancer with pembrolizumab	• Adverse effects overlap with those of anti-VEGF: hypertension, thrombotic and hemorrhagic events, GI perforation • In addition: hepatotoxicity, QT prolongation • CYP3A substrate
Pazopanib	• Inhibition of VEGFR1–3, FGFRs, KIT, PDGFR • Advanced RCC and advanced soft-tissue sarcoma after prior chemotherapy	• Adverse effects overlap with those of anti-VEGF: hypertension, thrombotic and hemorrhagic events, GI perforation • In addition: hepatotoxicity, QT prolongation • CYP3A substrate
Regorafenib	• Inhibition of RET, VEGFR1, PDGFR, FGFR, TIE2, RAF1, BRAF, ABL • Metastatic colorectal cancer after previous chemotherapy and anti-VEGF or anti-EGFR • Advanced GIST after imatinib or sunitinib	• Major adverse effects: hepatotoxicity, hypertension, thrombotic and hemorrhagic events, GI perforation, and wound-healing complications • CYP3A substrate
Section IV: PARP Inhibitors		
Poly(ADP-Ribose) Polymerase Inhibitor		
Olaparib	• Ovarian cancer after three or more prior lines of treatment in patients with germline mutant BRCA	• Adverse effects: nausea, vomiting, loss of appetite, muscle and joint pain, anemia; leukemia (rare), potentially fatal MDS (rare), pneumonitis • CYP3A4 substrate
Section V: Modulators of Protein Degradation		
Thalidomide and Congeners		
Thalidomide	• Newly diagnosed multiple myeloma • Relapsed or refractory, pretreated multiple myeloma	• Most serious adverse effect: sensory neuropathy in 10%–30% of patients; may not be reversible after discontinuation of treatment; patients with preexisting neuropathy at higher risk • Teratogenic; do not use in pregnancy • Causes sedation (enhanced by CNS depressants), fatigue, constipation
Lenalidomide	• Multiple myeloma • MDS (5q– MDS) • Chronic lymphocytic leukemia (CLL)	• Marrow suppression and leukopenia (20% of patients), rare hepatic or renal toxicity • Tumor lysis in some patients with CLL → lymph node swelling and tumor flare: start at lower dose in patients with CLL • Downregulates CD20, a target for monoclonal antibody therapy • In contrast to thalidomide: little neuropathy, sedation, or constipation; lack of teratogenicity • Reduce dose in patients with reduced renal function
Pomalidomide	• AIDS-related Kaposi sarcoma	• Teratogenic; available only under a pregnancy control system
Proteasome Inhibitors		
Bortezomib	• Multiple myeloma: initial therapy and after relapse • Mantle cell lymphoma: relapsed or refractory	• Thrombocytopenia (28%), fatigue (12%), peripheral neuropathy (12%) • Neutropenia, anemia, vomiting, diarrhea, limb pain, weakness • Rare: congestive heart failure and prolonged QT intervals • Metabolized by CYP3A4; beware of drug interactions
Section VI: Epigenetic Modulators		
Panobinostat	• Patients with MM who have received at least two previous treatments, including bortezomib and an immunomodulatory agent	• Fatal cardiac ischemic events, severe arrhythmias, close cardiac monitoring during treatment • CYP3A substrate
Section VII: Other Inhibitors		
BCL2 (Antiapoptotic Protein): Orally Available Inhibitor		
Venetoclax	• CLL with 17p deletion (poor prognosis)	• Neutropenia, thrombocytopenia, diarrhea, nausea • Absorption ↑ 3- to 5-fold with a meal • CYP3A substrate

Note: CYP3A4 substrate: For drugs that are subject to hepatic metabolism by CYP enzymes, drug exposure of a patient can be affected by coadministration of inhibitors or inducers of CYP3A4 and can then reduce efficacy or increase side effects.
Embryo-fetal toxicity: Consider that all of these drugs can cause fetal harm. Advise women of the potential risk to a fetus and to avoid pregnancy while taking the drug and for 1 month after cessation of therapy. Advise men to avoid fathering a child during the same time period. Avoid lactation during therapies.

Bibliography

Amakye D, et al. Unraveling the therapeutic potential of the Hedgehog pathway in cancer. *Nat Med*, **2013**, *19*:1410–1422.

André F, et al. Alpelisib for PIK3CA-mutated, hormone receptor–positive advanced breast cancer. *N Engl J Med*, **2019**, *380*:1929–1940.

Awad MM, et al. Acquired resistance to KRASG12C inhibition in cancer. *N Engl J Med*, **2021**, *384*:2382–2393.

Balogh E, Nass SJ, eds. *Advancing Progress in the Development of Combination Cancer Therapies with Immune Checkpoint Inhibitors. Proceedings of a Workshop of the National Academies of Sciences Engineering and Medicine*. The National Academies Press, Washington, DC, **2019**, 1–89. Available at: http://nap.edu/25405.

Bartlett JB, et al. The evolution of thalidomide and its IMiD derivatives as anticancer agents. *Nat Rev Cancer*, **2004**, *4*:314–322.

Bates SE. Epigenetic therapies for cancer. *N Engl J Med*, **2020**, *383*:650–663.

Besten W den, Lipford JR. Prospecting for molecular glues. *Nat Chem Biol*, **2020**, *16*:1157–1158.

Braun TP, et al. Response and resistance to BCR-ABL1-targeted therapies. *Cancer Cell*, **2020**, *37*:530–542.

Burger JA, O'Brien S. Evolution of CLL treatment—from chemoimmunotherapy to targeted and individualized therapy. *Nat Rev Clin Oncol*, **2018**, *15*:1–18.

Camidge DR, et al. Brigatinib versus crizotinib in ALK-positive non–small-cell lung cancer. *N Engl J Med*, **2018**, *379*:2027–2039.

Castel P, et al. The present and future of PI3K inhibitors for cancer therapy. *Nat Cancer*, **2021**, *2*:587–597.

Chapman PB, et al. Improved survival with vemurafenib in melanoma with BRAF V600E mutation. *N Engl J Med*, **2011**, *364*:2507–2516.

Choueiri TK, Kaelin WG. Targeting the HIF2–VEGF axis in renal cell carcinoma. *Nat Med*, **2020**, *26*:1519–1530.

Choueiri TK, et al. Inhibition of hypoxia-inducible factor-2α in renal cell carcinoma with belzutifan: a phase 1 trial and biomarker analysis. *Nat Med*, **2021**, *27*:802–805.

Cohen P, et al. Kinase drug discovery 20 years after imatinib: progress and future directions. *Nat Rev Drug Discov*, **2021**, *20*:551–569.

Cristofanilli M, et al. Fulvestrant plus palbociclib versus fulvestrant plus placebo for treatment of hormone-receptor-positive, HER2-negative metastatic breast cancer that progressed on previous endocrine therapy (PALOMA-3): final analysis of the multicentre, double-blind, phase 3 randomised controlled trial. *Lancet Oncol*, **2016**, *17*:425–439.

Curti BD, Faries MB. Recent advances in the treatment of melanoma. *N Engl J Med*, **2021**, *384*:2229–2240.

Curtin NJ, Szabo C. Poly(ADP-ribose) polymerase inhibition: past, present and future. *Nat Rev Drug Discov*, **2020**, *19*:711–736.

Dale B, et al. Advancing targeted protein degradation for cancer therapy. *Nat Rev Cancer*, **2021**, *21*:638–654.

Dibble CC, Cantley LC. Regulation of mTORC1 by PI3K signaling. *Trends Cell Biol*, **2015**, *25*:545–555.

DiNardo CD, et al. Durable remissions with ivosidenib in IDH1-mutated relapsed or refractory AML. *N Engl J Med*, **2018**, *378*:2386–2398.

Downward J, et al. Close similarity of epidermal growth factor receptor and v-erb-B oncogene protein sequences. *Nature*, **1984**, *307*:521–527.

Drilon A, et al. Efficacy of larotrectinib in TRK fusion–positive cancers in adults and children. *N Engl J Med*, **2018**, *378*:731–739.

Drilon A, et al. ROS1-dependent cancers—biology, diagnostics and therapeutics. *Nat Rev Clin Oncol*, **2021**, *18*:35–55.

Dyson NJ. RB1: a prototype tumor suppressor and an enigma. *Genes Dev*, **2016**, *30*:1492–1502.

Elkins JM, et al. Comprehensive characterization of the published kinase inhibitor set. *Nat Biotechnol*, **2015**, *34*:95–103.

Facchinetti F, et al. Facts and new hopes on selective FGFR inhibitors in solid tumors. *Clin Cancer Res*, **2020**, *26*:764–774.

Fink EC, Ebert BL. The novel mechanism of lenalidomide activity. *Blood*, **2015**, *126*:2366–2369.

Finn RS, et al. Palbociclib and letrozole in advanced breast cancer. *N Engl J Med*, **2016**, *375*:1925–1936.

Flaherty KT, et al. Combined BRAF and MEK inhibition in melanoma with BRAF V600 mutation. *N Engl J Med*, **2012**, *367*:1694–1703.

Folkman J. Tumor angiogenesis: therapeutic implications. *N Engl J Med*, **1971**, *285*:1182–1186.

Fruman DA, Rommel C. PI3K and cancer: lessons, challenges and opportunities. *Nat Rev Drug Discov*, **2014**, *13*:140–156.

Goel S, Tolaney SM. CDK4/6 inhibitors in breast cancer: a role in triple-negative disease? *Lancet Oncol*, **2019**, *20*:1479–1481.

Guo R, et al. MET-dependent solid tumours—molecular diagnosis and targeted therapy. *Nat Rev Clin Oncol*, **2020**, *17*:569–587.

Guri Y, Hall MN. mTOR signaling confers resistance to targeted cancer drugs. *Trends Cancer*, **2016**, *2*:688–697.

Hall MN. TOR and paradigm change: cell growth is controlled. *Mol Biol Cell*, **2016**, *27*:2804–2806.

Hammarlund EU, et al. Oxygen-sensing mechanisms across eukaryotic kingdoms and their roles in complex multicellularity. *Science*, **2020**, *370*:eaba3512.

Hanahan D, Weinberg RA. Hallmarks of cancer: the next generation. *Cell*, **2011**, *144*:646–674.

Hogg SJ, et al. Targeting the epigenetic regulation of antitumour immunity. *Nat Rev Drug Discov*, **2020**, *19*:776–800.

Howlader N, et al. The effect of advances in lung-cancer treatment on population mortality. *N Engl J Med*, **2020**, *383*:640–649.

Hughes PE, et al. Targeted therapy and checkpoint immunotherapy combinations for the treatment of cancer. *Trends Immunol*, **2016**, *37*:462–476.

Huinen ZR, et al. Anti-angiogenic agents—overcoming tumour endothelial cell anergy and improving immunotherapy outcomes. *Nat Rev Clin Oncol*, **2021**, *18*:527–540.

Issa GC, DiNardo CD. Acute myeloid leukemia with *IDH1* and *IDH2* mutations: 2021 treatment algorithm. *Blood Cancer J*, **2021**, *11*, 107.

Jamieson C, et al. Hedgehog pathway inhibitors: a new therapeutic class for the treatment of acute myeloid leukemia. *Blood Cancer Discov*, **2020**, *1*:134–145.

Jan M, et al. Cancer therapies based on targeted protein degradation—lessons learned with lenalidomide. *Nat Rev Clin Oncol*, **2021**, *18*:401–417.

Jia Y, et al. Overcoming EGFR(T790M) and EGFR(C797S) resistance with mutant-selective allosteric inhibitors. *Nature*, **2016**, *534*:129–132.

Joshi SK, et al. The AML microenvironment catalyzes a stepwise evolution to gilteritinib resistance. *Cancer Cell*, **2021**, *39*:999–1014.

Kannt A, Đikić I. Expanding the arsenal of E3 ubiquitin ligases for proximity-induced protein degradation. *Cell Chem Biol*, **2021**, *28*:1014–1031.

Katayama R, et al. Therapeutic targeting of anaplastic lymphoma kinase in lung cancer: a paradigm for precision cancer medicine. *Clin Cancer Res*, **2015**, *21*:2227–2235.

Kim KB, et al. Phase II study of the MEK1/MEK2 inhibitor trametinib in patients with metastatic BRAF-mutant cutaneous melanoma previously treated with or without a BRAF inhibitor. *J Clin Oncol*, **2013**, *31*:482–489.

King CR, et al. Amplification of a novel v-erbB-related gene in a human mammary carcinoma. *Science (New York, NY)*, **1985**, *229*:974–976.

Krönke J, et al. Lenalidomide induces ubiquitination and degradation of CK1α in del(5q) MDS. *Nature*, **2015**, *523*:183–188.

Lu G, et al. The myeloma drug lenalidomide promotes the cereblon-dependent destruction of Ikaros proteins. *Science (New York, NY)*, **2014**, *343*:305–309.

Martelli MP, et al. EML4-ALK rearrangement in non-small cell lung cancer and non-tumor lung tissues. *Am J Pathol*, **2010**, *174*:661–670.

Moore AR, et al. RAS-targeted therapies: is the undruggable drugged? *Nat Rev Drug Discov*, **2020**, *19*:533–552.

Moreau P, et al. Oral ixazomib, lenalidomide, and dexamethasone for multiple myeloma. *N Engl J Med*, **2016**, *374*:1621–1634.

Murthy RK, et al. Tucatinib, trastuzumab, and capecitabine for HER2-positive metastatic breast cancer. *N Engl J Med*, **2020**, *382*:597–609.

Nagy JA, et al. VEGF-A and the induction of pathological angiogenesis. *Ann Rev Pathol*, **2007**, *2*:251–275.

Nokin M-J, et al. Targeting infrequent driver alterations in non-small cell lung cancer. *Trends Cancer*, **2020**, *7*:410–429.

O'Leary B, et al. Treating cancer with selective CDK4/6 inhibitors. *Nat Rev Clin Oncol*, **2016**, *13*:470–430.
Ostrem JM, et al. K-Ras(G12C) inhibitors allosterically control GTP affinity and effector interactions. *Nature*, **2013**, *503*:548–551.
Oxnard GR, et al. Association between plasma genotyping and outcomes of treatment with osimertinib (AZD9291) in advanced non-small-cell lung cancer. *J Clin Oncol*, **2016**, *34*:3375–3382.
Paik PK, et al. Response to MET inhibitors in patients with stage IV lung adenocarcinomas harboring MET mutations causing exon 14 skipping. *Cancer Discov*, **2015**, *5*:842–849.
Park JW, et al. Adaptive randomization of neratinib in early breast cancer. *N Engl J Med*, **2016**, *375*:11–22.
Petroni G, et al. Immunomodulation by targeted anticancer agents. *Cancer Cell*, **2021**, *39*:310–345.
Robert C, et al. Improved overall survival in melanoma with combined dabrafenib and trametinib. *N Engl J Med*, **2015**, *372*:30–39.
Rosen N. Finally, effective inhibitors of mutant KRAS. *N Engl J Med*, **2021**, *384*:2447–2449.
Roskoski R Jr. Anaplastic lymphoma kinase (ALK) inhibitors in the treatment of ALK-driven lung cancers. *Pharmacol Res*, **2017**, *117*: 343–356.
Schechter A, et al. The neu gene: an erbB-homologous gene distinct from and unlinked to the gene encoding the EGF receptor. *Science*, **1985**, *229*:976–978.
Skoulidis F, et al. Sotorasib for lung cancers with KRAS p.G12C mutation. *N Engl J Med*, **2021**, *384*:2371–2381.
Slade D. PARP and PARG inhibitors in cancer treatment. *Genes Dev*, **2020**, *34*:360–394.
Slamon DJ, et al. Human breast cancer: correlation of relapse and survival with amplification of the HER-2/neu oncogene. *Science (New York, NY)*, **1987**, *235*:177–182.
Smit EF, Baas P. Lung cancer in 2015: bypassing checkpoints, overcoming resistance, and honing in on new targets. *Nat Rev Clin Oncol*, **2015**, *13*:75–76.
Tallis M, et al. Poly(ADP-ribosyl)ation in regulation of chromatin structure and the DNA damage response. *Chromosoma*, **2014**, *123*:79–90.
Tan AR, et al. Trilaciclib plus chemotherapy versus chemotherapy alone in patients with metastatic triple-negative breast cancer: a multicentre, randomised, open-label, phase 2 trial. *Lancet Oncol*, **2019**, *20*:1587–1601.
Thein KZ, et al. Precision therapy for RET-altered cancers with RET inhibitors. *Trends Cancer*, **2021**, *7*:1074–1088.
Thomas A, et al. Refining the treatment of NSCLC according to histological and molecular subtypes. *Nat Rev Clin Oncol*, **2015**, *12*:511–526.
Thress KS, et al. Acquired EGFR C797S mutation mediates resistance to AZD9291 in non-small cell lung cancer harboring EGFR T790M. *Nat Med*, **2015**, *21*:560–562.
Ullrich A, et al. Human epidermal growth factor receptor cDNA sequence and aberrant expression of the amplified gene in A431 epidermoid carcinoma cells. *Nature*, **1984**, *309*:418–425.
Vultur A, Herlyn M. SnapShot: melanoma. *Cancer Cell*, **2013**, *23*:706.e1.
Wei H, Yu X. Functions of PARylation in DNA damage repair pathways. *Genom Proteom Bioinform*, **2016**, *14*:131–139.
Yaeger R, Corcoran RB. Targeting alterations in the RAF–MEK pathway. *Cancer Discov*, **2019**, *9*:329–341.
Yang H, et al. IDH1 and IDH2 mutations in tumorigenesis: mechanistic insights and clinical perspectives. *Clin Cancer Res*, **2012**, *18*:5562–5571.

This page intentionally left blank

Chapter 72

Antibodies, CAR T Cells, and Proteins to Treat Cancer

Anton Wellstein and Michael B. Atkins

I. INHIBITING GROWTH FACTOR RECEPTORS IN CANCER CELLS

INHIBITORS OF EPIDERMAL GROWTH FACTOR RECEPTOR

HER2/NEU INHIBITORS

INHIBITORS OF PLATELET-DERIVED GROWTH FACTOR RECEPTOR

II. INHIBITING TUMOR ANGIOGENESIS

INHIBITION OF VEGF AND THE VEGF RECEPTOR PATHWAY

III. ACTIVATING IMMUNE CELLS

IMMUNE CHECKPOINT INHIBITORS

INHIBITORS OF CTLA-4

INHIBITORS OF PD-1

ANTAGONISTS OF PD-L1

COMBINATION OF ANTI–PD-1 AND ANTI–CTLA-4

IMMUNE CHECKPOINT TARGETING BEYOND CTLA-4 AND PD-1: ANTI-LAG3

IMPACT OF THE MICROBIOME ON IMMUNE CHECKPOINT INHIBITOR THERAPY

CYTOKINES TO STIMULATE IMMUNE RESPONSES

IV. TARGETING CANCER CELL SURFACE MOLECULES TO ENGAGE IMMUNE CELLS

CD19, CD20, CD52, CD38, CCR4, GD2, AND SLAMF7

BISPECIFIC ANTIBODIES: CD19 AND CD3; EGFR AND MET

V. CYTOTOXIN CONJUGATES WITH ANTIBODIES OR CYTOKINES

ANTIBODY-DRUG CONJUGATES (ADCS)

CYTOKINE-CYTOTOXIN CONJUGATES

VI. CELLULAR THERAPY AND VACCINES

CAR T CELLS

CANCER VACCINES

VII. OTHER PROTEINS

COLONY-STIMULATING FACTORS

ASPARAGINASE

A Note on Treatment Regimens

Cancer treatment regimens change to reflect continuous advances in basic and clinical science: new drugs, both small molecules and biologicals; improved methods of targeting and timing of drug delivery; agents with altered pharmacokinetic properties and selectivities; the use of rational multidrug combinations; and greater knowledge of the basic cell biology of tumorigenesis, metastasis, and immune function, among other advances. As a consequence, this chapter presents only a few detailed treatment regimens; rather, we refer the reader to the web-based resources of the U.S. FDA (drugs@fda) and the NCCN (National Comprehensive Cancer Network). Table 71–1 provides two examples of therapeutic regimens that illustrate the complexity of current cancer drug therapy.

Development of targeted therapies for treating patients with cancer relies on the continuing process of discovery of molecular changes that drive malignant progression of human cancers. A growing number of drugs are being developed to block oncogenic pathways that lead to dysregulated cancer cell growth and survival. Targeted therapy may be combined with classical cytotoxic cancer drugs described in Chapter 70 for improved efficacy. Growth factor receptors and downstream signaling molecules are among the most actively explored targets for cancer drug discovery. The drivers of cancer growth are oncogenic pathways in malignant cells themselves (e.g., mutant or overexpressed receptors), the reaction of the tumor microenvironment (e.g., angiogenesis), and the escape of malignant cells from the host's immune surveillance (Hanahan and Weinberg, 2011). Chapter 71 describes small-molecule drugs used in targeting pathways altered in cancer. The discussion of drugs in this chapter focuses on large molecules and cell-based therapeutics of cancer and is organized into sections:

I. **Inhibiting growth factors receptors in cancer cells**
II. **Inhibiting tumor angiogenesis**
III. **Activating immune cells**
IV. **Targeting cancer cell surface molecules to engage immune cells**
V. **Cytotoxin conjugates with antibodies or cytokines**
VI. **Cellular therapy and vaccines**
VII. **Other proteins**

The proteins discussed here target oncogenic pathways that contribute to cancer growth and metastatic spread as well as immune evasion and for the most part consist of *monoclonal antibodies* or their derivatives that

Abbreviations

ADC: antibody-drug conjugate
ADCC: antibody-dependent cellular cytotoxicity
ADCP: antibody-dependent cellular phagocytosis
ALK: anaplastic lymphoma kinase
ALL: acute lymphoblastic leukemia
AML: acute myelocytic leukemia
APC: antigen-presenting cell
BCMA: B-cell maturation antigen, tumor necrosis factor receptor superfamily member 17
BiTE: bispecific T-cell–engaging antibody
CAR T: chimeric antigen receptor T cells
CCR: CC motif chemokine receptor
CDC: complement-dependent cytotoxicity
CDR: complementarity-determining regions
CLL: chronic lymphocytic leukemia
CTLA-4: cytotoxic T lymphocyte–associated protein 4
EGF(R): (human) epidermal growth factor (receptor) = HER1
FOLFIRI: folinic acid (leucovorin), 5-fluorouracil, irinotecan
FOLFOX: folinic acid (leucovorin), 5-fluorouracil, oxaliplatin
5FU: 5-fluorouracil
G(M)-CSF: granulocyte(-macrophage) colony-stimulating factor
HER1 or 2: *h*uman *E*GF*R* 1 or 2
HNSCC: head and neck squamous cell carcinoma
IFN: interferon
IL(-2R): interleukin (2 receptor)
JC: John Cunningham
KDR: kinase insert domain receptor = VEGFR2
KIT: feline sarcoma virus oncogene homolog
LAG3: lymphocyte activation gene 3
LVEF: left ventricular ejection fraction
mCRC: metastatic colorectal cancer
MET: mesenchymal-epithelial transition factor (= HGFR)
MHC: major histocompatibility complex protein
MM: multiple myeloma
MMAE/F: monomethyl auristatin E/F
MSI-H: microsatellite instability high
NHL: non-Hodgkin lymphoma
NK: natural killer
NSCLC: non-small cell lung cancer
PAP: prostate acidic phosphatase
PD-1: programmed cell death protein 1
PDGF(R): platelet-derived growth factor (receptor)
PD-L1: programmed cell death ligand 1
RCC: renal cell carcinoma
TCR: T-cell receptor
VEGF(R): vascular endothelial growth factor (receptor)

recognize cell surface proteins on cancer cells or on host cells or antigens shed from cancer cells (see box Antibodies as Cancer Drugs).

Over 40 different antibody-based therapeutics are currently FDA-approved for cancer treatment, many added in the past 5 years (see Figure 1–7 and Table 72–1). Notably, small-molecule inhibitors used to treat cancer impact the efficacy of antibody therapy and are used in treatment combinations that may target the same pathway or parallel and escape pathways. Indeed, triggering cell death in tumors by chemotherapy, pathway-targeted inhibitors, or radiotherapy can promote cytotoxic T-cell infiltration. The efficacy of such combinations is discussed below. Ongoing combination trials have been recently reviewed (Balogh and Nass, 2019; Petroni et al., 2021).

Antibodies as Cancer Drugs

The general concept of harnessing the immune system to treat cancer originated with William B. Coley, a professor of clinical surgery at Cornell University. He suggested that natural immunity becomes lowered during malignancy and thus thought to raise the immune reaction by injecting bacterial toxins into sarcoma (Coley, 1910) with some successes, although also with serious adverse events. The foundation for contemporary antibody-based drugs was laid by Köhler and Milstein who, in 1975, generated the first hybridoma cell lines capable of producing monoclonal antibodies. Georges Köhler, César Milstein, and Niels Jerne shared the 1984 Nobel Prize in Physiology/Medicine "for theories concerning the specificity in development and control of the immune system and the discovery of the principle for production of monoclonal antibodies."

A monoclonal antibody is derived from a single B cell, recognizes a specific antigen, and can mediate cancer cell eradication by different mechanisms: (1) blocking ligands or function of cell surface receptors, (2) recruiting immune cells and complement to an antigen-antibody complex formed, (3) modulating immune cell function, or (4) carrying toxins or radionuclides as cytotoxic payloads to the cells of interest. A monoclonal antibody is generally specific for a single antigen, such as an epitope in a growth factor receptor; has a long plasma $t_{1/2}$ of days to weeks; and requires only intermittent parenteral administration. Immunoglobulin G1 (IgG1), the most frequently approved antibody isotype used in cancer treatment, can activate the complement pathway and interacts with Fcγ receptors on immune cells more potently than IgG2 or IgG4 (see Figure 72–1). Protein or carbohydrate engineering of the Fc region can alter the effector functions and mitigate complement-dependent cytotoxicity (CDC), antibody-dependent cellular cytotoxicity (ADCC), and antibody-dependent cellular phagocytosis (ADCP) (Goydel and Rader, 2021). Further background on monoclonal antibodies is provided in Chapters 38 and 39.

The naming of monoclonal antibodies (Table 72–1): *M*onoclonal *anti*body names end with the syllable "mab" preceded by different morphemes that indicate the characteristics of a given antibody, mostly based on its molecular design: "xi"—fusion of human constant (Fc) and mouse variable (Fv) antibody domains to generate chimeric proteins; "zu"—insertion of antigen-binding regions (complementarity-determining regions [CDRs]) into a human IgG protein scaffold, known as humanization; and "u"—amino acid sequence fully matching the human IgG and generated in transgenic mice that carry human Ig gene loci or *in vitro* (e.g., by phage display of human antibody libraries and their affinity for the target, improved by randomization of amino acid sequences in the CDRs) (see Figure 72–1).

I. Inhibiting Growth Factor Receptors in Cancer Cells

Inhibitors of Epidermal Growth Factor Receptor

Background

The epidermal growth factor receptor (EGFR) belongs to the ErbB family of transmembrane receptor tyrosine kinases also known as ErbB1 or HER1 (*h*uman *EGFR 1*). The EGFR is expressed in epithelial tissues, including the skin, and essential for the growth and differentiation of epithelial cells. Ligand binding to the extracellular domain of EGFR family members causes receptor dimerization and stimulates the protein

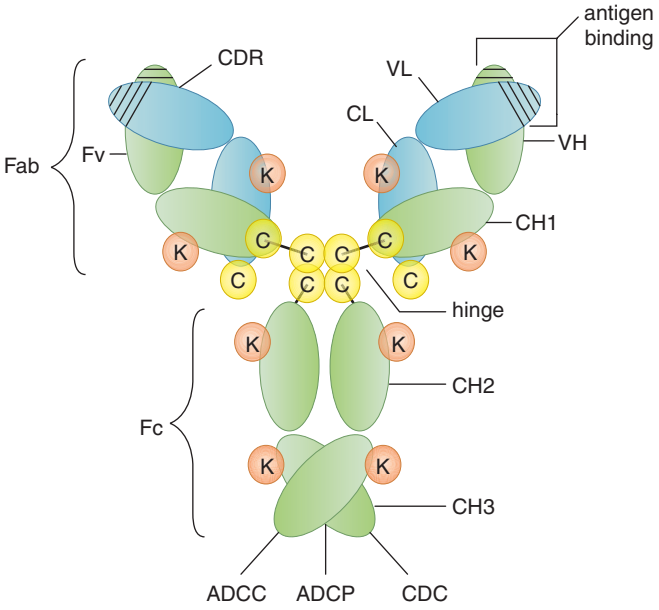


Figure 72–1 *IgG1 elements relevant for their use in cancer treatment.* Constant and variable heavy and light chains (CH, CL, VH, VL) are shown. The Fab domain contains the three complementarity-determining regions (CDR) in each of the variable chains that constitute the antigen binding site (paratope). The Fc domain modulates the effector functions, i.e., antibody-dependent cellular cytotoxicity (ADCC), complement-dependent cytotoxicity (CDC), and antibody-dependent cellular phagocytosis (ADCP). Six disulfide bridges containing cysteine residues (C, yellow) connect the chains. The molecular mass of an IgG1 is approximately 150 kDa. Cytotoxic payloads can be linked to an IgG through the cysteine residues or through a multitude of lysine (K) residues in the different domains.

tyrosine kinase activity of the intracellular domain, resulting in autophosphorylation of several Tyr residues located in the C-terminal tail of the receptor monomers. These phosphotyrosines provide interaction sites for a variety of adapter proteins, resulting in stimulation of signaling pathways, including MAPK and PI3K/Akt pathways (Figure 72–2).

In epithelial cancers, overexpression of the EGFR is a common finding. Monoclonal antibodies (*cetuximab*, *panitumumab*, *necitumumab*) recognize the extracellular domain of the EGFR, inhibit ligand-induced signaling, and can elicit an immune response.

HISTORY

EGFR/HER PATHWAY-TARGETED CANCER THERAPY

The discovery and analysis of the EGFR (= HER1) in 1984 revealed a remarkable homology of its protein kinase domain to an oncogenic protein in the avian erythroblastosis retrovirus, v-erb. This oncogenic protein was the second one isolated from v-erb, resulting in "erbB" as an alternate name for "HER" gene family members. The HER2 gene was discovered based on its homology to v-erbB in human breast cancer and its oncogenic function in rat ***neu***ral tumors, hence the different names HER2/ErbB2/Neu (King et al., 1985; Schechter et al., 1985). The association of HER2 overexpression with poor prognosis in breast cancer (Slamon et al., 1987) provided the impetus for the development of HER2-targeted antibodies and a new concept in the treatment of HER2-positive breast cancers. *Trastuzumab* was the first anti-HER2 monoclonal antibody developed based on these discoveries; it was approved in the U.S. in 1998 and in Europe in 2000 for the treatment of patients with HER2-positive breast cancers. A monoclonal antibody targeting EGFR, *cetuximab*, was approved in Switzerland in 2003 and in the U.S. in 2004.

TABLE 72–1 ■ NAMING AND FORMAT OF ANTIBODY-BASED CANCER THERAPEUTICS

Please note that the nomenclature below applies to antibody drug names that were assigned by the WHO through the year 2021.

The nomenclature for future monoclonal antibodies was changed by the WHO in Oct 2021 to accommodate the expansion of this class of drugs that contains 880 "mab" entities. In newly named antibodies the stem "mab" will be replaced by four distinct stems to identify the composition of the antibody and will be indicated in future editions of this textbook and on *AccessMedicine.com* and *AccessPharmacy.com*.

ANTIBODY NOMENCLATURE	DEFINITION	EXAMPLE
"mab"	Names end with this syllable	Bevacizu*mab*
"xi"	Chimeric mAb consisting of human constant (Fc) and mouse variable (Fv) antibody domains	Cetu*xi*mab
"zu"	Humanization by insertion of antigen-binding regions (CDRs) into a human IgG protein scaffold	Trastu*zu*mab
"u"	Fully human amino acid sequence match with human IgG	Nivol*u*mab
"li"	Indicates immunomodulatory function	Cemip*li*mab
ANTIBODY FORMAT	**EXAMPLES (TARGETS)**	**NUMBER BY 2021**
IgG1	Bevacizumab (VEGF), cetuximab (EGFR), ipilimumab (CTLA-4), trastuzumab (HER2)	30
IgG2	Panitumumab (EGFR)	1
IgG4	Pembrolizumab, nivolumab (PD-1)	5
Bispecific, linked scFvs	Blinatumomab (CD3 and CD19)	2
CAR T cells	Tisagenlecleucel (CD19)	5

scFv, single-chain antibody fragment of the variable domain.

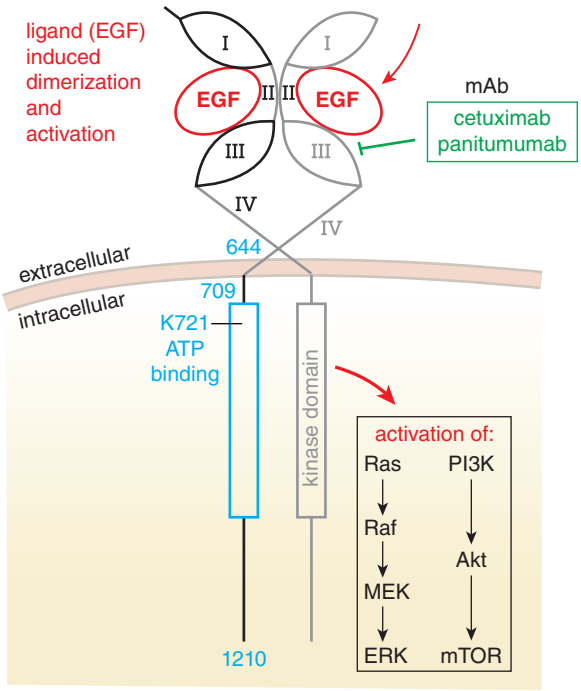


Figure 72–2 *Antibody targeting and domains of EGFR (HER1).* The extracellular portion of the EGFR contains the binding domains for growth factors that include EGF, transforming growth factor-α, and amphiregulin. Ligand binding to extracellular domains I and III induces conformational change and receptor dimerization through domains II and IV and activates intracellular signaling by cross-phosphorylation. The monoclonal antibodies *cetuximab*, *panitumumab*, and *necitumumab* block ligand binding. HER2, HER3, and HER4 share the extracellular domain organization of EGFR. mAb, monoclonal antibody. Amino acid residue numbers are in blue.

Antibody Inhibitors of EGFRs

Cetuximab

Mechanism of Action. *Cetuximab* is a chimeric human/mouse IgG1 antibody composed of the Fv regions of a murine anti-EGFR antibody with human IgG1 heavy and kappa light chain constant regions that bind to the extracellular domain III of EGFR (see Figure 72–2). Binding to EGFR prevents ligand-dependent signaling and receptor dimerization, thereby blocking growth and survival signals in normal and tumor cells. *Cetuximab* can also mediate ADCC against tumor cells that express high levels of EGFR.

ADME. *Cetuximab* exhibits nonlinear pharmacokinetics. A single loading dose of *cetuximab* intravenously is followed by weekly maintenance doses for the duration of treatment. Following intravenous administration, steady-state levels are achieved by the third weekly infusion. The volume of distribution approximates the intravascular space. The half-life of *cetuximab* is about 5 days.

Therapeutic Uses. Antibodies targeting EGFR show a different spectrum of antitumor activity than small-molecule EGFR kinase inhibitors (see Chapter 71), in part due to the additional ADCC of antibodies toward tumor cells. *Cetuximab* and related antibodies are used to treat patients with metastatic colon cancers and head and neck squamous cell carcinoma (HNSCC).

In colon cancer, *cetuximab* enhances the effectiveness of chemotherapy in patients with *KRAS* wild-type but not *KRAS*-mutant tumors (Van Cutsem et al., 2009). A combination of *cetuximab* with FOLFIRI (*folinic acid* [*leucovorin*], *5-fluorouracil* [5FU], and *irinotecan*; see Chapter 70) is typically used to treat patients with *KRAS* wild-type tumors that express EGFR. *Cetuximab* is also used as a single agent in patients who cannot tolerate *irinotecan*-based therapy and in patients with cancers that are resistant to *oxaliplatin*, *irinotecan*, and *5FU*.

In HNSCC, *cetuximab* is used in combination with radiation or with platinum complex–based chemotherapy (e.g., *cisplatin*; see Chapter 70) for locally or regionally advanced HNSCC. *Cetuximab* is also indicated as a monotherapy for patients with metastatic or recurrent HNSCC whose disease progresses on platinum-based chemotherapy.

Adverse Effects. Antibodies targeting EGFR have an adverse effect profile similar to that of first-generation EGFR protein tyrosine kinase inhibitors (see Chapter 71). Acneiform rash in the majority of patients, pruritus, nail changes, headache, and less frequently diarrhea are the most common adverse reactions. Rare but serious adverse effects include infusion reactions (~3% of patients), cardiopulmonary arrest in patients with HNSCC receiving radiation or chemotherapy (2%–3%), interstitial lung disease, and serum electrolyte imbalances, including hypomagnesemia. During pregnancy, *cetuximab* may be transmitted to the developing fetus and has the potential to cause fetal harm.

Panitumumab

Panitumumab is a human IgG2κ antibody that binds to the extracellular domain III of the EGFR and prevents ligand-dependent signaling. It is FDA approved for the treatment of EGFR-expressing metastatic colorectal cancer (mCRC) in combination with chemotherapy and as monotherapy after disease progression following chemotherapy regimens containing *fluoropyrimidine*, *oxaliplatin*, and *irinotecan* (Saltz et al., 2006). Although *panitumumab* effectively inhibits EGFR signaling similarly to *cetuximab*, it is less effective at triggering cellular immune mechanisms that may be crucial for effective therapy of HNSCC (Trivedi et al., 2016).

ADME; Adverse Effects. *Panitumumab* exhibits nonlinear pharmacokinetic characteristics. Following intravenous administration every 2 weeks, steady-state levels are achieved by the third infusion. The mean $t_{1/2}$ is 7.5 days. Adverse effects of *panitumumab* are similar to those of *cetuximab* and include rash and dermatological toxicity, infusion reactions, pulmonary fibrosis, and electrolyte abnormalities. Dermatological toxicities were reported in 90% of patients and were severe in 15% of patients receiving monotherapy.

Necitumumab

Necitumumab is a human IgG1 monoclonal antibody that binds to the extracellular domain of the EGFR. It is approved for first-line treatment of patients with metastatic, squamous NSCLC in combination with *gemcitabine* and *cisplatin*. *Necitumumab* is administered intravenously every 3 weeks prior to chemotherapy. Adverse effects are similar to those of *cetuximab*.

HER2/Neu Inhibitors

Human epidermal growth factor receptor 2 (HER2, also named Neu or ErbB2; see box on History) is a member of the HER1 to HER4 family, which shares a common extracellular domain organization that is shown in Figure 72–2 for EGFR (HER1). Of note, the fixed conformation of the extracellular domain of HER2 resembles the ligand-activated state of the other HER family members and explains the unique function of HER2 as a coreceptor that does not require ligand activation. In addition, due to this conformation, the overexpression of wild-type HER2 is sufficient to activate the intracellular tyrosine kinase and oncogenic signaling even in the absence of activating mutations, coreceptors, or ligands. Overexpression of HER2 is found in 20% to 30% of human breast cancers due to gene amplification on chromosome 17 and results in more aggressive tumors, lower response rates to hormonal therapies, and higher risk of disease recurrence after treatment. Antibodies targeting HER2 are discussed below; small-molecule kinase inhibitors of HER2 (*lapatinib*, *neratinib*, and *tucatinib*) are discussed in Chapter 71.

Trastuzumab

Mechanism of Action

Trastuzumab is a humanized IgG1 monoclonal antibody that binds to the extracellular domain IV of HER2 (see domain organization in

Figure 72–2), reduces HER2 signaling, and can induce antibody-dependent, immune cell–mediated cytotoxicity. HER2 protein overexpression or gene amplification is predictive of response to HER2-targeted therapies (Wolff et al., 2013).

ADME

Trastuzumab has dose-dependent pharmacokinetics with a mean $t_{1/2}$ of 5.8 days on a weekly maintenance dose with alternative dosing every 3 weeks. Steady-state levels are achieved between 16 and 32 weeks.

Therapeutic Uses

Trastuzumab is approved for HER2-overexpressing breast and gastric cancer. It is used in combination with cytotoxic chemotherapeutics such as taxanes as initial treatment or as a single agent following relapse of disease after cytotoxic chemotherapy (see Chapter 70).

Adverse Effects and Precautions

Acute adverse effects after infusion of *trastuzumab* are typical for monoclonal antibodies and can include fever, chills, nausea, dyspnea, and rashes. The most serious toxicity of *trastuzumab* is heart failure. Cardiotoxicity is caused by interruption of HER2/4 heterodimer signaling in cardiomyocytes, signaling that is essential for the maintenance of contractile function. The cardiotoxic potential was predicted from gene inactivation studies ("knockouts") in mice, which showed that mice lacking HER2, HER4, or HER ligands developed dilated cardiomyopathy and heart failure. Baseline electrocardiogram and cardiac ejection fraction measurement should be obtained before initiating treatment with *trastuzumab* to rule out underlying heart disease. Clinical monitoring for symptoms of congestive heart failure as well as periodic determination of left ventricular ejection fraction (LVEF) during and after the course of therapy are recommended. When *trastuzumab* is used as a single agent, fewer than 5% of patients will experience a decrease in LVEF, and about 1% will have clinical signs of congestive failure. However, left ventricular dysfunction occurs in up to 20% of patients who receive a combination of *doxorubicin* and *trastuzumab*, reflecting the added cardiotoxicity of *doxorubicin* (see Chapter 70). In contrast, the risk of cardiac toxicity is greatly reduced with the recommended combination of *trastuzumab* with taxanes.

Pertuzumab

Mechanism of Action

Pertuzumab is a humanized IgG1 monoclonal antibody directed against the extracellular receptor dimerization domain II of HER2 that is distinct from domain IV targeted by *trastuzumab* (see domain organization in Figure 72–2). *Pertuzumab*, in contrast to *trastuzumab*, prevents ligand-dependent heterodimerization of other HER family members with HER2 and thus inhibits ligand-induced signaling, cell growth, and survival. In addition to the inhibition of receptor heterodimerization, binding of *pertuzumab* to HER2-overexpressing cells can induce ADCC.

ADME

The median half-life of *pertuzumab* is 18 days; after an initial loading dose, maintenance doses are administered every 3 weeks.

Therapeutic Use

Preclinical studies with HER2-overexpressing human tumor cells grown as xenograft tumors in mice showed that the combination of *pertuzumab* and *trastuzumab* enhanced antitumor activity. In patients with HER2-positive metastatic breast cancer, the addition of *pertuzumab* to *trastuzumab* and *docetaxel* increases the median overall survival by over 1 year from 40.8 months to 56.5 months (Swain et al., 2015). Use of *pertuzumab* is approved for the treatment of patients with HER2-positive, locally advanced, inflammatory, or early-stage breast cancer in combination with *trastuzumab* and *docetaxel*.

Adverse Effects

Cardiotoxicity is similar to that of *trastuzumab*, and combinations with cardiotoxic anthracyclines are not indicated (Slamon et al., 2001). On the other hand, the combination of *pertuzumab* and *trastuzumab* does not cause any increase in cardiac toxicity (Swain et al., 2015). Based on its mechanism of action, *pertuzumab* can cause fetal harm when administered to a pregnant woman, and this risk should be weighed against the potential benefit.

Margetuximab

Margetuximab is a chimeric IgG1-targeting HER2 that was approved in 2020 for the treatment of patients with metastatic HER2-positive breast cancer who had received two or more prior anti-HER2 regimens. It is used in combination with chemotherapy and shares the risk of left ventricular dysfunction with *trastuzumab* and *pertuzumab*.

Inhibitors of Platelet-Derived Growth Factor Receptor

Signaling by the platelet-derived growth factor receptor (PDGFR) plays a significant part in mesenchymal biology, including stem cell growth, and is involved in oncogenesis through aberrant cancer cell signaling, modulation of the tumor microenvironment, and facilitation of angiogenesis and metastasis. Small-molecule protein kinase inhibitors, including *imatinib* (see discussion below), have been used in the treatment of patients with gastrointestinal stromal tumors that carry mutations in *KIT* or *PDGFRA* (see Figure 71–2).

Olaratumab

Mechanism of Action

Olaratumab is a human IgG1 monoclonal antibody that binds to PDGFRα and blocks ligand-mediated receptor activation (Tap et al., 2016).

Therapeutic Uses

Olaratumab is indicated, in combination with *doxorubicin*, for the treatment of adult patients with soft-tissue sarcoma with a histological subtype for which an anthracycline-containing regimen is appropriate and which is not amenable to curative treatment with radiotherapy or surgery. Recent studies showed that addition of *olaratumab* to *doxorubicin* doubled the median survival of patients with metastatic soft-tissue sarcoma to 26 months (Judson and van der Graaf, 2016).

Adverse Effects

The most common (≥20%) adverse effects of treatment with *olaratumab* are nausea, fatigue, neutropenia, musculoskeletal pain, inflammation of the mucous membranes (mucositis), alopecia, vomiting, diarrhea, decreased appetite, abdominal pain, neuropathy, and headache. Laboratory abnormalities are lymphopenia, neutropenia, thrombocytopenia, hyperglycemia, elevated activated partial thromboplastin time, hypokalemia, and hypophosphatemia. Other adverse effects include infusion-related reactions (low blood pressure, fever, chills, rashes) and embryofetal harm.

II. Inhibiting Tumor Angiogenesis

HISTORY

Cancer cells secrete angiogenic factors that induce the formation of new blood vessels and guarantee the flow of nutrients to the tumor cells to permit growth and metastasis. Many tumor types overexpress these angiogenic factors, turning on an "angiogenic switch" whereby the tumor cells adopt an invasive phenotype favoring proliferation of endothelial cells and neovascularization. In 1971, Judah Folkman hypothesized that the growth of solid tumors was dependent on angiogenesis and that blockade of the effects of putative angiogenic factors would be a good treatment modality for human cancers (Folkman, 1971). Folkman's hypothesis proved to be correct and led to the characterization of a number of secreted angiogenic factors, including

vascular endothelial growth factor (VEGF), fibroblast growth factor, transforming growth factor-β, and PDGF. Furthermore, inhibitors of these angiogenic factors have, indeed, become useful therapeutic agents against certain cancers. VEGF is a major driver of angiogenesis, and inhibitors of VEGF signaling comprise an important class of antitumor agents. Most recently, a better understanding of the immunosuppressive features of angiogenesis has prompted the combination of antiangiogenic therapy with immune checkpoint inhibitors to enhance their efficacy, as discussed below (Huinen et al., 2021).

Inhibition of VEGF and the VEGF Receptor Pathway

Vascular endothelial growth factor initiates endothelial cell proliferation and vascular permeability when it binds to a member of the VEGF receptor (VEGFR) family, a group of highly homologous receptors with intracellular tyrosine kinase domains; these receptors include VEGFR1 (FLT1), VEGFR2 (KDR), and VEGFR3 (FLT4). The binding of VEGF to its receptors activates the intracellular VEGFR tyrosine kinase activity and initiates mitogenic and antiapoptotic signaling pathways (Nagy et al., 2007). Antibodies targeting VEGF, such as *bevacizumab*, sterically hinder the interaction of VEGF with its receptor. *Aflibercept* acts as a VEGF trap; it is a recombinant molecule that uses the VEGFR1-binding domain to sequester VEGF, basically acting as a "soluble decoy receptor" for VEGF. Several small-molecule drugs that inhibit the protein tyrosine kinase of VEGFR (*axitinib, cabozantinib, lenvatinib, pazopanib*, *sorafenib*, *sunitinib*, and *tivozanib*; see Chapter 71) as well as monoclonal antibodies that target the receptor (*ramucirumab*) have been approved for clinical use. The inhibition of endothelial function by these different approaches results in a similar spectrum of mostly cardiovascular adverse effects.

Bevacizumab

Mechanism of Action

Bevacizumab is a humanized monoclonal IgG1 antibody that binds to VEGF; the antibody contains the antigen-recognition domain of a murine antibody inserted into a human IgG1. *Bevacizumab* prevents the interaction of VEGF with its receptors on the surface of endothelial cells and inhibits receptor signaling that normally increases vascular permeability and angiogenesis. *Bevacizumab* delays progression of renal cell carcinoma (RCC) and, in combination with cytotoxic chemotherapy, is approved in the treatment of patients with mCRC, non–small cell lung cancer (NSCLC), and ovarian or cervical cancer, and also following initial therapy, in patients with glioblastoma, and in combination with *atezolizumab* (anti-PD-L1, see below) for the treatment of unresectable or metastatic hepatocellular carcinoma.

ADME

Bevacizumab is administered intravenously as a 30- to 90-min infusion every 2 weeks in patients with metastatic colon cancer and, in conjunction with combination chemotherapy, in patients with nonsquamous metastatic NSCLC every 3 weeks with chemotherapy. The antibody has a plasma half-life of about 20 days (range, 11–50 days).

Therapeutic Use in Cancer

Bevacizumab is indicated as treatment of various cancers, frequently in combination with other drugs. In metastatic colon cancer, the addition of *bevacizumab* to either FOLFOX (*folinic acid* [*leucovorin*], *5FU*, and *oxaliplatin*) or FOLFIRI (see Chapter 70) increases patient median survival by about 5 months (Hurwitz et al., 2004). In NSCLC, the addition of *bevacizumab* to *carboplatin* and *paclitaxel* increases median survival by about 2 months. Also approved are combinations of *bevacizumab* with cytotoxic chemotherapy in patients with cervical or ovarian cancer and used rarely as a combination with interferon (IFN)-α for the treatment of metastatic RCC. For glioblastoma, *bevacizumab* is approved as a single agent following prior therapy.

Other Therapeutic Uses: Wet Macular Degeneration

Vascular endothelial growth factor is an important mediator of pathologic vascular permeability. Intravitreous administration of anti-VEGF–targeted therapy has become a standard treatment of wet macular degeneration (see also Chapter 74). *Bevacizumab* and a modified fragment of *bevacizumab*, *ranibizumab*, are used. *Ranibizumab* is derived from the *bevacizumab* IgG1 by deletion of the Fc domain and changes in six amino acids in the antigen-recognition domain to increase affinity for VEGF. *Brolucizumab*, a humanized single-chain antibody fragment against VEGF, is also used for the treatment of wet macular degeneration and delivered by intravitreal injection.

Because these drugs are administered by injection into the vitreous cavity of the eye, few systemic adverse effects are observed. A direct comparison of intravitreous *bevacizumab*, *ranibizumab*, and *aflibercept* (the VEGF trap protein; see discussion that follows) in patients with diabetic macular edema showed that all three treatments improved vision, although *aflibercept* appeared more effective in patients with worse baseline levels of visual impairment (Wells et al., 2015).

Adverse Effects

Bevacizumab causes a wide range of serious class-related adverse effects that include hypertension, gastrointestinal perforation, thromboembolic events, and hemorrhage including epistaxis. A major concern is the potential for vessel injury and bleeding in patients with lung cancer. *Bevacizumab* is contraindicated for patients with a history of hemoptysis, brain metastasis, or a bleeding diathesis. In appropriately selected patients, the predicted rate of life-threatening pulmonary hemorrhage is less than 2%, and for arterial thromboembolism (stroke, myocardial infarction) observed during treatments that include *bevacizumab*, the predicted rate is less than 4%.

Other adverse effect characteristics of drugs that target the VEGF pathway include hypertension and proteinuria. A majority of patients require antihypertensive therapy, particularly those receiving higher doses and more prolonged treatment. *Bevacizumab* can be associated with congestive heart failure, probably secondary to hypertension, and with reversible posterior leukoencephalopathy in patients with poorly controlled hypertension. Gastrointestinal perforation, a potentially life-threatening complication, has been observed in up to 11% of patients with ovarian cancer. In patients with colon cancer, colonic perforation occurs infrequently (<1%) during *bevacizumab* treatment but increases in frequency in patients with intact primary colonic tumors, peritoneal carcinomatosis, peptic ulcer disease, chemotherapy-associated colitis, diverticulitis, or prior abdominal radiation treatment. Following colon cancer surgery, patients on *bevacizumab* have a higher rate (13% vs. 3.4%) of serious wound-healing complications. Because of the long $t_{1/2}$ of *bevacizumab*, elective surgery should be delayed for at least 4 weeks from the last dose of antibody, and treatment should not be resumed for at least 4 weeks after surgery.

Ramucirumab

Mechanism of Action

Ramucirumab is a human IgG1 monoclonal antibody that binds to VEGFR2, blocking the binding of VEGFR ligands and thereby inhibiting ligand-induced activity in endothelial cells.

ADME

Ramucirumab is administered intravenously every 2 to 3 weeks. Its mean half-life is 14 days.

Therapeutic Uses

Ramucirumab is used in combination with chemotherapy for the treatment of patients with mCRC with disease progression on or after prior therapy; as a single agent or in combination with *paclitaxel* for treatment of patients with advanced gastric adenocarcinoma with disease progression on or after prior chemotherapy; and in combination with *docetaxel* for treatment of patients with metastatic NSCLC with disease progression on or after chemotherapy.

Adverse Effects

The most common adverse effects are hypertension and diarrhea. Other severe adverse effects overlap with those of bevacizumab (anti-VEGF; see previous discussion): increased risk of hemorrhage, gastrointestinal perforation, and impaired wound healing.

Aflibercept

Mechanism of Action

Ziv-aflibercept acts as a trap for VEGFR ligands. It is a recombinant fusion protein that contains the extracellular VEGF-binding domain of human VEGFR1/2 fused to the Fc portion of human IgG1. *Ziv-aflibercept* acts as a soluble receptor that binds to human VEGFR ligands with high affinity (K_ds of 0.5–40 pM), reducing their plasma concentrations to levels that are too low for significant activation of their cognate receptors.

ADME

The elimination half-life is about 6 days (range, 4–7 days). *Ziv-aflibercept* is administered by intravenous infusion over 1 h every 2 weeks.

Therapeutic Uses

In combination with FOLFIRI, *ziv-aflibercept* is indicated for patients with mCRC that is resistant to or has progressed following an *oxaliplatin*-containing regimen.

Adverse Effects

The most common adverse effects are hypertension and diarrhea. Other severe adverse effects overlap with those of *bevacizumab* (anti-VEGF; see previous discussion): increased risk of hemorrhage, gastrointestinal perforation, and impaired wound healing.

III. Activating Immune Cells

HISTORY

The discovery of the T-cell receptor (TCR) and of costimulatory signal pathways in the period from 1982 to 1992 defined essential requirements for T-cell activation during antigen presentation: In addition to the antigen-TCR interaction ("signal 1"), CD28 on T cells binds to B7 proteins on antigen-presenting cells (APCs) ("signal 2"; Figure 72–3; see also Figures 38–2, 38–3, and 38–4). The finding, in the mid-1990s, of coinhibitory molecules that can dampen T-cell activation revealed a potential mechanism for drugs to modulate immune checkpoints and alter anticancer immune activity. Tasuku Honjo and James Allison shared the 2018 Nobel Prize in Physiology or Medicine for "for their discovery of cancer therapy by inhibition of negative immune regulation." TCR activation induces expression of the inhibitory cytotoxic T lymphocyte–associated protein 4 (CTLA-4), a CD28 homolog with higher affinity for the B7 ligands than CD28. CTLA-4 inhibits T-cell activation to control autoimmune tissue damage. Programmed cell death 1 (PD-1), an inhibitory molecule expressed on T cells after activation, and PD-L1, the ligand for PD-1, were described in the early 2000s. It became obvious that T cells present in human tumors were being inhibited by factors in the tumor tissues: T cells isolated from cancer patients and activated exogenously showed antitumor efficacy, paving the way for targeting immune inhibition. Improved survival of patients with metastatic melanoma after anti–CTLA-4 treatment led to FDA approval of *ipilimumab* in 2011; anti–PD-1 drugs *pembrolizumab* and *nivolumab* were approved in 2014 for the treatment of patients with metastatic melanoma. Clinical trials in multiple types of cancer have been ongoing since then and have led to the approval of immune checkpoint inhibitors for more than 20 additional indications as described in this section. Milestones in the discoveries and more details can be found in a review by Korman et al. (2021).

Immune Checkpoint Inhibitors

Recent striking successes of cancer immunotherapy have given credence to the long-held hope that the immune system may be unleashed to treat cancer. The conceptual breakthrough is based on the discovery of inhibitory signals that limit T cell activation, so-called immune checkpoints. The discovery of receptors, ligands, and their function in the control of immune cell activity during the past two decades led to the development of clinically effective monoclonal antibodies that enable antitumor T cells to eradicate cancer cells with acceptable adverse effects (Figure 72–3).

Blocking immune checkpoints on cytotoxic T cells enables them to eradicate cancer cells that express antigens recognized by the T cells. Sustained eradication of metastatic melanoma in a significant number of patients was an impressive outcome of initial clinical trials with a monoclonal antibody, *ipilimumab*, which is an immune checkpoint inhibitor (Robert et al., 2013). Note that immune checkpoint inhibition is distinct from conventional approaches such as vaccination with a known antigen (to generate T cells that recognize cells expressing the antigen); nonspecific stimulation of T cells; isolation, *ex vivo* expansion, and administration of tumor-infiltrating lymphocytes; or approaches using antigen-responsive T cells that express a specific chimeric antigen receptor (CAR). The different approaches are described next.

Mechanisms of T Cell Activation

Antigen-mediated activation of T cells is initiated by engagement of the TCR with antigen presented on major histocompatibility complex (MHC) protein on the surface of an APC, "signal 1." In addition, engagement of CD28 costimulatory molecules on T cells is required for effective T-cell activation that includes cell proliferation, migration, and production of cytokines. This "signal 2" is provided by B7 surface proteins on APCs, for example, CD80 (B7-1) or CD86 (B7-2) (see Figures 38–4 and 72–3). T-cell activation is tightly controlled by immune-suppressive cells and cytokines as well as by coinhibitory molecules present on T cells themselves, such as CTLA-4 or PD-1 (see Figure 72–3). CTLA-4 is expressed by activated CD4 and CD8 T cells and competes with costimulatory CD28 for binding to the B7 protein ligands. Binding of CTLA-4 to B7 proteins interrupts the initial CD28 costimulatory signals and serves as an early negative regulator of the T-cell activation responses, that is, an *immune checkpoint*. Additional control of T-cell activity is provided by later inhibitory signals through other molecules, such as PD-1 that binds the ligand PD-L1. PD-1 is mainly expressed by activated CD4 and CD8 T cells as well as B cells and macrophages.

Blocking monoclonal antibodies to CTLA-4, PD-1, or PD-L1 are currently used to neutralize coinhibitory receptors during T-cell priming (CTLA-4), to maintain cancer antigen-specific T-cell function, and to provide a sustained T-cell response (PD-1) that includes increased production of cytokines such as tumor necrosis factor-α and IFN-γ, as well as granzyme B. Antibodies that target different checkpoint molecules are discussed in the following sections: *ipilimumab* (anti–CTLA-4); *atezolizumab, avelumab, durvalumab* (anti–PD-L1); *nivolumab*, *pembrolizumab*, *cemiplimab, dostarlimab* (anti–PD-1); and *relatlimab* (anti-LAG3).

Inhibitors of CTLA-4

CTLA-4 is upregulated during the antigen priming of T cells and binds B7 on APCs to attenuate the T-cell response and thus reduce the risk for chronic autoimmune-dependent inflammation (see Figure 72–3).

Ipilimumab

Ipilimumab is a human IgG1 monoclonal antibody that binds to CTLA-4 and was the first immune checkpoint inhibitor approved for cancer treatment.

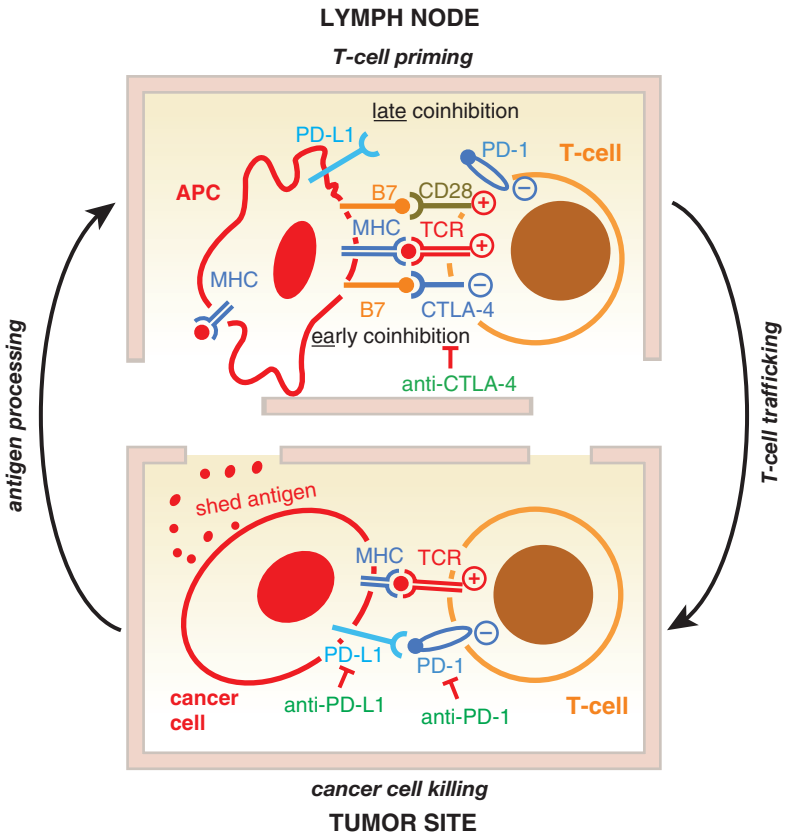


Figure 72–3 *Targeting of immune checkpoints.* Antigens shed from dead cancer cells are captured and processed by APCs. APCs travel to the regional lymph nodes and present the antigen bound to MHC to stimulate TCRs (+). Costimulatory signals leading to T-cell activation are provided by the interaction between CD28 on T cells and B7 (CD80, CD86) on APCs. Inhibitory molecules, including cytotoxic T lymphocyte–associated protein 4 (CTLA-4) and programmed cell death 1 (PD-1), are induced during immune responses and represent a "checkpoint" to dampen T-cell hyperactivation. Sequence homology within the extracellular binding domains of CTLA-4 and CD28 allows binding to the B7 ligands. Higher affinity and avidity of CTLA-4 for the ligands leads to a dampening of the immune response. Human PD-1 is expressed on T cells after TCR stimulation and binds the B7 homologues PD-L1 (B7-H1) and PD-L2 (B7-DC) on APCs but can be induced in nonhematopoietic cells (e.g., by cytokines). Activated T cells also upregulate PD-1 on their surface. (Figures 38–3 and 38–4 show additional details of T-cell activation.)

Activated effector T cells capable of killing cancer cells traffic from the lymph node to the tumor site, where they recognize cancer cells by the interaction of the TCR with the MHC-antigen complex and kill cancer cells through cytolysis (release of perforin and granzyme). This triggers additional antigen release that can induce subsequent rounds of anticancer immunity. Release of IFN-γ upon TCR recognition of the MHC-neoantigen complex induces upregulation of PD-L1 on the surface of tumor cells. PD-L1 binds to PD-1 on the surface of activated T cells, dampening the immune response in the tumor microenvironment.

T-cell checkpoint inhibitors can enhance T-cell priming through blockade of CTLA-4 (*ipilimumab*) or by T-cell activation in the tumor via blocking the PD-1/PD-L1 interaction with antibodies to PD-1 (*cemiplimab, dostarlimab, nivolumab, pembrolizumab*) or PD-L1 (*atezolizumab, avelumab, durvalumab*). Antibodies that block inhibitory immune checkpoints allow a sustained T-cell response that also includes an increased production of cytokines and T-cell proliferation. (Reviewed in Waldman et al. [2020].)

Mechanism of Action

Ipilimumab blocks the interaction of CTLA-4 with B7 ligands on APCs and thereby augments T-cell activation (see Figure 72–3). Inhibition of CTLA-4 signaling can also expand the T-cell repertoire (Cha et al., 2014) and inhibit regulatory T-cell function that dampens cytotoxic T-cell activity and thus increases the antitumor immune response further (Mellman et al., 2011) as well as induce T-cell memory (Felix et al., 2016) that prolongs the immune response.

ADME

Ipilimumab is administered intravenously and shows linear pharmacokinetics in the dose range of 0.3 to 10 mg/kg, with a terminal half-life of 15.4 days.

Therapeutic Uses

A pivotal trial in patients with metastatic melanoma showed improved median overall survival after *ipilimumab* treatment (Hodi et al., 2010). This trial led to FDA approval of *ipilimumab* for the treatment of patients with metastatic melanoma. Further studies showed that some patients had clinical responses that lasted for 10 or more years (Schadendorf et al., 2015). The tumor response can take several months to manifest, and tumors may increase in size during this period, in part due to the evolving inflammatory reaction (Mellman et al., 2011). *Ipilimumab* is approved for treatment of patients with advanced melanoma as a single agent or in combination with *nivolumab*; other approved indications in combination with *nivolumab* are advanced RCC; colorectal cancer with high microsatellite instability (MSI-H) or mismatch repair deficient; pretreated hepatocellular carcinoma; metastatic NSCLC with no EGFR or anaplastic lymphoma kinase (ALK) genomic aberrations; and malignant pleural mesothelioma. Recommended dosing of *ipilimumab* is 3 mg/kg intravenously over 90 min every 3 weeks for a maximum of four doses.

Adverse Effects

Blockade of CTLA-4 compromises immune tolerance to some normal tissue antigens and provokes inflammatory toxicities, mostly in the skin, pituitary gland, thyroid, intestine, and liver. The activation of T-cell responses by *ipilimumab* treatment leads to immune-related toxicities in

the majority of patients (73.6%), with 18.6% reaching grade 3 to 4 toxicity (Callahan et al., 2016). Adverse effects on the skin (pruritus, rash, vitiligo) and gastrointestinal tract (diarrhea, colitis) are the most frequent, whereas immune-mediated hepatitis, hypophysitis, and hypo- or hyperthyroidism are less frequent. Withholding treatment because of moderate adverse reactions until return to baseline and high-dose corticosteroids for severe immune-mediated reactions are recommended (Dougan et al., 2021; Esfahani et al., 2020).

Tremelimumab

Tremelimumab (formerly *ticilimumab*) is a human IgG2 monoclonal antibody that targets CTLA-4; it is under investigation in several clinical trials and received orphan drug designation by the FDA for the treatment of malignant mesothelioma.

Inhibitors of PD-1

Activation of the PD-1 checkpoint pathway in T cells by PD-L1 or PD-L2 evokes a negative regulatory immune response and inactivates T cells (see Figure 72–3). Tumor cells can exploit this pathway by presenting the PD-L1 ligand to tumor-infiltrating T cells and thus shield the tumor from T cell–mediated destruction. Presentation of PD-L1 to activated T cells results in T-cell exhaustion, whereas antibody-mediated blockade of PD-1 can restore or maintain T-cell antitumor response.

Nivolumab

Mechanism of Action

Nivolumab is a human monoclonal IgG4 antibody that blocks the interaction between PD-1 and its ligands.

ADME; Clinical Use

Nivolumab is administered as an intravenous infusion every 2 (or 4) weeks until the time of disease progression or unacceptable toxicity. The drug has an elimination half-life of 26.7 days. *Nivolumab* is FDA approved as a single agent for the treatment of a variety of cancers, including advanced or resected high-risk melanoma, previously treated advanced NSCLC, RCC, advanced head and neck cancer, MSI-H or mismatch repair–deficient colorectal cancer, hepatocellular carcinoma, and relapsed/refractory classical Hodgkin lymphoma. Quite strikingly, *nivolumab* provided an overall response rate of 87% in patients with Hodgkin lymphoma who had failed treatment with *brentuximab vedotin* (see discussion that follows).

Adverse Effects

The most common adverse effect of treatment in patients with melanoma is a rash (>20%); in patients with advanced squamous NSCLC, the adverse effects are fatigue, dyspnea, pneumonitis, musculoskeletal pain, decreased appetite, cough, nausea, and constipation. Immune-mediated adverse reactions are much less common than with CTLA-4 antibodies; corticosteroids are recommended based on the severity of the reaction. For pneumonitis, colitis, hepatitis, nephritis, and renal dysfunction, withhold treatment if the adverse effects are moderate and discontinue treatment if the effects are severe (Dougan et al., 2021; Esfahani et al., 2020).

Pembrolizumab

Pembrolizumab (formerly called *lambrolizumab* or MK-3475) is a humanized monoclonal IgG4κ isotype antibody that blocks interaction between PD-1 and its ligands. This humanized antibody has a mouse variable region grafted onto a human antibody framework and shows high affinity toward PD-1 receptors of humans and other primates but no appreciable affinity for mouse or rat PD-1.

ADME; Clinical Use

The recommended treatment schedule for *pembrolizumab* is 2 mg/kg as an intravenous infusion over 30 min every 3 weeks (or a flat dose of 200 mg every 3 weeks or 400 mg every 6 weeks). The elimination half-life is 26 days. In clinical trials, *pembrolizumab* demonstrated an overall response rate of 26% in patients with *ipilimumab*-pretreated and refractory advanced melanoma (Hamid et al., 2013; Robert et al., 2014). *Pembrolizumab* is more effective and less toxic than *ipilimumab* in patients with either treatment-naïve or BRAF inhibitor–pretreated melanoma. *Pembrolizumab* is currently approved for the treatment of 18 different cancers that include advanced and high-risk and intermediate risk resected melanoma, NSCLC (either prior to or after platinum-containing chemotherapy), head and neck cancer, urothelial cancer, Merkel cell carcinoma, Hodgkin lymphoma, triple-negative breast cancer and intermediate-high risk RCC. The approval for *pembrolizumab* includes any cancer with DNA mismatch repair deficiency or MSI-H that leads to a higher mutation load followed by the generation of mutant protein antigens by cancer cells that can activate the immune system. This first cancer tissue–agnostic approval extends immune checkpoint inhibitor treatment to subsets of patients with hereditary colorectal cancer due to Lynch syndrome (MSI-H) and up to 20% of patients with sporadic colon cancer. The frequency is less in endometrial, ovarian, and pancreatic cancers and requires testing of each patient's tumor to assess suitability of the treatment. More recently, the tissue-agnostic approval for *pembrolizumab* was broadened to advanced cancers with high tumor mutational burdens of 10 or more mutations per megabase, which includes a large fraction of solid tumors, as described in Chapter 69 (see Figure 69–3).

Adverse Effects

The most common adverse effects, experienced by more than 20% of patients, are fatigue, cough, nausea, pruritus, rash, decreased appetite, constipation, arthralgia, and diarrhea. Serious adverse events include immune-mediated inflammation, specifically pneumonitis, colitis, hepatitis, hypophysitis, and both hyper- and hypothyroidism. Withholding treatment and administration of systemic corticosteroids for grade 2 or higher grade adverse events or discontinuing treatment with more severe adverse effects is recommended. Because human IgG4 can cross the placenta, the benefit from *pembrolizumab* treatment should be balanced with the potential fetal risk in a pregnant woman.

Cemiplimab

Cemiplimab is a human IgG4 monoclonal antibody that binds to PD-1 and blocks the pathway. It was approved in 2018 for treating patients with advanced cutaneous squamous and in 2021 for the treatment of patients with basal cell carcinoma as well as first-line treatment of those with locally advanced or metastatic NSCLC with high PD-L1. Adverse effects are typical for PD-1 blockade and include immune-related inflammation of different organs.

Dostarlimab

Dostarlimab is a humanized IgG4 monoclonal PD-1–blocking antibody approved in 2021 for the treatment of patients with mismatch repair–deficient chemotherapy-resistant endometrial cancer.

Antagonists of PD-L1

Programmed cell death 1 has two known ligands, PD-L1 (B7-H1) and PD-L2 (B7-DC), each with a distinct expression profile. The PD-L1 ligand is expressed on APCs, T cells, B cells, and nonhematopoietic cells, which can include cancer cells (see Figure 72–3). Expression of PD-L2 is restricted to APCs, macrophages, myeloid dendritic cells, and mast cells. PD-L1 is expressed in many cancers and thus can suppress the activation of cytotoxic T cells that enter the tumor. Anti–PD-L1 antibodies can block this inhibitory effect and promote an effective antitumor response. *Atezolizumab, durvalumab*, and *avelumab* are approved antibody inhibitors of PD-L1 with similar profiles.

Atezolizumab

Mechanism of Action

Atezolizumab, also known as MPDL3280A, is a humanized IgG1 monoclonal antibody to PD-L1 approved in 2016.

ADME

Atezolizumab is administered as an intravenous infusion over 60 min every 3 weeks. The terminal half-life is 27 days.

Clinical Use

Atezolizumab is approved for the treatment of different advanced or metastatic cancers: urothelial cancer, NSCLC, small cell lung cancer (with *carboplatin* and *etoposide*, see Chapter 70), triple-negative breast cancer (with *paclitaxel*, see Chapter 70), hepatocellular carcinoma (with *bevacizumab*), and melanoma (with *cobimetinib* and *vemurafenib*, see Chapter 71). High PD-L1 expression is a prerequisite for several of these indications.

Adverse Effects

The most common adverse effects in 20% or more of patients with metastatic NSCLC are fatigue, decreased appetite, dyspnea, cough, nausea, musculoskeletal pain, and constipation. Patients with urothelial carcinoma may report urinary tract infection. Possible immune-related adverse effects include hepatitis, colitis, hypophysitis, thyroid disorders, adrenal insufficiency, and rarely, type 1 diabetes mellitus, pancreatitis, myasthenia gravis, Guillain-Barré syndrome, and ocular inflammation; these may require discontinuation of treatment. Due to potential embryofetal toxicity, female patients should be advised accordingly.

Durvalumab

Durvalumab is a human IgG1κ monoclonal antibody to PD-L1 and was initially approved in 2017 for the treatment of metastatic urothelial cancer. It is currently approved for advanced NSCLC and for first-line treatment of small cell lung cancer (with *etoposide* and either *carboplatin* or *cisplatin*, see Chapter 70).

Avelumab

Avelumab is a human IgG1κ monoclonal antibody to PD-L1 and was approved in 2017 for the treatment of metastatic urothelial and Merkel cell cancer. It is also approved for RCC as first-line treatment in combination with *axitinib*, a VEGFR kinase inhibitor (see Chapter 71).

Combination of Anti–PD-1 and Anti–CTLA-4

Because anti–CTLA-4 and anti–PD-1 target distinct immune checkpoints during T-cell activation, preclinical studies have shown that concurrent targeting of CTLA-4 and PD-1 significantly improves therapeutic efficacy when compared to effects of single agents (Curran et al., 2010). In patients with advanced melanoma, the combination of *ipilimumab* and *nivolumab* produces a tumor response in more than 50% of the patients, and 5-year overall survival is highest in the combination treatment group, with 52% survival versus 44% with *nivolumab* only and 26% with *ipilimumab* only (Larkin et al., 2019). Immune-related adverse effects are similar, but more frequent than with the single agent, although combination therapy increases the rate of rare but fatal myocarditis (combination 0.17%; nivolumab <0.01%) (Johnson et al., 2016). Autoimmune reactions can be managed by dose delays, glucocorticoid treatment, and immunosuppressant treatment reviewed by Dougan et al. (2021) and Esfahani et al. (2020). Immune checkpoint inhibitors also induce regression of melanoma metastases at intracranial and visceral sites in over half of the patients, and integration of immune checkpoint inhibitors with other treatment approaches for melanoma can further improve this (Atkins et al., 2021; Curti and Faries, 2021). Combinations of immune checkpoint inhibitors with other cancer therapies are being evaluated in numerous international clinical trials and are becoming the standard of care in solid and hematological malignancies (Balogh and Nass, 2019).

Immune Checkpoint Targeting Beyond CTLA-4 and PD-1: Anti-LAG3

LAG3 (lymphocyte activation gene 3), TIM3 (T cell immunoglobulin and mucin domain-3), and TIGIT (T cell immunoreceptor with Ig and ITIM domains) are other cell surface receptors that negatively regulate the responsiveness of immune cells and are thus potential targets for immune modulatory therapies (Kraehenbuehl et al., 2022). LAG3 targeting enhanced anti-PD-1 efficacy in preclinical models and was evaluated recently in a Phase III trial in patients with untreated advanced melanoma. Addition of *relatlimab* (anti-LAG3, human IgG4) to *nivolumab* (anti-PD-1) treatment improved median progression-free survival of the patients from 4.6 to 10.1 months. Adverse events increased in frequency and matched those known for anti-PD-1 treatment (Tawbi et al., 2022). This establishes the validity of dual immune checkpoint inhibition beyond the combination of anti-PD-1 and anti-CTLA-4 discussed above.

Impact of the Microbiome on Immune Checkpoint Inhibitor Therapy

Most recently, the impact of the composition of the gut microbiome on the efficacy of immune checkpoint inhibitor treatment of cancer has become an active area of research. Fecal transplants are one tool applied in ongoing clinical trials and are reviewed by Woelk and Snyder (2021); see also Chapter 6.

Cytokines to Stimulate Immune Responses

Interleukin-2

Of the 70 or so proteins and glycoproteins that are classified as cytokines, only IFN (see Chapters 38 and 39) and interleukin (IL)-2 are in routine clinical use, often for actions against cancer.

Activated $CD4^+$ T cells are the primary producers of IL-2. The functions of IL-2 are myriad; among other roles, IL-2 promotes the growth of T cells and natural killer (NK) cells, stimulates the proliferation of and antibody production by B cells, and drives the development of regulatory T cells (mediators of tolerance and suppression). As noted by Liao and colleagues (2013), "IL-2 has broad essential biological actions, not only driving T cell proliferation and modulating effector cell differentiation, but also limiting potentially dangerous autoimmune reactions." This statement provides a general rationale for the clinical use of IL-2 and the likelihood that IL-2 administration might also engender adverse effects.

Mechanism of Action

Interleukin-2 stimulates the proliferation of activated T cells and the secretion of cytokines from NK cells and monocytes. IL-2 stimulation increases cytotoxic killing by T cells and NK cells.

Interleukin-2 is a 133–amino acid glycoprotein (molecular weight ~15 kDa) produced by activated T cells and NK cells; it promotes activated T-cell proliferation and enhanced killing by NK cells. Responsiveness depends on expression of the IL-2 receptor (IL-2R). The receptor comprises three independently expressed membrane monospans, α, β, and γ, which can assemble in several combinations to produce working receptors of varying affinities. These monospans are variously expressed in lymphohematopoietic cells, lack C-terminal protein kinase activities, and transduce signals via coupling to Jak proteins (Figure 72–4). In contrast, nearly all types of cancer cells lack receptor expression and are unresponsive to IL-2.

ADME

Interleukin-2 (as recombinant IL-2, *aldesleukin*) is administered intravenously. The serum $t_{1/2}$ of IL-2 after intravenous administration has an

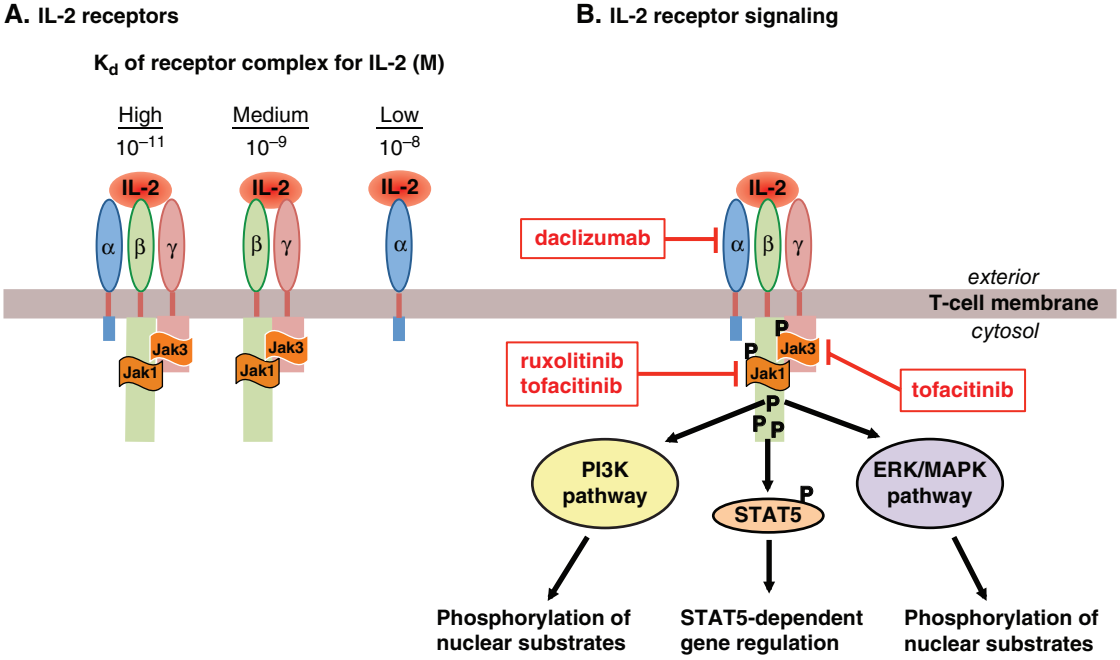


Figure 72–4 *A pharmacologist's view of IL-2 receptors, their cellular signaling pathways, and their inhibition.* **A.** *IL-2 receptors.* The IL-2 receptor has three components: an α chain, a 55-kDa protein (CD25) involved mainly in IL-2 binding; a β chain, a 75-kDa protein that binds to Jak1; and a γ chain, a 64-kDa protein that signals via Jak3. These components combine as shown to produce receptors of varying affinities and with different capacities for intracellular signaling. The γ chain is a component of many cytokine receptors (IL-2, IL-4, IL-7, IL-9, IL-15, and IL-21). Resting T cells have a low density of IL-2Rα; however, activation of the TCR (see Figure 38–2) and IL-2, itself, induce IL-2Rα expression. IL-2Rα can also be induced by a large variety of cytokines, viruses, and activators of protein kinase C. The β and γ monospans are constitutively expressed on many lymphohematopoietic cells, and their expression can also be regulated. Tumor cells generally lack receptor expression and are unresponsive to IL-2. **B.** *IL-2 receptor signaling.* The formation of the quaternary complex of IL-2 and the α, β, and γ units of IL-2R suffice to activate intracellular signaling. The IL-2R monospans shown in **A** lack the protein tyrosine kinase activities common to many monospanning hormone receptors. Rather, these receptors respond to IL-2 binding at the extracellular domain by differentially recruiting Jak1 and Jak3 to their cytosolic domains, Jak1 at IL-2Rβ and Jak3 at IL-2Rγ. Heterodimerization of β and γ cytosolic domains leads to activation of Jak1 and Jak3. The Jaks transactivate and also phosphorylate key tyrosines on IL-2Rβ. These phosphotyrosines (**P**) permit important protein-protein interactions that direct downstream signaling:

- Binding of an SHC adapter/scaffolding protein that is the platform for activation of the Ras/MAPK pathway and the PI3K pathway
- Recruitment and phosphorylation of STAT5 (to a lesser extent STATs 1 and 3), leading to STAT-dependent gene regulation

IL-2/IL-2R signaling can be therapeutically targeted by inhibitors that interact with IL-2Rα, Jak1, and Jak3 (red boxes and T bars), as is discussed in Chapters 39, 71, and 75.

α phase of about 13 min and a β phase of about 90 min. IL-2 is excreted in the urine as an inactive metabolite.

Therapeutic Uses

Aldesleukin possesses the biological activities of native human IL-2. The drug is approved for use in patients with metastatic RCC and metastatic melanoma. High-dose IL-2 produces an overall response rate of 15% to 25% in patients with RCC; 8% achieve a complete response. In patients with metastatic melanoma, high-dose IL-2 induces an overall response rate of 16% to 22%, and 6% achieve a complete response. In both instances, responses last a median of about 5 years. Low-dose IL-2 also produces responses, but few are complete responses, and the duration appears less than with high-dose IL-2.

Adverse Effects

Interleukin-2 toxicities are dominated by the capillary leak syndrome, in which intravascular fluid leaks into the extravascular space, producing hypotension, edema, respiratory difficulties, confusion, tachycardia, oliguric renal failure, and electrolyte problems. Other adverse effects include fever, chills, malaise, nausea, vomiting, and diarrhea. Laboratory abnormalities include thrombocytopenia, abnormal liver function tests, lymphopenia and eosinophilia. Most patients develop a pruritic skin rash. Hypothyroidism may occur. Cardiac arrhythmias are a rare complication. These toxicities can be life threatening, yet nearly all are reversible within 24 to 48 h of discontinuing therapy. Patients should have normal renal and hepatic function, adequate respiratory reserve (FEV1 > 2 L), and a normal exercise tolerance test before beginning therapy and should be closely supervised in an inpatient facility capable of care at the intensive care unit level during drug administration.

IV. Targeting Cancer Cell Surface Molecules to Engage Immune Cells

HISTORY

The first therapeutic antibody approved for oncology patients in 1997, *rituximab*, is a chimeric mouse/human monoclonal antibody that binds to the CD20 surface antigen on B cells and has improved outcomes for patients with B-cell malignancies. CD20 is a glycosylated tetra-transmembrane phosphoprotein with intracellular N- and C-terminal regions and a small and a large extracellular loop expressed on developing B cell and many B-cell malignancies. *Rituximab* binds to the large extracellular loop, whereas *ofatumumab*, approved in 2009, binds to a distinct epitope on the large extracellular loop as well as sites on the small loops. *Obinutuzumab*, approved in 2013, is derived from *rituximab* with modifications of its glycosylation to

improve immune effector effects. *Bi*specific *T*-cell–*e*ngaging (BiTE) antibody targeting of cancer cell surface molecules is based on antibody engineering concepts that won first approval with *blinatumomab* in 2014: *Blinatumomab* bridges cytotoxic T cells and cancer cells independent of MHC-bound antigen recognition. BiTE antibodies are under development for the treatment of a range of cancers; other bifunctional antibody approaches under development have been reviewed by Goebeler and Bargou (2020). See also Chapter 38.

CD19, CD20, CD52, CD38, CCR4, GD2, and SLAMF7

Rituximab (CD20)

Rituximab is a chimeric murine/human monoclonal IgG1 antibody that targets the CD20 B-cell surface antigen. On binding to the large extracellular loop of CD20, *rituximab* mediates B-cell lysis mostly through CDC and ADCC. In patients with non-Hodgkin lymphoma (NHL), administration of *rituximab* depletes circulating and tissue-based B cells within the first 3 weeks, with sustained depletion for 6 to 9 months after treatment in a majority of patients. Median B-cell levels return to normal by 12 months.

ADME

The drug is administered by intravenous infusion both as a single agent and in combination with chemotherapy. *Rituximab* has a $t_{1/2}$ of about 22 days. As a single agent, it is given weekly for 4 weeks, with maintenance dosing every 3 to 6 months. In combination regimens, the drug may be administered every 3 to 4 weeks, with chemotherapy, for up to eight doses. The rate of infusion should be increased slowly to prevent serious hypersensitivity reactions. Pretreatment with antihistamines, *acetaminophen*, and glucocorticoids decreases the risk of hypersensitivity reactions. Patients with large numbers of circulating tumor cells (as in chronic lymphocytic leukemia [CLL]) are at increased risk for tumor lysis syndrome; in these patients, the initial dose should be no more than 50 mg/m^2 on day 1 of treatment, and patients should receive standard tumor lysis prophylaxis.

Therapeutic Uses

Rituximab is approved as a single agent for relapsed indolent lymphomas and significantly enhances response and survival in combination with chemotherapy for the initial treatment of diffuse large B-cell lymphoma. *Rituximab* improves response rates when added to combination chemotherapy for other indolent B-cell NHLs, including CLL, mantle cell lymphoma, Waldenström macroglobulinemia, and marginal zone lymphomas. Maintenance of remission with *rituximab* delays time to progression and improves overall survival in indolent NHL. It is increasingly used for treatment of autoimmune diseases such as rheumatological disease, thrombotic thrombocytopenic purpura, autoimmune hemolytic anemias, cryoglobulin-induced renal disease, and multiple sclerosis.

Resistance and Adverse Effects

Resistance to *rituximab* may emerge through downregulation of CD20, impaired antibody-dependent cellular cytotoxicity, decreased complement activation, limited effects on signaling and induction of apoptosis, and inadequate blood levels. Polymorphisms in two of the receptors for the antibody Fc region responsible for complement activation may predict the clinical response to *rituximab* monotherapy in patients with follicular lymphoma but not in CLL. *Rituximab* infusion reactions can be life threatening, but with pretreatment, reactions are usually mild and limited to fever, chills, throat itching, urticaria, and mild hypotension. All respond to decreased infusion rates and antihistamines. Uncommonly, patients may develop severe mucocutaneous skin reactions, including Stevens-Johnson syndrome. *Rituximab* may cause reactivation of hepatitis B virus or, rarely, JC (John Cunningham) virus (with progressive multifocal leukoencephalopathy). Hypogammaglobulinemia and autoimmune syndromes (idiopathic thrombocytopenic purpura, thrombotic thrombocytopenic purpura, autoimmune hemolytic anemia, pure red cell aplasia, and delayed neutropenia) may appear 1 to 5 months after administration.

Ofatumumab (CD20)

Ofatumumab is a human IgG1 that binds to CD20 at sites on the large and small extracellular loops of CD20, distinct from the site targeted by *rituximab*. Antibody binding results in B-cell lysis via CDC and ADCC. *Ofatumumab* is approved for treating patients with CLL after failure of *fludarabine* and *alemtuzumab* (see below). A complex dosing scheme is used, beginning with small (300-mg) doses on day 1, followed by higher doses (up to 2 g per week) later. *Ofatumumab*'s primary toxicities consist of immunosuppression and opportunistic infection, hypersensitivity reactions during antibody infusion, and myelosuppression. Blood counts should be monitored during treatment. Rarely, patients may develop reactivation of viral infections. The drug should not be administered to patients with active hepatitis B infection; liver function should be monitored in hepatitis B carriers.

Obinutuzumab (CD20)

Obinutuzumab is a humanized IgG1 that binds to CD20. Antibody binding mediates B-cell lysis through CDC and ADCC. Glycosylation of the Fc portion of *obinutuzumab* is altered to enhance binding of effector immune cells. This is the first glycoengineered antibody approved by the FDA and is used for the treatment of patients with CLL and follicular lymphoma in combination with chemotherapy.

Alemtuzumab (CD52)

Alemtuzumab is a humanized IgG1 that binds to CD52 found on the surface of a subset of normal neutrophils and on all B and T lymphocytes, on testicular elements and sperm, and on most B and T cell lymphomas. Consistently high levels of CD52 expression on lymphoid tumor cells and the lack of CD52 modulation with antibody binding make this antigen a potential target for unconjugated monoclonal antibodies. *Alemtuzumab* can induce tumor cell death through ADCC and CDC.

ADME

Alemtuzumab is administered intravenously in dosages of 30 mg/day, three times per week. Premedication with *diphenhydramine* and *acetaminophen* should precede drug infusion. Dosing should begin with a low-dose infusion, followed by an increased dose 2 days later and, if well tolerated, the highest dose 2 days later. The drug has an initial mean $t_{1/2}$ of 1 h, but after multiple doses, the $t_{1/2}$ extends to 12 days, and steady-state plasma levels are reached at approximately week 6 of treatment.

Therapeutic Uses

Alemtuzumab is approved as a single agent for the treatment of B-cell CLL. Clinical activity has been demonstrated in both B and T cell low-grade lymphomas and CLL, including patients with disease refractory to purine analogues. In chemotherapy-refractory CLL, overall response rates are about 40%, with complete responses of 6% in multiple series. Response rates in treatment-naïve patients with CLL are higher (overall response rates of 83% and complete responses of 24%). *Alemtuzumab* is also approved for the treatment of patients with relapsing forms of multiple sclerosis under a different label.

Adverse Effects

Serious toxicities include acute infusion reactions and depletion of normal neutrophils and T cells. Myelosuppression, with depletion of all blood lineages, occurs in the majority of patients and may represent either direct marrow toxicity or autoimmune responses. Immunosuppression leads to a significant risk of fungal, viral, and other opportunistic infections, particularly in patients who have previously received purine analogues. Patients should receive prophylaxis against *Pneumocystis carinii* and herpesvirus during treatment and for at least 2 months following therapy with *alemtuzumab*. Because reactivation of cytomegalovirus infections

may follow antibody use, patients should be monitored for symptoms and signs of viremia, hepatitis, and pneumonia. $CD4^+$ T-cell counts may remain profoundly depleted (<200 cells/μL) for 1 year.

Tafasitamab (CD19)

Tafasitamab is directed to CD19, a B-cell surface antigen, and carries a modified Fc domain (IgG1/2 hybrid) to enhance Fc-mediated cytotoxic function (ADCC and ADCP) toward B-cell malignancies. It was approved in 2020 for use in combination with *lenalidomide* (see Chapter 71).

Naxitamab (GD2)

Naxitamab is a humanized IgG1 that binds to the sialic acid–containing glycosphingolipid GD2 (disialoganglioside) overexpressed on the surface of neuroblastoma and other peripheral and central nervous system neuroectodermal cells. Binding to GD2 induces cell death through CDC and ADCC. *Naxitamab* was approved in 2020 for the treatment of relapsed neuroblastoma in combination with granulocyte-macrophage colony-stimulating factor (GM-CSF). Severe neurotoxicity, including neuropathic pain, is a potential adverse effect.

Dinutuximab (GD2)

Dinutuximab is a chimeric human/mouse IgG1 that targets GD2 and has a similar activity and adverse effect profile as *naxitamab* (above). It is used in combination with GM-CSF, IL-2, and retinoic acid (RA) for the treatment of neuroblastoma.

Daratumumab (CD38)

Daratumumab is a human IgG1 that binds to CD38, a transmembrane glycoprotein expressed on the surface of hematopoietic cells, including multiple myeloma (MM); binding to CD38-expressing tumor cells induces cell death through CDC and ADCC. *Daratumumab* was approved for the treatment of patients with MM in combination with other agents such as *lenalidomide* or *bortezomib* and *dexamethasone* (see Chapter 71). The most frequent adverse effects are upper respiratory tract infection, neutropenia, and thrombocytopenia.

Isatuximab (CD38)

Isatuximab is a chimeric human/mouse IgG1 that binds to CD38 with an activity profile and adverse effects similar to *daratumumab* (see above). It was approved in 2020 for the treatment of patients with MM.

Elotuzumab (SLAMF7)

Elotuzumab is a humanized IgG1 that targets SLAMF7 (CD319), a cell surface glycoprotein expressed on MM and NK cells. Binding of *elotuzumab* to the antigen on myeloma cells induces ADCC-dependent tumor cell lysis. ADCC is enhanced by stimulation of the SLAMF7 pathway in NK cells by *elotuzumab*. It is approved for the treatment of patients with MM who have received one to three prior therapies and used in combination with *lenalidomide* and *dexamethasone*.

Mogamulizumab (CCR4)

Mogamulizumab is a humanized IgG1 antibody that targets CCR4, a G protein–coupled receptor that regulates trafficking of lymphocytes and is expressed in some T-cell malignancies. It was approved in 2018 for treatment of relapsed or refractory mycosis fungoides and Sézary disease, types of cutaneous T-cell lymphomas. Adverse effects include moderate to severe rashes and potential bone marrow suppression.

Bispecific Antibodies: CD19 and CD3; EGFR and MET

The introduction of *blinatumomab*, discussed below, opened a distinct category—BiTE antibody targeting of cancer cell surface molecules. Notably, *blinatumomab* bridges cytotoxic T cells and cancer cells independent of MHC-bound antigen recognition. *Amivantamab* is a recently approved bispecific antibody that targets two receptors on cancer cells, EGFR and MET (mesenchymal-epithelial transition factor), and is discussed further below. Bifunctional antibody approaches under development are reviewed by Goebeler and Bargou (2020).

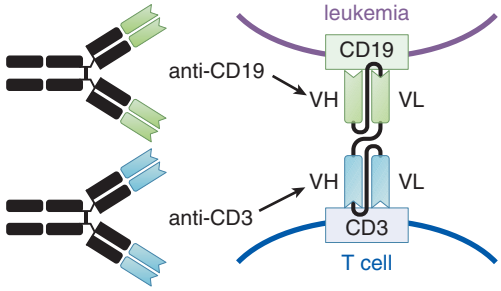


Figure 72–5 *The bispecific T-cell–engaging (BiTE) fusion antibody, blinatumomab.* *BiTE* consists of the VL and VH domains of an anti-CD19 IgG joined via an amino acid link with the VL and VH domains of an anti-CD3 IgG, resulting in a molecular mass of 54 kDa (IgGs are ~150 kDa). The bispecific antibody generates a "synapse" between cytotoxic T cells and leukemia cells that express CD19 on their surface.

Blinatumomab (CD19 and CD3)

Blinatumomab is a first-in-class bispecific antibody that recognizes sites on the surfaces of B cells and T cells, thereby enabling a patient's T cells to recognize malignant B cells (Figure 72–5). To generate this fusion protein, the VH and VL antigen binding domains in IgG1s raised against CD3 (part of the TCR) and against CD19 were isolated. The domains that bind to CD3 were linked to the domains that recognize CD19 on B cells. Due to this bispecific targeting, *blinatumomab* can mediate the formation of a cytotoxic connection between CD19-expressing tumor cells and T cells, resulting in the lysis of CD19-positive cells, release of inflammatory cytokines, and proliferation of T cells. *Blinatumomab* was approved in 2014 for the treatment of acute lymphoblastic leukemia (ALL) including minimal residual disease. Major adverse effects are cytokine release syndrome, neurological toxicities, neutropenic fever, and sepsis.

Amivantamab (EGFR and MET)

Amivantamab is a bispecific IgG1 that binds to the extracellular domains of EGFR and MET. Binding can disrupt ligand signaling through the receptors and initiate targeting of cancer cells by the immune system through ADCC. *Amivantamab* was approved in 2021 for the treatment of patients with locally advanced or metastatic NSCLC with EGFR exon 20 insertion mutations. Adverse effects include pneumonitis, dermatological reactions, and ocular toxicity.

V. Cytotoxin Conjugates With Antibodies or Cytokines

HISTORY

The concept of a "magic bullet," the translation of "Zauberkugel," was introduced by Paul Ehrlich in 1913 to describe his goal of developing cytotoxic drugs that selectively target and destroy cancer cells without impacting healthy tissues. Anticancer drugs developed in the century since Ehrlich's work still aim for this goal (Strebhardt and Ullrich, 2008). The drugs discussed below follow this concept: Cytotoxic drugs are linked to monoclonal antibodies to form antibody-drug conjugates (ADCs) that direct toxic payloads to cancer cells that overexpress the antibody target. Binding of an ADC to the surface antigen

on targeted cells can lead to its internalization, intracellular cleavage of the conjugate, and the release of the cytotoxic payload. This will result in the death of the targeted cancer cells. As a "bystander" effect, the release of the cytotoxic payload may impact cells in the immediate tumor microenvironment for additional therapeutic benefit (Jabbour et al., 2021). For details of ADCs, see Figure 38-10 and Table 38-2.

Antibody-Drug Conjugates (ADCs)

Gemtuzumab Ozogamicin (CD33)

Mechanism of Action

Gemtuzumab ozogamicin is a first-generation humanized monoclonal antibody against CD33 covalently linked to a semisynthetic derivative of *calicheamicin*, a potent antitumor antibiotic that induces double-strand DNA breaks and causes cell death. The CD33 antigen is present on most hematopoietic cells, on more than 80% of acute myelocytic leukemia (AML) cells, and on most myeloid cells in patients with myelodysplastic syndrome. Other normal cell types lack CD33 expression, making this attractive for targeted therapy. CD33 has no known biological function, although monoclonal antibody cross-linking inhibits normal and myeloid leukemia cell proliferation. Following its binding to CD33, *gemtuzumab ozogamicin* undergoes endocytosis; cleavage of *calicheamicin* from the antibody takes place within the lysosome. The potent toxin then enters the nucleus, binds in the minor groove of DNA, and causes double-strand DNA breaks and cell death. *Ozogamicin* is conjugated to *gemtuzumab* through surface lysine residues (see Figure 72–1) at an average ratio of two to three cytotoxic molecules per antibody.

ADME

The antibody conjugate produces a 30% complete response rate in patients with relapsed AML when administered at a dose of 9 mg/m^2 for up to three doses at 2-week intervals. The $t_{1/2}$ of total and unconjugated *calicheamicin* are 41 and 143 h, respectively. Following a second dose, the $t_{1/2}$ of drug-antibody conjugate increases to 64 h. Most patients require two to three doses to achieve remission.

Therapeutic Use

The drug was initially approved in 2000, withdrawn due to toxicity, and approved in 2020 for CD33-positive AML. Primary toxicities include myelosuppression and hepatocellular damage in 30% to 40% of patients, manifested by hyperbilirubinemia and enzyme elevations. This agent also causes a syndrome that resembles hepatic veno-occlusive disease when patients subsequently undergo myeloablative therapy or when treatment follows high-dose chemotherapy.

Ado-trastuzumab Emtansine (HER2)

Ado-trastuzumab emtansine is a first-generation antibody-cytotoxic drug conjugate that was approved in 2013. *Ado-trastuzumab-DM1* combines the HER2-targeted properties of *trastuzumab* with the antimicrotubule agent DM1 (derived from *maytansine*), allowing preferential intracellular cytotoxic drug delivery to cancer cells in the treatment of HER2-positive breast cancer. The complex binds to HER2 and enters the cell by receptor-mediated endocytosis. The DM1 (*emtansine*) is conjugated to surface lysine residues on the antibody (see Figure 72–1) and released into the cytosol as the complex is cleaved by proteases in lysosomes. The DM1 disrupts microtubule-dependent events, causing mitotic arrest, disruption of intracellular trafficking, and apoptosis. The adverse effects of treatment include cardiac dysfunction as described for unconjugated *trastuzumab*. Hepatotoxicity is an additional risk as well as bone marrow suppression.

Fam-trastuzumab Deruxtecan (HER2)

Fam-trastuzumab deruxtecan is an HER2-targeted antibody conjugated to a *camptothecin* derivative (topoisomerase I inhibitor) via the hinge cysteine residues (see Figure 72–1). It was approved for the treatment of patients with locally advanced or metastatic HER2-positive breast or gastric cancer in 2019 and 2021, respectively. Interstitial lung disease and pneumonitis, including fatal cases, have been reported as serious adverse effects as well as cardiotoxicity.

Inotuzumab Ozogamicin (CD22)

Inotuzumab ozogamicin is a humanized anti-CD22 antibody-cytotoxic drug conjugate approved in 2017 for the treatment of relapsed or refractory ALL. After binding to CD22-positive tumor cells and cellular uptake, the cytotoxic is released via hydrolytic cleavage. Ozogamicin, a derivative of calicheamicin, binds to the minor grove in DNA, induces double-strand breaks, and causes cell death; conjugation to the antibody is via surface lysine residues (see Figure 72–1). Adverse effects include hepatotoxicity, including hepatic veno-occlusive disease, and bone marrow suppression.

Brentuximab Vedotin (CD30)

Brentuximab vedotin is an anti-CD30 IgG1 monoclonal antibody linked with the microtubule-disrupting agent monomethyl auristatin E (MMAE). CD30 is expressed on a number of malignant cells and is especially prevalent in Hodgkin and anaplastic lymphoma. Binding of the antibody to CD30-expressing cells is followed by internalization and the intracellular release of MMAE via proteolytic cleavage. MMAE disrupts the microtubule network, inducing cell cycle arrest and apoptotic cell death. MMAE is mostly metabolized by CYP3A, and coadministration with CYP3A4 inhibitors will increase exposure to the MMAE toxin. Coadministration with CYP3A4 inducers (e.g., *rifampicin*) reduces exposure. The most common adverse reactions are neutropenia, peripheral sensory neuropathy, fatigue, nausea, anemia, upper respiratory tract infection, diarrhea, pyrexia, rash, thrombocytopenia, cough, and vomiting.

Belantamab Mafodotin (BCMA)

Belantamab mafodotin is a humanized IgG1 targeted to B-cell maturation antigen (BCMA) and conjugated via cysteine residues (see Figure 72–1) to the cytotoxic monomethyl auristatin F (MMAF). Upon binding to BCMA, *belantamab mafodotin* is internalized and MMAF is released. MMAF can bind to tubulin and disrupt the microtubule network and cause cell death. It was approved 2020 for the treatment of patients with relapsed or refractory MM. Adverse effects include damage to the corneal epithelium that impacts vision and may cause loss of vision.

Polatuzumab Vedotin (CD79b)

Polatuzumab vedotin is a CD79b-directed antibody conjugated with the antimitotic MMAE via cysteine residues (see Figure 72–1). It was approved in 2019 for the treatment of patients with relapsed or refractory diffuse large B-cell lymphoma in combination with *bendamustine* (see Chapter 70) and *rituximab*.

Enfortumab Vedotin (Nectin4)

Enfortumab vedotin is a Nectin4-directed antibody conjugated with the antimitotic MMAE via cysteine residues (see Figure 72–1). It was approved in 2019 for the treatment of locally advanced or metastatic urothelial cancer.

Sacituzumab Govitecan (Trop-2)

Sacituzumab govitecan is a Trop-2–directed antibody conjugated to a *camptothecin* derivative (topoisomerase I inhibitor) via the hinge cysteine residues (see Figure 72–1). It was approved in 2020 for the treatment of metastatic triple-negative breast cancer and metastatic urothelial cancer. It can cause severe neutropenia and severe diarrhea.

Moxetumomab Pasudotox (CD22)

Moxetumomab pasudotox is a CD22-directed antibody fused to a truncated form of the *Pseudomonas* exotoxin A. Upon entering cells, the exotoxin inhibits protein synthesis through ADP-ribosylation of elongation

factor 2 and causes cell death. It was approved in 2018 for the treatment of relapsed or refractory hairy cell leukemia.

Radioimmunoconjugates (CD20)

Radioimmunoconjugates provide targeted delivery of radionuclides to tumor cells. ^{131}Iodine (^{131}I) is a favored radioisotope because it is readily available, relatively inexpensive, and easily conjugated to a monoclonal antibody. The γ particles emitted by ^{131}I can be used for both imaging and therapy, but protein-iodine conjugates have the drawback of releasing free ^{131}I and ^{131}I-tyrosine into the blood and thus present a health hazard to people in contact with the patient. The β-emitter ^{90}Y (yttrium) has emerged as an alternative to ^{131}I, based on its higher energy and longer path length. Thus, it may be more effective in tumors with larger diameters. It also has a short $t_{1/2}$ and remains conjugated, even after endocytosis, providing a safer profile for outpatient use. Currently available radioimmunoconjugates consist of murine monoclonal antibodies against CD20 conjugated with ^{131}I (tositumomab) or ^{90}Y (ibritumomab). Both drugs have shown response rates in relapsed lymphoma of 65% to 80%. Adverse effects include antibody-related hypersensitivity, bone marrow suppression, and secondary leukemias. Both radioconjugates targeting CD20 on lymphomas have been approved; however, concerns over toxicity have limited their use.

Cytokine-Cytotoxin Conjugates

Tagraxofusp

Tagraxofusp is a fusion protein of human IL-3 and truncated diphtheria toxin that is directed to cells expressing the IL-3 receptor (CD123), where it will inhibit protein synthesis and cause cell death. CD123 is overexpressed in leukemic stem and more differentiated cells from ALL, AML, hairy cell leukemia, and Hodgkin lymphoma as well as plasmacytoid dendritic cell neoplasm, the malignancy it was FDA-approved in 2018. Potentially life-threatening capillary leak syndrome is the major adverse event.

VI. Cellular Therapy and Vaccines

RATIONALE

To meet the challenge of poorly immunogenic cancers, the potential of T-cell–based therapies is harnessed by genetically engineering T cells to redirect them to tumor cell surface antigens (Carpenito et al., 2009). The engineered T cells contain a chimeric antigen receptor (CAR) that consists of the antigen-binding domain of a monoclonal antibody to confer recognition of the targeted tumor antigen coupled to intracellular domains capable of activating T cells (Figure 72–6). When expressed in T cells harvested from the patient, these CARs recognize cell surface antigens and activate T cells independent of antigen presentation by an MHC molecule. In initial studies, CAR-expressing T cells were targeted to NY-ESO-1, a tumor antigen of the cancer/testis family that is highly expressed on many poor-prognosis melanoma cells. The CAR T cells showed long-lasting efficacy (>3 years) in some patients with metastatic melanoma. CAR targeting CD19, a B-cell antigen, also resulted in striking efficacy in patients with B-cell leukemias (Klebanoff et al., 2016). The FDA approved CD19 CAR T cells (*tisagenlecleucel-T*; previously named CTL019) in 2017 based on data showing >80% remission rate for relapsed or refractory B-cell ALL during treatment. Modification of CAR constructs to improve antigen recognition, efficacy, and safety of engineered T cells is an area of active research (Hou, 2021). In addition to this genetic engineering of T cells, the technique of isolating and expanding immune cells that recognize cancer antigens is used in adoptive, cell-based immunotherapy (Restifo et al., 2012).

The names for CAR T-cell therapies contain the syllables "gen" for genetic modification, "leu" for leukocytes as the targeted cell type, and "cel" for cell-based therapy (e.g., tisa*gen*lec*leucel*).

CAR T Cells

Five CAR T therapies have been approved by the FDA, with four of them targeting CD19 in relapsed or refractory B-cell ALL or B-cell NHL and one targeting BCMA. CAR T-cell treatment is not approved for the treatment of central nervous system lymphoma.

Tisagenlecleucel

Tisagenlecleucel, FDA-approved in 2017, was the first CD19 CAR-expressing autologous T cell–based immunotherapy (see Figure 72–6) for the treatment of patients with ALL that is refractory or in second or later relapse. The CAR is composed of a murine single-chain antibody fragment that recognizes CD19 and is fused to intracellular signaling domains from 4-1BB (CD137) and CD3 zeta. The CD3 zeta component is critical for initiating T-cell activation and antitumor activity, while 4-1BB enhances the cell expansion and survival (Ellis et al., 2021). Other CD19-directed CAR T immunotherapies approved since 2017 are *lisocabtagene maraleucel*, *brexucabtagene autoleucel*, and *axicabtagene ciloleucel.*

Adverse Effects

The most common adverse effects of CAR T therapy are cytokine release syndrome (indicated by fever, nausea, headache, rapid heartbeat, low blood pressure, and dyspnea) and potentially life-threatening neurological toxicities. *Tocilizumab*, a humanized IgG1 against the IL-6 receptor, blocks signaling of the IL-6 cytokine that is implicated in the pathogenesis of cytokine release syndrome. IL-6 plays an important role in inflammatory disease and is discussed in Chapter 39; *tocilizumab* is approved for treatment of CAR T cell–induced severe cytokine release syndrome and may be used in combination with corticosteroids.

Resistance to Anti-CD19 CAR T Treatment

Loss of the CD19 antigen is a frequent cause of treatment resistance. Because CD22 expression is typically retained after the loss of CD19 from the tumor cell therapy with anti-CD19, CAR T cells can be used to overcome the resistance (Fry et al., 2018).

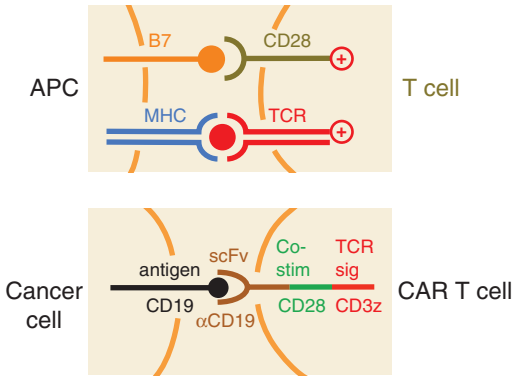


Figure 72–6 *Chimeric antigen receptor (CAR) T-cell activation by a target antigen on cancer cells.* T cells are harvested from a cancer patient and transduced with a CAR expression vector. Elements in a CAR include an extracellular, single-chain antibody fragment of the variable domain (scFv) that recognizes the targeted antigen; intracellular costimulatory domain (e.g., from CD28); and intracellular signaling domain of the TCR. CD19 is shown as an antigen example. APC-mediated stimulation of T cells is depicted for comparison (see Figure 72–3).

Idecabtagene Vicleucel

Idecabtagene vicleucel is an anti-BCMA CAR T immunotherapy that was FDA-approved in 2021 for the therapy of MM.

Sipuleucel-T

Sipuleucel-T is a cell-based approach designed to induce an immune cell response against prostate acidic phosphatase (PAP), which is expressed in most prostate cancers. This cellular immunotherapy is designed to harness a T-cell response from a patient's peripheral blood cells, which are isolated by leukapheresis. The cells are exposed to a human recombinant protein, PAP–GM-CSF. APCs present among the blood cells are thought to take up and process PAP and direct the immune response toward the antigen. A patient receives the treated cells by infusion on three occasions at 2-week intervals. The approach is modeled on adoptive immunotherapy (Restifo et al., 2012). The treatment is approved for patients with minimally symptomatic, hormone-refractory, metastatic prostate cancer. Adverse effects include acute infusion reactions, which can be reduced by premedication with acetaminophen and antihistamine.

Lifileucel

Lifileucel is an adoptive cellular therapy approach that uses tumor-infiltrating lymphocytes isolated from patients. The approach is technically challenging, and clinical trials in patients with melanoma and other cancers in combination with immune checkpoint inhibitors as well as IL-2 are ongoing (Curti and Faries, 2021).

Cancer Vaccines

Vaccination against cancer antigens remains an area of active research. mRNA-based vaccines that code for neoantigens prevalent in a given cancer type or are tailored to the neoantigen composition of each patient's cancer hold significant promise, especially in combination with immune checkpoint inhibitor treatment (Sahin and Türeci, 2018; Sahin et al., 2020). It is noteworthy that the development of mRNA-based approaches in cancer immunotherapy over the past decade (Pastor et al., 2018) was the basis for the rapid generation of highly effective vaccines against COVID-19 that helped to blunt the worldwide pandemic. Currently, only an oncolytic herpesvirus, *T-VEC*, that functions as a cancer vaccine is FDA approved, in this case for the treatment of patients with metastatic melanoma.

T-VEC

T-VEC (*talimogene laherparepvec*) is an oncolytic herpesvirus that replicates within tumors and expresses GM-CSF. Tumor antigens are released after virally induced cell death, and the presence of GM-CSF can promote an antitumor immune response. T-VEC was approved in 2015 for the local treatment of unresectable but injectable cutaneous and nodal lesions in patients with melanoma. T-VEC may synergize with immune checkpoint inhibitor treatment; the combination therapy is under study. Adverse effects are influenza-like symptoms and injection site pain.

VII. Other Proteins

Colony-Stimulating Factors

Many agents used for cancer chemotherapy suppress the production of multiple types of hematopoietic cells, and bone marrow suppression can limit the delivery of chemotherapy on schedule and at prescribed doses. The availability of recombinant growth factors for erythrocytes (i.e., erythropoietin), granulocytes (i.e., G-CSF), and granulocytes and macrophages (i.e., GM-CSF) has advanced the ability to use combination therapy or high-dose cytotoxic therapy with diminished complications such as febrile neutropenia (see Chapter 70) or erythropenia when treating von Hippel-Lindau (VHL)-related disease using the HIF-2 inhibitor *belzutifan* (see Chapter 71). Chapter 45 presents the basic pharmacology of hematopoietic growth factors.

Asparaginase

L-Asparaginase

Bacterial *L-asparaginase* hydrolyzes the amino acid asparagine. It has been used in the treatment of ALL since initial approval in 1978. Some leukemic cells depend on exogenous asparagine because they lack asparagine synthase and cannot survive asparagine depletion, whereas most normal cells can synthesize asparagine from glutamine. Treatment reduces circulating asparagine levels by greater than 90%. Available formulations of *asparaginase* are isolated from *Escherichia coli*, including a polyethylene glycol modified enzyme with prolonged duration of activity, and the enzyme from *Erwinia chrysanthemi* (approved June 2021) for patients who have developed hypersensitivity to *E. coli*–derived *asparaginase*. Adverse reactions include hepatic toxicity and pancreatitis.

Drug Facts for Your Personal Formulary: *Pathway-Targeted Therapies*

Drug	Therapeutic Use	Clinical Pharmacology and Tips
Section I: Growth Factors and Receptors		
Epidermal Growth Factor Receptor (EGFR)		
Monoclonal Antibody EGFR Inhibitors: Intravenous Administration		
Cetuximab *(chimeric human/mouse IgG1)*	• Metastatic colorectal cancer with wild-type KRAS in combination with chemotherapy • HNSCC in combination with radiation or cisplatin	• Skin rash, diarrhea, interstitial lung disease • Rare: infusion reaction, cardiopulmonary arrest, hypomagnesemia
Panitumumab *(human IgG2)*	• Metastatic colorectal cancer with wild-type KRAS in combination with chemotherapy	• Side effects similar to those of cetuximab
Necitumumab *(human IgG1)*	• Metastatic NSCLC in combination with chemotherapy	• Side effects similar to those of cetuximab
Human Epidermal Growth Factor Receptor 2 (HER2)		
Monoclonal Antibody HER2 Inhibitors: Intravenous Administration		
Trastuzumab *(humanized IgG1)*	• HER2-positive breast cancer and gastric cancer • Combination with taxanes possible as chemotherapy	• Congestive heart failure (<5% reduced LVEF; <1% symptomatic); ↑ to 20% in combination with doxorubicin due to cardiotoxicity; monitor LVEF during and after • Acute infusion reaction, nausea, dyspnea, rashes
Pertuzumab *(humanized IgG1)*	• HER2-positive breast cancer in combination with trastuzumab and taxane	• Targets different HER2 domain than trastuzumab; prevents dimerization with other HERs • Side effects similar to those of trastuzumab
Platelet-Derived Growth Factor Receptor Inhibitors		
Olaratumab *(human IgG1)*	• Soft-tissue sarcoma in combination with doxorubicin	• Nausea, fatigue, gastrointestinal toxicity • Neutropenia, thrombocytopenia, elevated activated partial thromboplastin time, hypokalemia, hypophosphatemia
Section II: Tumor Angiogenesis		
Vascular Endothelial Growth Factor Inhibitors		
Bevacizumab *(humanized IgG1)*	• Metastatic colorectal cancer combined with chemotherapy (FOLFOX or FOLFIRI) • NSCLC combined with carboplatin and paclitaxel • Ovarian cancer combined with chemotherapy • RCC combined with interferon-α • Glioblastoma following prior therapy	• Hypertension, related congestive heart failure: monitor blood pressure and treat hypertension • Impaired wound healing: delay elective surgery for 1 month after the last dose; do not resume treatment for at least 1 month after surgery • Spontaneous gastrointestinal perforation
Ramucirumab *(human IgG1 to VEGFR2)*	• Metastatic colorectal cancer, advanced gastric adenocarcinoma, and NSCLC with disease progression on or after prior therapy as a single drug or in combination with chemotherapy	• Hypertension, diarrhea • Hemorrhage, gastrointestinal perforations • Impaired wound healing
Aflibercept *(extracellular domain of VEGFR1/2 fused to Fc portion of human IgG1)*	• Metastatic colorectal cancer in combination with FOLFIRI chemotherapy following FOLFOX	• Soluble trap for VEGF receptor ligands • Hypertension, diarrhea, impaired wound healing • Increased risk of hemorrhage, gastrointestinal perforation • Pathophysiology: increased circulating extracellular domain of VEGFR and decreased VEGF are hallmarks of preeclampsia
Section III: Activating Immune Cells		
Immune Checkpoint Inhibitors		
Ipilimumab *(anti–CTLA-4 human IgG1)*	• Metastatic melanoma as single agent or in combination with nivolumab (anti-PD-1) • Clinical trials with different cancers are ongoing	• Autoimmune inflammatory toxicities in majority of patients (>70%) • Adverse effects: skin (pruritus, rash, vitiligo), GI tract (diarrhea, colitis) • Less frequent: hepatitis, pneumonitis, hypophysitis, hypo- or hyperthyroidism, myocarditis
Tremelimumab *(anti–CTLA-4 human IgG2)*	• Clinical trials with different cancers are ongoing	• See above

Drug Facts for Your Personal Formulary: *Pathway-Targeted Therapies (continued)*

Drug	Therapeutic Use	Clinical Pharmacology and Tips
Nivolumab *(anti–PD-1 human IgG4)*	• Advanced melanoma that progressed after ipilimumab (anti–CTLA-4) • Previously treated NSCLC • Advanced RCC • Relapsed/refractory Hodgkin lymphoma	• Adverse effects: rash, fatigue, dyspnea, musculoskeletal pain, decreased appetite, cough, nausea, constipation • Immune-related serious adverse effects include pneumonitis, colitis, hepatitis, nephritis, renal dysfunction, hypophysitis, hypo- and hyperthyroidism
Pembrolizumab *(anti–PD-1 humanized IgG4)*	• Advanced melanoma that progressed after ipilimumab (anti–CTLA-4) • NSCLC that expresses PD-L1 and progressed under chemotherapy • NSCLC with wild-type EGFR and ALK and disease progression after chemotherapy • HNSCC with disease progression after chemotherapy	• Adverse effects: rash, fatigue, dyspnea, musculoskeletal pain, decreased appetite, cough, nausea, constipation • Immune-related serious adverse effects include pneumonitis, colitis, hepatitis, nephritis, renal dysfunction, hypophysitis, hypo- and hyperthyroidism
Atezolizumab *(anti–PD-L1 human IgG1)*	• NSCLC that is treatment resistant • Urothelial cancer that is locally advanced or metastatic	• Adverse effects: fatigue, decreased appetite, dyspnea, cough, nausea, musculoskeletal pain, constipation • In patients with urothelial cancer: urinary tract infections • Immune-related pneumonitis, colitis, hepatitis, nephritis, renal dysfunction, hypo- and hyperthyroidism, hypophysitis, adrenal insufficiency, pancreatitis, Guillain-Barré syndrome, severe infections
Section IV: Targeting Cancer Cells to Engage Immune Cells		
Antibodies Targeting Cell Surface Antigens		
Rituximab *(chimeric human/mouse IgG1 anti-CD20)*	• NHL • CLL • Rheumatological and other autoimmune disease, including multiple sclerosis	• Infusion-related toxicity with fever, rash, and dyspnea; B-cell depletion; late-onset neutropenia; risk of hypersensitivity reaction: use slow increase in infusion rate and antihistamines • Rare: severe mucocutaneous skin reaction, including Stevens-Johnson syndrome • Risk of tumor lysis syndrome in patients with high tumor burden in the circulation: use lower dose initially • Reactivation of hepatitis B virus or JC polyomavirus
Ofatumumab *(human IgG1 anti-CD20)*	• CLL after treatment failure	• Immunosuppression and opportunistic infections, hypersensitivity reaction during infusion and myelosuppression: monitor blood counts during treatment
Obinutuzumab *(humanized IgG1 anti-CD20)*	• CLL in combination with chemotherapy	• Frequent adverse effects: cytopenia, fever, cough, musculoskeletal disorders
Alemtuzumab *(Campath* or *Lemtrada; humanized IgG1 anti-CD52)*	• CLL (label: *Campath*) • Multiple sclerosis (label: *Lemtrada*)	• Infusion-related toxicity, T-cell depletion with increased infection; myelosuppression with pancytopenia • Antibiotic prophylaxis
Dinutuximab *(chimeric human/mouse anti-GD2)*	• High-risk neuroblastoma	• Infusion reaction • Nerve damage
Daratumumab *(human IgG1 anti-CD38)*	• MM in combination with lenalidomide or bortezomib	• Infusion reactions • Peripheral sensory neuropathy, upper respiratory tract infection
Elotuzumab *(humanized IgG1 anti-CD319)* (SLAMF7)	• MM after one to three prior therapies	• Infusion reaction
Blinatumomab *(bispecific anti-CD19 and anti-CD3)*	• Ph-negative relapsed or refractory B-cell precursor ALL	• Cytokine release syndrome, neurological toxicity, neutropenic fever

Bibliography

Atkins MB, et al. The state of melanoma: emergent challenges and opportunities. *Clin Cancer Res*, **2021**, *27*:2678–2697.

Balogh E, Nass SJ. Advancing progress in the development of combination cancer therapies with immune checkpoint inhibitors. Proceedings of a workshop of the National Academies of Sciences Engineering and Medicine. The National Academies Press, Washington, DC, **2019**, 1–89.

Callahan MK, et al. Targeting T cell co-receptors for cancer therapy. *Immunity*, **2016**, *44*:1069–1078.

Carpenito C, et al. Control of large, established tumor xenografts with genetically retargeted human T cells containing CD28 and CD137 domains. *Proc Natl Acad Sci USA*, **2009**, *106*:3360–3365.

Cha E, et al. Improved survival with T cell clonotype stability after anti–CTLA-4 treatment in cancer patients. *Sci Transl Med*, **2014**, *6*:238ra70.

Coley WB. The treatment of inoperable sarcoma by bacterial toxins (the mixed toxins of the *Streptococcus erysipelas* and the *Bacillus prodigiosus*). *J R Soc Med*, **1910**, *3*:1–48.

Curran MA, et al. PD-1 and CTLA-4 combination blockade expands infiltrating T cells and reduces regulatory T and myeloid cells within B16 melanoma tumors. *Proc Natl Acad Sci USA*, **2010**, *107*:4275–4280.

Curti BD, Faries MB. Recent advances in the treatment of melanoma. *N Engl J Med*, **2021**, *384*:2229–2240.

Dougan M, et al. Understanding and treating the inflammatory adverse events of cancer immunotherapy. *Cell*, **2021**, *184*:1575–1588.

Ellis GI, et al. Genetic engineering of T cells for immunotherapy. *Nat Rev Genet*, **2021**, *22*:427–447.

Esfahani K, et al. Moving towards personalized treatments of immune-related adverse events. *Nat Rev Clin Oncol*, **2020**, *17*:504–515.

Felix J, et al. Ipilimumab reshapes T cell memory subsets in melanoma patients with clinical response. *Oncoimmunology*, **2016**, *5*:1136045.

Folkman J. Tumor angiogenesis: therapeutic implications. *N Engl J Med*, **1971**, *285*:1182–1186.

Fry TJ, et al. CD22-targeted CAR T cells induce remission in B-ALL that is naive or resistant to CD19-targeted CAR immunotherapy. *Nat Med*, **2018**, *24*:20–28.

Goebeler ME, Bargou RC. T cell-engaging therapies—BiTEs and beyond. *Nat Rev Clin Oncol*, **2020**, *17*:418–434.

Goydel RS, Rader C. Antibody-based cancer therapy. *Oncogene*, **2021**, *40*:3655–3664.

Hamid O, et al. Safety and tumor responses with lambrolizumab (anti-PD-1) in melanoma. *N Engl J Med*, **2013**, *369*:134–144.

Hanahan D, Weinberg RA. Hallmarks of cancer: the next generation. *Cell*, **2011**, *144*:646–674.

Hodi FS, et al. Improved survival with ipilimumab in patients with metastatic melanoma. *N Engl J Med*, **2010**, *363*:711–723.

Hou AJ, et al. Navigating CAR-T cells through the solid-tumour microenvironment. *Nat Rev Drug Discov*, **2021**, *20*:531–550.

Huinen ZR, et al. Anti-angiogenic agents—overcoming tumour endothelial cell anergy and improving immunotherapy outcomes. *Nat Rev Clin Oncol*, **2021**, *18*:527–540.

Hurwitz H, et al. Bevacizumab plus irinotecan, fluorouracil, and leucovorin for metastatic colorectal cancer. *N Engl J Med*, **2004**, *350*:2335–2342.

Jabbour E, et al. The clinical development of antibody–drug conjugates—lessons from leukaemia. *Nat Rev Clin Oncol*, **2021**, *18*:418–433.

Johnson DB, et al. Fulminant myocarditis with combination immune checkpoint blockade. *N Engl J Med*, **2016**, *375*:1749–1755.

Judson I, van der Graaf WT. Sarcoma: olaratumab—really a breakthrough for soft-tissue sarcomas? *Nat Rev Clin Oncol*, **2016**, *13*:534–536.

King CR, et al. Amplification of a novel v-erbB-related gene in a human mammary carcinoma. *Science (New York, NY)*, **1985**, *229*:974–976.

Klebanoff CA, et al. Prospects for gene-engineered T cell immunotherapy for solid cancers. *Nat Med*, **2016**, *22*:26–36.

Korman AJ, et al. The foundations of immune checkpoint blockade and the ipilimumab approval decennial. *Nat Rev Drug Discov*, **2021**, *20*:xx–yy.

Kraehenbuehl L, et al. Enhancing immunotherapy in cancer by targeting emerging immunomodulatory pathways. *Nat Rev Clin Oncol*, **2022**, *19*:37–50.

Larkin J, et al. Five-year survival with combined nivolumab and ipilimumab in advanced melanoma. *N Engl J Med*, **2019**, *381*:1535–1546.

Liao W, et al. Interleukin-2 at the crossroads of effector responses, tolerance, and immunotherapy. *Immunity*, **2013**, *38*:13–25.

Mellman I, et al. Cancer immunotherapy comes of age. *Nature*, **2011**, *480*:480–489.

Nagy JA, et al. VEGF-A and the induction of pathological angiogenesis. *Ann Rev Pathol*, **2007**, *2*:251–275.

Pastor F, et al. An RNA toolbox for cancer immunotherapy. *Nat Rev Drug Discov*, **2018**, *17*:751–767.

Petroni G, et al. Immunomodulation by targeted anticancer agents. *Cancer Cell*, **2021**, *39*:310–345.

Restifo NP, et al. Adoptive immunotherapy for cancer: harnessing the T cell response. *Nat Rev Immunol*, **2012**, *12*:269–281.

Robert C, et al. Drug of the year: programmed death-1 receptor/programmed death-1 ligand-1 receptor monoclonal antibodies. *Eur J Cancer*, **2013**, *49*:2968–2971.

Robert C, et al. Anti-programmed-death-receptor-1 treatment with pembrolizumab in ipilimumab-refractory advanced melanoma: a randomised dose-comparison cohort of a phase 1 trial. *Lancet*, **2014**, *384*:1109–1117.

Sahin U, et al. An RNA vaccine drives immunity in checkpoint-inhibitor-treated melanoma. *Nature*, **2020**, *585*:107–112.

Sahin U, Türeci Ö. Personalized vaccines for cancer immunotherapy. *Science*, **2018**, *359*:1355–1360.

Saltz L, et al. Panitumumab. *Nat Rev Drug Discov*, **2006**, *5*:987–988.

Schadendorf D, et al. Pooled analysis of long-term survival data from phase II and phase III trials of ipilimumab in unresectable or metastatic melanoma. *J Clin Oncol*, **2015**, *33*:1889–1894.

Schechter A, et al. The neu gene: an erbB-homologous gene distinct from and unlinked to the gene encoding the EGF receptor. *Science*, **1985**, *229*:976–978.

Slamon DJ, et al. Human breast cancer: correlation of relapse and survival with amplification of the HER-2/neu oncogene. *Science (New York, NY)*, **1987**, *235*:177–182.

Slamon DJ, et al. Use of chemotherapy plus a monoclonal antibody against HER2 for metastatic breast cancer that overexpresses HER2. *N Engl J Med*, **2001**, *344*:783–792.

Strebhardt K, Ullrich A. Paul Ehrlich's magic bullet concept: 100 years of progress. *Nat Rev Cancer*, **2008**, *8*:473–480.

Swain SM, et al. Pertuzumab, trastuzumab, and docetaxel in HER2-positive metastatic breast cancer. *N Engl J Med*, **2015**, *372*:724–734.

Tap WD, et al. Olaratumab and doxorubicin versus doxorubicin alone for treatment of soft-tissue sarcoma: an open-label phase 1b and randomised phase 2 trial. *Lancet (London, England)*, **2016**, *388*:488–497.

Tawbi HA, et al. Relatlimab and Nivolumab versus Nivolumab in Untreated Advanced Melanoma. *New Engl J Med*, **2022**, *386*:24–34.

Trivedi S, et al. Anti-EGFR targeted monoclonal antibody isotype influences antitumor cellular immunity in head and neck cancer patients. *Clin Cancer Res*, **2016**, *22*:5229–5237.

Van Cutsem E, et al. Cetuximab and chemotherapy as initial treatment for metastatic colorectal cancer. *N Engl J Med*, **2009**, *360*:1408–1417.

Waldman AD, et al. A guide to cancer immunotherapy: from T cell basic science to clinical practice. *Nat Rev Immunol*, **2020**, *20*:651–668.

Wells JA, et al. Aflibercept, bevacizumab, or ranibizumab for diabetic macular edema. *N Engl J Med*, **2015**, *372*:1193–1203.

Woelk CH, Snyder A. Modulating gut microbiota to treat cancer. *Science*, **2021**, *371*:573–574.

Wolff AC, et al. Recommendations for human epidermal growth factor receptor 2 testing in breast cancer: American Society of Clinical Oncology/College of American Pathologists clinical practice guideline update. *J Clin Oncol*, **2013**, *31*:3997–4013.

This page intentionally left blank

Chapter 73

Hormones, Hormone Receptor Antagonists, and Related Agents in the Therapy of Cancer

Claudine Isaacs, Kerry L. Burnstein, and Anna T. Riegel

INTRODUCTION TO HORMONE-REGULATED CANCERS

ENDOCRINE THERAPY OF BREAST CANCER

ANDROGEN DEPRIVATION THERAPY OF PROSTATE CANCER

USE OF GLUCOCORTICOIDS IN HEMATOLOGICAL MALIGNANCIES

ENDOCRINE THERAPY FOR OTHER MALIGNANCIES

DRUGS THAT TARGET GNRH/LHRH
- GnRH Agonists
- GnRH Antagonists

DRUGS THAT TARGET THE ESTROGEN RECEPTOR
- Estrogen Antagonists
- Drugs That Decrease Estrogen Levels

DRUGS THAT TARGET THE PROGESTERONE RECEPTOR
- Drugs Combined With Agents That Antagonize the Estrogen Receptor

DRUGS THAT TARGET THE ANDROGEN RECEPTOR
- AR Antagonists
- Drugs That Inhibit Androgen Synthesis
- Prostate Cancer Chemoprevention (5α-Reductase Inhibitors)
- Bipolar Androgen Deprivation Therapy

DRUGS THAT TARGET THE GLUCOCORTICOID RECEPTOR

A Note on Treatment Regimens

Cancer treatment regimens change to reflect continuous advances in basic and clinical science: new drugs, both small molecules and biologicals; improved methods of targeting and timing of drug delivery; agents with altered pharmacokinetic properties and selectivities; the use of rational multidrug combinations; and greater knowledge of the basic cell biology of tumorigenesis, metastasis, and immune function, among other advances. As a consequence, this chapter presents relatively few detailed treatment regimens; rather, we refer the reader to the web-based resources of the U.S. FDA (drugs@fda) and the NCCN (National Comprehensive Cancer Network). Table 71–1 provides the details and focuses on the complexities of treatment of two cancers.

Introduction to Hormone-Regulated Cancers

The growth of a number of cancers is hormone-dependent or regulated by hormones. Therapies that employ estrogen and androgen receptor antagonists, steroid hormone synthesis inhibitors, and gonadotropin-releasing hormone (GnRH) analogues or antagonists extend survival and delay or prevent tumor recurrence of both breast and prostate cancer. These molecules interrupt the normal feedback controls regulating steroid hormone synthesis, inhibit androgen and estrogen production, or inhibit binding of these hormones to their cognate receptors, which are ligand-activated transcription factors. By inhibiting the activation and actions of androgen and estrogen receptors, these drugs block or reduce expression of genes and gene networks that ultimately promote tumor growth and survival. Glucocorticoids are used for their antiproliferative and lympholytic properties in hematologic malignancies and, in other oncological settings, to mitigate untoward responses to other treatments as well as some cancer-related symptoms.

The pharmacology of the estrogens and androgens is described in detail in Chapters 48 and 49. Decreasing the actions of these steroid hormones is the therapeutic goal in certain cancers, most notably those of the prostate and breast, because these organs are dependent on steroid hormones for their growth, function, and morphological integrity.

Endocrine Therapy of Breast Cancer

The presence of the estrogen receptor (ER) and/or progesterone receptor (PR) (ER+/PR+) in female breast cancer tissue identifies the subset of hormone receptor–positive cancers with a greater than 60% likelihood of responding to endocrine therapy. The response rate to antiestrogen treatment is somewhat lower in the subset of patients with tumors that are ER+ or PR+ but also positive for human epidermal growth factor receptor (HER2)/neu amplification. In contrast, ER-negative and PR-negative (ER–/PR–) carcinomas do not respond to endocrine therapy. Antiestrogen approaches for the therapy of ER+/PR+ breast cancer include the use of selective estrogen receptor modulator (SERMs), selective estrogen receptor downregulators (SERDs), and aromatase inhibitors (AIs) (Table 73–1 and Figure 73–1A). Historically, high doses of estrogen have been recognized as effective treatment of breast cancer. The growth inhibitory effect of estrogens may be related to their ability to induce apoptosis in endocrine-resistant breast cancer (reviewed in Jordan, 2015). However, interruption of estrogen-induced signaling with SERMs, SERDs, and drugs that reduce estrogen production such as AIs and GnRH/luteinizing hormone–releasing hormone (LHRH) analogues are more effective and better tolerated. These drugs have largely replaced estrogens or progestins for the treatment of breast cancer, although estrogen and progesterone are used occasionally. Male breast cancer is rare and predominantly (>90%) ER+/PR+. Treatment is directed at inhibiting ER with *tamoxifen*. There is a paucity of data on the use of AIs or SERDs in males.

Initial or acquired resistance to endocrine therapies (*tamoxifen* or AIs) frequently occurs. Multiple mechanisms contribute to endocrine resistance in breast cancer (reviewed in Fan and Jordan, 2019); these mechanisms include loss of ER expression, changes in transcriptional coregulator expression or activity, and hormone-independent activation of the ER by stress kinase or growth factor–activated cellular kinase

Abbreviations

ADT: androgen deprivation therapy
AI: aromatase inhibitor
ALL: acute lymphoblastic leukemia
AR: androgen receptor
AUC: area under the curve
BAT: bipolar androgen deprivation therapy
BCRP: breast cancer resistance protein
CDK: cyclin-dependent kinase
CLL: chronic lymphocytic leukemia
CRPC: castration-resistant prostate cancer
CYP: cytochrome P450
ER: estrogen receptor
ERE: estrogen-response element
FSH: follicle-stimulating hormone
GnRH: gonadotropin-releasing hormone
GR: glucocorticoid receptor
HER: human epidermal growth factor
HL: Hodgkin lymphoma
HR: hormone receptor
LH(RH): luteinizing hormone (–releasing hormone)
MM: multiple myeloma
mTOR: mechanistic (or mammalian) target of rapamycin
NHL: non-Hodgkin lymphoma
OATP: organic anion transporting polypeptide
Pgp: P-glycoprotein
PR: progesterone receptor
PROTAC: proteolysis targeting chimeras
SERD: selective estrogen receptor downregulator
SERM: selective estrogen receptor modulator
UGT: UDP-glucuronosyltransferase

pathways. In particular, cross talk between the ER and the HER2/neu pathway has been implicated in *tamoxifen* resistance. In a significant portion of metastatic breast cancers, estrogen deprivation therapy can lead to selection of cancer cells expressing an ER with mutations that allow hormone-independent activation. Acquired ER mutations also contribute to endocrine therapy–resistant breast cancer growth (reviewed in Jeselsohn et al., 2017). Targeting endocrine-resistant disease harboring ER mutations with SERDs and SERMs is an active area of drug investigation and development (reviewed in McDonnell et al., 2021).

Androgen Deprivation Therapy of Prostate Cancer

Androgens are essential for the development and maintenance of the prostate gland. The critical role of androgens in promoting prostate cancer growth was demonstrated in 1941 and led to the awarding of a Nobel Prize in 1966 to Dr. Charles Huggins. Huggins's findings established androgen deprivation therapy (ADT) as the mainstay of treatment for patients with advanced prostate cancer, an approach still used today. ADT is accomplished by surgical castration (bilateral orchiectomy) or "medical" castration, using long-acting GnRH agonist formulations to downregulate GnRH receptors in the anterior pituitary or with GnRH antagonists that directly block GnRH receptor-ligand interaction to achieve more rapid suppression of the major circulating androgen, testosterone (Table 73–2 and Figure 73–1B). ADT results in castrate levels (≤50 ng/dL) of testosterone and hence subsequent reduction of the testosterone metabolite, dihydrotestosterone, which binds with highest affinity to the androgen receptor (AR). Thus, ADT limits the action of the AR in prostate and other sensitive tissues (see Chapter 49).

In localized, high-risk or advanced prostate cancer (prior to metastasis or after metastatic progression), the combination of ADT and AR antagonists or androgen synthesis inhibitors is the standard of care but not a curative treatment. Disease progression commonly occurs despite this "combined androgen blockade" and signifies the development of castration-resistant prostate cancer (CRPC). In CRPC, the target of combined androgen blockade, AR, is typically "reactivated" through a variety of mechanisms (reviewed in Desai et al., 2021; Einstein et al., 2019; Luo et al., 2018; Nakazawa et al., 2017) including increased AR levels, expression of constitutively active AR splice variants that lack the ligand binding domain, or acquisition of somatic *AR* gene gain-of-function mutations that render AR promiscuous for activation by other steroid hormones (including weak adrenal androgens or glucocorticoids) and, in rare cases, by AR antagonists. Increased expression or activity of AR coactivators and pioneer transcription factors (e.g., FOXA1) or decreased AR corepressor proteins may also participate in continued AR activity in CRPC. Another mechanism that may drive persistent AR signaling in CRPC is local production of sufficient quantities of androgens by CRPC tumors to activate AR. Therefore, second-generation, highly potent AR antagonists and inhibitors of androgen synthesis are employed concurrently with GnRH agonists or antagonists as frontline or second-line therapies following progression to CRPC (reviewed in Mitsiades and Kaochar, 2021). Combining ADT with a newer generation AR antagonist or an androgen synthesis–directed drug is associated with improvements in symptoms and prolongation of survival.

Use of Glucocorticoids in Hematological Malignancies

Glucocorticoids act by binding to a specific glucocorticoid receptor (GR) that is a member of the nuclear receptor family of transcription factors. Agonist-liganded GR translocates to the nucleus and induces complex gene expression changes (see Chapter 50) that lead to antiproliferative and apoptotic responses in sensitive cells. Because of their lympholytic effects and their ability to suppress mitosis in lymphocytes, glucocorticoids are used as cytotoxic agents in the treatment of acute leukemia in children and malignant lymphoma in children and adults. In acute lymphoblastic or undifferentiated leukemia of childhood, glucocorticoids may produce prompt clinical improvement and objective hematological remissions in 30% of children. However, the duration of remission is brief. Remissions occur more rapidly with glucocorticoids than with antimetabolites, and there is no evidence of cross-resistance to unrelated agents. Thus, therapy is initiated with *prednisone* and *vincristine* (see Chapter 70), often followed by an anthracycline or *methotrexate* (see Chapter 70), and *L-asparaginase* (see Chapter 72). Glucocorticoids are a valuable component of curative regimens for other lymphoid malignancies, including Hodgkin disease, non-Hodgkin lymphoma, multiple myeloma, and chronic lymphocytic leukemia (CLL). Glucocorticoids are extremely helpful in controlling autoimmune hemolytic anemia and thrombocytopenia associated with CLL.

Endocrine Therapy for Other Malignancies

Endocrine therapies are occasionally used in other malignancies. *Tamoxifen* and *medroxyprogesterone acetate* have activity in metastatic endometrial carcinoma, particularly in low-grade tumors. Use of endocrine therapy in ovarian cancer has also been considered.

Drugs That Target GnRH/LHRH

GnRH Agonists

Synthetic GnRH analogues (e.g., *triptorelin*, *goserelin*, *leuprolide*, *histrelin*, and *nafarelin*) have amino acid substitutions at key residues of the

TABLE 73–1 ■ ESTROGEN RECEPTOR (ER)–TARGETED THERAPY IN ER+ BREAST CANCER

DRUG (DAILY STANDARD DOSE)	THERAPEUTIC APPROACH IN DISEASE SETTING					
	CHEMOPREVENTION		ADJUVANT THERAPY		METASTATIC DISEASE	
	PREMENOPAUSAL	POSTMENOPAUSAL	PREMENOPAUSAL	POSTMENOPAUSAL	PREMENOPAUSAL	POSTMENOPAUSAL
Tamoxifen (20 mg PO)	Yes (5 years)	Yes (5 years)	Yes (5–10 years)	Yes (before AI for 2–5 years)	Yes	Yes
Raloxifene (60 mg PO)		Yes (5 years)				
Fulvestrant (500 mg IM day 1, day 15, day 29, then once per month)						Yes
Anastrozole (1 mg PO)		Yes (5 years)		Yes (5–10 years) (up front or after tamoxifen)		Yes
Letrozole (2.5 mg PO)				Yes (5–10 years) (up front or after tamoxifen)		Yes
Exemestane (25 mg PO)		Yes (5 years)		Yes (5–10 years) (up front or after tamoxifen)		Yes
GnRH agonist in addition to one of above given monthly			Yes (with tamoxifen or AI)		Yes (with tamoxifen, AI, or fulvestrant)	

naturally occurring decapeptide GnRH that increases analogue binding affinity to the GnRH receptor and reduces susceptibility to enzymatic degradation. These GnRH agonists are 100-fold more potent than the native decapeptide. Administration of long-acting GnRH agonists downregulates GnRH receptors in the anterior pituitary gland in both women and men.

In women, the downregulation of GnRH receptors suppresses the release of the gonadotropins follicle-stimulating hormone (FSH) and LH from the pituitary and prevents follicular maturation in the ovary. Serum estrogen levels are reduced to those seen in postmenopausal women or in women after oophorectomy. GnRH agonists are used in the adjuvant treatment of breast cancer or metastatic disease for women who have functioning ovaries; typically, they are used in combination with either *tamoxifen* or AIs.

In treating prostate cancer, the most common form of ADT involves chemical suppression of pituitary gland function with long-acting GnRH agonists (*leuprolide*, *goserelin*, *buserelin*). ADT is given to patients with localized intermediate- to high-risk prostate cancer in conjunction with radiation therapy or in some cases following surgical removal of the prostate gland. GnRH agonists cause an initial release of both LH and FSH and a subsequent increase in testosterone production termed "testosterone flare" from testicular Leydig cells. After 1 week of therapy, GnRH receptors are downregulated on the gonadotropin-producing cells, causing a decline in the pituitary response. The fall in serum LH leads to a decrease in testosterone production to castrate levels within 3 to 4 weeks of the first treatment. Subsequent treatments maintain testosterone at castrate levels.

ADME

GnRH agonists are administered as an injection once a month or every 3, 4, or 6 months. There is in an initial rise in LH and FSH levels, but after 14 to 21 days of therapy, a sustained decrease in serum LH and estrogen n wom en or testosterone in m en is obser ed.

Therapeutic Use in Breast and Prostate Cancer. In the adjuvant setting, studies have demonstrated a further reduction in the risk of recurrence when GnRH agonists are given in conjunction with either AIs or *tamoxifen*, when compared to *tamoxifen* alone, in very young or higher-risk premenopausal women (Pagani et al., 2020). In addition, these drugs can be given with *tamoxifen*, AIs, or *fulvestrant* in premenopausal women with metastatic breast cancer. In men with metastatic prostate cancer, ADT may be used in conjunction with chemotherapy (e.g., *docetaxel*). Alternatively, ADT is typically combined with potent newer generation AR antagonists or the androgen synthesis inhibitor *abiraterone acetate* in what is referred to as "combined androgen blockade." The advantage of combined androgen blockade is that the GnRH agonist (or GnRH antagonist) will deplete testicular androgens, while the competitive AR antagonist component competes at the AR with residual androgens made by the adrenal glands. When the steroid synthesis inhibitor *abiraterone* is used for combined androgen blockade, there is a decrease in androgens made by the adrenal glands. Numerous trials showed a benefit in survival in metastatic castration-sensitive prostate cancer following ADT combined with either *abiraterone* or *docetaxel*, but these treatments are associated with higher costs (and, in the case of *docetaxel*, significantly higher toxicities) compared to ADT alone (Wang L et al., 2021).

Toxicity

Side effects in women are generally related to hypoestrogenism (i.e., hot flashes, vaginal dryness, decreased libido, osteoporosis, amenorrhea, and dyspareunia). The side effects of GnRH agonists are often reversible on cessation of therapy. AI use in premenopausal women combined with ovarian suppression with GnRH analogues increases menopausal symptoms and sexual dysfunction, and thus, benefits and risks should be carefully considered with each patient (Burstein et al., 2016).

In men, during the transient rise in LH, the resultant testosterone surge or "flare" m y induce acute stimulation of prostate canc r gr owth

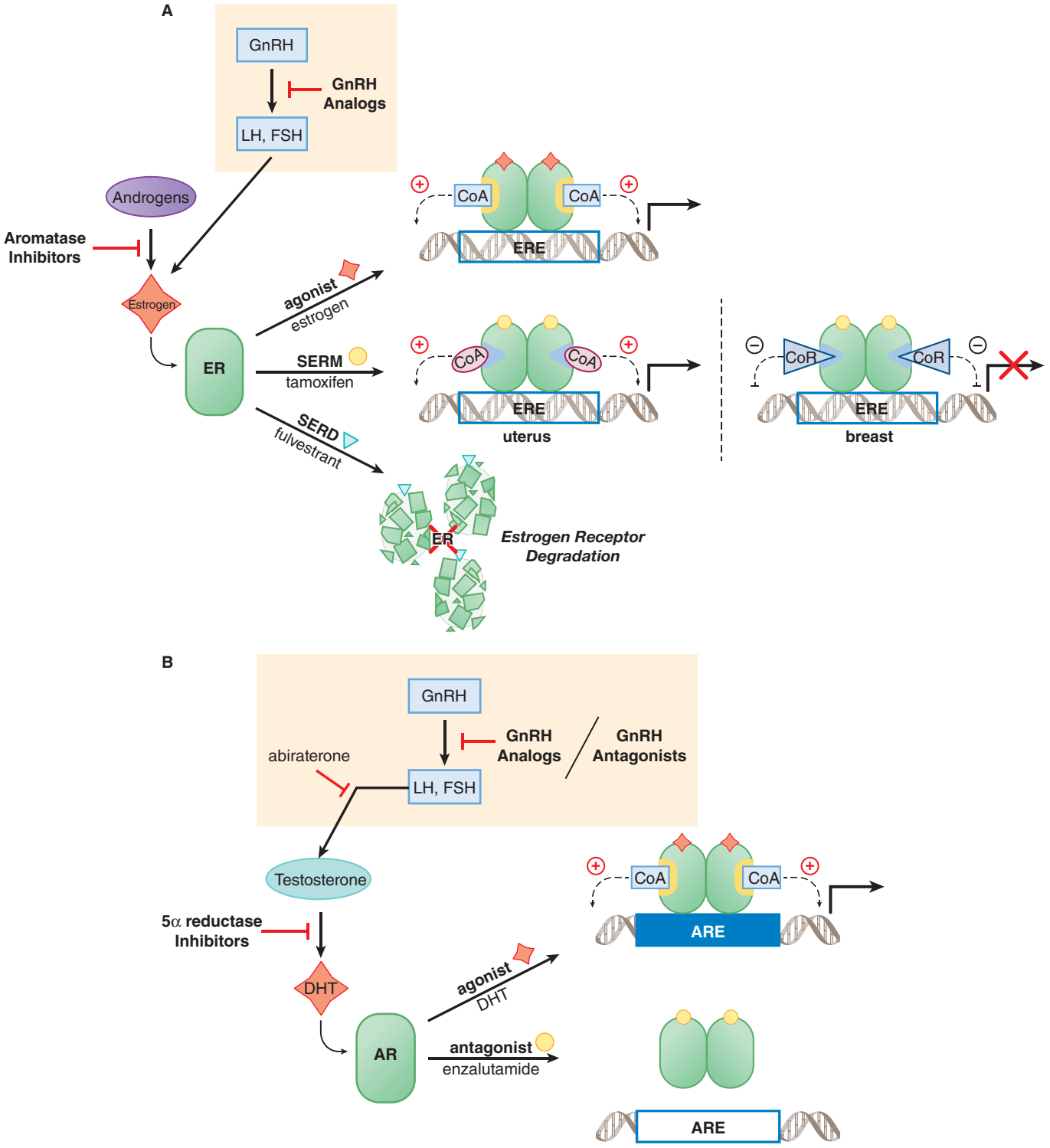


Figure 73–1 **A.** *Targets of agents that interrupt estrogen receptor signaling in females.* GnRH promotes the release of LH and FSH (Figure 46–1), which regulate ovarian production of estrogen and progesterone and can stimulate estrogen production. Aromatase converts the estrogen precursors androstenedione, dihydroepiandrostenedione, and testosterone to estrogens (see Figures 48–1 and 48–2). Estrogen signaling can be interrupted by inhibiting hormone production (GnRH analogues, aromatase inhibitors), by antagonizing the effects of estrogen at the estrogen receptor (ER) with a SERM, or by destabilizing the ER and promoting its degradation with a SERD (e.g., *fulvestrant*). The effects of SERMs (e.g., *tamoxifen*) on organs are dependent on a number of factors including the activity and relative expression levels of transcriptional coactivators (CoA) and corepressors (CoR) in a particular tissue. In the breast, *tamoxifen* is an antagonist; a corepressor effect on gene transcription predominates. On the other hand, in the uterus, *tamoxifen* can drive estrogenic effects, after binding ER, by utilizing coactivators of gene transcription. **B.** *Targets of agents that interrupt androgen receptor signaling in males.* GnRH promotes the release of LH and FSH, which stimulates adrenal and testicular androgen production. GnRH receptors can be inhibited by long-acting GnRH analogues that downregulate GnRH receptors or by GnRH antagonists. *Abiraterone acetate* is a steroid synthesis inhibitor that blocks the conversion of pregnenolone and progesterone to androgen precursors. Androgen signaling is inhibited by androgen receptor (AR) antagonists (e.g., *enzalutamide*), which competitively bind the AR and inhibit binding and activation by androgens, particularly dihydrotestosterone (DHT). AR bound to *enzalutamide* (instead of DHT) has diminished interaction with coactivators and androgen response elements (AREs) on DNA.

TABLE 73–2 ■ ANDROGEN RECEPTOR–TARGETED THERAPY FOR PROSTATE CANCER

DRUG (DAILY STANDARD DOSE)	CASTRATION-SENSITIVE		CASTRATION-RESISTANT	
	NON-METAST	METAST	NON-METAST	METAST
Bicalutamide (one 50-mg tablet daily in combination with ADT)[a]	Yes	Yes	Yes	Yes
Enzalutamide (four 40-mg tablets once daily)		Yes	Yes	Yes
Apalutamide (four 60-mg tablets once daily)		Yes	Yes	
Darolutamide (two 300-mg tablets twice daily)			Yes	
Abiraterone acetate (4×250 mg or 2×500 mg tablets daily) plus prednisone: 5-mg tablet once daily (metastatic castration-sensitive prostate cancer) or 5-mg tablet twice daily (metastatic CRPC)		Yes		Yes
ADT consisting of bilateral orchiectomy, GnRH agonist, or GnRH antagonist in addition to one of above agents	Yes	Yes	Yes	Yes

Metast, metastatic.

[a]Bicalutamide dose as monotherapy is higher.

and symptoms from stimulation of metastatic deposits in bone. Patients may experience an increase in bone pain, spinal cord compression, or obstructive bladder symptoms lasting for 2 to 3 weeks. The flare phenomenon can be effectively counteracted with concurrent administration of 2 to 4 weeks of oral AR antagonist therapy, which may inhibit the action of the increased serum testosterone levels. Common side effects of ADT include vasomotor instability, loss of libido, impotence, gynecomastia, fatigue, anemia, weight gain, decreased insulin sensitivity, altered lipid profiles, osteoporosis and fractures, and loss of muscle mass (Formenti et al., 2021). Skeletal-related events due to ADT may be mitigated by bisphosphonate therapy (see Chapter 52). Studies have shown a small but significant increase in the risk of diabetes and coronary heart disease in patients taking GnRH agonists (Gupta et al., 2018). Because of these concerns, GnRH agonists have undergone FDA review since 2010.

GnRH Antagonists

GnRH antagonists are a class of agents used in the treatment of prostate cancer. They antagonize the GnRH receptor by reversible competition with GnRH, resulting in rapid reduction in LH and FSH secretion from the pituitary and concomitant loss of testosterone production to castrate levels from the testes. *Degarelix* is a small peptide-based GnRH antagonist used in prostate cancer treatment. The recently FDA-approved *relugolix* is an orally available, nonpeptide antagonist used for treatment of prostate cancer (Shore et al., 2020).

ADME

After an initial subcutaneous loading dose (240 mg in two doses of 120 mg), *degarelix* is given as a monthly injection (80 mg). C_{max} is achieved after 2 days, and excretion is mainly through the hepatobiliary system. *Relugolix* is given orally as an initial loading dose (360 mg), followed by one tablet (120 mg) daily for prostate cancer. Steady-state levels are achieved after 7 days. Food can diminish oral bioavailability by 50%.

Absorption and distribution of *relugolix* are influenced by its interaction with the efflux transporter P-glycoprotein (Pgp) and CYP3A4 (the predominant metabolizer of the drug). Oral coadministration with agents that inhibit Pgp (e.g., *erythromycin*) increases the C_{Pmax} and area under the curve (AUC) of *relugolix*; coadministration with *rifampicin* (an inducer of intestinal Pgp and hepatic CYP3A4) will decrease C_{Pmax} and AUC. Metabolites of *relugolix* are excreted mainly in feces.

Degarelix distributes slowly from a subcutaneous depot to total body water. As a peptide, *degarelix* does not interact with Pgp and is not metabolized by hepatic CYPs. Rather, hepatic/biliary hydrolysis of the drug produces peptide fragments that are excreted in the bile/feces (~75%), with approximately 25% of the original dose appearing in the urine.

Therapeutic Use

GnRH antagonists are used in the treatment of advanced and metastatic prostate cancer. Besides avoidance of the initial increase in testosterone seen with GnRH agonists, GnRH antagonists are reported to be more effective at suppressing FSH and have fewer cardiovascular side effects than GnRH agonists (Dearnaley et al., 2020; Shore et al., 2020).

Toxicity

Degarelix is associated with higher rates of injection site reactions compared with GnRH agonists. ADT may cause prolongation of the cardiac QT/QTc interval, a serious side effect of both *relugolix* and *degarelix*. The most common side effects of *relugolix* include hot flushes, increased blood glucose levels, increased triglyceride levels, musculoskeletal pain, anemia, increased liver enzymes, tiredness, constipation, and diarrhea. *Relugolix* causes reproductive and fetal harm in animal toxicity tests; thus, caution is warranted, and male patients with female partners of reproductive potential should use contraception during and for 2 weeks following the last dose of *relugolix*.

The early GnRH antagonists *cetrorelix* and *abarelix* (no longer marketed), although effective, are now rarely used for prostate cancer because of the risk for severe systemic allergic reactions. Later generation GnRH peptide antagonists are not associated with systemic allergic reactions.

Drugs That Target the Estrogen Receptor

Estrogen Antagonists

Selective Estrogen Receptor Modulators

The SERMs bind to the ER and exert either estrogenic or antiestrogenic effects, depending on the specific organ (see Chapter 48). *Tamoxifen* is one of the most widely studied antiestrogenic drugs used in the treatment of breast cancer. *Tamoxifen* also exerts estrogenic agonist effects on nonbreast tissues, which influences the side effect profile of the drug. Therefore, several novel antiestrogen compounds that offer the potential for enhanced efficacy and reduced toxicity compared with *tamoxifen* have been developed. These novel antiestrogens can be divided into *tamoxifen* analogues (e.g., *toremifene*, *droloxifene*, *idoxifene*); "fixed-ring" compounds (e.g., *raloxifene*, *bazedoxifene*, *lasofoxifene*, *arzoxifene*, *miproxifene*, *levormeloxifene*); and the SERDs (e.g., *fulvestrant*), the latter also termed "pure antiestrogens" (McDonnell et al., 2021).

Tamoxifen. *Tamoxifen* was developed as an oral contraceptive but instead was found to induce ovulation and to have antiproliferative effects on estrogen-dependent breast cancer cell lines (reviewed in Abderrahman and Jordan, 2019). *Tamoxifen* is prescribed for the adjuvant therapy of

early-stage breast cancer and for the therapy of advanced breast cancer. *Tamoxifen* and other SERMs such as *raloxifene* are also used for the prevention of breast cancer in high-risk patients such as those with a strong family history or prior nonmalignant breast pathology (Visvanathan et al., 2013). The uses, pharmacology, and mechanism of action of *raloxifene* are discussed in Chapter 48.

Mechanism of Action. *Tamoxifen* is a competitive inhibitor of the binding of estrogens (e.g., 17β-estradiol) to the ER and antagonizes estrogen-induced proliferation of human breast cancer.

There are two subtypes of ERs: ERα and ERβ, which have different tissue distributions and can either homo- or heterodimerize. ERα plays a major role in breast cancer progression and is a prognostic marker; the role of ERβ is unclear. Binding of estrogen and SERMs to the estrogen-binding sites of the ERs initiates a change in conformation of the ER, dissociation of the ER from heat-shock proteins, and ER dimerization. Dimerization facilitates the binding of the ER to specific DNA estrogen-response elements (EREs) in the vicinity of estrogen-regulated genes. Coregulator proteins interact with the liganded receptor to act as corepressors or coactivators of gene expression (see Chapter 48). Elegant studies of the crystal structure of ERα bound to different ligands indicate that, when an ER agonist is bound to the ER, a conformational change occurs in the ligand-binding pocket that enables helix 12 in this region to provide a docking site for p160 transcriptional coactivators, thereby increasing transcription of target genes.

Conversely, the interaction of an estrogen antagonist such as 4-hydroxytamoxifen bound to the ligand binding domain of ER causes a structural rearrangement of helix 12. This alteration in structure reduces transcriptional coactivator binding and favors binding of transcriptional corepressors. The net result is inhibition of estrogen-induced gene transcription (reviewed in Legare and Basik, 2016; Nettles and Greene, 2005). Differences in tissue distribution of the ER subtypes and the relative abundance and activities of different transcriptional coactivators and corepressors likely explain the antagonist response to tamoxifen in ER+ breast cancer and its partial agonist activities in noncancerous tissues (see Figure 73–1A) (reviewed in Abderrahman and Jordan, 2019; Green and Carroll, 2007).

Organs displaying agonist effects of *tamoxifen* include the uterine endometrium (endometrial hypertrophy, vaginal bleeding, and endometrial cancer); the coagulation system (thromboembolism); bone metabolism (increase in bone mineral density, which can slow development of osteoporosis); and liver (*tamoxifen* lowers total serum cholesterol, low-density lipoprotein cholesterol, and lipoproteins and raises apolipoprotein A1 levels).

ADME. *Tamoxifen* is given orally once (20 mg) per day. It is readily absorbed following oral administration, with peak concentrations measurable after 3 to 7 h and steady-state levels reached at 4 to 6 weeks. Metabolism of *tamoxifen* is complex and principally involves CYPs 3A4/5 and 2D6 in the formation of *N*-desmethyl tamoxifen and CYP2D6 to form 4-hydroxytamoxifen, a more potent metabolite (Figure 73–2). Both metabolites can be further converted to 4-hydroxy-*N*-desmethyltamoxifen (endoxifen), which retains high affinity for the ER. The parent drug has a terminal $t_{1/2}$ of 7 days. After enterohepatic circulation, glucuronides and other metabolites are excreted in the stool; excretion in the urine is minimal. Polymorphisms in CYP2D6 that reduce its activity lead to lower plasma levels of the potent metabolites 4-OH tamoxifen and endoxifen, but whether this leads to inferior efficacy of *tamoxifen* treatment and higher risk of disease relapse is unclear (reviewed in Hertz and Rae, 2016; Tamura et al., 2020). While drugs that inhibit CYP2D6 activity, such as antidepressants, have been postulated to minimize *tamoxifen* activity in breast cancer, more recent studies do not suggest a clinically significant impact (Haque et al., 2016).

Therapeutic Uses. *Tamoxifen* is used for the treatment of women with ER+ metastatic breast cancer or following primary excision of an ER+ tumor as an adjuvant treatment to prevent recurrence and extend overall survival. For the adjuvant treatment of premenopausal women, *tamoxifen* is given for at least 5 years (see Table 73–1). *Tamoxifen* may also be used for postmenopausal women, but AIs are preferred as they are associated with further reductions in the risk of recurrence (Early Breast

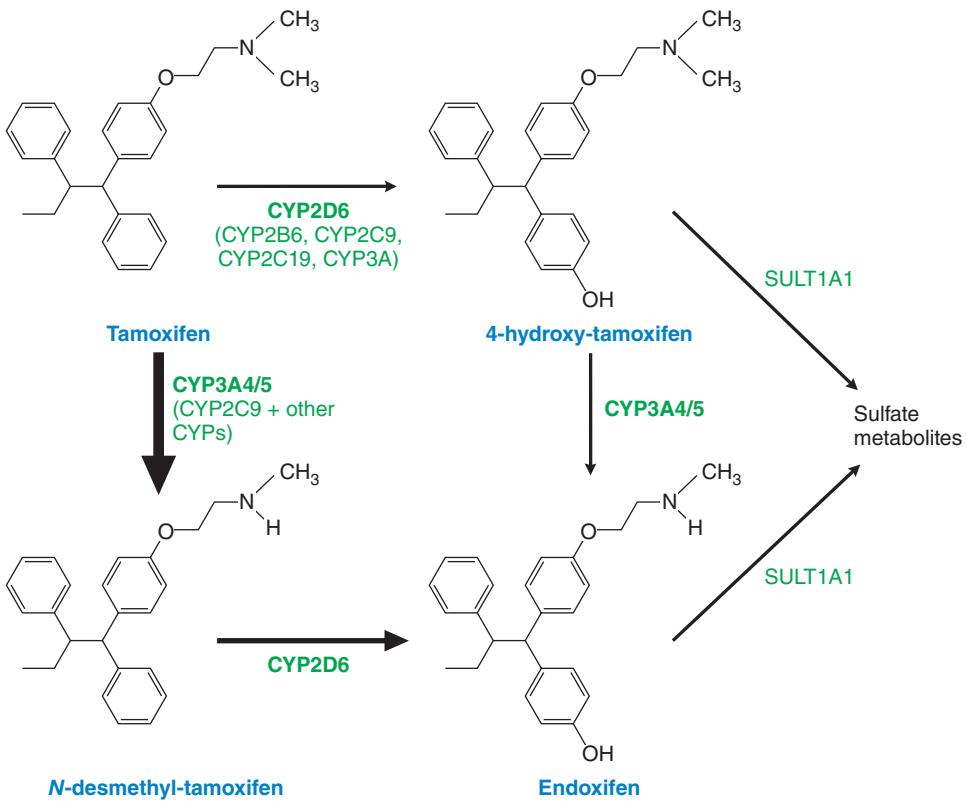


Figure 73–2 *Tamoxifen and its metabolites.*

Cancer Trialists' Collaborative Group, 2015). Recent studies indicate that patients with breast cancer derive modest additional benefit, in terms of disease-free survival, overall survival, and decrease in contralateral breast cancer risk, if *tamoxifen* is taken for up to 10 years or AIs are continued for 5 to 10 years after completion of 5 years of *tamoxifen* (Burstein, 2020; Davies et al., 2013; Goss et al., 2016). Although taken for a finite time, *tamoxifen* has persisting long-term benefits (Ekholm et al., 2016). *Tamoxifen* may be taken as sole adjuvant therapy or after adjuvant chemotherapy (see Table 73–1). Alternative or additional antiestrogen strategies in the adjuvant treatment of premenopausal women with ER+ breast cancer include oophorectomy or suppression of ovarian function with GnRH analogues in combination with either *tamoxifen* or an AI. These combinations, used in premenopausal women, further reduce estrogen stimulation of breast cancer and result in lower rates of disease recurrence in very young women and in higher risk patients receiving chemotherapy (Francis et al., 2015; Pagani et al., 2020). Some studies suggest improved response rates with these combinations in patients with metastatic disease. *Tamoxifen* also is effective (a 40%–50% reduction in tumor incidence) in preventing breast cancer in women at increased risk. In the preventive setting, *tamoxifen* reduces only ER+ tumors, not ER– tumors and does not affect overall mortality. In patients with metastatic cancer, responses to hormonal therapy may not be apparent clinically or by imaging for 8 to 12 weeks. The medication typically should be continued until the disease progresses or unwanted toxicities develop.

Adverse Effects. The common adverse reactions to *tamoxifen* include vasomotor symptoms (hot flashes), atrophy of the lining of the vagina, menstrual irregularities, vaginal bleeding and discharge and pruritus vulvae, these occur with increasing severity in postmenopausal women. The partial agonist activity of *tamoxifen* increases the incidence of endometrial cancer by 2- to 3-fold, particularly in postmenopausal women who receive *tamoxifen* for more than 2 years, although the absolute risk remains low. The increase in risk of thromboembolic events is slight and increases with the age of a patient and during the perioperative period. Hence, it often is advisable to temporarily halt *tamoxifen* prior to elective surgery. *Tamoxifen* is associated with fatty liver disease (although rarely clinically significant), a small increase in risk of cataracts, and, rarely, retinal deposits and decreased visual acuity. The FDA has assigned a boxed warning to *tamoxifen* based on uterine malignancies and thromboembolic events associated with its use, noting that for breast cancer patients, the benefits of *tamoxifen* outweigh these risks.

Toremifene. *Toremifene* is a triphenylethylene derivative of *tamoxifen* and has a similar pharmacological profile, clinical efficacy, and safety. *Toremifene* is used occasionally in the metastatic setting for the treatment of breast cancer in postmenopausal women with tumors that are ER+ or of unknown receptor status. It can also be used for the treatment of desmoid tumors. In rare cases, *toremifene* can cause heart rhythm problems through prolongation of the QT interval.

Selective Estrogen Receptor Downregulators

The SERDS, also termed *pure antiestrogens*, include *fulvestrant* and a number of agents in experimental trials. SERDs, unlike SERMs, are devoid of any estrogen agonist activity.

Fulvestrant. *Fulvestrant* is currently the only FDA-approved SERD, either as a single agent or in combination with *palbociclib*, a CDK4/6 inhibitor (see Chapter 71), for postmenopausal women with ER+/PR+ metastatic breast cancer that has progressed on antiestrogen therapy.

Mechanism of Action. *Fulvestrant* is a steroidal antiestrogen that binds to the ER with an affinity more than 100 times that of *tamoxifen*. The drug not only inhibits the binding of estrogen but also alters the receptor structure and exposes a region that leads to targeting of the protein for proteasomal degradation (see Figure 73–1A); *fulvestrant* also may inhibit receptor dimerization. Unlike *tamoxifen*, which stabilizes or even increases ER expression, *fulvestrant* reduces the number of ER molecules in cells; as a consequence of this downregulation, the drug abolishes ER-mediated transcription of estrogen-dependent genes.

ADME. *Fulvestrant* (500-mg dose) is given intramuscularly with initial biweekly loading doses in the first month followed by once-monthly injections thereafter. Using this dosing regimen, steady-state levels are achieved within the first month (see Table 73–1). C_{Pmax} is reached approximately 7 days after intramuscular administration; the plasma $t_{1/2}$ is approximately 40 days. There is rapid distribution and extensive protein binding of this highly lipophilic drug. Various pathways, similar to those of steroid metabolism (oxidation, aromatic hydroxylation, and conjugation), metabolize *fulvestrant*. CYP3A4 appears to be the major CYP isoenzyme involved in the metabolism of *fulvestrant*. The putative metabolites possess no estrogenic activity, and only the 17-keto compound demonstrates a level of antiestrogenic activity (~22% that of *fulvestrant*). Less than 1% of the parent drug is excreted intact in the urine.

Therapeutic Uses. *Fulvestrant* is used in postmenopausal women or in premenopausal women receiving GnRH agonists as antiestrogen therapy of ER+/PR+ metastatic breast cancer typically after progression on first-line antiestrogen therapy such as *tamoxifen* or an AI. *Fulvestrant* is at least as effective in this setting as the third-generation AI *anastrozole* (Robertson et al., 2016). Endocrine-resistant breast cancer that harbors ERα mutations retains some sensitivity to *fulvestrant*. Newer SERDs are also being developed for the treatment of advanced endocrine-resistant disease (McDonnell et al., 2021). A variety of agents that promote ER degradation such as proteolysis targeting chimeras (PROTACs) are also active areas of investigation (Wang Z et al., 2021).

Toxicity and Adverse Effects. Fulvestrant generally is well tolerated, the most common adverse effects being nausea, asthenia, pain, hot flashes, arthralgias, and headache. The risk of injection site reactions, seen in almost 10% patients, is reduced by giving the injection slowly.

Drugs That Decrease Estrogen Levels

Aromatase Inhibitors

Aromatase converts androgens to estrogens (e.g., androstenedione to estrone). AIs (Figure 73–3) block this enzymatic activity, thereby reducing estrogen production (Figure 73–4). AIs are now considered the standard of care for adjuvant treatment of postmenopausal women with ER+ breast cancer, either as initial therapy or after *tamoxifen* (Dowsett et al., 2010) as well as in higher-risk premenopausal women in combination with GnRH agonists. AIs are also approved in the initial treatment of metastatic ER+/PR+ breast cancer, often in combination with CDK4/6 inhibitors, in postmenopausal women and in combination with GnRH agonists in premenopausal women.

Aromatase (CYP19A1) converts adrenal androgens and gonadal androstenedione and testosterone to the estrogens estrone (E1) and estradiol (E2), respectively (see Figures 73–3 and 73–4; reactions catalyzed by aromatase are noted by a green "A" beside the reaction arrow in Figure 73–4). In postmenopausal women, this conversion occurs in nonovarian tissues (fat, liver, muscle, brain, breast, and breast tumors) and is the primary source of circulating estrogens. In premenopausal women, estrogen is primarily produced in the ovaries. AIs increase gonadotropin production in premenopausal women, which reduces their ability to inhibit ovarian estrogen production. As a result, AIs are not effective in premenopausal women without additional ovarian suppression (e.g., with GnRH analogues; see previous section). In postmenopausal women, AIs suppress most peripheral aromatase activity, leading to profound estrogen deprivation. AIs are classified as first, second, or third generation. In addition, they are further classified as type 1 (steroidal) or type 2 (nonsteroidal) AIs according to their structure and mechanism of action (see Figure 73–3). Type 1 inhibitors are steroidal analogues of androstenedione that bind covalently and irreversibly to the same site on the aromatase molecule. Thus, they commonly are known as aromatase inactivators. Type 2 inhibitors are nonsteroidal and bind reversibly to the heme group of the enzyme, producing reversible inhibition.

Third-Generation Aromatase Inhibitors. First- and second-generation (e.g., *aminoglutethimide*, *formestane*) AIs are no longer used for breast

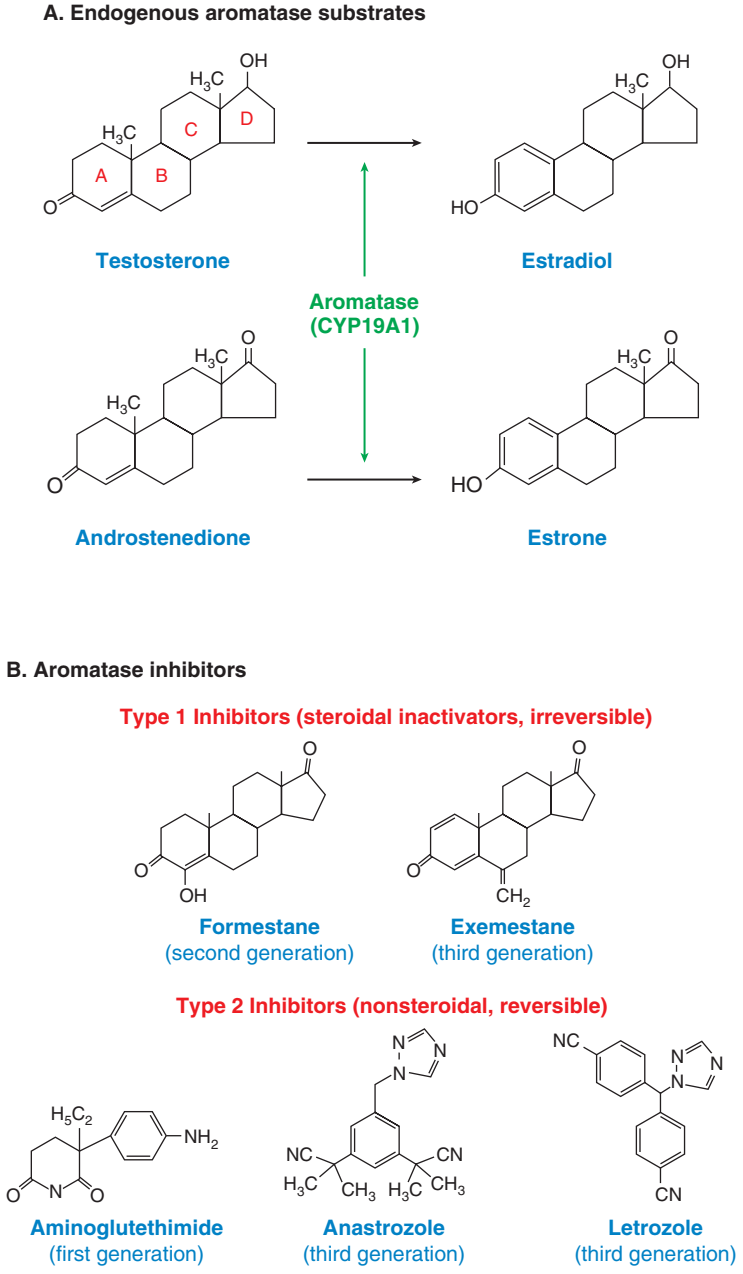


Figure 73–3 *Aromatase and its endogenous substrates and inhibitors.* **A.** Aromatization of endogenous substrates. In a multistep reaction, aromatase trihydroxylates the methyl group at C19, eliminating it as formate and aromatizing the A ring of the androgen substrate. Not shown are cofactors (NADPH, O_2) and other reaction products (H_2O, formate). **B.** Aromatase inhibitors (AIs). Type 1 AIs are steroidal analogues of androstenedione that bind covalently and irreversibly to the steroid substrate site on the enzyme and are known as aromatase inactivators. Type 2 inhibitors are nonsteroidal, bind reversibly to the heme group of the enzyme, and produce reversible inhibition.

cancer treatment because of their side effects. Third-generation inhibitors include the type 1 steroidal agent *exemestane* and the type 2 nonsteroidal imidazoles *anastrozole* and *letrozole*; they are FDA approved for use in postmenopausal women. Third-generation AIs are used as part of the treatment of early-stage and advanced breast cancer in postmenopausal women and for chemoprevention (see Table 73–1). The type 1 and 2 AIs have similar clinical efficacy and toxicity profiles (Goss et al., 2013), and these are summarized for AIs using *anastrozole* as a prototype. For *letrozole* and *exemestane*, additional drug-specific information is given. Daily administration of AIs (see dosages in Table 73–1) reduces total-body androgen aromatization by more than 95% after 1 month of treatment. AIs also reduce aromatization within large ER+ breast tumors.

Anastrozole. *Anastrozole* is a potent and selective triazole AI. *Anastrozole*, like *letrozole*, binds competitively and specifically to the heme of the CYP19.

ADME. *Anastrozole* is absorbed rapidly after oral administration; food slows the rate but not the overall extent of absorption. Steady state is attained after 7 days of repeated dosing. *Anastrozole* is metabolized by hepatic *N*-dealkylation, hydroxylation (CYP3A4 mainly), and glucuronidation (UGT1A4); the main circulating metabolite of *anastrozole*, triazole, is inactive. The hepatic metabolites account for 85% of the administered dose and are excreted via bile/feces; approximately 10% of the drug is excreted in the urine as the unmetabolized parent compound. The elimination $t_{1/2}$ is approximately 50 h. The pharmacokinetics of

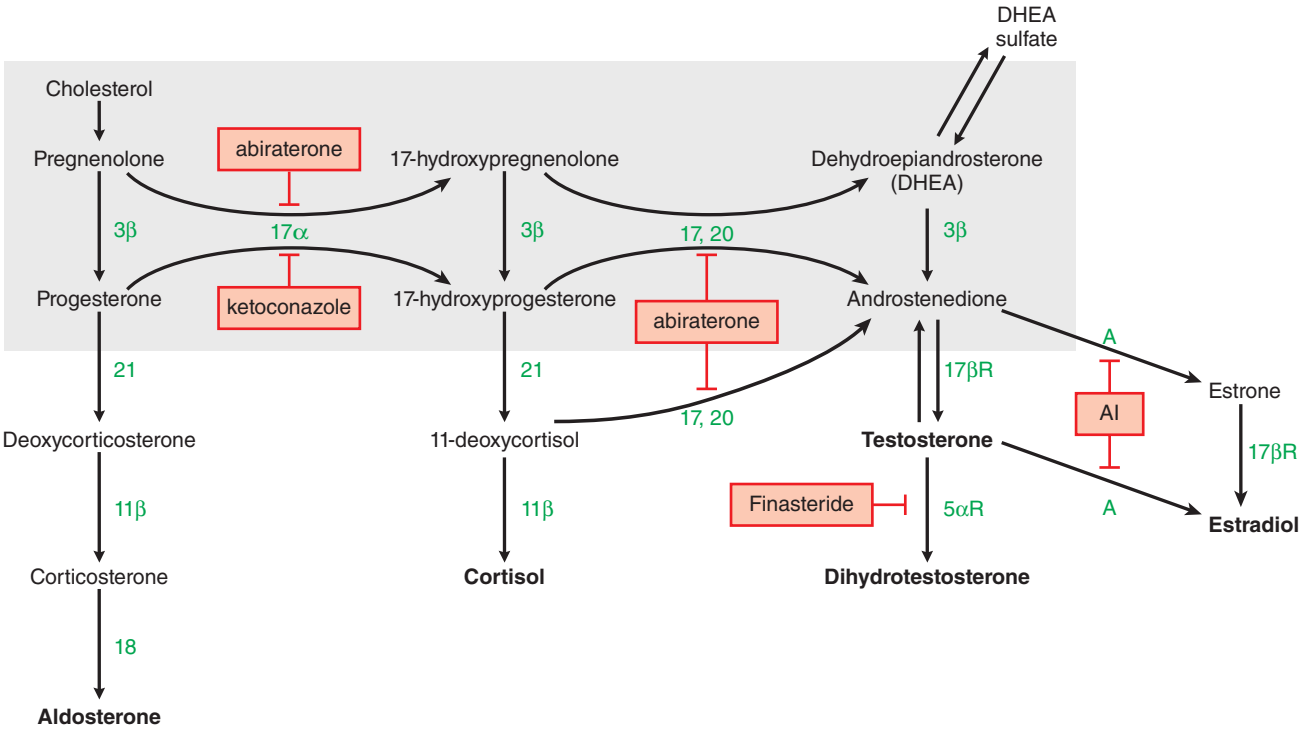


Figure 73–4 *Steroid synthesis pathways.* The shaded area contains the pathways used by the adrenal glands and gonads and clinically used agents that inhibit the pathways. Enzymes are labeled in green, inhibitors in red. Abbreviations pertaining to this figure only: *A*, aromatase; AI, aromatase inhibitor; *3β*, 3β-hydroxysteroid dehydrogenase; *5αR*, 5α-reductase; *11β*, 11β-hydroxylase; *17,20*, 17,20-lyase (CYP17A1); *17α*, 17α-hydroxylase (also CYP17A1); *17βR*, 17β-reductase; *18*, aldosterone synthase; *21*, 21-hydroxylase.

anastrozole, which can be affected by drug interactions via CYP3A4, are decreased 27% by coadministration of *tamoxifen* or *cimetidine*. *Anastrozole* is a substrate for the multidrug resistance transporter (Pgp/ABCB1) but not for the breast cancer resistance transporter (BCRP/ABCG2) (Miyajima et al., 2013).

Therapeutic Uses. AIs are approved for up-front adjuvant hormonal therapy for 5 to 10 years or following prior *tamoxifen* in postmenopausal women with early-stage breast cancer and as treatment of advanced and metastatic breast cancer. In early-stage breast cancer, *anastrozole* is significantly more effective than *tamoxifen* in delaying time to tumor recurrence and decreasing the odds of a new primary contralateral tumor. AIs are combined with ovarian suppression as adjuvant treatment of breast cancer in premenopausal women less than 35 years of age and in those requiring chemotherapy. This combination is associated with significant reduction in risk of recurrence compared to *tamoxifen* alone (Pagani et al., 2020). In postmenopausal women with ER+ or PR+ metastatic breast cancer, AIs are typically used as first-line therapy, as studies have shown them to be superior to *tamoxifen* and *megestrol acetate*. In premenopausal women with advanced disease, AIs are also used but must be given with a GnRH agonist to suppress ovarian function.

Adverse Effects and Toxicity. Most adverse effects are related to estrogen depletion. In postmenopausal women, compared to *tamoxifen*, *anastrozole* causes a lower incidence of hot flashes, vaginal bleeding, vaginal discharge, endometrial cancer, ischemic cerebrovascular events, venous thromboembolic events, and deep venous thrombosis/pulmonary embolism. *Anastrozole* is associated with a higher incidence of symptomatic arthralgias, vaginal dryness, and sexual dysfunction than *tamoxifen*. In addition, the estrogen depletion caused by AIs results in loss of bone mineral density. Compared with *tamoxifen*, treatment with *anastrozole* results in significantly lower bone mineral density in the lumbar spine and total hip and has been associated with increased fracture risk. Bisphosphonates (see Chapter 52) prevent AI-induced loss of bone mineral density in postmenopausal women.

Letrozole. The clinical uses and side effect profile of the type 2 AI *letrozole* are similar to those detailed in the previous section for *anastrozole* (see Table 73–1). *Letrozole*-specific information is as follows.

ADME. *Letrozole* is rapidly absorbed after oral administration, with a bioavailability of 99.9%. Steady-state plasma concentrations of *letrozole* are reached after 2 to 6 weeks of treatment. Following metabolism by CYP2A6 and CYP3A4, *letrozole* is eliminated as an inactive metabolite mainly via the kidneys and has a $t_{1/2}$ of approximately 41 h.

Therapeutic Uses. *Letrozole* is used to treat breast cancer in the same manner as *anastrozole* (see Table 73–1). In addition, improved progression-free survival is observed when ER+/PR+ advanced-stage breast cancer is treated with *letrozole* in combination with a CDK4/6 inhibitor (see Chapter 71) (Finn et al., 2016; Hortobagyi et al., 2016).

Exemestane. *Exemestane* is a more potent, orally administered analogue of the endogenous aromatase substrate androstenedione and lowers estrogen levels more effectively than does its predecessor, *formestane*. *Exemestane* irreversibly inactivates aromatase and is a "suicide substrate" type 1 inhibitor of aromatase. The clinical uses and side effect profile of *exemestane* are similar to those detailed in the section for the AI *anastrozole*.

ADME. Orally administered *exemestane* is rapidly absorbed from the gastrointestinal tract; its absorption is increased by 40% after a high-fat meal. *Exemestane* has a terminal $t_{1/2}$ of approximately 24 h. It is extensively metabolized in the liver, ultimately to inactive metabolites; one metabolite, 17-hydroxyexemestane, has weak androgenic activity that may contribute to antitumor activity. Although metabolites are excreted in the urine, no dosage adjustments are recommended in patients with renal dysfunction.

Therapeutic Uses. The clinical indications for use of *exemestane* in breast cancer treatment are the same as for *anastrozole* (see Table 73–1). In addition, the use of the mechanistic (or mammalian) target of rapamycin (mTOR) inhibitor *everolimus* (see Chapter 71) with *exemestane* has

been approved for treatment of advanced-stage breast cancer that has progressed on nonsteroidal type 2 AIs (Piccart et al., 2014).

Toxicity and Adverse Reactions. The side effects are similar to those described for *anastrozole*. *Exemestane* causes fetal toxicities and abortion in preclinical animal experiments and is considered unsafe for administration to pregnant women; women of childbearing potential should use contraception during AI treatment and for a month thereafter.

Drugs That Target the Progesterone Receptor

Progestational agents are mainly used as secondary agents in hormonal therapy for metastatic hormone-dependent breast cancer and are also used in the management of endometrial carcinoma previously treated by surgery and radiotherapy. Progesterone binds to the PR present in target tissues, such as breast and the endometrium. Activation of the PR by progestins in the endometrium is antiproliferative. Chapter 48 covers the mechanisms and uses of hormones and drugs that interact with the PR.

Medroxyprogesterone acetate is available for oral administration; an alternative oral progestational agent is *megestrol acetate.* These agents provide beneficial effects in one-third of patients with endometrial cancer. The response of breast cancer to *megestrol* is predicted by both the presence of ER and PR and the evidence of response to a prior hormonal treatment. The effect of progestin therapy in breast cancer appears to be dose-dependent, with some patients demonstrating second responses following dose escalation of *megestrol*. Clinical use of progestins in breast cancer has been largely superseded by *tamoxifen* and AIs. Additional uses of progestins are as treatment of metastatic endometrial cancer and to stimulate appetite and restore a sense of well-being in cachectic patients with advanced stages of cancer and AIDS.

Drugs Combined With Agents That Antagonize the Estrogen Receptor

The addition of several agents used in cancer (see Chapter 71) when combined with SERMs, SERDs, or AIs improves the therapeutic outcome of breast cancer patients.

CDK4/6 Inhibitors

There are currently three CDK4/6 inhibitors—*palbociclib*, *ribociclib*, and *abemaciclib*—that have been approved for use in combination with ER antagonists (SERDs and AIs). These combinations result in significantly improved outcomes in ER+ HER2-advanced breast cancer (Finn et al., 2016; Goetz et al., 2017; Hortobagyi et al., 2016; Slamon et al., 2020; Sledge et al., 2020; Turner et al., 2018).

A major effect of CDK4/6 inhibitors is to prevent progression through the G_1/S cell cycle checkpoint by inhibiting the activation of D-type cyclins and thereby reducing the production of phosphorylated retinoblastoma-associated protein and E2F transcriptional activity (see Figure 71–3). In ER+ breast cancer, the G_1-to-S checkpoint is a therapeutic target since cyclin D is frequently overexpressed. The presence of the ER in a breast cancer is currently the best predictive marker of response to CDK4/6 inhibitors. While there are questions about the optimal sequence and choice of ER antagonists with CDK4/6 inhibitors, recent trials demonstrate no difference in outcome in the first-line setting whether an AI or *fulvestrant* is used with a CDK4/6 inhibitor (Slamon et al., 2020; Sledge et al., 2020; Turner et al., 2018).

PI3 Kinase (PI3K) and mTOR Inhibitors

The PI3K inhibitor *alpelisib* and the mTOR inhibitor *everolimus* (see Chapter 71) are approved for use in combination with *fulvestrant* and *exemestane*, respectively, in treating advanced-stage breast cancer (Andre et al., 2019; Piccart et al., 2014). The mTOR, a kinase in the phosphoinositide-3-kinase (PI3K)/AKT signaling pathway, controls cell growth and proliferation (see Figure 71–5). In breast cancer, the PI3K/AKT/mTOR pathway modulates the growth of breast cancer through ER-mediated signaling as well as through the HERs (human epidermal growth factor receptors) and mediates the clinical sensitivity to ER antagonists. Breast cancers that harbor activating mutations in PI3K can be resensitized to endocrine therapies by treatment with inhibitors of PI3K or mTOR.

Drugs That Target the Androgen Receptor

AR Antagonists

AR antagonists competitively inhibit the binding of testosterone and dihydrotestosterone to the AR, thereby reducing AR nuclear translocation, chromatin binding, and AR-mediated gene transcription (see Figure 73–1B). Unlike castration, AR antagonist therapy by itself does not decrease LH production; therefore, testosterone levels are normal or increased. Men treated with AR antagonists maintain some degree of potency and libido and do not have the same spectrum of side effects seen with castration. However, AR antagonist therapy is typically given in combination with ADT (described above). AR antagonists include the newer second-generation drugs *enzalutamide, apalutamide*, and *darolutamide*, and the earlier generation antagonists *bicalutamide, flutamide*, and *nilutamide*. Table 73–2 summarizes dosing and conditions for the use of these agents in treating prostate cancer.

Enzalutamide

Enzalutamide is a second-generation, nonsteroidal competitive AR antagonist that has 5- to 8-fold higher binding affinity for the AR compared to the older antagonist *bicalutamide* (Tran et al., 2009). Similar to other AR antagonists, *enzalutamide* prevents binding of androgens to the AR, thereby decreasing receptor-mediated transcriptional activity. *Enzalutamide* is approved for use in nonmetastatic and metastatic CRPC as well as in metastatic castration-sensitive prostate cancer. *Enzalutamide* prolongs survival in patients with metastatic CRPC when given to chemotherapy-naïve patients or after *docetaxel* therapy (Paschalis and de Bono, 2020). *Enzalutamide* combined with ADT, compared with ADT alone, also prolongs metastasis-free and overall survival in men with nonmetastatic castration-resistant disease and rapidly rising circulating prostate-specific antigen (indicative of more aggressive disease and increased risk of progression) (Hussain et al., 2018; Sternberg et al., 2020).

ADME. *Enzalutamide* has improved efficacy and potency compared to older, first-generation AR antagonists. *Enzalutamide* is given orally once daily. Median time to reach C_{Pmax} is 1 h. Its $t_{1/2}$ is approximately 6 days; steady-state levels are achieved in 28 days. CYP2C8 and CYP3A4 metabolize *enzalutamide*. Therefore, concomitant treatment with *enzalutamide* and strong CYP2C8 inhibitors such as *gemfibrozil* or *pioglitazone* may raise levels of *enzalutamide*. Conversely, strong inducers of CYP2C8 or CYP3A4 (e.g., *rifampin*) may reduce plasma *enzalutamide*. *Enzalutamide* can induce multiple hepatic CYPs, including 3A4, 2C9, and 2C19 and thereby has the potential to affect the metabolism of up to 50% of drugs (reviewed in Del Re et al., 2017). Coadministration with substrates of these CYPs (including *fentanyl*, *nifedipine*, *disopyramide*, *quetiapine*, *quinidine*, and *warfarin*) should be avoided as *enzalutamide* may decrease levels of these drugs (Benoist et al., 2018).

Adverse Effects and Toxicity. The most frequently reported side effects are fatigue, back pain, hot flush, constipation, arthralgia, decreased appetite, diarrhea, and hypertension. Fractures, falls, and cognitive impairment have been reported (Fizazi et al., 2020a; Sternberg et al., 2020). A population-based retrospective study showed that *enzalutamide* use is associated with an increased risk of short-term mortality in elderly prostate cancer patients with preexisting cardiovascular comorbidities compared to similar patients without cardiovascular disease (Lu-Yao et al., 2020). *Enzalutamide* crosses the blood-brain barrier; seizures occur infrequently (0.5%–2.2% of patients).

Apalutamide

Apalutamide is a second-generation, nonsteroidal AR antagonist that is approved for use in nonmetastatic castration-resistant and metastatic castration-sensitive prostate cancer (Smith et al., 2021).

ADME. *Apalutamide* is given orally once daily as four tablets, taken with or without food. Mean absolute oral bioavailability is approximately

100%. Median time to achieve peak plasma concentration (t_{max}) is 2 h and may be increased with a high-fat meal. Mean elimination $t_{1/2}$ is 3 to 4 days. Steady state is achieved by approximately 4 weeks. *Apalutamide* is a strong inducer of CYP3A4 and CYP2C19 and a weak inducer of CYP2C9. Concomitant use of *apalutamide* and drugs that are substrates of these CYPs should be avoided as lower exposure to these CYP substrates may result. Coadministration of *apalutamide* and substrates of UDP-glucuronosyltransferase (UGT), Pgp (e.g., *fexofenadine*), breast cancer resistance protein (BCRP), or organic anion transporting polypeptide 1B1 (OAT1B1) (e.g., *rosuvastatin*) may result in decreased exposure to these medications.

Toxicity. *Apalutamide* has a similar toxicity profile as *enzalutamide* and can cause fatigue, hypertension, rash, nausea, hot flush, weight loss, arthralgia, falls, decreased appetite, diarrhea, fracture, and peripheral edema. Hypothyroidism has been reported. *Apalutamide* can increase the QT interval. Seizures occurred in 0.2% of patients receiving *apalutamide*, possibly related to interaction of the drug with the $GABA_A$ receptor.

Darolutamide

Darolutamide is a second-generation, nonsteroidal AR antagonist that is used in the treatment of nonmetastatic CRPC. Addition of *darolutamide* to ongoing ADT significantly increases time to metastasis-free survival in men with nonmetastatic prostate cancer (Fizazi et al., 2020b).

ADME. *Darolutamide* is given orally twice daily and should be given with food. The $t_{1/2}$ is approximately 20 h. C_{Pmax} is reached approximately 4 h after administration of a single 600-mg oral dose. In contrast to *enzalutamide* and *apalutamide*, *darolutamide* does not induce CYPs. Use of *darolutamide* concurrently with inhibitors of Pgp and strong-to-moderate CYP3A inducers may decrease *darolutamide* exposure. Concurrent use of *darolutamide* with inhibitors of Pgp and strong CYP3A4 inhibitors may increase *darolutamide* exposure. *Darolutamide* inhibits BCRP transporter, and therefore, concurrent use of *darolutamide* and BCRP substrates (e.g., *prazosin*, *glyburide*, *cimetidine*, *sulfasalazine*, *rosuvastatin*, and nucleoside and nucleotide analogues such as *zidovudine* and *lamivudine*) may increase BCRP substrate toxicity. *Darolutamide* inhibits OATP1B1 and OATP1B3 transporters. Concomitant use of *darolutamide* and OAT1B1 or OATP1B3 substrates (e.g., statins, taxanes, *cisplatin*) may increase concentrations of drugs that are substrates of OAT1B1 and OATP1B3. Post-hoc analysis of a phase III trial of *darolutamide* reported a low incidence of clinically relevant drug-drug interactions (Shore et al., 2019).

Adverse Effects and Toxicity. *Darolutamide* is structurally distinct from *enzalutamide* and *apalutamide* and is not associated with an increase in adverse events such as falls, seizures, cognitive disorders, mental impairment disorders, and hypertension compared to ADT alone (Fizazi et al., 2020b). The most common adverse reactions are fatigue, pain in extremities, and rash. Significant adverse reactions in patients taking *darolutamide* include ischemic heart disease, heart failure, urinary retention, pneumonia, and hematuria. Of the newer generation AR antagonists, seizures appear to be least frequent with *darolutamide*, which does not penetrate the blood-brain barrier.

Older AR Antagonists

These nonsteroidal agents are still in use but are being supplanted by the more potent second-generation AR antagonists.

Bicalutamide. The agent *bicalutamide* is given once daily in conjunction with a GnRH agonist, usually to combat tumor flare. *Bicalutamide* has a $t_{1/2}$ of 5 to 6 days; it undergoes glucuronidation to inactive metabolites, and the parent compound and metabolites are eliminated in bile and urine. The $t_{1/2}$ of *bicalutamide* is increased in severe hepatic insufficiency and is unchanged in renal insufficiency. *Bicalutamide* is well tolerated at higher doses and has reduced toxicity and improved tolerability and pharmacokinetic profiles relative to *flutamide* and *nilutamide*. Daily *bicalutamide* is significantly inferior compared to surgical or medical castration and should not be used as monotherapy of prostate cancer. *Bicalutamide* inhibits CYP3A4, and therefore, coadministration of *bicalutamide* and CYP3A4 substrates should be monitored. Use of *bicalutamide* in patients receiving anticoagulants requires monitoring since *bicalutamide* may displace coumarin anticoagulants from binding sites.

Nilutamide. *Nilutamide* is given orally once daily. It has an elimination $t_{1/2}$ of 45 h and is metabolized to five products that are all excreted in the urine. Common side effects include mild nausea, alcohol intolerance (5%–20%), and diminished ocular adaptation to darkness (25%–40%); rarely, interstitial pneumonitis occurs.

Flutamide. *Flutamide* is given orally three times per day. It has a $t_{1/2}$ of 5 h; its major metabolite, hydroxyflutamide, is biologically active. Common side effects include diarrhea, breast tenderness, and nipple tenderness. Less commonly, nausea, vomiting, and hepatotoxicity occur. Rare reports of fatal hepatotoxicity have been observed. It is used infrequently as it has the least favorable toxicity profile of the antiandrogens.

Resistance to AR Antagonists, Including to Second-Generation Drugs

Resistance to these agents develops frequently through mechanisms that primarily but not exclusively enhance AR activity, as noted above (reviewed in Schmidt et al., 2021). In an effort to overcome resistance to current antiandrogen therapies, studies are ongoing with a variety of new agents, including those that promote AR degradation such as PROTACs (Salami et al., 2018) and small-molecule receptor degraders (Mohler et al., 2021).

Drugs That Inhibit Androgen Synthesis

As discussed above, even in conditions of castration levels of circulating testosterone, persistent AR activity in CRPC cells may support tumor growth. Synthesis of extragonadal androgens occurs in the adrenal glands or CRPC tumors themselves (see Figure 73–4). Androstenedione, produced by the adrenal glands, is converted to testosterone and dihydrotestosterone in peripheral tissues and prostate tumors. Intratumoral *de novo* androgen synthesis also may provide sufficient androgen for AR-driven cell proliferation. Thus, inhibitors of androgen synthesis combined with ADT have proven useful in reducing AR signaling in CRPC.

Abiraterone Acetate

Abiraterone, with *prednisone*, is used for the treatment of metastatic prostate cancer in patients who are chemotherapy naïve or in those who have received previous *docetaxel*. In both settings, *abiraterone* prolongs survival (reviewed in Paschalis and de Bono, 2020). The combination of *abiraterone* and *prednisone*, in conjunction with ADT, also prolongs overall survival in men with metastatic castration-sensitive prostate cancer (Virgo et al., 2021; Wang L et al., 2021). Resistance to *abiraterone*, similar to that to *enzalutamide*, can occur through a variety of mechanisms mainly involving the AR (discussed above). AR antagonists and abiraterone in additional prostate cancer clinical settings is being evaluated in ongoing clinical trials.

Mechanism of Action. *Abiraterone* is an irreversible inhibitor of 17α-hydroxylase and 17,20-lyase (CYP17A1) activity in testicular, adrenal, and prostatic cancer tissue (see Figure 73–3). Inhibition of CYP17A1 reduces the conversion of pregnenolone and progesterone to their 17α-OH derivatives and their subsequent enzymatic conversion to dehydroepiandrosterone and androstenedione. Thus, circulating levels of testosterone drop to almost undetectable levels after *abiraterone* administration. *Abiraterone* also has some activity as an AR antagonist and inhibitor of other steroid synthetic enzymes and CYPs. Overall, *abiraterone* has greater potency and selectivity than *ketoconazole*, which is described below.

ADME. Abiraterone is the active metabolite of *abiraterone acetate*. With continuous administration, *abiraterone* increases adrenocorticotropic hormone levels, resulting in mineralocorticoid excess. Oral *abiraterone* acetate is administered with *prednisone* to counteract adrenal suppression. *Abiraterone* should be taken on an empty stomach due to the effect of food to increase both C_{max} and AUC of the drug. Alternatively, the National Comprehensive Cancer Network lists low-dose *abiraterone* (250 mg/day) with a low-fat breakfast as an alternative treatment option to 1000 mg/day of *abiraterone* taken on an empty stomach (Szmulewitz et al., 2018)

Abiraterone is metabolized by CYP3A4; thus, concomitant use of drugs that are strong CYP3A4 inducers (e.g., *carbamazepine*, *rifampicin*, St. John's wort, several reverse transcriptase inhibitors) should be avoided. *Abiraterone acetate* inhibits the highly polymorphic CYP2D6 and CYP2C8; thus, concomitant use of drugs that are substrates of these CYPs (e.g., 2D6 substrates: *risperidone*, *metoprolol*, *nebivolol*, *imipramine*-based antidepressants, *fluoxetine*, *tamoxifen*, *codeine*, *hydrocodone*, *oxycodone*, *tramadol*; 2C8 substrates: *amiodarone*, *carbamazepine*, *cerivastatin*, *diclofenac*, *ibuprofen*, *paclitaxel*, *rosiglitazone*) requires careful monitoring for toxicity and possibly dose adjustment.

Toxicity. The most common side effects include fatigue, arthralgia, hypertension, nausea, edema, hypokalemia, hot flush, diarrhea, vomiting, upper respiratory tract infection, cough, and headache. As was observed with enzalutamide in the same study, use of *abiraterone* is associated with an increased risk of short-term mortality in elderly men with preexisting cardiovascular comorbidities (Lu-Yao et al., 2020). Laboratory abnormalities that are most common include anemia, elevated alkaline phosphatase, hypertriglyceridemia, lymphopenia, hypercholesterolemia, hyperglycemia, and hypokalemia.

Ketoconazole

Ketoconazole is an antifungal agent that also inhibits both testicular and adrenal steroidogenesis by blocking CYP17 (17α-hydroxylase), CYP11A, and other P450 cytochrome enzymes. *Ketoconazole* can be administered off-label in conjunction with ADT to reduce adrenal androgen synthesis in CRPC. Oral *ketoconazole* is coadministered with *hydrocortisone* to compensate for inhibition of adrenal steroidogenesis. *Ketoconazole* has limited use in practice due to its toxicity and inferiority compared to *abiraterone acetate*. Chapter 61 presents the basic and clinical pharmacology of the azoles.

Prostate Cancer Chemoprevention (5α-Reductase Inhibitors)

5α-Reductase catalyzes the conversion of testosterone to dihydrotestosterone, the most potent endogenous androgen. Two inhibitors of 5α-reductase, *finasteride* and *dutasteride*, have been studied for prostate cancer chemoprevention. The results have been controversial: two clinical trials indicated decreased risk of low-grade prostate cancer but increased incidence of high-grade disease in the drug treatment arms. However, longer-term follow-up and other studies (Goodman et al., 2019) revealed no elevated incidence for high-grade prostate cancer or mortality from prostate cancer in the *finasteride*-treated subjects; the initial results in higher grade prostate cancer may have resulted from detection bias or may have been related to the study design (reviewed in Chau and Figg, 2018).

Bipolar Androgen Deprivation Therapy

Since CRPC typically becomes resistant to AR-directed therapies such as *enzalutamide* and *abiraterone*, investigators have examined whether bipolar androgen deprivation therapy (BAT) might resensitize tumors to AR-directed drugs. With prolonged deprivation of stimulation by androgens, prostate cancer cells may adapt via upregulation of the AR, thereby becoming resistant to treatment. BAT is rapid cycling between extremes of high and low (castrate) circulating testosterone in order to disrupt the process by which prostate cancer cells adapt to a low androgen milieu. A randomized study compared BAT versus *enzalutamide* in men with CRPC whose disease had progressed following *abiraterone*. Although there was no significant difference between the treatment arms in terms of progression-free survival, BAT did improve the extent and duration of response in the crossover arm in which BAT was the intervening therapy between *abiraterone* and *enzalutamide*. This study underscores the need to define optimal sequencing of BAT and different AR-targeted therapies (Denmeade et al., 2021).

Drugs That Target the Glucocorticoid Receptor

The pharmacology, major therapeutic uses, and toxic effects of the glucocorticoids are discussed in Chapters 50, 74, and 75. Only the applications of these drugs in the treatment of neoplastic disease are considered here.

A number of glucocorticoids are available and at equivalent dosages exert similar effects (see Chapter 50). *Prednisone*, for example, usually is administered orally in doses up to 100 mg for the first few days and gradually reduced to the lowest possible effective dose. Side effects of these agents include glucose intolerance, immunosuppression, osteoporosis, and psychosis (see Chapter 50). *Dexamethasone* is one of the preferred agents for remission induction in multiple myeloma, typically in combination with *bortezomib*, *anthracyclines*, or *lenalidomide*. Glucocorticoids, particularly *dexamethasone*, are used in conjunction with radiotherapy to reduce edema related to tumors in critical areas such as the superior mediastinum, brain, and spinal cord. Frequent lower doses (4–6 mg every 6 h) can have dramatic effects in restoring neurological function in patients with cerebral metastases, but these effects are temporary. However, acute changes in *dexamethasone* dosage can lead to a rapid recrudescence of symptoms. *Dexamethasone* should not be discontinued abruptly in patients receiving radiotherapy or chemotherapy for brain metastases. *Dexamethasone* is also frequently used as part of an antiemetic regimen in patients receiving chemotherapy.

Acknowledgment: *Anton Wellstein contributed to this chapter in the previous edition of this book. We have retained some of his text in the current edition.*

Drug Facts for Your Personal Formulary: *Hormones and Related Agents in Cancer Therapy*

Drug	Therapeutic Use	Clinical Pharmacology and Tips
Glucocorticoid Receptor Agonists		
Dexamethasone Prednisone Others	• Treatment of malignant hematologic disorders (ALL, CLL, MM, HL, NHL) • Symptom palliation in various cancer types (antiemetic; ↓ edema due to spinal cord compression, brain metastases)	• Major toxicities: Cushing syndrome, glucose intolerance, immunosuppression, osteoporosis, psychosis, insomnia • Acute reduction in dosing can lead to recurrence of symptoms • Drug interactions: concomitant use of CYP3A4 inhibitors or inducers. Concomitant use of erythropoietin-stimulating agents or estrogens increases risk of thromboembolism
Selective Estrogen Receptor Modulators: Antiestrogens in breast cancer therapy		
Tamoxifen	• Adjuvant therapy for pre- and postmenopausal women with HR+ breast cancer • Treatment of advanced or metastatic HR+ breast cancer in pre- and postmenopausal women • Breast cancer prevention in pre- and postmenopausal women • Rarely used to treat metastatic endometrial cancer	• SERM with partial agonist and antagonist action. Antagonist of ER in breast. Long $t_{1/2}$. Steady-state levels reached in 3–4 weeks • Some major toxicities due to ER agonism (endometrial carcinoma, thromboembolic events) or ER antagonism (vasomotor symptoms, menstrual irregularities) • Other adverse effects: cataracts • Drug interactions: strong CYP3A/4 inducers (rifampin) reduce AUC and C_{max}. Inhibitors of CYP2D6 increase metabolism (e.g., paroxetine) but it has not been established that this changes clinical efficacy. Increases anticoagulant effect of warfarin
Toremifene	• Postmenopausal metastatic HR+ breast cancer	• Pharmacology, clinical efficacy, and adverse effects similar to those of tamoxifen • Rare: prolongs QT interval, increased risk of torsades de pointes • Drug interactions: with agents that prolong QT. CYP3A/4 inducers/inhibitors change AUC and C_{max}. Use with caution with CYP2C9 substrates (warfarin, phenytoin)
Selective Estrogen Receptor Downregulators: Antiestrogens in breast cancer therapy		
Fulvestrant	• Advanced or metastatic HR+ breast cancer (+/– CDK4/6 inhibitors or +/– PI3K inhibitor alpelisib) in postmenopausal women who have disease progression after antiestrogen therapy	• Binds to ER, blocks estrogen action, and causes degradation of the ER • No estrogen agonist effects • IM loading then monthly dosing; steady state achieved in first month • Side effects: injection site reaction, nausea, weakness, bone and back pain, fatigue, vasomotor symptoms, headache • Drug interactions: no major interactions noted
Aromatase Inhibitors: Antiestrogen in breast cancer therapy		
Anastrozole, letrozole (nonsteroidal, competitive inhibitors) Exemestane (steroidal, irreversible inhibitor)	• Adjuvant treatment of postmenopausal (natural or in conjunction with OFS) women with HR+ breast cancer • Treatment of postmenopausal women (natural or in conjunction with OFS) with HR+ advanced and metastatic breast cancer (+/– CDK4/6 inhibitors) (exemestane +/– mTOR inhibitor) • Breast cancer prevention in postmenopausal women	• AIs significantly lower serum estrogens • Contraindicated in premenopausal women with ovarian function • Major side effects: vasomotor symptoms, arthralgia, loss of bone mineral density, osteoporosis, fractures, vaginal dryness, dyspareunia • Drug interactions: concomitant use with tamoxifen reduces plasma levels 27% (anastrazole). Estrogen coadministration reduces AI efficacy. CYP3A/4 inducers decrease exemestane exposure
Progesterone Receptor Agonists		
Megestrol acetate	• Treatment of endometrial cancer and rarely of breast and prostate cancer • Appetite stimulant in patients with AIDS or cancer-associated cachexia	• Adverse effects: weight gain, nausea, vomiting, edema, breakthrough bleeding, shortness of breath, thrombophlebitis, pulmonary embolism • Drug interactions: coadministration reduces exposure to HIV protease inhibitor indinavir
Medroxyprogesterone acetate	• Management of advanced-stage endometrial carcinoma • Therapy of metastatic hormone-dependent breast cancer	• Adverse effects: hot flashes, weight gain, depression, amenorrhea • With long-term use, bone loss is possible • Drug interactions: CYP3A/4 inducers reduce plasma levels
Gonadotropin-Releasing Hormone Analogues: Chemical castration in cancer therapy		
Prostate Cancer *GnRH agonists*: Leuprolide Goserelin Buserelin Histrelin Triptorelin	• Androgen deprivation therapy: ↓ pituitary release of LH and FSH, ↓ testicular testosterone production • Treatment of all forms of advanced prostate cancer • In combination with radiation therapy or surgery for management of moderate-/high-risk locally confined prostate cancer • In combination with androgen receptor antagonists or androgen synthesis inhibitors for combined androgen blockade	• Can cause initial testosterone surge and tumor flare. Administered with androgen receptor antagonists to reduce initial side effects from testosterone surge • Side effects related to low testosterone: vasomotor symptoms, loss of libido, osteoporosis, fatigue, impotence, gynecomastia, loss of muscle mass • Small but significant increase in risk of diabetes or development of cardiovascular disease • Drug interactions: no major interactions noted

Drug Facts for Your Personal Formulary: *Hormones and Related Agents in Cancer Therapy (continued)*

Drug	Therapeutic Use	Clinical Pharmacology and Tips
GnRH antagonists: Degarelix (cetrorelix) Relugolix	• Treatment of all forms of advanced prostate cancer • Cetrorelix less used, requires more frequent injection, higher cost	• No initial testosterone surge; rapid suppression of serum testosterone and prostate-specific antigen levels • Side effect profile in men similar to GnRH agonists above • More effectively decreases FSH compared to GnRH agonists • Relugolix is given orally • Fewer major cardiovascular events (relugolix compared to leuprolide) • Prolonged QT interval with relugolix (rare) • Drug interactions: relugolix interacts with Pgp and CYP3A4. Coadministration with strong CYP3A/4 inducers (rifampin) reduces AUC and C_{max}; Pgp inhibitors (erythromycin) increase AUC and C_{max}
Breast Cancer *GnRH agonist*: Goserelin Leuprolide	• Suppression of ovarian estrogen and progesterone production in pre- and perimenopausal women • With antiestrogens as adjuvant therapy or for metastatic disease	• Adverse effects due to hypoestrogenism: vasomotor symptoms, ↓ libido, osteoporosis, tumor flare, fatigue, vaginal dryness, dyspareunia • Drug interactions: No major interactions noted
Nonsteroidal Androgen Receptor Antagonists: Antiandrogens in prostate cancer therapy		
Enzalutamide	• In conjunction with ADT (or bilateral orchiectomy) for treatment of metastatic castration-sensitive prostate cancer, nonmetastatic and metastatic CRPC	• Adverse effects related to AR antagonism: sexual dysfunction, gynecomastia, breast pain, fatigue, diarrhea, headache, musculoskeletal pain, vasomotor symptoms, hot flashes, hypertension • Increase risk of cardiovascular disease (particularly in elderly men with preexisting conditions) • Rare: seizures (likely due to central "off-target" effects) • Drug interactions: enzalutamide may affect metabolism of many drugs through induction of multiple CYPs. Strong CYP2C8 inhibitors (e.g., the fibrate gemfibrozil) will raise enzalutamide levels. Strong inducers of CYP2C8 or CYP3A4 (e.g., rifampin) will reduce plasma enzalutamide
Apalutamide	• In conjunction with ADT (or bilateral orchiectomy), for treatment of metastatic castration-sensitive prostate cancer and nonmetastatic CRPC (not yet FDA approved for metastatic CRPC)	• Adverse effects related to AR antagonism • Drug interactions: apalutamide strongly induces CYP3A4 and CYP2C19. Concomitant use with substrates of these CYPs should be avoided. Coadministration of apalutamide with substrates of UGT, Pgp, BCRP, and OATP1B1 may decrease exposure to these medications
Darolutamide	• In conjunction with ADT (or bilateral orchiectomy), for treatment of nonmetastatic CRPC	• Adverse effects related to AR antagonism • Drug interactions: concomitant use with Pgp inhibitors and CYP3A inducers may decrease darolutamide exposure. Use with Pgp inhibitors and CYP3A inhibitors may increase darolutamide exposure. BCRP transporter inhibited by darolutamide; coadministration with BCRP substrates may increase substrate toxicity. Darolutamide inhibits OATP1B1 and OATP1B3; concomitant use with OATP substrates may increase concentration of these medications
Bicalutamide	• Older generation AR antagonist used with GnRH analogues to treat all forms of advanced prostate cancer. Current use mainly to ameliorate GnRH agonist flare	• Adverse effects related to AR antagonism • Favorable toxicity and pharmacokinetic profile relative to flutamide or nilutamide • Drug interactions: coadministration of bicalutamide and CYP3A4 substrates should be monitored. Bicalutamide may displace coumarin anticoagulants from binding sites, and therefore, use in patients receiving anticoagulants requires caution
Inhibitors of Steroidogenesis: Antiandrogens in prostate cancer therapy		
Abiraterone	• Treatment of metastatic prostate cancer (castration-sensitive and castration-resistant) • Used in combination with prednisone or dexamethasone (to compensate for adrenal insufficiency induced by abiraterone)	• Irreversibly inhibits CYP17A1, ↓ testosterone and other androgens • Fluid retention, hypertension, hypokalemia, hepatotoxicity, fatigue, joint swelling, vasomotor symptoms, diarrhea, arrhythmia • Increase risk of cardiovascular disease (particularly in elderly men with preexisting conditions) • Should be taken on empty stomach; or reduced dose with low-fat breakfast • Drug interactions: strong CYP3A4 inducers decrease abiraterone plasma levels. Plasma levels of substrates of CYP2D6 and CYP2C8 may increase by coadministration with abiraterone

OFS, ovarian function suppression.

Bibliography

Abderrahman B, Jordan VC. A novel strategy to improve women's health: selective estrogen receptor modulators. In: Zhang X, ed. *Estrogen Receptor and Breast Cancer. Cancer Drug Discovery and Development.* Humana Press, Cham, **2019**, 189–213.

Andre F, et al. Alpelisib for *PIK3CA*-mutated, hormone receptor–positive advanced breast cancer. *N Engl J Med*, **2019**, *380*:1929–1940.

Benoist GE, et al. Drug-drug interaction potential in men treated with enzalutamide: mind the gap. *Br J Clin Pharmacol,* **2018**: *84*:122–129.

Burstein HJ. Systemic therapy for estrogen receptor–positive, HER2-negative breast cancer. *N Engl J Med,* **2020**, *383*:2557–2570.

Burstein HJ, et al. Adjuvant endocrine therapy for women with hormone receptor-positive breast cancer: American Society of Clinical Oncology clinical practice guideline update on ovarian suppression. *J Clin Oncol*, **2016**, *34*:1689–1701.

Chau CH, Figg WD. Revisiting 5α-reductase inhibitors and the risk of prostate cancer. *Nat Rev Urol*, **2018**, *7*:400–401.

Davies C, et al. Long-term effects of continuing adjuvant tamoxifen to 10 years versus stopping at 5 years after diagnosis of oestrogen receptor-positive breast cancer: ATLAS, a randomised trial. *Lancet*, **2013**, *381*:805–816.

Dearnaley DP, et al. The oral gonadotropin-releasing hormone receptor antagonist relugolix as neoadjuvant/adjuvant androgen deprivation therapy to external beam radiotherapy in patients with localised intermediate-risk prostate cancer: a randomised, open-label, parallel-group phase 2 trial. *Eur Urol,* **2020**, *78*:184–192.

Del Re M, et al. The role of drug-drug interactions in prostate cancer treatment: focus on abiraterone acetate/prednisone and enzalutamide. *Cancer Treat Rev*, **2017**, *55*:71–82.

Denmeade SR, et al. TRANSFORMER: a randomized phase II study comparing bipolar androgen therapy versus enzalutamide in asymptomatic men with castration-resistant metastatic prostate cancer. *J Clin Oncol,* **2021**, *39*:1371–1382.

Desai K, et al. Hormonal therapy for prostate cancer. *Endocr Rev*, **2021**, *42*:354–373.

Dowsett M, et al. Meta-analysis of breast cancer outcomes in adjuvant trials of aromatase inhibitors versus tamoxifen. *J Clin Oncol*, **2010**, *28*:509–518.

Early Breast Cancer Trialists' Collaborative Group. Aromatase inhibitors versus tamoxifen in early breast cancer: patient-level meta-analysis of the randomised trials. *Lancet*, **2015**, *386*:1341–1352.

Einstein DJ, et al. Targeting the androgen receptor and overcoming resistance in prostate cancer. *Curr Opin Oncol*, **2019**, *3*:175–182.

Ekholm M, et al. Two years of adjuvant tamoxifen provides a survival benefit compared with no systemic treatment in premenopausal patients with primary breast cancer: long-term follow-up (>25 years) of the phase III SBII:2pre Trial. *J Clin Oncol*, **2016**, *34*:2232–2238.

Fan P, Jordan VC. New insights into acquired endocrine resistance of breast cancer. *Cancer Drug Resist*, **2019**, *2*:198–209.

Finn RS, et al. Palbociclib and letrozole in advanced breast cancer. *N Engl J Med*, **2016**, *375*:1925–1936.

Fizazi K, et al. Quality of life in patients with metastatic prostate cancer following treatment with cabazitaxel versus abiraterone or enzalutamide (CARD): an analysis of a randomised, multicentre, open-label, phase 4 study. *Lancet Oncol*, **2020a**, *11*:1513–1525.

Fizazi K, et al. Nonmetastatic, castration-resistant prostate cancer and survival with darolutamide. *N Engl J Med*, **2020b**, *383*:1040–1049.

Formenti AM, et al. Effects of medical treatment of prostate cancer on bone health. *Trends Endocrinol Metab,* **2021**, *32*:135–158.

Francis PA, et al. Adjuvant ovarian suppression in premenopausal breast cancer. *N Engl J Med*, **2015**, *372*:436–446.

Goetz MP, et al. MONARCH 3: abemaciclib as initial therapy for advanced breast cancer. *J Clin Oncol*, **2017**, *35*:3638–3646.

Goodman PJ, et al. Long-term effects of finasteride on prostate cancer mortality. *N Engl J Med,* **2019**, *380*:393–394.

Goss PE, et al. Exemestane versus anastrozole in postmenopausal women with early breast cancer: NCIC CTG MA.27—a randomized controlled phase III trial. *J Clin Oncol*, **2013**, *31*:1398–1404.

Goss PE, et al. Extending aromatase-inhibitor adjuvant therapy to 10 years. *N Engl J Med*, **2016**, *375*:209–219.

Green KA, Carroll JS. Oestrogen-receptor-mediated transcription and the influence of co-factors and chromatin state. *Nat Rev Cancer*, **2007**, *9*:713–722.

Gupta D, et al. Cardiovascular and metabolic effects of androgen-deprivation therapy for prostate cancer. *J Oncol Pract*, **2018**, *10*:580–587.

Haque R, et al. Tamoxifen and antidepressant drug interaction in a cohort of 16,887 breast cancer survivors. *J Natl Cancer Inst*, **2016**, *108*:djv337.

Hertz DL, Rae JM. One step at a time: CYP2D6 guided tamoxifen treatment awaits convincing evidence of clinical validity. *Pharmacogenomics*, **2016**, *17*:823–826.

Hortobagyi GN, et al. Ribociclib as first-line therapy for HR-positive, advanced breast cancer. *N Engl J Med*, **2016**, *375*:1738–1748.

Hussain M, et al. Enzalutamide in men with nonmetastatic, castration-resistant prostate cancer. *N Engl J Med,* **2018**, *378*:2465–2474.

Jeselsohn R, et al. The evolving role of the estrogen receptor mutations in endocrine therapy-resistant breast cancer. *Curr Oncol Rep*, **2017**, *5*:35.

Jordan VC. The new biology of estrogen-induced apoptosis applied to treat and prevent breast cancer. *Endocr Relat Cancer*, **2015**, *22*:R1–R31.

Legare S, Basik M. The link between ERα corepressors and histone deacetylases in tamoxifen resistance in breast cancer. *Mol Endocrinol*, **2016**, *9*:965–976.

Luo J, et al. Role of androgen receptor variants in prostate cancer. *Eur Urol*, **2018**, *5*:715–723.

Lu-Yao G, et al. Mortality and hospitalization risk following oral androgen signaling inhibitors among men with advanced prostate cancer by pre-existing cardiovascular comorbidities. *Eur Urol*, **2020**, *2*:158–166.

McDonnell DP, et al. Next-generation endocrine therapies for breast cancer. *J Clin Oncol*, **2021**, *39*:1383–1388.

Mitsiades N, Kaochar S. Androgen receptor signaling inhibitors: post-chemotherapy, pre-chemotherapy and now in castration-sensitive prostate cancer. *Endocr Relat Cancer*, **2021**, *28*:T19–T38.

Miyajima M, et al. Investigation of the effect of active efflux at the blood-brain barrier on the distribution of nonsteroidal aromatase inhibitors in the central nervous system. *J Pharm Sci*, **2013**, *102*:3309–3319.

Mohler ML, et al. An overview of next-generation androgen receptor-targeted therapeutics in development for the treatment of prostate cancer. *Int J Mol Sci*, **2021**, *22*:2124.

Nakazawa M, et al. Mechanisms of therapeutic resistance in prostate cancer. *Curr Oncol Rep*, **2017**, *2*:13.

Nettles KW, Greene GL. Ligand control of coregulator recruitment to nuclear receptors. *Annu Rev Physiol*, **2005**, *67*:309–333.

Pagani O, et al. Absolute improvements in freedom from distant recurrence to tailor adjuvant endocrine therapies for premenopausal women: results from TEXT and SOFT. *J Clin Oncol*, **2020**, *38*:1293–1303.

Paschalis A, de Bono JS. Prostate cancer 2020: "the times they are a 'changing." *Cancer Cell*, **2020**, *38*(1):25–27.

Piccart M, et al. Everolimus plus exemestane for hormone-receptor-positive, human epidermal growth factor receptor-2-negative advanced breast cancer: overall survival results from BOLERO-2. *Ann Oncol*, **2014**, *12*:2357–2362.

Robertson JFR, et al. Fulvestrant 500 mg versus anastrozole 1 mg for hormone receptor-positive advanced breast cancer (FALCON): an international, randomized, double-blind, phase 3 trial. *Lancet*, **2016**, *388*:2997–3005.

Salami J, et al. Androgen receptor degradation by the proteolysis-targeting chimera ARCC-4 outperforms enzalutamide in cellular models of prostate cancer drug resistance. *Commun Biol,* **2018**, *1*:100.

Schmidt KT, et al. Resistance to second-generation androgen receptor antagonists in prostate cancer. *Nat Rev Urol*, **2021**, *18*:209–226.

Shore N, et al. Evaluation of clinically relevant drug-drug interactions and population pharmacokinetics of darolutamide in patients with nonmetastatic castration-resistant prostate cancer: results of pre-specified and post hoc analyses of the phase III ARAMIS trial. *Target Oncol*, **2019**, *14*:527–539.

Shore ND, et al. Oral relugolix for androgen-deprivation therapy in advanced prostate cancer. *N Engl J Med*, **2020**, *382*:2187–2196.

Slamon DJ, et al. Overall survival with ribociclib plus fulvestrant in advanced breast cancer. *N Engl J Med*, **2020**, *382*:514–524.

Sledge GW, et al. The effect of abemaciclib plus fulvestrant on overall survival in hormone receptor–positive, ERBB2-negative breast cancer that progressed on endocrine therapy—MONARCH 2. *JAMA Oncol*, **2020**, *6*:116–124.

Smith MR, et al. Apalutamide and overall survival in prostate cancer. *Eur Urol,* **2021**, *79*:150–158.

Sternberg CN, et al. Enzalutamide and survival in nonmetastatic, castration-resistant prostate cancer. *N Engl J Med*, **2020**, *382*:2197–2206.

Szmulewitz RZ, et al. Prospective international randomized phase II study of low-dose abiraterone with food versus standard dose abiraterone in castration-resistant prostate cancer. *J Clin Oncol,* **2018**, *36*:1389–1395.

Tamura K, et al. *CYP2D6* genotype-guided tamoxifen dosing in hormone receptor-positive metastatic breast cancer (TARGET-1): a randomized, open-label, phase II study. *J Clin Oncol,* **2020**, *38*:558–566.

Tran C, et al. Development of a second-generation antiandrogen for treatment of advanced prostate cancer. *Science*, **2009**, *324*:787–790.

Turner NC, et al. Overall survival with palbociclib and fulvestrant in advanced breast cancer. *N Engl J Med*, **2018**, *379*:1926–1936.

Virgo KS, et al. Initial Management of noncastrate advanced, recurrent, or metastatic prostate cancer: ASCO guideline update. *J Clin Oncol*, **2021**, *39*:1274–1305.

Visvanathan K, et al. Use of pharmacologic interventions for breast cancer risk reduction. *J Clin Oncol*, **2013**, *31*:2942–2962.

Wang L, et al. Comparison of systemic treatments for metastatic castration-sensitive prostate cancer: a systematic review and network meta-analysis. *JAMA Oncol,* **2021**, *7*:412–420.

Wang Z, et al. Selective degradation of the estrogen receptor in the treatment of cancers. *J Steroid Biochem Mol* Biol, **2021**, *209*:10548.